

SECTION 00 31 32
Geotechnical Data Report
For
U.S. Naval Station Mayport
Maintenance Dredging
Duval County, Florida

Prepared by
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May 2025

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GEOTECHNICAL DATA REPORT
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SECTION 00 31 32.10 GEOTECHNICAL DATA REPORT

1 SCOPE

The information provided in this section encompasses the geotechnical field investigations available for this project. The investigations consist of vibracores; the associated logs and laboratory data are presented in paragraphs 4.5 and 4.6, respectively. A character of materials paragraph is included to provide a comprehensive description of the materials utilizing both recent and historical knowledge of the project area. Also included in this section are definitions of terms and boring log notes, which provide additional explanation of the vibracore logs and drilling techniques.

Items discussed in the character of materials paragraph may not appear explicitly on the boring logs. Based on historic knowledge of the project area, the character of materials paragraph includes items that supplement the data documented by the vibracore logs. When reviewing the boring logs, use all the data on the logs, including the materials description, legend, and blow counts. When evaluating the subsurface conditions, use all the data, including the character of materials paragraph and vibracore logs.

During the solicitation period, any questions that pertain to the information provided in this section should be addressed to the contract specialist identified in Block 9 of SF1442. After contract award, questions should be addressed to the appointed ACO/COR, which will coordinate with Geosystems Branch.

2 CHARACTER OF MATERIALS

2.1 THE LOCAL GEOLOGY

The local geology of the project area is characterized by Pleistocene and Holocene age, marine terrace, and beach ridge sediments, which were deposited on an irregular and undulating surface. The thickness of these deposits can be up to 10 feet in the St. Johns River valley. The deposits are thickest below the ridges and in places where they overlie depressions. The deposits consist primarily of tan to yellow, medium to fine grained quartz sands, which are locally stained rusty brown or red due to iron oxide. Locally thin gray sandy clay beds, which contain occasional mollusk shells, are present, particularly near the coast.

Beneath these sediments, the Pliocene age Cypresshead Formation is encountered which typically consists of tan or light gray quartz sand, shell, and clayey sand/sandy clay. The Miocene age Charlton Member of the Coosawhatchie Formation underlies the Pliocene age sands in the project area. Contained within the Hawthorn Group, the Charlton Member includes limestone, dolostone, clay, and phosphates. The thickness of the Miocene age Charlton Member and the Pliocene age deposits ranges from as little as 25 feet to more than 130 feet in the project area.

The Miocene age Hawthorn Group consists mainly of dark gray and olive green sandy to silty clay, clayey sand, clay, sandy limestone, and dolostone. The lithology of the unit is distinguished by the presence of significant percentages of black phosphate sand, granules, and pebbles. The Hawthorn Group ranges in thickness from about 250 feet to as much as 500 feet in the project area.

2.2 MATERIALS ENCOUNTERED

Naval Station Mayport and the surrounding vicinity have been dredged to the maximum authorized depth with allowable over-depth several times; the most recent dredging event occurred in 2025.

Therefore, most of the shoaled materials are newly deposited in the channel through normal coastal processes occurring in and around inlets. As such, all the vibracore logs collected prior to the last dredging event are representative of the shoaled material, except for the vibracore logs from 2008 (VB-MPH08- series). Material above -50 ft MLLW on the 2008 boring logs do not represent shoaling; these borings were collected prior to deepening of the channel in 2011 to -50 ft MLLW (material removed to -53 ft MLLW), and show the material still present beneath the over depth and in the side slopes which can slough into the channel. Thus, those boring logs give important information for the dredging.

New vibracores were collected after the deepening in 2013, 2014, 2016, and 2022 to characterize the recently shoaled material. The shoaling material above the design depth shown on the boring logs was removed during the last maintenance dredging event, but similar material is expected to have shoaled in since.

The materials to be encountered within the dredging template consist of interbedded layers of clay, silt, silty sand, sand with silt, poorly graded sand, and shell and oyster shell ranging in size from sand to gravel, including both whole and broken shell fragments.

Construction debris, trash, coquina rock, hard and stiff clays have been encountered during previous dredge events; this type of material can be encountered in the over depth and below. In situ rock, if encountered, is not required to be dredged.

This is an active navigation channel. Trash and debris, such as ropes, anchors, aluminum cans, plastic bottles, etc. from navigation activities are present in the materials to be dredged.

3 DEFINITIONS

Definitions not explicitly indicated in the sections below are typical industry standard definitions from their respective ASTMs.

3.1 DEFINITIONS OF BASIC TERMS

Carbonate – Soil component that reacts with HCl of an indeterminate origin (shell, rock, etc.).

Fill – Material that has been placed by man, described with all soil characteristics.

Layer – Rock or soil with a thickness of 6 inches or less.

Lens – A geologic deposit of variable thickness, which disappears laterally in all directions and cannot be correlated to adjacent borings.

Rock – A naturally occurring substance composed of one or more minerals bound together. This geologic term includes a range of engineering properties: strength, hardness, permeability, weathering, and discontinuity. These properties are noted or can be inferred from the boring logs as blow counts, penetration rate, RQD, hardness, etc.

Shell – Material composed of predominantly (>75%) coarse-grained sand to gravel-sized whole or broken shell.

3.2 CURRENT LOGGING TERMS

Definition of terms used on boring logs dated since 2000:

Banded - 0.02' (6 mm) to 0.1' (3.0 cm).

Boulder-Sized – Particles greater than 12 inches in diameter.

Cavity - Voids greater than the diameter of the core.

Coarse Grained - Grain diameter greater than 0.079" (2 mm) for sedimentary rocks or 0.197" (5 mm) for igneous or metamorphic rocks.

Coarse Grained Sand-Sized – Less than 10 percent of fine and medium grained sand sizes are present.

Coarse Gravel-Sized – Particles greater than 3/4 of an inch but less than 3 inches in diameter.

Cobble-Sized – Particles greater than 3 inches but less than 12 inches in diameter.

Decomposed – Applicable to saprolitic rock; rock is essentially reduced to a soil with a relic rock texture; can be molded or crumbled by hand.

Dipping (Dip) - 20 to 45 degrees.

Discontinue – Particles were present within the unit above but are no longer present.

Fine Grained - Grain diameter between 0.004" (0.1 mm) and 0.016" (0.4 mm) for sedimentary rocks or 0.039" (1 mm) for igneous or metamorphic rocks.

Fine Grained Sand-Sized - Less than 10 percent of medium and coarse grained sand sizes are present.

Fine Gravel-Sized – Particles greater than No. 4 sieve but less than 3/4 of an inch in diameter.

Flat (Dip) - 0 to 20 degrees.

Fossiliferous – Greater than 40 percent fossils.

Hard (Hardness) - Difficult to scratch with a knife (cannot be pitted with a geology hammer, but can be chipped with moderate blows of the hammer).

Highly Fractured - Spacing 0.3' (9.1 cm) to 1' (30.5 cm).

Highly Weathered - Entire section is discolored; alteration is greater than 50%; some areas of slightly weathered rock are present; some minerals are leached away; retains only a fraction of its original strength (wet strength usually lower than dry strength).

Intact – Spacing greater than 6' (1.8 m).

Intensely Fractured - Spacing less than 0.3' (9.1 cm).

Massive – Over 3' (0.9 m) thick.

Medium-Bedded - 0.3' (9.1 cm) to 1' (30.5 cm) thick.

Medium Grained - Grain diameters between 0.016" (0.4 mm) to 0.079" (2 mm) for sedimentary rocks or 0.039" (1 mm) to 0.197" (5 mm) for igneous or metamorphic rocks.

Medium Grained Sand-Sized - Less than 10 percent of fine and coarse grained sand sizes are present.

Moderately Fractured - Spacing 1' (30.5 cm) to 3' (0.9 m).

Moderately Hard (Hardness) - Can be scratched easily with a knife, but cannot be scratched by a fingernail (can be pitted with moderate blows of a geology hammer).

Moderately Open (Fracture Aperture) - 0.020" (0.5 mm) to 0.098" (2.5 mm).

Moderately Weathered - Discoloration is evident; surface is pitted and altered, with alterations penetrating well below rock surfaces; 10% to 50% of the rock is altered; strength is noticeably less than unweathered rock.

Non-fossiliferous – No observed fossils.

Open (Fracture Aperture) - 0.098" (2.5 mm) to 0.394" (10 mm).

Pitted - Voids 0.039" (1mm) to 0.236" (6mm) in diameter.

Porous - Voids less than 0.039" (1mm) in diameter.

Slightly Fractured - Spacing 3' (0.9 m) to 6' (1.8 m).

Slightly Weathered - Superficial discoloration, alteration and/or discoloration along discontinuities; less than 10% of the rock volume is altered; strength is essentially unaffected.

Soft (Hardness) - Can be scratched with a fingernail (cannot be crumbled between fingers, but can be easily pitted with light blows of a geology hammer).

Solid - Absence of voids.

Sparsely Fossiliferous – Less than 40 percent fossils.

Steeply Dipping (Dip) - 45 to 90 degrees.

Thick-Bedded – 1' (30.5 cm) to 3' (0.9 m) thick.

Thin-Bedded - 0.1' (3.0 cm) to 0.3' (9.1 cm) thick.

Thin Parting - Paper thin to 0.002' (0.6 mm).

Tight (Fracture Aperture) - 0.004" (0.1 mm) to 0.020" (0.5 mm).

Unweathered - No evidence of any mechanical or chemical alteration.

Very Fine Grained - Grain diameter less than 0.004" (0.1 mm); individual grains or crystals are too small to be seen with the naked eye.

Very Hard (Hardness) - Cannot be scratched with a knife (chips can be broken off only with heavy blows of a geology hammer).

Very Soft (Hardness) - Can be deformed by hand (has a rock-like character, but can be easily broken by hand).

Very Tight (Fracture Aperture) - less than 0.004" (0.1 mm).

Very Wide (Fracture Aperture) - 0.394" (10 mm) to 0.984" (25 mm).

Vuggy - Voids 0.236" (6mm) to the diameter of the core.

3.3 PREVIOUS LOGGING TERMS

Definition of terms used on boring logs dated before 2000:

Dense – Equivalent to SPT N-value of 30 to 50.

Incompetent – Rock that disintegrates while coring; weak.

Indurated – Rock or soil hardened or consolidated by pressure or cementation. Very difficult to break by hand.

Poorly-Indurated – See semi-indurated.

Seam – Rock or soil with average thickness of 2 to 3 inches.

Semi-Indurated – Rock or soil with a lesser degree of hardening or consolidation by pressure or cementation. Crumbles with little effort by hand.

3.4 TESTING AND PROCEDURE METHODS

Test/Procedure	Method
Abrasion Resistance of Large-Size Coarse Aggregate, Los Angeles Machine	ASTM C535
Abrasion Resistance of Small-Size Coarse Aggregate, Los Angeles Machine	ASTM C131
Air Content by the Pressure Method	ASTM C231
Air Content by the Volumetric Method	ASTM C173
Air-Entraining Admixtures for Concrete	ASTM C233
Atterberg Limits, wet preparation method, test method a multipoint	ASTM D4318
Bulk Specific Gravity of Bituminous Mixtures	ASTM D2726
Capping Cylindrical Concrete Specimens	ASTM C617
Carbonate Content	ASTM D4373
Compaction: 4" Mold, Modified Proctor	ASTM D1557
Compaction: 4" Mold, Standard Proctor	ASTM D698
Compaction: 6" Mold, Modified Proctor	ASTM D1557
Compaction: 6" Mold, Standard Proctor	ASTM D698
Compressive Strength of Cast in Place Concrete Cylinders	ASTM C873
Compressive Strength of Cylindrical Concrete Specimens (Set of 4)	ASTM C39
Compressive Strength of Lightweight Concrete	ASTM C495
Consolidation, each load of rebound increment with complete time	ASTM D2435
Density and Water Content of Soil by Nuclear Methods (Minimum 4)	ASTM D2922/D3017
Density of Bituminous Concrete in Place by Nuclear Methods (Minimum 4)	ASTM D2950
Density of Soil in Place by the Sand-Cone Method	ASTM D1556
Direct Shear	ASTM D3080
Drilled Cores and Sawed Beams of Concrete	ASTM C42
Extraction of Bituminous Paving Mixtures	ASTM D2172
Freezing and Thawing (up to 55 cycles)	ASTM D5312
Grain size sieve Analysis	ASTM D422
Hydrometer Analysis	ASTM D422

Test/Procedure	Method
Making and Curing Concrete Test Specimens in the Field (Set of 6)	ASTM C31
Making and Curing Concrete Test Specimens in the Lab (Set of 4)	ASTM C192
Marshall Resistance to Plastic Flow (Set of 3)	ASTM D1559
Material Passing No. 200 Sieve	ASTM D1140
Modulus of Elasticity	ASTM D3148
Moisture Content	ASTM D2216
Munsell Color	Munsell Soil Color Charts
Organic Content, Test Method C	ASTM D2974
Organic Impurities in Fine Aggregate	ASTM C40
Permeability: Constant Head Permeability	ASTM D2434
Permeability: Falling Head Permeability	ASTM D5084
Petrographic Examination	ASTM C295
Sampling Aggregate (Minimum 4 hours)	ASTM D75
Sampling Freshly Mixed Concrete	ASTM C172
Sedimentation Rate	No ASTM
Sieve Analysis of Aggregate	ASTM C136
Slump of Hydraulic-Cement Concrete	ASTM C143
Soil Classification (Boring logs)	ASTM D 2488
Specific Gravity and Absorption of Fine Aggregate	ASTM C128
Specific Gravity for Rock	ASTM C127
Specific Gravity for Soil	ASTM D854
SPLITTING TENSILE STRENGTH	ASTM D3967
Splitting Tensile Strength of Concrete Cylinders	ASTM C496
Subsample Preparation ASTM D4220, Group B	ASTM D4220,
Sulfate Soundness	ASTM C88
Total Evaporable Moisture Content	ASTM C566
Triaxial Compression Test for Rock with Strain Gages or LVDTs (per confining pressure)	ASTM D2664
Triaxial compression: Consolidated Undrained (CU) Triaxial Compression Test R Test with Pore Pressure Measurements (per confining pressure)	ASTM D4767
Triaxial compression: Unconsolidated Undrained (UU) Triaxial Compression Test Q Test with Pore Pressure Measurements (per confining pressure)	ASTM D2850
Unbonded Caps for Compressive Strength	ASTM C1231
Unconfined Compressive Strength for Rock Greater than 9,000 psi with Strain	ASTM D2938
Unconfined Compressive Strength for Rock up to 9,000 psi with Strain	ASTM D2938
Unconfined Compressive Strength for Soil	ASTM D2166
Unit Weight and Absorption	ASTM C127
Unit Weight of Aggregate	ASTM C29 -
Visual Percent Shell	No ASTM
Wetting and Drying	ASTM D5313

4 GEOTECHNICAL DATA

4.1 SUMMARY OF FIELD INVESTIGATIONS

The table below summarizes the field investigations data set available for this project.

Table 1. Available Vibracore Data

Boring Designation	State Plane, FL-East, NAD83		Project Feature	Available for Inspection
	X	Y		
VB-MP2022-01	526302	2204221	Turning Basin	Yes
VB-MP2022-02	527484	2204396		Yes
VB-MP2022-03	527211	2203362		Yes
VB-MP2022-04	528615	2203917		Yes
VB-MP2022-05	529446	2204124	Entrance Channel	Yes
VB-MP2022-06	530523	2204752		Yes
VB-MP2022-07	531442	2204574		Yes
VB-MP2022-08	534164	2205763		Yes
VB-MP2022-09	534242	2205060		Yes
VB-MP2022-10	540527	2204358	Bar Cut 3	Yes
VB-MP2022-11	540710	2205142		Yes
VB-MP2022-12	543032	2204862		Yes
VB-MP2022-13	544336	2203899		Yes
VB-MP2022-14	544730	2204657		Yes
VB-MP2022-15	548397	2203444		Yes
VB-MP16-01	525757	2204607	Turning Basin	No
VB-MP16-02	527330	2204375		No
VB-MP16-03	526777	2204007		No
VB-MP16-04	526528	2203090		No
VB-MP16-05	527170	2203434		No
VB-MP16-06	529485	2204078	Entrance Channel	No
VB-MP16-07	531031	2204488		No
VB-MP16-08	542499	2204164	Bar Cut 3	No
VB-MP16-09	543449	2204070		No
VB-MP16-10	544313	2203961		No
VB-MP16-11	545542	2203779		No
VB-MP16-12	544268	2204667		No
VB-MPB2014M-1	531259	2204554	Entrance Channel	No
VB-MPB2014M-2	530528	2204435		No
VB-MPB2014M-3	529679	2204476		No
VB-MPB2014M-4	529109	2204021		No
VB-MPB2014M-5	533297	2205009		No

Boring Designation	State Plane, FL-East, NAD83		Project Feature	Available for Inspection
	X	Y		
VB-MP13-1	525631	2203592	Turning Basin	No
VB-MP13-2	526278	2203678		No
VB-MP13-3	526398	2204432		No
VB-MP13-4	528806	2203965	Entrance Channel	No
VB-MP13-5	535014	2205087	Bar Cut 3	No
VB-MP13-6	534612	2205642		No
VB-MP13-7	542969	2204844		No
VB-MP13-8	545468	2204540		No
VB-MP13-9	548323	2203510		No
VB-MPH08-01	525630	2204943	Turning Basin	No
VB-MPH08-02	525443	2204136		No
VB-MPH08-03	525105	2203200		No
VB-MPH08-06	526513	2204686		No
VB-MPH08-08	526009	2202959		No
VB-MPH08-13	527020	2202950		No
VB-MPH08-14	527840	2203920		No
VB-MPH08-15	528386	2203815	Entrance Channel	No
VB-MPH08-16	528809	2204311		No
VB-MPH08-18	529795	2204614		No
VB-MPH08-19	530353	2204305		No
VB-MPH08-20	530731	2204819		No
VB-MPH08-22	531676	2205049		No
VB-MPH08-24	532983	2204998		No
VB-MPH08-25	533761	2205012		No
VB-MPH08-26	532651	2205401	Bar Cut 3	No
VB-MPH08-38	535770	2205586		No
VB-MPH08-40	537692	2204768		No
VB-MPH08-41	539213	2205270		No
VB-MPH08-45	540244	2205165		No
VB-MPH08-49	541181	2205080		No
VB-MPH08-51	541612	2204235		No
VB-MPH08-53	542205	2204909		No
VB-MPH08-57	543175	2204834		No
VB-MPH08-69	546162	2204453		No
VB-MPH08-71	546576	2203739		No
VB-MPH08-73	547216	2204329		No
VB-MPH08-81	549173	2204040		No
VB-MPH08-83	549557	2203268		No

* Coordinates presented correspond to the project coordinate system and datum

4.2 SUMMARY OF INDEX TESTING DATA

The table below summarizes the index testing available for this project.

Table 2: Index Testing Available for this Project

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MP2022-01	1	MH				15					10YR 3/1
	2	MH									10YR 3/1
	3	SM					0				10YR 4/1
	4	LS					0				10YR 6/1
VB-MP2022-02	1	MH	140	47	93	14					10YR 3/1
	2	MH									10YR 3/1
	3	SP-SM					0				10YR 3/1
	4	SP					0				10YR 3/1
	5	SC					0				10YR 6/1
VB-MP2022-03	1	MH				14					10YR 3/1
	2	MH									10YR 3/1
	3	MH									10YR 3/1
	4	MH					0				10YR 3/1
	5	SP-SM					0				10YR 3/1
VB-MP2022-04	1	MH				12					10YR 3/1
	2	MH									10YR 3/1
	3	SP-SM					0				10YR 4/1
	4	SP-SM					0				10YR 4/1
VB-MP2022-05	1	MH	158	48	110	11					10YR 3/1
	2	MH									10YR 3/1
	3	MH									10YR 3/1
VB-MP2022-06	1	SP				1	0				10YR 5/1
	2	SP-SC					0				10YR 5/1
	3	SP-SC									10YR 5/1
	4	SC					0				10YR 5/1
VB-MP2022-07	1	MH	94	30	64	6					10YR 5/1
	2	SC					0				10YR 5/1
	3	SP					20				10YR 5/1
VB-MP2022-08	1	SHELL				1	40				10YR 5/1
	2	SP					30				10YR 5/1
	3	SM					0				5Y 3/2
	4	SP-SM					0				5Y 3/2

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MP2022-09	1	SP					15				10YR 5/1
	2	SP-SM					5				10YR 5/1
	3	SM					0				5Y 3/2
VB-MP2022-10	1	SP					5				10YR 5/1
	2	SM									10YR 3/1
	3	ML									10YR 3/1
	4	SP-SM					0				10YR 5/1
	5	SP-SM									10YR 5/1
	6	SM									10YR 4/1
VB-MP2022-11	1	SP				1	0				10YR 5/1
	2	SC					0				10YR 4/1
	3	SC					0				10YR 3/1
	4	SP-SM					0				10YR 4/1
VB-MP2022-12	1	SP				1	0				10YR 4/1
	2	SP-SM					0				10YR 4/1
	3	SP-SM					0				10YR 4/1
	4	SM					0				10YR 4/1
	5	SP					0				10YR 4/1
	6	SP					0				10YR 4/1
VB-MP2022-13	1	MH	142	44	98	10					10YR 3/1
	2	MH									10YR 3/1
VB-MP2022-14	1	SP-SM				1	0				10YR 3/1
	2	SM									10YR 3/1
	3	SP-SM					0				10YR 3/1
	4	SP					0				10YR 4/1
VB-MP2022-15	1	ML				5					10YR 3/1
	2	SM									10YR 3/1
VB-MP16-01	1	MH					0.1		237		N 3/1
	2	MH					0		205		N 3/1
	3	MH					0		167		N 3/1
	4	MH					0		194		N 3/1
	5	SM					0		22.4		5Y 4/3
	6	SP-SM					0		22.9		5Y 4/3
VB-MP16-02	1	MH					0		207		N 3/1
	2	MH					0		245		N 3/1
	3	MH					0		188		N 3/1
	4	SP-SM					0		27.1		5Y 4/3

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MP16-03	1	MH					0		211		N 3/1
	2	MH					0		259		N 3/1
	3	MH					0		172		N 3/1
	4	SM					0		21.6		N 3/1
VB-MP16-04	1	MH					0		318		N 3/1
	2	MH					0		325		N 3/1
	3	MH					0		286		N 3/1
	4	SM					0.4		24.5		5Y 4/3
VB-MP16-05	1	MH					0		280		N 3/1
	2	OH					0.1		214		N 3/1
	3	SM					0		21.7		5Y 4/3
	4	SP-SM					0		22.7		5Y 4/3
VB-MP16-06	1	MH					0		195		N 3/1
	2	OH					0		119		N 3/1
	3	MH					0		82.6		N 3/1
	4	MH					0		69.5		N 3/1
VB-MP16-07	1	SP					6.2		49		10YR 3/1
	2	SP					3.4		20.1		10YR 5/1
VB-MP16-08	1	MH					0		229		N 3/1
	2	MH					0		229		N 3/1
	3	MH					0		159		N 3/1
	4	MH					0		146		N 3/1
VB-MP16-09	1	MH					0		179		N 3/1
	2	MH					0		177		N 3/1
	3	SM					0		82.9		N 3/1
	4	MH					0		792		N 3/1
VB-MP16-10	1	MH					0		198		N 3/1
	2	MH					0		241		N 3/1
	3	MH					0		104		N 3/1
	4	MH					0		145		N 3/1
VB-MP16-11	1	MH					0		261		N 3/1
	2	SM					0.1		76.3		5GY 3/1
	3	SM					2.1		28		5GY 4/1
	4	MH					1.4		105		10Y 4/1
VB-MP16-12	1	SM					0		59.8		N 3/1
	2	MH					0		103		N 3/1
	3	MH					0		112		N 3/1

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MP16-12	4	SM					0		36		N 3/1
VB-MPB2014M-02	1	SP-SM					0				5Y 4/2
	2	SM					0				N 2.5/
	3	SM					0				N 4/
VB-MPB2014M-03	1	SP-SM					1				5Y 5/2
	2	SP-SM					1				5Y 5/2
	3	SP-SM					1				N 3/
VB-MPB2014M-05	1	SP					54				5Y 6/2
	2	SP-SM					1				5Y 4/2
	3	SP-SM					0				5Y 4/2
VB-MP13-1	1	MH					0				N 4/
	2	MH					0				N 4/
	3	SP-SM					52				2.5Y 6/1
VB-MP13-2	1	MH					0				N 4/
	2	SP-SM					8				10YR 5/1
	3	MH					0				N 4/
	4	SM					0				2.5Y 4/3
VB-MP13-3	1	MH					0				N 4/
	2	MH					0				N 4/
VB-MP13-4	1	MH					0				N 4/
	2	SP					0				2.5Y 3/1
VB-MP13-5	1	SP					62				2.5Y 5/2
	2	SP-SM					5				N 5/
VB-MP13-6	1	SP					29				10YR 6/1
	2	CH					0				10YR 4/1
VB-MP13-7	1	SM					7				5Y 5/2
	2	SP					15				5Y 7/2
VB-MP13-8	1	SM					3				5Y 4/3
	2	SM					5				5Y 4/1
VB-MP13-9	1	ML					4				N 4/1
	2	SM					1				N 4/1
VB-MPH08-1	1	SM					5				2.5Y 5/2
	2	SM					0				2.5Y 4/2
	3	SP					0				2.5Y 3/2
	4	SM					0				2.5Y 3/2
VB-MPH08-2	1	ML	70	20	50		5			2.4	2.5Y 2.5/1
	2	SM					0				2.5Y 6/2

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MPH08-2	3	SM					0				2.5Y 6/2
	4	SP-SM					0				2.5Y 4/2
	5	SP-SM					0				2.5Y 4/2
VB-MPH08-3	1	ML					50				2.5Y 2.5/1
	2	SP-SM					75				2.5Y 6/1
	3	SP-SM					5				2.5Y 6/1
	4	SM					5				2.5Y 5/2
	5	SP-SM					0				2.5Y 5/2
VB-MPH08-6	1	CH	65	21	44		0			2.8	5Y 3/2
	2	SM					0				5Y 4/2
	3	SP					0				5Y 4/2
VB-MPH08-8	1	CH					0				5Y 3/2
	2	CH					5				5Y 3/2
	3	SP-SM					25				2.5Y 8/4
	4	SP-SM					5				2.5Y 8/4
VB-MPH08-13	1	SM					5				2.5Y 7/3
	2	SP-SM					0				2.5Y 3/2
	3	SP					5				2.5Y 3/2
	4	SP					0				2.5Y 3/2
VB-MPH08-14	1	CH					0				5Y 3/2
	2	SP					80				2.5Y 8/2
	3	SP-SM					0				2.5Y 6/6
	4	SP					0				2.5Y 3/2
	5	SP					0				2.5Y 3/2
VB-MPH08-15	1	CH	59	16	43		0			2.2	5Y 3/2
	2	SM					10				2.5Y 7/3
	3	SP					0				2.5Y 6/3
	4	SP					0				5Y 4/2
	5	SP					0				5Y 4/2
VB-MPH08-16	1	SM					0				5Y 4/3
	2	SM					0				5Y 4/3
	3	SP					90			2.7	5Y 6/2
	4	SP					90				5Y 8/1
	5	SP					0				5Y 8/2
VB-MPH08-18	1	SP-SM					0	2.1			5Y 8/1
	2	SP					0	2.3		2.6	5Y 8/1
	3	SP					0	1.4			5Y 8/1

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MPH08-18	4	SP					0	1.8		2.6	5Y 8/1
	5	SP					0				5Y 8/1
VB-MPH08-19	1	CL					0	5.9		2.6	5Y 4/2
	2	SP					0	1.1			5Y 8/1
	3	SP					0	0.9		2.5	5Y 8/1
	4	SP					0	2.6			5Y 8/1
	5	SP					10				5Y 8/1
VB-MPH08-20	1	CH					0				5Y 3/2
	2	CH	60	31	29		0			2.4	5Y 3/2
VB-MPH08-22	1	SM					0	1.6		2.6	2.5Y 8/1
	2	SM					0	1.1			2.5Y 8/1
	3	SM					0				2.5Y 8/1
VB-MPH08-24	1	SP-SM					5				2.5Y 3/2
	2	SM					0	3		2.5	5Y 3/2
	3	SP-SM					0				5Y 3/2
VB-MPH08-25	1	SP					50	30.7		2.7	5Y 8/2
	2	SP					0	3.1			5Y 8/1
	3	SP-SM					0				5Y 4/2
VB-MPH08-26	1	SP-SM					5	2			5Y 4/2
	2	SP-SM					0				5Y 4/2
	3	SP-SM					5				5Y 4/2
VB-MPH08-38	1	SM					15	8			5Y 7/1
	2	SP-SM					0	3		2.4	5Y 4/3
	3	SP-SM					0	5			5Y 4/3
VB-MPH08-39	1	SP-SM					0				5Y 4/3
	2	SP-SM					0				5Y 4/3
	3	SP-SM					0				5Y 4/3
VB-MPH08-40	1	CH	66	28	38		10			2.3	5Y 3/1
	2	CH					5				5Y 3/1
VB-MPH08-41	1	CH	57	29	28		0			2.2	5Y 3/2
	2	SM					0				5Y 8/1
	3	SP					0				5Y 8/1
VB-MPH08-45	1	SM					5				5Y 5/2
	2	SP-SM					0				5Y 8/1
	3	SP-SM					0				5Y 8/1
	4	SM					0				2.5YR 6/3
VB-MPH08-49	1	SP					0	3.3		2.7	5Y 8/1

Boring Designation	Sample Number	USCS	LL	PL	PI	Org %	Visual Shell %	CO ₃ %	W _n	G _s	Munsell Color
VB-MPH08-49	2	SP					0				5Y 8/1
	3	SM					0				5Y 3/2
	4	SM					0				5Y 3/2
	5	SM					0				10Y 4/1
VB-MPH08-51	1	SM					5				5Y 5/1
	2	SM					15				5Y 5/1
	3	SP-SM					5			2.6	5Y 5/1
	4	ML					0				5Y 7/2
VB-MPH08-53	1	SP					0	2.6			5Y 7/1
	2	SM					5				5Y 5/2
	3	SP					0				5Y 5/2
	4	SM					0	2.1			5Y 5/2
	5	SP-SM					0				5Y 8/1
VB-MPH08-57	1	SP					0	3.1			5Y 7/1
	2	SM					20				5Y 4/2
	3	SM					5				5Y 4/2
	4	SM					5				5Y 4/2
VB-MPH08-69	1	SM					0				5Y 6/2
	2	CL					0				5Y 5/1
	3	SM					15				5Y 5/1
VB-MPH08-71	1	CH					0				5Y 2.5/1
	2	SM					15				5Y 4/1
	3	CL					10				5Y 4/1
VB-MPH08-73	1	CH	34	10	24		0			2.5	5Y 2.5/1
	2	SP-SM					5				5Y 3/2
	3	CH					0				5Y 2.5/1
VB-MPH08-81	1	SM					10				5Y 3/2
	2	SM					5				5Y 3/2
	3	CH					0				5Y 3/1
VB-MPH08-83	1	SM					10				5Y 3/1
	2	CL					5				5Y 3/1
USCS: Unified Soil Classification System; LL: liquid limit; PL: plastic limit; PI: Plastic Index; Org %: Organic Content; CO3%: Carbonate Content; Gs: specific gravity, Wn: Moisture Content											

4.3 BORING LOG NOTES

All vibracore borings were sampled using an Alpine style, pneumatic vertical hammer, vibracore drilling apparatus having a continuous 20-foot by 3 or 3 1/2-inch core barrel with clear plastic core liner until refusal was encountered. Refusal is defined as no additional penetration of the vibracore with one minute of additional effort applied to penetrate through the resistant layer at maximum vibration. The vibracore apparatus is equipped with a graph-type depth of penetration recorder.

4.4 INSPECTION OF RECOVERED MATERIALS

The material recovered from borings marked with a checkmark or yes in Table 1 is available for inspection by prospective offerors at the Corps of Engineers District warehouse. The address and hours of operation for the warehouse are listed below:

Address: 3077 Talleyrand Avenue
Jacksonville, 32206 Florida
Hours: 7:00 am to 2:30 pm

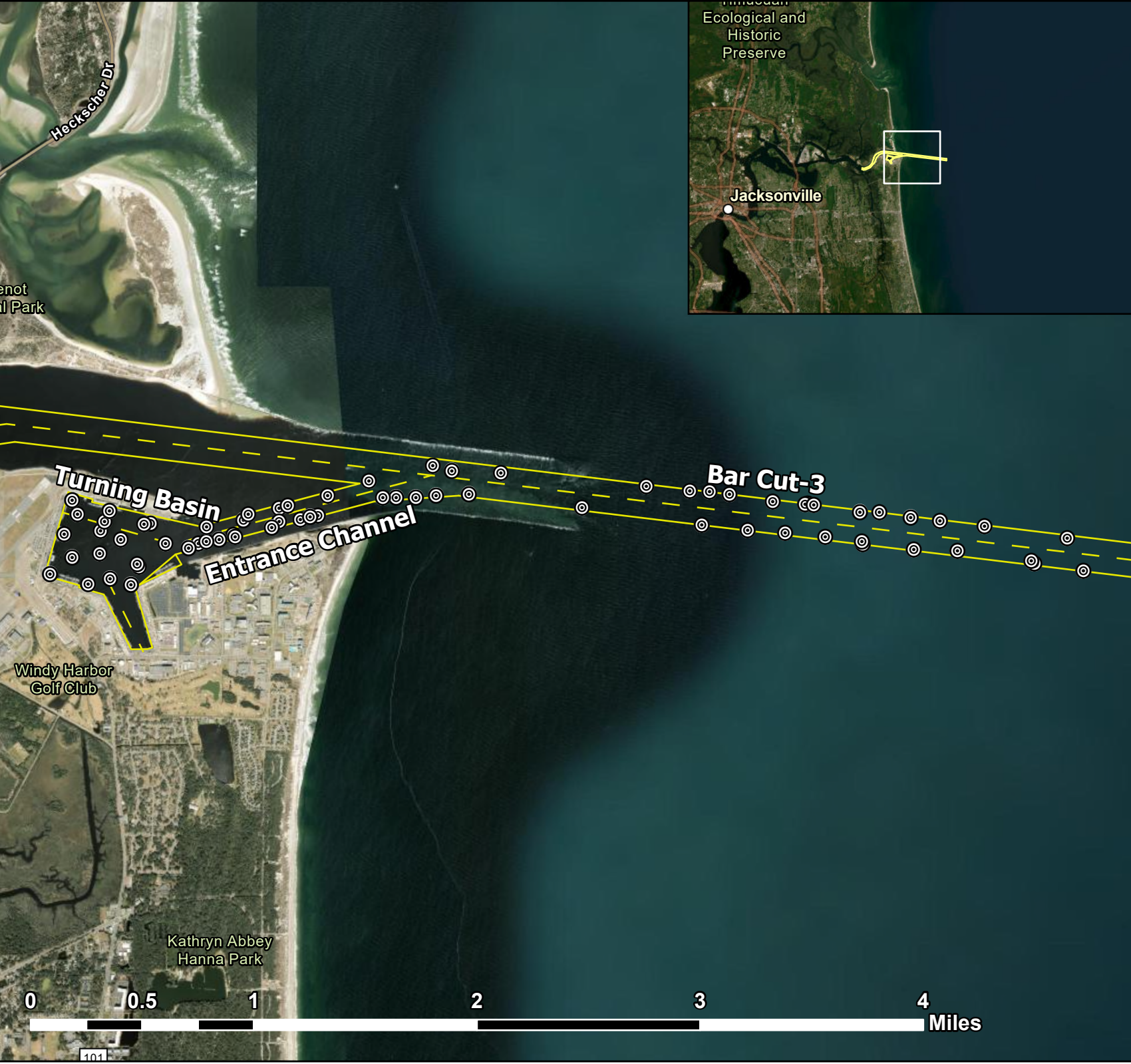
The recovered materials will be available for inspection during normal business hours as noted above, except Federal holidays, during the entire bid period. Prospective offerors shall notify the project geologist at 904-232-1406 or Chief, Geology and Explorations Section at 904-232-1617 at least four (4) working days before the visit.

The following information will be required to schedule the visit:

- (1) Project title;
- (2) Specific borings or entire set which are to be viewed;
- (3) Date, time, and duration of the visit;
- (4) Name of the person(s) and company to view the borings; and
- (5) Point of contact and phone number regarding the visit.

Prospective offerors shall record their material examination visit in a record book maintained at the inspection site.

It is strongly suggested that prospective offerors view the samples before submitting their bid.



**US Army Corps
of Engineers** ®
Jacksonville District

GEOTECHNICAL DRAWINGS

NOT FOR CONSTRUCTION

U.S. Naval Station
Mayport O&M
Dredging 50 Foot
Project

Legend

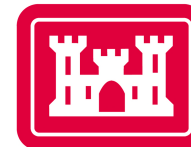
- ⊙ Vibracores
- Channel Outline
- - Channel Midline



Author: Aaron Seitz	Date: 5/14/2025
------------------------	--------------------

Plate 1





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GEOTECHNICAL DRAWINGS

NOT FOR CONSTRUCTION

U.S. Naval Station
Mayport O&M
Dredging 50 Foot
Project

Legend

- ⊙ Vibracores
- Channel Outline
- - - Channel Midline



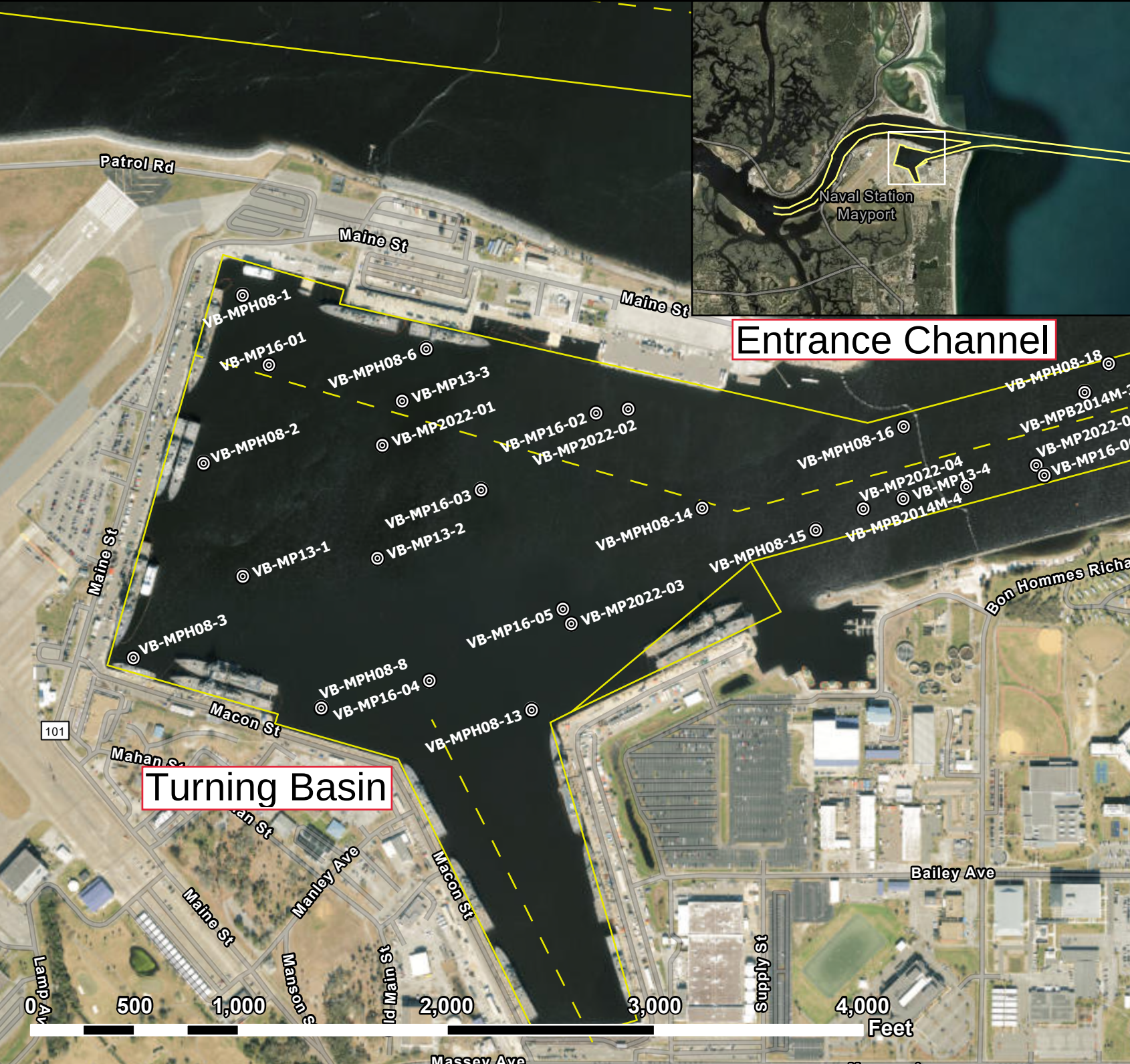
Author: Aaron Seitz	Date: 5/14/2025
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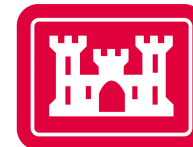
Plate 2



Entrance Channel

Turning Basin





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Jacksonville District

GEOTECHNICAL DRAWINGS

NOT FOR CONSTRUCTION

U.S. Naval Station
Mayport O&M
Dredging 50 Foot
Project

Legend

- ⊙ Vibracores
- Channel Outline
- - - Channel Midline

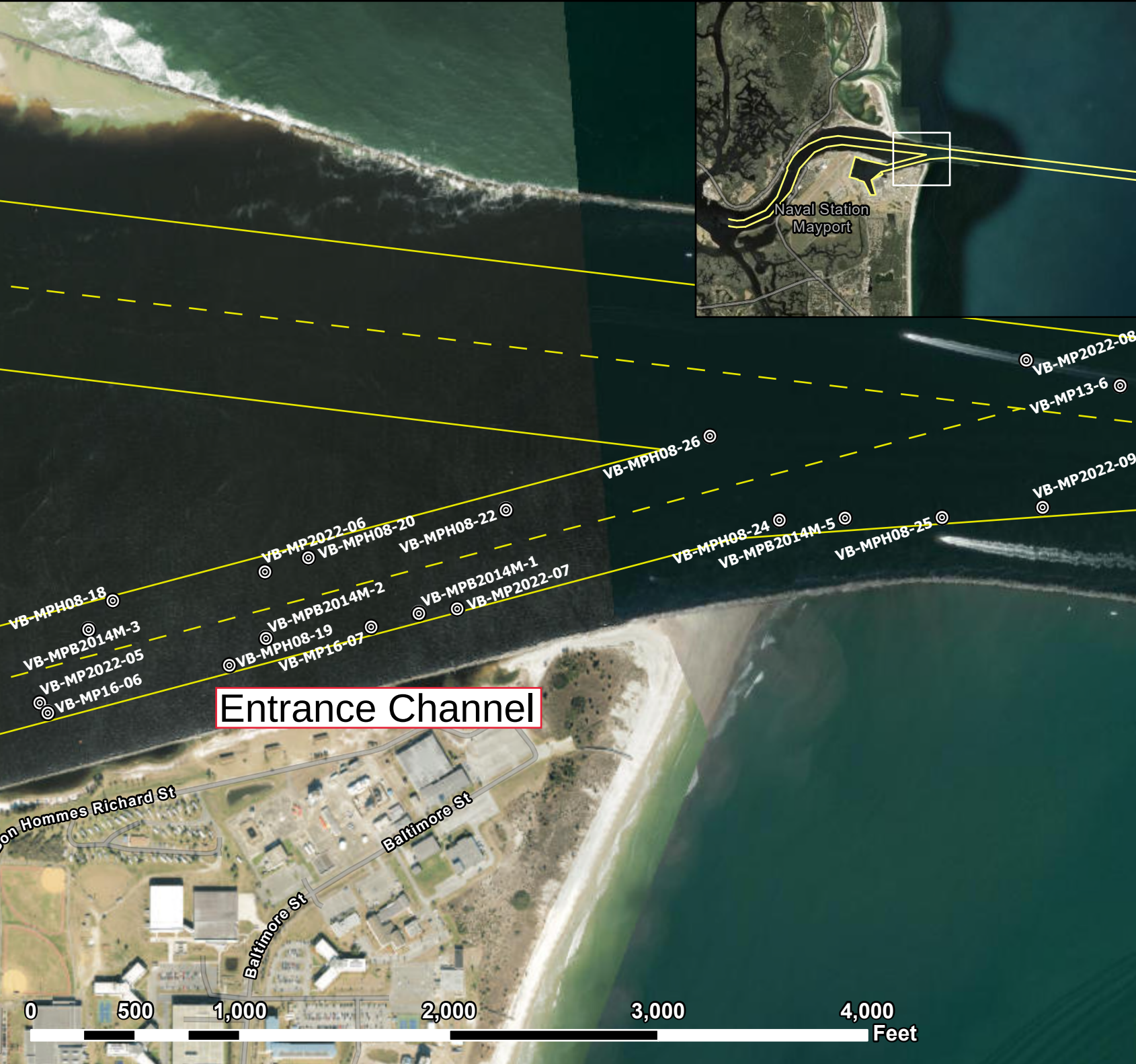


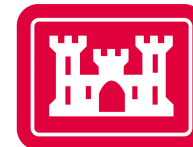
Author: Aaron Seitz	Date: 5/14/2025
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Plate 3



Entrance Channel





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GEOTECHNICAL DRAWINGS

NOT FOR CONSTRUCTION

U.S. Naval Station
Mayport O&M
Dredging 50 Foot
Project

Legend

- ⊙ Vibracores
- Channel Outline
- - - Channel Midline



Author: Aaron Seitz	Date: 5/14/2025
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Plate 4



Bar Cut 3

⊙ VB-MP2022-08

⊙ VB-MP13-6

⊙ VB-MPH08-38

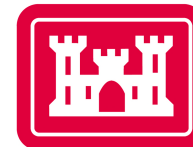
⊙ VB-MP2022-09

⊙ VB-MP13-5

⊙ VB-MPH08-41

⊙ VB-MPH08-40

0 500 1,000 2,000 3,000 4,000 Feet



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GEOTECHNICAL DRAWINGS

NOT FOR CONSTRUCTION

U.S. Naval Station
Mayport O&M
Dredging 50 Foot
Project

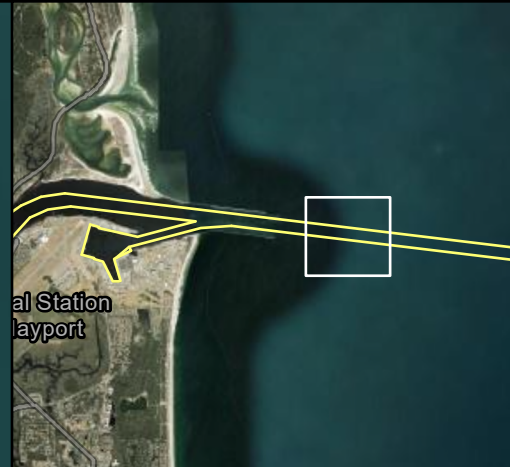
Legend

- ⊙ Vibracores
- Channel Outline
- - - Channel Midline

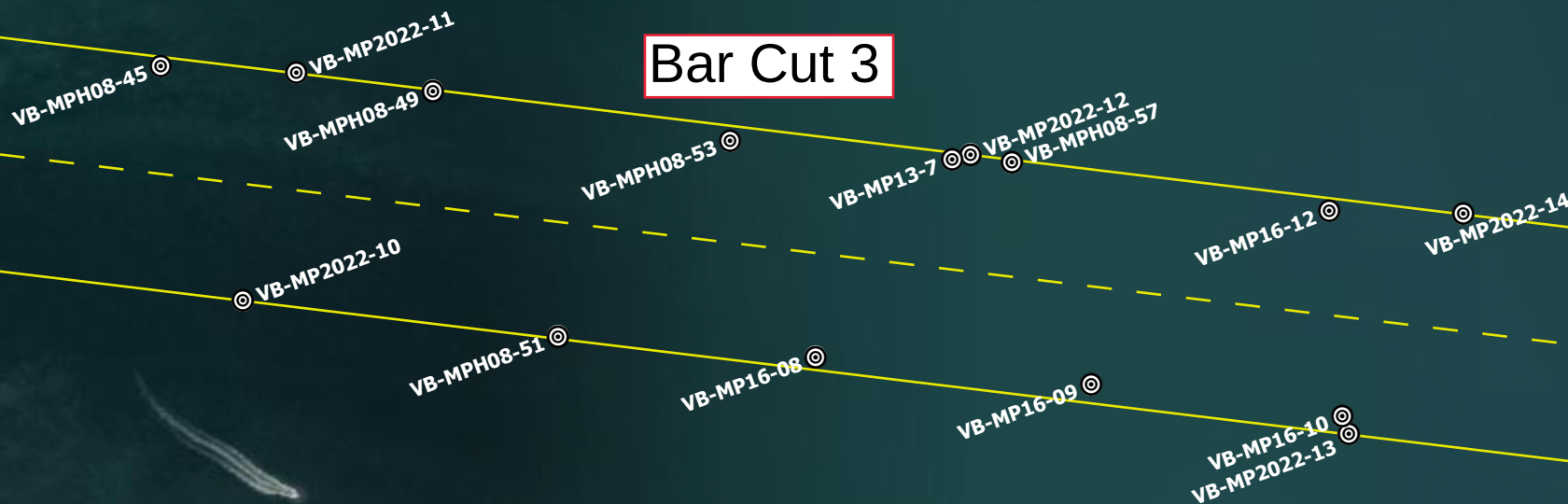


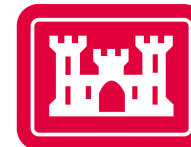
Author: Aaron Seitz	Date: 5/14/2025
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Plate 5



Bar Cut 3





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GEOTECHNICAL DRAWINGS

NOT FOR CONSTRUCTION

U.S. Naval Station
Mayport O&M
Dredging 50 Foot
Project

Legend

- ⊙ Vibracores
- Channel Outline
- - - Channel Midline

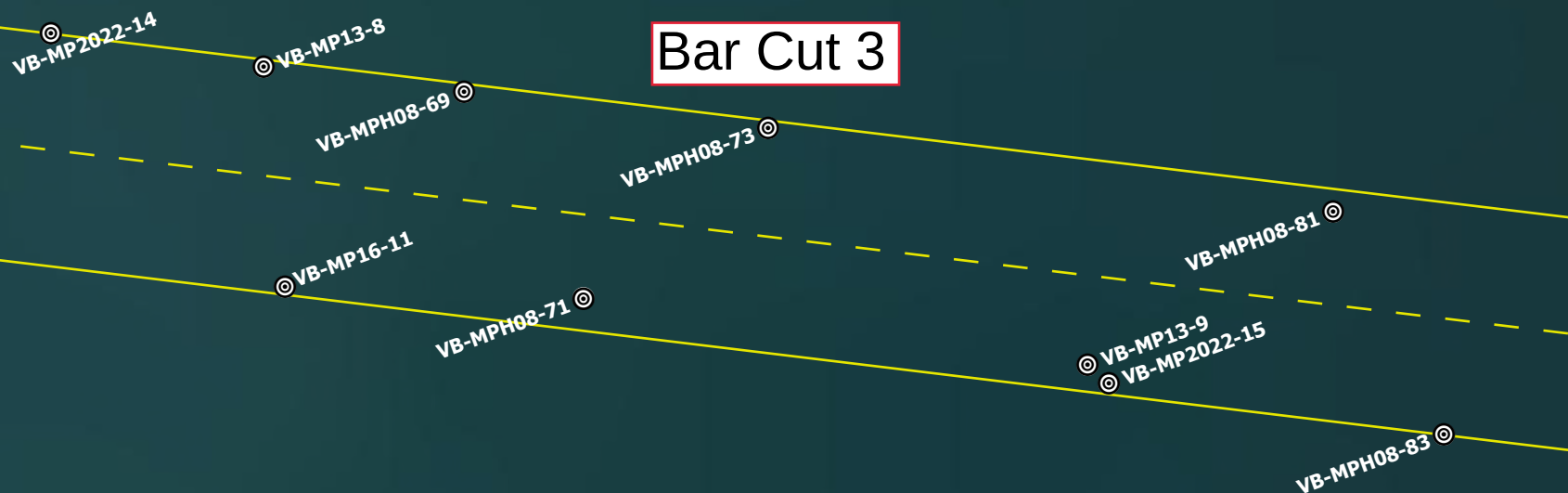


Author: Aaron Seitz	Date: 5/14/2025
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Plate 6



Bar Cut 3



4.5 BORING LOGS

Applicable boring logs are presented on the following pages.

The borings presented in this section are available historic borings within the project area. Most of the materials within the project template presented on the boring logs have been removed by previous dredging events with subsequent shoaling of similar materials. The current elevations of the materials at these locations are shown on the contract drawings.

While the Government's borings are representative of subsurface conditions at their respective locations and vertical reaches, local variations in the characteristics of the subsurface materials of this region are to be expected. Accordingly, prospective offerors shall form their own conclusions from the examination of the recovered materials prior to submission of their offer.

SAJ FORM 1836
JUN 02

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																		
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 526,302 Y = 2,204,221			ELEVATION TOP OF BORING -46.4 Ft.																					
ELEV. (ft)	DEPTH (ft)	REMARKS	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-63.0	16.6	NO RECOVER		NR			Vibracore																	
			BORING TERMINATED IN REFUSAL				-63.0																	
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.9/1.1</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>2.4/2.6</td> <td>MH*</td> </tr> <tr> <td>3</td> <td>3.9/4.1</td> <td>SM*</td> </tr> <tr> <td>4</td> <td>4.9/5.1</td> <td>LIMESTONE*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.9/1.1	MH*	2	2.4/2.6	MH*	3	3.9/4.1	SM*	4	4.9/5.1	LIMESTONE*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	0.9/1.1	MH*																						
2	2.4/2.6	MH*																						
3	3.9/4.1	SM*																						
4	4.9/5.1	LIMESTONE*																						

Boring Designation VB-MP2022-02

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-02				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 5		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22		COMPLETED 06-30-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -48.3 Ft.					
				17. TOTAL RECOVERY FOR BORING 85 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.3	0.0		SILT, inorganic-H, mostly silt, trace fine-grained sand-sized quartz, 10YR 3/1 very dark gray (MH)	100			-48.3		
				100	1		Vibracore		
				100	2		-49.2		
				100	3		Vibracore		
-50.8	2.5		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 3/1 very dark gray (SP-SM)	100			-50.2		
				100	4		Vibracore		
-51.8	3.5		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace silt, 10YR 3/1 very dark gray (SP)	100			-50.8		
				100	5		Vibracore		
				100	6		-51.2		
				100	7		Vibracore		
				100	8		-51.8		
				100	9		Vibracore		
				100	10		-52.2		
				100	11		Vibracore		
				100	12		-55.3		
				100	13		Vibracore		
				100	14		-56.8		
-56.8	8.5		SAND, clayey, mostly fine to medium-grained sand-sized quartz, little clay, 10YR 6/1 gray (SC)	100			-57.2		
				100	15		Vibracore		
				100	16		-60.3		
-60.3	12.0		LIMESTONE, mostly highly weathered soft fine-grained sand-sized limestone, some carbonated clay, 10YR 6/1 gray	100			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																										
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																										
LOCATION COORDINATES X = 527,484 Y = 2,204,396			ELEVATION TOP OF BORING -48.3 Ft.																													
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																							
-65.3	17.0			100			Vibracore																									
-68.3	20.0	NO RECOVERY		NR			Vibracore																									
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>0.9/1.1</td><td>MH</td></tr> <tr><td>2</td><td>1.9/2.1</td><td>MH*</td></tr> <tr><td>3</td><td>2.9/3.1</td><td>SP-SM*</td></tr> <tr><td>4</td><td>3.9/4.1</td><td>SP*</td></tr> <tr><td>5</td><td>8.9/9.1</td><td>SC*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>1.0</td> <td>Atterberg (PI)</td> <td>93</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.9/1.1	MH	2	1.9/2.1	MH*	3	2.9/3.1	SP-SM*	4	3.9/4.1	SP*	5	8.9/9.1	SC*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	1.0	Atterberg (PI)	93			Abbreviations:			
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																														
1	0.9/1.1	MH																														
2	1.9/2.1	MH*																														
3	2.9/3.1	SP-SM*																														
4	3.9/4.1	SP*																														
5	8.9/9.1	SC*																														
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																														
1.0	Atterberg (PI)	93																														

Boring Designation VB-MP2022-03

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-03				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 5		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22		COMPLETED 06-30-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -46.8 Ft.					
				17. TOTAL RECOVERY FOR BORING 80 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-46.8	0.0		SILT, inorganic-H, mostly silt, trace fine-grained sand-sized quartz, 10YR 3/1 very dark gray (MH)	100			-46.8		
					1		Vibracore		
				100			-47.7		
					2		Vibracore		
				100			-48.7		
					3		Vibracore		
				100			-49.7		
					4		Vibracore		
-51.3	4.5		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 3/1 very dark gray (SP-SM)	100			-50.7		
					5		Vibracore		
				100			-51.3		
							-51.7		
							Vibracore		
			From El. -52.8 to -58.3 Ft., mostly fine to medium-grained sand-sized quartz, few silt, 5Y 3/2 dark olive gray				-52.8		
				100			Vibracore		
-58.3	11.5		SAND, clayey, mostly fine to coarse-grained sand-sized quartz, some clay, 10YR 5/1 gray (SC)	100			-58.3		
							Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																				
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																				
LOCATION COORDINATES X = 527,211 Y = 2,203,362			ELEVATION TOP OF BORING -46.8 Ft.																							
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																	
-62.8	16.0			100			Vibracore																			
-66.8	20.0	NO RECOVERY		NR			Vibracore																			
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>0.9/1.1</td><td>MH*</td></tr> <tr><td>2</td><td>1.9/2.1</td><td>MH*</td></tr> <tr><td>3</td><td>2.9/3.1</td><td>MH*</td></tr> <tr><td>4</td><td>3.9/4.1</td><td>MH*</td></tr> <tr><td>5</td><td>4.9/5.1</td><td>SP-SM*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.9/1.1	MH*	2	1.9/2.1	MH*	3	2.9/3.1	MH*	4	3.9/4.1	MH*	5	4.9/5.1	SP-SM*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																								
1	0.9/1.1	MH*																								
2	1.9/2.1	MH*																								
3	2.9/3.1	MH*																								
4	3.9/4.1	MH*																								
5	4.9/5.1	SP-SM*																								

Boring Designation VB-MP2022-04

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS			
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks						
2. BORING DESIGNATION VB-MP2022-04				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW		
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER				
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0				
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER						
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22		COMPLETED 06-30-22		
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -47.9 Ft.						
				17. TOTAL RECOVERY FOR BORING 81 %						
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer						
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE	
-47.9	0.0		SILT, inorganic-H, mostly silt, trace fine-grained sand-sized quartz, 10YR 3/1 very dark gray (MH)	100	1		-47.9			
							-48.3			Vibracore
							-49.3			Vibracore
							-50.1			Vibracore
-50.1	2.2		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 4/1 dark gray (SP-SM)	100	3		-50.3			
							-51.3			Vibracore
-53.3	5.4		LIMESTONE, 10YR 6/1 gray	100	4		-53.3			
			From El. -58.1 to -60.3 Ft., 10YR 6/1 gray	100			-58.1			
			From El. -60.3 to -64.1 Ft., 10YR 6/1 gray	100			-60.3			

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																		
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 528,615 Y = 2,203,917			ELEVATION TOP OF BORING -47.9 Ft.																					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-64.1	16.2	NO RECOVERY		100			Vibracore																	
-67.9	20.0			NR			Vibracore																	
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.4/0.6</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>1.4/1.6</td> <td>MH*</td> </tr> <tr> <td>3</td> <td>2.4/2.6</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>3.4/3.6</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.4/0.6	MH*	2	1.4/1.6	MH*	3	2.4/2.6	SP-SM*	4	3.4/3.6	SP-SM*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	0.4/0.6	MH*																						
2	1.4/1.6	MH*																						
3	2.4/2.6	SP-SM*																						
4	3.4/3.6	SP-SM*																						

Boring Designation VB-MP2022-05

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District		SHEET 1 OF 2 SHEETS			
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-05		LOCATION COORDINATES X = 529,446 Y = 2,204,124		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83 VERTICAL MLLW			
3. DRILLING AGENCY CSI/TSF JV		CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 3 UNDISTURBED (UD) 0			
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED		DEG. FROM VERTICAL		13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22 COMPLETED 06-30-22			
8. TOTAL DEPTH OF BORING 11.8 Ft.				16. ELEVATION TOP OF BORING -49.2 Ft.					
				17. TOTAL RECOVERY FOR BORING 79 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-49.2	0.0		SILT, inorganic-H, mostly silt, some fine-grained sand-sized quartz, 10YR 3/1 very dark gray (MH)	100	1 2 3		-49.2		
				-49.6			Vibracore		
							Vibracore		
				-50.6			Vibracore		
							Vibracore		
				100			-52.1		
				100			Vibracore		
-53.5	4.3		CLAY, lean, mostly clay, few fine-grained sand-sized quartz, 10YR 3/1 very dark gray (CL)	100			-53.5		
					Vibracore				
					Vibracore				
-57.2	8.0		SILT, inorganic-L, mostly silt, few fine-grained sand-sized quartz, few sand to gravel-sized limestone, 10YR 3/1 very dark gray (ML)	100			-57.2		
					Vibracore				
-58.5	9.3			100			-58.5		
					Vibracore				
-61.0	11.8	NR		NR			Vibracore		
			BORING TERMINATED IN REFUSAL				Abbreviations:		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																				
			PROJECT U.S. Naval Station Mayport, 2022		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																			
LOCATION COORDINATES X = 529,446 Y = 2,204,124			ELEVATION TOP OF BORING -49.2 Ft.																								
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
			3. Laboratory Testing Results <table style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 15%;">SAMPLE ID</th> <th style="width: 15%;">SAMPLE DEPTH</th> <th style="width: 70%;">LABORATORY CLASSIFICATION</th> </tr> <tr> <td>1</td> <td>0.4/0.6</td> <td>MH</td> </tr> <tr> <td>2</td> <td>1.4/1.6</td> <td>MH*</td> </tr> <tr> <td>3</td> <td>2.9/3.1</td> <td>MH*</td> </tr> </table> *Lab visual classification based on gradation curve <table style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 15%;">SAMPLE DEPTH</th> <th style="width: 35%;">LABORATORY SOIL TESTING</th> <th style="width: 50%;">RESULT UNIT</th> </tr> <tr> <td>0.5</td> <td>Atterberg (PI)</td> <td>110</td> </tr> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.4/0.6	MH	2	1.4/1.6	MH*	3	2.9/3.1	MH*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	0.5	Atterberg (PI)	110						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	0.4/0.6	MH																									
2	1.4/1.6	MH*																									
3	2.9/3.1	MH*																									
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																									
0.5	Atterberg (PI)	110																									

Boring Designation VB-MP2022-06

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-06				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22		COMPLETED 06-30-22	
8. TOTAL DEPTH OF BORING 16.5 Ft.				16. ELEVATION TOP OF BORING		-50.0 Ft.			
				17. TOTAL RECOVERY FOR BORING		79 %			
				18. SIGNATURE AND TITLE OF INSPECTOR		Rashmi Kola, Geotechnical Engineer			
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-50.0	0.0						-50.0		
			SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace clay, 10YR 5/1 gray (SP)	100	1		-50.4 Vibracore		
-51.3	1.3			100			-51.3 Vibracore		
			SAND, poorly-graded with clay, mostly fine to medium-grained sand-sized quartz, few clay, 10YR 5/1 gray (SP-SC)	100	2		-51.9 Vibracore		
-52.5	2.5			100	3		-52.5 Vibracore		
			SAND, clayey, mostly fine to medium-grained sand-sized quartz, little clay, trace wood debris, 10YR 5/1 gray (SC)	100	4		-52.9 Vibracore		
-54.0	4.0			100			-54.0 Vibracore		
			SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, few sand to gravel-sized shell, trace clay, trace wood debris, 10YR 5/1 gray (SP)	100			-55.2 Vibracore		
			From El. -55.2 to -57.1 Ft., mostly fine to medium-grained sand-sized quartz, little sand to gravel-sized shell, trace clay, 10YR 5/1 gray	100			-57.1 Vibracore		
			From El. -57.1 to -63.0 Ft., mostly fine to medium-grained sand-sized quartz, trace silt, 10YR 6/1 gray	100			Vibracore		
-63.0	13.0						-63.0		
		NO RECOVERY		NR			Vibracore		

DRILLING LOG (Cont. Sheet)				INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																
				PROJECT U.S. Naval Station Mayport, 2022		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW														
LOCATION COORDINATES X = 530,523 Y = 2,204,752				ELEVATION TOP OF BORING -50.0 Ft.																				
ELEV. (ft)	DEPTH (ft)	REMARKS	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-66.5	16.5	NO RECOVERY		NR			Vibracore																	
			BORING TERMINATED IN REFUSAL NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.4/0.6</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>1.3/1.5</td> <td>SP-SC*</td> </tr> <tr> <td>3</td> <td>1.9/2.1</td> <td>SP-SC*</td> </tr> <tr> <td>4</td> <td>2.9/3.1</td> <td>SC*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.4/0.6	SP*	2	1.3/1.5	SP-SC*	3	1.9/2.1	SP-SC*	4	2.9/3.1	SC*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	0.4/0.6	SP*																						
2	1.3/1.5	SP-SC*																						
3	1.9/2.1	SP-SC*																						
4	2.9/3.1	SC*																						

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District		SHEET 1 OF 2 SHEETS	
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks			
2. BORING DESIGNATION VB-MP2022-07		LOCATION COORDINATES X = 531,442 Y = 2,204,574		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83 VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV		CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER	
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 3 UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED		DEG. FROM VERTICAL		13. TOTAL NUMBER CORE BOXES		0	
		BEARING		14. ELEVATION GROUND WATER			
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 06-30-22 COMPLETED 06-30-22	
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING		-49.3 Ft.	
8. TOTAL DEPTH OF BORING 15.7 Ft.				17. TOTAL RECOVERY FOR BORING		96 %	
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer			

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-49.3	0.0		SILT, inorganic-H, mostly silt, few fine-grained sand-sized quartz, 10YR 5/1 gray (MH)	100	1		-49.3 -49.7 Vibracore		
-52.0	2.7			100			Vibracore		
-52.6	3.3		SAND, clayey, mostly fine to medium-grained sand-sized quartz, some clay, 10YR 5/1 gray (SC)	100	2		-52.0 -52.6 Vibracore		
			SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, little sand to gravel-sized shell, trace clay, 10YR 5/1 gray (SP)	100	3		-53.2 Vibracore		
			From El. -54.3 to -59.3 Ft., mostly fine to medium-grained sand-sized quartz, few sand to gravel-sized shell, trace clay, 10YR 5/1 gray	100			-54.3 Vibracore		
-59.3	10.0						-59.3		
-59.8	10.5		SAND, poorly-graded with clay, mostly fine to medium-grained sand-sized quartz, few sand to gravel-sized shell, few clay, 10YR 5/1 gray (SP-SC)	100			-59.8 Vibracore		
			SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace clay, 10YR 6/1 gray (SP)	100			Vibracore		
-64.3	15.0						-64.3		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																				
			PROJECT U.S. Naval Station Mayport, 2022		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																			
LOCATION COORDINATES X = 531,442 Y = 2,204,574			ELEVATION TOP OF BORING -49.3 Ft.																								
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
-65.0	15.7	NR		NR			-65.0 Vibracore																				
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">SAMPLE ID</td> <td style="text-align: center;">SAMPLE DEPTH</td> <td style="text-align: center;">LABORATORY CLASSIFICATION</td> </tr> <tr> <td style="text-align: center;">1</td> <td style="text-align: center;">0.4/0.6</td> <td style="text-align: center;">MH</td> </tr> <tr> <td style="text-align: center;">2</td> <td style="text-align: center;">2.7/2.9</td> <td style="text-align: center;">SC*</td> </tr> <tr> <td style="text-align: center;">3</td> <td style="text-align: center;">3.9/4.1</td> <td style="text-align: center;">SP*</td> </tr> </table> *Lab visual classification based on gradation curve <table style="width:100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">SAMPLE DEPTH</td> <td style="text-align: center;">LABORATORY SOIL TESTING</td> <td style="text-align: center;">RESULT UNIT</td> </tr> <tr> <td style="text-align: center;">0.5</td> <td style="text-align: center;">Atterberg (PI)</td> <td style="text-align: center;">64</td> </tr> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.4/0.6	MH	2	2.7/2.9	SC*	3	3.9/4.1	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	0.5	Atterberg (PI)	64				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	0.4/0.6	MH																									
2	2.7/2.9	SC*																									
3	3.9/4.1	SP*																									
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																									
0.5	Atterberg (PI)	64																									

Boring Designation VB-MP2022-08

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-08				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22		COMPLETED 06-30-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-52.6 Ft.			
				17. TOTAL RECOVERY FOR BORING		100 %			
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-52.6	0.0			100	1		-52.6		
-53.4	0.8		SHELL, mostly sand to gravel-sized shell, some medium to coarse-grained sand-sized quartz, trace silt, 10YR 5/1 gray	100			-52.8 Vibracore		
					2		-53.4 Vibracore		
-54.6	2.0		SAND, poorly-graded, mostly fine to coarse-grained sand-sized quartz, some sand to gravel-sized shell, trace silt, 10YR 5/1 gray (SP)	100			Vibracore		
				100			-54.6		
			SAND, silty, mostly fine to medium-grained sand-sized quartz, few silt, 5Y 3/2 dark olive gray (SM)	100			Vibracore		
					3		-55.5		
				100			Vibracore		
-58.6	6.0						-58.6		
			SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 5Y 3/2 dark olive gray (SP-SM)	100			Vibracore		
					4		-59.5		
				100			Vibracore		
-62.6	10.0						-62.6		
			SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, 10YR 4/1 dark gray (SM)	100			Vibracore		
-67.6	15.0						-67.6		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																		
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 534,164 Y = 2,205,763			ELEVATION TOP OF BORING -52.6 Ft.																					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-72.6	20.0		CLAY, lean, mostly clay, few sand to gravel-sized shells at 15', 10YR 4/1 dark gray (CL)	100			Vibracore																	
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.0/0.2</td> <td>SHELL*</td> </tr> <tr> <td>2</td> <td>0.8/1.0</td> <td>SP*</td> </tr> <tr> <td>3</td> <td>2.9/3.1</td> <td>SM*</td> </tr> <tr> <td>4</td> <td>6.9/7.1</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	SHELL*	2	0.8/1.0	SP*	3	2.9/3.1	SM*	4	6.9/7.1	SP-SM*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	0.0/0.2	SHELL*																						
2	0.8/1.0	SP*																						
3	2.9/3.1	SM*																						
4	6.9/7.1	SP-SM*																						

Boring Designation VB-MP2022-09

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-09				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-30-22		COMPLETED 06-30-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-53.9 Ft.			
				17. TOTAL RECOVERY FOR BORING		95 %			
				18. SIGNATURE AND TITLE OF INSPECTOR		Rashmi Kola, Geotechnical Engineer			
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-53.9	0.0		SAND, poorly-graded, mostly fine to coarse-grained sand-sized quartz, little sand to gravel-sized shell, trace silt, 10YR 5/1 gray (SP)	100	1		-53.9		
				100			-54.1 Vibracore		
-55.2	1.3		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few sand to gravel-sized shell, few silt, 10YR 5/1 gray (SP-SM)	100			-55.2 Vibracore		
				100	2		-55.8 Vibracore		
-57.8	3.9		SAND, silty, mostly fine to medium-grained sand-sized quartz, few silt, 5Y 3/2 dark olive gray (SM)	100			-57.8 Vibracore		
				100	3				
-65.9	12.0		SAND, clayey, mostly fine to medium-grained sand-sized quartz, little clay, few sand to gravel-sized shell, few sand to gravel-sized limestone, 10YR 4/1 dark gray (SC)	100			-65.9 Vibracore		
-67.9	14.0		CLAY, lean, mostly clay, trace fine-grained sand-sized quartz, 10YR 6/1 gray (CL)	100			-67.9 Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS														
			PROJECT U.S. Naval Station Mayport, 2022		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW													
LOCATION COORDINATES X = 534,242 Y = 2,205,060			ELEVATION TOP OF BORING -53.9 Ft.																		
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE												
-72.9	19.0			100			Vibracore														
-73.9	20.0	NR		NR			Vibracore														
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width: 100%; border-collapse: collapse;"> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> <tr> <td>1</td> <td>0.0/0.2</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>1.9/2.1</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>3.9/4.1</td> <td>SM*</td> </tr> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	SP*	2	1.9/2.1	SP-SM*	3	3.9/4.1	SM*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																			
1	0.0/0.2	SP*																			
2	1.9/2.1	SP-SM*																			
3	3.9/4.1	SM*																			

Boring Designation VB-MP2022-10

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MP2022-10				LOCATION COORDINATES X = 540,527 Y = 2,204,358		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY CSI/TSF JV				CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 6		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 0
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		15. DATE BORING		STARTED 07-02-22
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -50.1 Ft.		COMPLETED 07-02-22		
8. TOTAL DEPTH OF BORING 20.0 Ft.				17. TOTAL RECOVERY FOR BORING 100 %		18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer		

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-50.1	0.0		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, few sand to gravel-sized shell, trace silt, 10YR 5/1 gray (SP) SAND, silty, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 3/1 very dark gray (SM) SILT, inorganic-L, mostly silt, some fine to medium-grained sand-sized quartz, 10YR 3/1 very dark gray (ML) SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 5/1 gray (SP-SM) SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, 10YR 4/1 dark gray (SM) SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few intermittent silt layers, 5Y 3/2 dark olive gray (SP-SM)	100	1		-50.1		
-50.8	0.7			100	2		-50.3 Vibracore -50.8 Vibracore		
-51.8	1.7			100	3		-51.8 Vibracore -52.0 Vibracore		
-53.2	3.1			100	4		-53.2 Vibracore -53.5 Vibracore		
				100	5		-55.0 Vibracore		
-57.3	7.2			100	6		-57.3 Vibracore -58.0 Vibracore		
-60.1	10.0			100			-60.1 Vibracore		
-65.1	15.0			100					

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																								
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																								
LOCATION COORDINATES X = 540,527 Y = 2,204,358			ELEVATION TOP OF BORING -50.1 Ft.																											
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																					
-70.1	20.0		SAND, silty, mostly fine to medium-grained sand-sized quartz, little intermittent silt layers, trace clay, 5Y 3/2 dark olive gray (SM)	100			Vibracore																							
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.0/0.2</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>0.7/0.8</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>1.9/2.1</td> <td>ML*</td> </tr> <tr> <td>4</td> <td>3.4/3.6</td> <td>SP-SM*</td> </tr> <tr> <td>5</td> <td>4.9/5.1</td> <td>SP-SM*</td> </tr> <tr> <td>6</td> <td>7.9/8.1</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	SP*	2	0.7/0.8	SM*	3	1.9/2.1	ML*	4	3.4/3.6	SP-SM*	5	4.9/5.1	SP-SM*	6	7.9/8.1	SM*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																												
1	0.0/0.2	SP*																												
2	0.7/0.8	SM*																												
3	1.9/2.1	ML*																												
4	3.4/3.6	SP-SM*																												
5	4.9/5.1	SP-SM*																												
6	7.9/8.1	SM*																												

Boring Designation VB-MP2022-11

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-11				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 07-02-22		COMPLETED 07-02-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-48.4 Ft.			
				17. TOTAL RECOVERY FOR BORING		82 %			
				18. SIGNATURE AND TITLE OF INSPECTOR		Rashmi Kola, Geotechnical Engineer			
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.4	0.0		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace silt, 10YR 5/1 gray (SP)	100	1		-48.4 -48.6 Vibracore		
-49.8	1.4		SAND, clayey, mostly fine to medium-grained sand-sized quartz, some clay, 10YR 4/1 dark gray (SC)	100	2		-49.8 Vibracore		
-52.7	4.3		From El. -51.8 to -52.7 Ft., mostly fine-grained sand-sized quartz, little clay, 10YR 3/1 very dark gray	100	3		-51.8 -52.7 Vibracore		
-54.9	6.5		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 4/1 dark gray (SP-SM)	100	4		-53.3 -54.9 Vibracore		
-57.2	8.8		CLAY, lean, mostly clay, trace fine-grained sand-sized quartz, 10YR 4/1 dark gray (CL)	100			-57.2 Vibracore		
-59.7	11.3		SAND, poorly-graded with clay, mostly fine to medium-grained sand-sized quartz, few intermittent clay layers, 10YR 4/1 dark gray (SP-SC)	100			-59.7 Vibracore		
			SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, 5Y 3/2 dark olive gray (SM)	100			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																		
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 540,710 Y = 2,205,142			ELEVATION TOP OF BORING -48.4 Ft.																					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-64.7	16.3	NO RECOVERY		100			Vibracore																	
-68.4	20.0			NR			Vibracore																	
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.0/0.2</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>1.4/1.6</td> <td>SC*</td> </tr> <tr> <td>3</td> <td>3.4/3.6</td> <td>SC*</td> </tr> <tr> <td>4</td> <td>4.9/5.1</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	SP*	2	1.4/1.6	SC*	3	3.4/3.6	SC*	4	4.9/5.1	SP-SM*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	0.0/0.2	SP*																						
2	1.4/1.6	SC*																						
3	3.4/3.6	SC*																						
4	4.9/5.1	SP-SM*																						

Boring Designation VB-MP2022-12

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District		SHEET 1 OF 2 SHEETS				
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks						
2. BORING DESIGNATION VB-MP2022-12		LOCATION COORDINATES X = 543,032 Y = 2,204,862		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83 VERTICAL MLLW				
3. DRILLING AGENCY CSI/TSF JV		CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER				
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 6 UNDISTURBED (UD) 0				
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED		DEG. FROM VERTICAL		13. TOTAL NUMBER CORE BOXES		0				
		BEARING		14. ELEVATION GROUND WATER						
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 07-02-22 COMPLETED 07-02-22				
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING		-46.8 Ft.				
8. TOTAL DEPTH OF BORING 20.0 Ft.				17. TOTAL RECOVERY FOR BORING		80 %				
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer						
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE	
-46.8	0.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, 10YR 4/1 dark gray (SP)	100	1		-46.8			
-47.7	0.9			100			-47.0			Vibracore
					2		-47.7			Vibracore
				100			-48.7			Vibracore
-49.3	2.5			100	3		-49.3			Vibracore
				100	4		-49.7			Vibracore
-50.7	3.9		SAND, silty, mostly fine-grained sand-sized quartz, some silt, 10YR 4/1 dark gray (SM)	100	5		-50.7			
				100	6		-51.7			Vibracore
				100						
-54.3	7.5			100			-54.3			Vibracore
-55.6	8.8			100			-55.6			Vibracore
			SAND, silty, mostly fine to medium-grained sand-sized quartz, some intermittent silt seams, 10YR 4/1 dark gray (SM)	100						
				100						Vibracore

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																								
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																								
LOCATION COORDINATES X = 543,032 Y = 2,204,862			ELEVATION TOP OF BORING -46.8 Ft.																											
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																					
-62.5	15.7			100			-62.5 Vibracore																							
-62.8	16.0		CLAY, fat, mostly clay, trace fine-grained sand-sized quartz, 10YR 4/1 dark gray (CH)	100			-62.8 Vibracore																							
				NR			Vibracore																							
-66.8	20.0	NO RECOVERY																												
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>0.0/0.2</td><td>SP*</td></tr> <tr><td>2</td><td>0.9/1.1</td><td>SP-SM*</td></tr> <tr><td>3</td><td>1.9/2.1</td><td>SP-SM*</td></tr> <tr><td>4</td><td>2.9/3.1</td><td>SM*</td></tr> <tr><td>5</td><td>3.9/4.1</td><td>SP*</td></tr> <tr><td>6</td><td>4.9/5.1</td><td>SP*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	SP*	2	0.9/1.1	SP-SM*	3	1.9/2.1	SP-SM*	4	2.9/3.1	SM*	5	3.9/4.1	SP*	6	4.9/5.1	SP*					Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																												
1	0.0/0.2	SP*																												
2	0.9/1.1	SP-SM*																												
3	1.9/2.1	SP-SM*																												
4	2.9/3.1	SM*																												
5	3.9/4.1	SP*																												
6	4.9/5.1	SP*																												

Boring Designation VB-MP2022-13

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-13				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 07-02-22		COMPLETED 07-02-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-48.8 Ft.			
				17. TOTAL RECOVERY FOR BORING		93 %			
				18. SIGNATURE AND TITLE OF INSPECTOR		Rashmi Kola, Geotechnical Engineer			
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.8	0.0		SILT, inorganic-H, mostly silt, some fine-grained sand-sized quartz, 10YR 3/1 very dark gray (MH)	100			-48.8		
					1		Vibracore		
				100			Vibracore		
					2		-51.7		
				100			Vibracore		
-55.8	7.0		SILT, inorganic-H, mostly silt, some fine-grained sand-sized quartz, trace sand to gravel-sized shell, trace wood debris, 10YR 3/1 very dark gray (MH)	100			-55.8		
-57.3	8.5						Vibracore		
			SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, trace silt, 10YR 4/1 dark gray (SP-SM)	100			-57.3		
							Vibracore		
-60.8	12.0		SAND, clayey, mostly fine to medium-grained sand-sized quartz, little clay, 10YR 4/1 dark gray (SC)	100			-60.8		
-61.7	12.9		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace silt, 10YR 5/1 gray (SP)	100			Vibracore		
							Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																		
			PROJECT U.S. Naval Station Mayport, 2022		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																	
LOCATION COORDINATES X = 544,336 Y = 2,203,899			ELEVATION TOP OF BORING -48.8 Ft.																						
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																
-67.3	18.5	•••••		100			Vibracore																		
-68.8	20.0	NR		NR			Vibracore																		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> <tr> <td style="border-top: 1px solid black;">1</td> <td style="border-top: 1px solid black;">0.9/1.1</td> <td style="border-top: 1px solid black;">MH</td> </tr> <tr> <td style="border-bottom: 1px solid black;">2</td> <td style="border-bottom: 1px solid black;">2.9/3.1</td> <td style="border-bottom: 1px solid black;">MH*</td> </tr> </table> *Lab visual classification based on gradation curve <table style="width:100%; border-collapse: collapse;"> <tr> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY SOIL TESTING</th> <th style="text-align: left;">RESULT UNIT</th> </tr> <tr> <td style="border-top: 1px solid black;">1.0</td> <td style="border-top: 1px solid black;">Atterberg (PI)</td> <td style="border-top: 1px solid black;">98</td> </tr> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.9/1.1	MH	2	2.9/3.1	MH*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	1.0	Atterberg (PI)	98	Abbreviations:				
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																							
1	0.9/1.1	MH																							
2	2.9/3.1	MH*																							
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																							
1.0	Atterberg (PI)	98																							

Boring Designation VB-MP2022-14

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-14		LOCATION COORDINATES X = 544,730 Y = 2,204,657		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV		CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 07-02-22		COMPLETED 07-02-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -47.3 Ft.					
				17. TOTAL RECOVERY FOR BORING 71 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.3	0.0	[Symbol]	SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, 10YR 3/1 very dark gray (SP-SM)	100	1		-47.3		0
							-47.5	Vibracore	
-48.3	1.0	[Symbol]	SAND, silty, mostly fine-grained sand-sized quartz, some silt, 10YR 3/1 very dark gray (SM)	100			-48.3	Vibracore	
							-48.7	Vibracore	
-49.8	2.5	[Symbol]	SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, 10YR 3/1 very dark gray (SP-SM)	100	2		-49.8	Vibracore	
							-50.2	Vibracore	
-51.5	4.2	[Symbol]	SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, 10YR 4/1 dark gray (SP)	100	3		-51.5	Vibracore	
							-51.7	Vibracore	
-56.1	8.8	[Symbol]	SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 4/1 dark gray (SP-SM)	100	4				5
-56.5	9.2								
-58.7	11.4	[Symbol]	SILT, inorganic-L, mostly silt, trace fine-grained sand-sized quartz, 10YR 4/1 dark gray (ML)	100					10
-61.5	14.2	[Symbol]	SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace silt, 10YR 5/1 gray (SP)	100					15
		NO RECOVERY		NR			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																		
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 544,730 Y = 2,204,657			ELEVATION TOP OF BORING -47.3 Ft.																					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-67.3	20.0	NO RECOVERY		NR			Vibracore																	
			BORING TERMINATED IN REFUSAL NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.0/0.2</td> <td>SP-SM*</td> </tr> <tr> <td>2</td> <td>1.4/1.6</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>2.9/3.1</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>4.4/4.6</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	SP-SM*	2	1.4/1.6	SM*	3	2.9/3.1	SP-SM*	4	4.4/4.6	SP*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	0.0/0.2	SP-SM*																						
2	1.4/1.6	SM*																						
3	2.9/3.1	SP-SM*																						
4	4.4/4.6	SP*																						

Boring Designation VB-MP2022-15

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT U.S. Naval Station Mayport, 2022				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP2022-15				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY CSI/TSF JV				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Brian Simmons (Amdrill)				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 07-02-22		COMPLETED 07-02-22	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -50.8 Ft.					
				17. TOTAL RECOVERY FOR BORING 84 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Rashmi Kola, Geotechnical Engineer					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-50.8	0.0		SILT, inorganic-L, mostly silt, some fine-grained sand-sized quartz, 10YR 3/1 very dark gray (ML)	100	1		-50.8		
				100			-51.0		
							Vibracore		
-52.7	1.9		SAND, silty, mostly fine-grained sand-sized quartz, little silt, 10YR 3/1 very dark gray (SM)	100	2		-52.7		
							Vibracore		
-54.8	4.0		SILT, inorganic-L, mostly silt, some sand to gravel-sized shell, trace fine-grained sand-sized quartz, 10YR 3/1 very dark gray (ML)	100			-54.8		
							Vibracore		
-57.5	6.7		SAND, silty, mostly fine to medium-grained sand-sized quartz, some fine gravel-sized shell, 10YR 4/1 dark gray (SM)	100			-57.5		
							Vibracore		
-59.8	9.0		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 10YR 5/1 gray (SP-SM)	100			-59.8		
							Vibracore		
-62.5	11.7		SAND, clayey, mostly fine to medium-grained sand-sized quartz, few clay, trace silt, mostly fine to medium-grained sand-sized quartz, few clay, trace silt, 10YR 5/1 gray (SC)	100			-62.5		
							Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS												
PROJECT U.S. Naval Station Mayport, 2022			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW												
LOCATION COORDINATES X = 548,397 Y = 2,203,444			ELEVATION TOP OF BORING -50.8 Ft.															
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE									
-67.5	16.7	NO RECOVERY		100			Vibracore											
-70.8	20.0			NR			Vibracore											
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.0/0.2</td> <td>ML*</td> </tr> <tr> <td>2</td> <td>1.9/2.1</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.0/0.2	ML*	2	1.9/2.1	SM*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																
1	0.0/0.2	ML*																
2	1.9/2.1	SM*																

Boring Designation VB-MP16-01

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT					
2. BORING DESIGNATION VC_MP16-01				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)			HORIZONTAL NAD83		
3. DRILLING AGENCY Corps of Engineers - CESAW				11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES			DISTURBED 6		
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES			2		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING			STARTED 07-19-16		
8. TOTAL DEPTH OF BORING 12.8 Ft.				16. ELEVATION TOP OF BORING			-35.2 Ft.		
				17. TOTAL RECOVERY FOR BORING			64.4 %		
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-35.2	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, trace organic matter, N 3/11 very dark gray (MH)						
					1		-36.2		
					2		-38.2		
			From El. -38.2 to -44.6 Ft., trace quartz, N 3/11 very dark gray			4A			
					3		-40.7		
					4		-43.2		
-44.6	9.4		SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, trace limestone, 5Y 4/3 olive (SM)						
					5	5A	-45.7		
-47.2	12.0		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, trace limestone, 5Y 4/3 olive (SP-SM)		6		-47.2		
-48.0	12.8						-48.0		
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in				Abbreviations:					

Boring Designation VB-MP16-01

DRILLING LOG (Cont. Sheet)			INSTALLATION			SHEET 2 OF 2 SHEETS																																																
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																																																
LOCATION COORDINATES X = 525,757 Y = 2,204,607			ELEVATION TOP OF BORING -35.2 Ft.																																																			
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																																													
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SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																																																				
1	1.0/1.2	MH*																																																				
2	3.0/3.2	MH*																																																				
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5	10.5/10.7	SM*																																																				
6	12.0/12.2	SP-SM*																																																				
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Boring Designation VB-MP16-02

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT				
2. BORING DESIGNATION VC_MP16-02				LOCATION COORDINATES X = 527,330 Y = 2,204,375		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAW				CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 3
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		15. DATE BORING 07-19-16		
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -44.5 Ft.		17. TOTAL RECOVERY FOR BORING 97.3 %		
8. TOTAL DEPTH OF BORING 15.8 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR				

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-44.5	0.0		SILT, inorganic-H, little fine-grained sand-sized quartz, N 3/11 very dark gray (MH)						
					1		-45.5		
					2		-47.0		
					3	11A	-49.5		
-50.5	6.0		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few siltstone, 5Y 4/2 olive gray (SP-SM)						
					4		-51.5		
-54.5	10.0		SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, silt seam (3") @ 15.4'- 5.65' depth., 5Y 4/2 olive gray (SM)						
						10A			

DRILLING LOG (Cont. Sheet)				INSTALLATION			SHEET 2 OF 2 SHEETS																	
PROJECT				COORDINATE SYSTEM/DATUM		HORIZONTAL	VERTICAL																	
Mayport Harbor Vibracores 2016				State Plane, FLE (U.S. Ft.)		NAD83	MLLW																	
LOCATION COORDINATES				ELEVATION TOP OF BORING																				
X = 527,330 Y = 2,204,375				-44.5 Ft.																				
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-60.3	15.8	↑↑↑↑				7A	-60.3																	
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>1.0/1.2</td><td>MH*</td></tr> <tr><td>2</td><td>2.5/2.7</td><td>MH*</td></tr> <tr><td>3</td><td>5.0/5.2</td><td>MH*</td></tr> <tr><td>4</td><td>7.0/7.2</td><td>SP-SM*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing 1 Moisture Content 1 Percent Visual Shell 2 Moisture Content 2 Percent Visual Shell 3 Moisture Content 3 Percent Visual Shell 4 Moisture Content 4 Percent Visual Shell	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.2	MH*	2	2.5/2.7	MH*	3	5.0/5.2	MH*	4	7.0/7.2	SP-SM*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	1.0/1.2	MH*																						
2	2.5/2.7	MH*																						
3	5.0/5.2	MH*																						
4	7.0/7.2	SP-SM*																						

Boring Designation VB-MP16-03

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT				
2. BORING DESIGNATION VC_MP16-03				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)			HORIZONTAL NAD83	VERTICAL MLLW
3. DRILLING AGENCY Corps of Engineers - CESAW				11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER	
4. NAME OF DRILLER				12. TOTAL SAMPLES			DISTURBED 4	UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 2			14. ELEVATION GROUND WATER	
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING			STARTED 07-19-16	COMPLETED 07-19-16
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -46.2 Ft.			17. TOTAL RECOVERY FOR BORING 100 %	
8. TOTAL DEPTH OF BORING 14.9 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR				

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-46.2	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, N 3/11 very dark gray (MH)						
					1		-47.2		
					2		-49.2		
			From El. -51.2 to -53.2 Ft., little fine-grained sand-sized quartz, N 3/11 very dark gray		3	9	-51.2		
-53.2	7.0		SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, 5Y 4/3 olive (SM)		4		-53.2		
			From El. -56.2 to -61.1 Ft., mostly fine to medium-grained sand-sized quartz, little silt, 5Y 4/2 olive gray			10A			
-61.1	14.9						-61.1		

DRILLING LOG (Cont. Sheet)			INSTALLATION				SHEET 2 OF 2 SHEETS																																	
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																																		
LOCATION COORDINATES X = 526,777 Y = 2,204,007			ELEVATION TOP OF BORING -46.2 Ft.																																					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																															
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>1.0/1.2</td><td>MH*</td></tr> <tr><td>2</td><td>3.0/3.2</td><td>MH*</td></tr> <tr><td>3</td><td>5.0/5.2</td><td>MH*</td></tr> <tr><td>4</td><td>7.0/7.2</td><td>SM*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table style="width: 100%; border-collapse: collapse;"> <tbody> <tr><td>1</td><td>Moisture Content</td></tr> <tr><td>1</td><td>Percent Visual Shell</td></tr> <tr><td>2</td><td>Moisture Content</td></tr> <tr><td>2</td><td>Percent Visual Shell</td></tr> <tr><td>3</td><td>Moisture Content</td></tr> <tr><td>3</td><td>Percent Visual Shell</td></tr> <tr><td>4</td><td>Moisture Content</td></tr> <tr><td>4</td><td>Percent Visual Shell</td></tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.2	MH*	2	3.0/3.2	MH*	3	5.0/5.2	MH*	4	7.0/7.2	SM*	1	Moisture Content	1	Percent Visual Shell	2	Moisture Content	2	Percent Visual Shell	3	Moisture Content	3	Percent Visual Shell	4	Moisture Content	4	Percent Visual Shell				Abbreviations:		
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3	Moisture Content																																							
3	Percent Visual Shell																																							
4	Moisture Content																																							
4	Percent Visual Shell																																							

15

SAJ FORM 1836-A
JUN 02

Boring Designation VB-MP16-05

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT				
2. BORING DESIGNATION VC_MP16-05				LOCATION COORDINATES X = 527,170 Y = 2,203,434		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAW				CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 2
6. THICKNESS OF OVERBURDEN N/A				7. DEPTH DRILLED INTO ROCK N/A		14. ELEVATION GROUND WATER		15. DATE BORING 07-19-16
8. TOTAL DEPTH OF BORING 18.0 Ft.				16. ELEVATION TOP OF BORING -43.6 Ft.		17. TOTAL RECOVERY FOR BORING 96 %		18. SIGNATURE AND TITLE OF INSPECTOR

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-43.6	0.0		SILT, inorganic-H, trace quartz, N 3/11 very dark gray (MH)		1	6A			
-48.6	5.0		SILT, organic-H, few organic matter, trace quartz, N 3/11 very dark gray (OH)		2				
-49.6	6.0		SAND, silty, mostly fine to medium-grained sand-sized quartz, little silt, 5Y 4/3 olive (SM)		3				
-52.6	9.0		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, 5Y 4/2 olive gray (SP-SM)		4				
						5A			

DRILLING LOG (Cont. Sheet)			INSTALLATION				SHEET 2 OF 2 SHEETS																																	
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																																		
LOCATION COORDINATES X = 527,170 Y = 2,203,434			ELEVATION TOP OF BORING -43.6 Ft.																																					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																															
-61.6	18.0		From El. -58.6 to -76.6 Ft., mostly fine to medium-grained sand-sized quartz, little silt, 5Y 3/1 very dark gray			5A																																		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.0/2.2</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>5.0/5.2</td> <td>OH*</td> </tr> <tr> <td>3</td> <td>7.0/7.2</td> <td>SM*</td> </tr> <tr> <td>4</td> <td>9.0/9.2</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Moisture Content</td> </tr> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Moisture Content</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> <tr> <td>3</td> <td>Moisture Content</td> </tr> <tr> <td>3</td> <td>Percent Visual Shell</td> </tr> <tr> <td>4</td> <td>Moisture Content</td> </tr> <tr> <td>4</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/2.2	MH*	2	5.0/5.2	OH*	3	7.0/7.2	SM*	4	9.0/9.2	SP-SM*	1	Moisture Content	1	Percent Visual Shell	2	Moisture Content	2	Percent Visual Shell	3	Moisture Content	3	Percent Visual Shell	4	Moisture Content	4	Percent Visual Shell				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																																						
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3	Moisture Content																																							
3	Percent Visual Shell																																							
4	Moisture Content																																							
4	Percent Visual Shell																																							

Boring Designation VB-MP16-06

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT					
2. BORING DESIGNATION VC_MP16-06		LOCATION COORDINATES X = 529,485 Y = 2,204,078		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW		
3. DRILLING AGENCY Corps of Engineers - CESAW		CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 4	UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED				13. TOTAL NUMBER CORE BOXES 2		14. ELEVATION GROUND WATER			
6. THICKNESS OF OVERBURDEN N/A		DEG. FROM VERTICAL		15. DATE BORING		STARTED 07-19-16	COMPLETED 07-19-16		
7. DEPTH DRILLED INTO ROCK N/A		BEARING		16. ELEVATION TOP OF BORING -47.5 Ft.		17. TOTAL RECOVERY FOR BORING 97 %			
8. TOTAL DEPTH OF BORING 12.1 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-47.5	0.0		SILT, inorganic-H, little fine-grained sand-sized quartz, weak reaction with HCl, N 3/11 very dark gray (MH)		1				
-51.5	4.0		SILT, organic-H, some fine-grained sand-sized quartz, few organic matter, weak reaction with HCl, N 3/11 very dark gray (OH)		2	3A			
-53.5	6.0		SILT, inorganic-H, some fine-grained sand-sized quartz, weak reaction with HCl, trace clay clods, N 3/11 very dark gray (MH)		3				
			From El. -54.7 to -57.5 Ft., few fine-grained sand-sized quartz, no reaction with HCl, 5B 5/1 bluish gray		4				
			From El. -57.5 to -59.6 Ft., little fine-grained sand-sized quartz, trace clay, weak reaction with HCl, trace clay clods, 10Y 4/1 dark greenish gray			2A			
-59.6	12.1		BORING TERMINATED IN REFUSAL				-59.6		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification	Abbreviations:					

Boring Designation VB-MP16-06

DRILLING LOG (Cont. Sheet)			INSTALLATION			SHEET 2 OF 2 SHEETS																		
PROJECT			COORDINATE SYSTEM/DATUM		HORIZONTAL	VERTICAL																		
Mayport Harbor Vibracores 2016			State Plane, FLE (U.S. Ft.)		NAD83	MLLW																		
LOCATION COORDINATES			ELEVATION TOP OF BORING																					
X = 529,485 Y = 2,204,078			-47.5 Ft.																					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
			System.																					
			3. Rock in Shoe																					
			4. Laboratory Testing Results																					
			<table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/1.2</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>4.0/4.2</td> <td>OH*</td> </tr> <tr> <td>3</td> <td>6.0/6.2</td> <td>MH*</td> </tr> <tr> <td>4</td> <td>8.0/8.2</td> <td>MH*</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.2	MH*	2	4.0/4.2	OH*	3	6.0/6.2	MH*	4	8.0/8.2	MH*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	1.0/1.2	MH*																						
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3	6.0/6.2	MH*																						
4	8.0/8.2	MH*																						
			*Lab visual classification based on gradation curve																					
			5. Additional Laboratory Testing																					
			1 Moisture Content																					
			1 Percent Visual Shell																					
			2 Moisture Content																					
			2 Percent Visual Shell																					
			3 Moisture Content																					
			3 Percent Visual Shell																					
			4 Moisture Content																					
			4 Percent Visual Shell																					

Boring Designation VB-MP16-07

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT					
2. BORING DESIGNATION VC_MP16-07		LOCATION COORDINATES X = 531,031 Y = 2,204,488		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAW		CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL ☐ AUTO HAMMER ☐ MANUAL HAMMER					
4. NAME OF DRILLER				12. TOTAL SAMPLES 2		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED				13. TOTAL NUMBER CORE BOXES 2		14. ELEVATION GROUND WATER			
6. THICKNESS OF OVERBURDEN N/A		DEG. FROM VERTICAL		BEARING		15. DATE BORING 07-22-16			
7. DEPTH DRILLED INTO ROCK N/A		16. ELEVATION TOP OF BORING -51.9 Ft.		17. TOTAL RECOVERY FOR BORING 93.6 %					
8. TOTAL DEPTH OF BORING 18.9 Ft.		18. SIGNATURE AND TITLE OF INSPECTOR							
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-51.9	0.0								
-52.6	0.7		SAND, poorly-graded, mostly fine-grained sand-sized quartz, few fine-grained sand-sized shell, trace silt, 10YR 3/1 very dark gray (SP)		1				
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace shell, trace silt, 10YR 5/1 gray (SP)		2				
			From El. -56.4 to -66.9 Ft., some fine to medium-grained sand-sized quartz, trace fine to coarse-grained sand-sized shell, trace silt, 10YR 6/1 gray			12A			
						13A			

DRILLING LOG (Cont. Sheet)			INSTALLATION				SHEET 2 OF 2 SHEETS																			
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																				
LOCATION COORDINATES X = 531,031 Y = 2,204,488			ELEVATION TOP OF BORING -51.9 Ft.																							
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																	
-70.8	18.9		From El. -66.9 to -70.8 Ft., some fine to medium-grained sand-sized quartz, little fine to coarse-grained sand-sized shell, trace silt, 10YR 6/1 gray			13A																				
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/0.7</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>2.0/2.2</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Moisture Content</td> </tr> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Moisture Content</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/0.7	SP*	2	2.0/2.2	SP*	1	Moisture Content	1	Percent Visual Shell	2	Moisture Content	2	Percent Visual Shell				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																								
1	0.5/0.7	SP*																								
2	2.0/2.2	SP*																								
1	Moisture Content																									
1	Percent Visual Shell																									
2	Moisture Content																									
2	Percent Visual Shell																									

Boring Designation VB-MP16-08

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT				
2. BORING DESIGNATION VC_MP16-08				LOCATION COORDINATES X = 542,499 Y = 2,204,164		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAW				CONTRACTOR FILE NO.		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 2
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		15. DATE BORING 07-22-16		STARTED 07-22-16
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -48.5 Ft.		17. TOTAL RECOVERY FOR BORING 95.1 %		COMPLETED 07-22-16
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR				

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-48.5	0.0		SILT, inorganic-H, some fine-grained sand-sized quartz, N 3/11 very dark gray (MH)						
					1				
			From El. -51.5 to -56.5 Ft., few fine-grained sand-sized quartz, N 3/11 very dark gray		2				
					3	21A			
					4				
-56.5	8.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, 10Y 4/1 dark greenish gray (SM)						
			From El. -58.1 to -61.0 Ft., mostly fine-grained sand-sized quartz, little silt, 10YR 6/1 gray						
			From El. -61.0 to -63.5 Ft., mostly fine-grained sand-sized quartz, some silt, 10YR 6/1 gray			20A			

DRILLING LOG (Cont. Sheet)			INSTALLATION				SHEET 2 OF 2 SHEETS																	
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 542,499 Y = 2,204,164			ELEVATION TOP OF BORING -48.5 Ft.																					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-68.5	20.0		From El. -63.5 to -64.0 Ft., mostly fine-grained sand-sized quartz, little silt, 5GY 5/1 greenish gray From El. -64.0 to -68.5 Ft., mostly fine-grained sand-sized quartz, some silt, silt seams, 5GY 5/1 greenish gray			20A																		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/1.2</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>3.0/3.2</td> <td>MH*</td> </tr> <tr> <td>3</td> <td>5.0/5.2</td> <td>MH*</td> </tr> <tr> <td>4</td> <td>7.0/7.2</td> <td>MH*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing 1 Moisture Content 1 Percent Visual Shell 2 Moisture Content 2 Percent Visual Shell 3 Moisture Content 3 Percent Visual Shell 4 Moisture Content 4 Percent Visual Shell	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.2	MH*	2	3.0/3.2	MH*	3	5.0/5.2	MH*	4	7.0/7.2	MH*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	1.0/1.2	MH*																						
2	3.0/3.2	MH*																						
3	5.0/5.2	MH*																						
4	7.0/7.2	MH*																						

Boring Designation VB-MP16-09

DRILLING LOG		DIVISION South Atlantic	INSTALLATION		SHEET 1 OF 2 SHEETS
1. PROJECT Mayport Harbor Vibracores 2016			9. SIZE AND TYPE OF BIT		
2. BORING DESIGNATION VC_MP16-09			10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAW			11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER			12. TOTAL SAMPLES		DISTURBED 4
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED			13. TOTAL NUMBER CORE BOXES		2
6. THICKNESS OF OVERBURDEN N/A			14. ELEVATION GROUND WATER		
7. DEPTH DRILLED INTO ROCK N/A			15. DATE BORING		STARTED 07-22-16
8. TOTAL DEPTH OF BORING 20.0 Ft.			16. ELEVATION TOP OF BORING		-46.6 Ft.
			17. TOTAL RECOVERY FOR BORING		96.6 %
			18. SIGNATURE AND TITLE OF INSPECTOR		

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-46.6	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, N 3/11 very dark gray (MH)						
					1				
			From El. -49.6 to -51.6 Ft., little fine-grained sand-sized quartz, N 3/11 very dark gray		2				
-51.6	5.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, N 3/11 very dark gray (SM)		3	16A			5
-53.6	7.0		SILT, inorganic-H, few fine-grained sand-sized quartz, N 3/11 very dark gray (MH)		4				
-56.6	10.0		From El. -56.2 to -56.6 Ft., some silt, little fine-grained sand-sized quartz, 5GY 4/1 dark greenish gray						10
			SAND, silty, mostly fine-grained sand-sized quartz, 10YR 5/1 gray (SM)			17A			
-60.1	13.5		CLAY, lean, little fine-grained sand-sized quartz, little silt, sand seam @ 3.7'-3.8' depth, 10Y 4/1 dark greenish gray (CL)						
-61.6	15.0								15

DRILLING LOG (Cont. Sheet)			INSTALLATION			SHEET 2 OF 2 SHEETS																																	
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																																	
LOCATION COORDINATES X = 543,449 Y = 2,204,070			ELEVATION TOP OF BORING -46.6 Ft.																																				
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																														
-66.3	19.7		SAND, silty, mostly fine-grained sand-sized quartz, little silt, 10YR 6/1 gray (SM)			17A																																	
-66.6	20.0		SILT, inorganic-H, little fine-grained sand-sized quartz, 10Y 4/1 dark greenish gray (MH)				-66.6																																
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/1.2</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>3.0/3.2</td> <td>MH*</td> </tr> <tr> <td>3</td> <td>5.0/5.2</td> <td>SM*</td> </tr> <tr> <td>4</td> <td>7.0/7.2</td> <td>MH*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Moisture Content</td> </tr> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Moisture Content</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> <tr> <td>3</td> <td>Moisture Content</td> </tr> <tr> <td>3</td> <td>Percent Visual Shell</td> </tr> <tr> <td>4</td> <td>Moisture Content</td> </tr> <tr> <td>4</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.2	MH*	2	3.0/3.2	MH*	3	5.0/5.2	SM*	4	7.0/7.2	MH*	1	Moisture Content	1	Percent Visual Shell	2	Moisture Content	2	Percent Visual Shell	3	Moisture Content	3	Percent Visual Shell	4	Moisture Content	4	Percent Visual Shell			Abbreviations:		
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4	Moisture Content																																						
4	Percent Visual Shell																																						

Boring Designation VB-MP16-10

DRILLING LOG		DIVISION South Atlantic	INSTALLATION		SHEET 1 OF 2 SHEETS
1. PROJECT Mayport Harbor Vibracores 2016			9. SIZE AND TYPE OF BIT		
2. BORING DESIGNATION VC_MP16-10			10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAW			11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER			12. TOTAL SAMPLES		DISTURBED 4
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED			13. TOTAL NUMBER CORE BOXES		2
6. THICKNESS OF OVERBURDEN N/A			14. ELEVATION GROUND WATER		
7. DEPTH DRILLED INTO ROCK N/A			15. DATE BORING		STARTED 07-22-16
8. TOTAL DEPTH OF BORING 18.0 Ft.			16. ELEVATION TOP OF BORING		-47.6 Ft.
			17. TOTAL RECOVERY FOR BORING		89.6 %
			18. SIGNATURE AND TITLE OF INSPECTOR		

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-47.6	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, N 3/11 very dark gray (MH)						
			From El. -50.6 to -52.6 Ft., trace quartz, N 3/11 very dark gray		1				
			From El. -52.6 to -60.6 Ft., few fine-grained sand-sized quartz, N 3/11 very dark gray		2				
					3	14A			5
					4				
-55.6	8.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, 10Y 6/1 greenish gray (SM)						
			From El. -57.2 to -58.8 Ft., mostly fine-grained sand-sized quartz, little silt, 10Y 6/1 greenish gray						
			CLAY, fat, few fine-grained sand-sized quartz, 10Y 4/1 dark greenish gray (CH)						
-59.9	12.3								
-60.1	12.5		SILT, organic-H, little organic matter, few fine gravel-sized quartz, 10YR 3/2 very dark grayish brown (OH)						
-60.7	13.1		SAND, silty, mostly fine-grained sand-sized quartz, little silt, 10YR 3/1 very dark gray (SM)						
			CLAY, lean, some fine-grained sand-sized quartz, 10Y 4/1 dark greenish gray (CL)						
-62.3	14.7		SAND, clayey, mostly fine-grained sand-sized			15A			

DRILLING LOG (Cont. Sheet)			INSTALLATION			SHEET 2 OF 2 SHEETS																		
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 544,313 Y = 2,203,961			ELEVATION TOP OF BORING -47.6 Ft.																					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-63.6	16.0		quartz, little clay, 10YR 5/1 gray (SC)																					
-64.2	16.6		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, 10YR 6/1 gray (SP)			15A																		
-65.6	18.0		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, 5Y 4/2 olive gray (SP-SM)																					
NOTES:																								
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4 Moisture Content																								
4 Percent Visual Shell																								
Abbreviations:																								

Boring Designation VB-MP16-11

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT					
2. BORING DESIGNATION VC_MP16-11				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)			HORIZONTAL NAD83 VERTICAL MLLW		
3. DRILLING AGENCY Corps of Engineers - CESAW				11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES			DISTURBED 4 UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED				13. TOTAL NUMBER CORE BOXES			2		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING			STARTED 07-22-16 COMPLETED 07-22-16		
8. TOTAL DEPTH OF BORING 15.0 Ft.				16. ELEVATION TOP OF BORING -47.6 Ft.					
				17. TOTAL RECOVERY FOR BORING 74.4 %					
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.6	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, N 3/11 very dark gray (MH)						
-50.4	2.8				1				
-52.8	5.2		SAND, silty, mostly fine-grained sand-sized quartz, some silt, 5GY 3/1 very dark greenish gray (SM) From El. -50.6 to -51.0 Ft., mostly fine-grained sand-sized quartz, some silt, 10Y 3/1 very dark greenish gray From El. -51.0 to -52.8 Ft., mostly fine-grained sand-sized quartz, little silt, 5GY 4/1 dark greenish gray		2				
			SILT, inorganic-H, little fine-grained sand-sized quartz, 10Y 4/1 dark greenish gray (MH) From El. -54.6 to -60.0 Ft., little fine-grained sand-sized quartz, trace shell, 10Y 4/1 dark greenish gray		3	16A			
-61.0	13.4				4				
-62.0	14.4		PEAT, some fine-grained sand-sized quartz, some organic matter, little silt, 10YR 3/1 very dark gray (PT)						
-62.6	15.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, 10YR 5/1 gray (SM)			17A			

DRILLING LOG (Cont. Sheet)			INSTALLATION			SHEET 2 OF 2 SHEETS																																		
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																																		
LOCATION COORDINATES X = 545,542 Y = 2,203,779			ELEVATION TOP OF BORING -47.6 Ft.																																					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																															
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4	Moisture Content																																							
4	Percent Visual Shell																																							

Boring Designation VB-MP16-12

DRILLING LOG		DIVISION South Atlantic		INSTALLATION			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Harbor Vibracores 2016				9. SIZE AND TYPE OF BIT					
2. BORING DESIGNATION VC_MP16-12				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)			HORIZONTAL NAD83		
3. DRILLING AGENCY Corps of Engineers - CESAW				11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES			DISTURBED 4		
5. DIRECTION OF BORING ☐ VERTICAL ☒ INCLINED				13. TOTAL NUMBER CORE BOXES			2		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING			STARTED 07-22-16		
8. TOTAL DEPTH OF BORING 17.0 Ft.				16. ELEVATION TOP OF BORING			-45.5 Ft.		
				17. TOTAL RECOVERY FOR BORING			84 %		
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-45.5	0.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, strong reaction with HCl, N 3/11 very dark gray (SM)		1				
-48.5	3.0		SILT, inorganic-H, little fine-grained sand-sized quartz, strong reaction with HCl, N 3/11 very dark gray (MH)		2				
-51.1	5.6		From El. -50.0 to -51.1 Ft., some fine-grained sand-sized quartz, strong reaction with HCl, N 3/11 very dark gray		3	1A			
-53.2	7.7		SAND, silty, mostly fine-grained sand-sized quartz, little silt, strong reaction with HCl, intermittent silty sand seams @ 1.5'-4.0', N 3/11 very dark gray (SM)		4				
			CLAY, lean, few fine-grained sand-sized quartz, weak reaction with HCl, 5B 5/1 bluish gray (CL)			2A			

DRILLING LOG (Cont. Sheet)			INSTALLATION				SHEET 2 OF 2 SHEETS																																
PROJECT Mayport Harbor Vibracores 2016			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																																	
LOCATION COORDINATES X = 544,268 Y = 2,204,667			ELEVATION TOP OF BORING -45.5 Ft.																																				
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																														
-62.5	17.0					2A	-62.5																																
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4	Moisture Content																																						
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Boring Designation VB-MPB2014M-1

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 1 SHEETS		
1. PROJECT Jacksonville Harbor O&M				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPB2014M-1		LOCATION COORDINATES X = 531,259 Y = 2,204,554		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ		CONTRACTOR FILE NO. 6738-15-5400		11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 0		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 0	
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 11-02-14		COMPLETED 11-02-14	
8. TOTAL DEPTH OF BORING 4.5 Ft.				16. ELEVATION TOP OF BORING -47.9 Ft.					
				17. TOTAL RECOVERY FOR BORING 84 %					
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-47.9	0.0						-47.9		
-48.4	0.5		SAND, clayey, mostly fine-grained sand-sized quartz, some clay, few medium-grained sand-sized shell, weak reaction with HCl, 10Y 5/1 greenish gray (SC)						
			CLAY, lean, few fine-grained sand-sized quartz, weak reaction with HCl, 5GY 4/1 dark greenish gray (CL)						
			At El. -50.7 Ft., few organic matter						
			At El. -51.1 Ft., discontinue organic matter						
-51.7	3.8		No Recovery						
-52.4	4.5		At El. -52.4 Ft. Tight, hard clay/silt at base. BORING TERMINATED IN REFUSAL				-52.4		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.						

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Jacksonville Harbor O&M				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPB2014M-2				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 10-29-14		COMPLETED 10-29-14	
8. TOTAL DEPTH OF BORING 18.2 Ft.				16. ELEVATION TOP OF BORING		-48.1 Ft.			
				17. TOTAL RECOVERY FOR BORING		64 %			
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.1	0.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, no reaction with HCl, 5Y 5/2 olive gray (SP)	2340			-48.1		
-48.6	0.5		At El. -48.4 Ft., N 3/ very dark gray		1		-48.6		
-48.9	0.8		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, trace organic matter, no reaction with HCl, 5Y 4/2 olive gray (SP-SM)						
-49.7	1.6		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, no reaction with HCl, N 3/ very dark gray (SP)		2		-50.1		
-51.8	3.7		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace organic matter, no reaction with HCl, N 2.5/ black (SM)		3		-51.1		
-52.6	4.5		At El. -50.4 Ft., discontinue organic matter, N 4/ dark gray						
-53.9	5.8		At El. -50.8 Ft., intermittent layers of silty fine sand seams						
-54.4	6.3		SILT, inorganic-L, some fine-grained sand-sized quartz, N 4/ dark gray (ML)						
			SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, no reaction with HCl, N 4/ dark gray (SP-SM)						
			CLAY, fat, few fine-grained sand-sized quartz, no reaction with HCl, 5G 4/1 dark greenish gray (CH)						
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace clay, trace shell, no reaction with HCl, 5Y 6/1 gray (SP)						
			At El. -57.6 Ft., 1" seam of sand with clay, 5Y 5/1 gray						
			At El. -57.7 Ft., weak reaction with HCl, 5Y 6/1 gray						
			At El. -58.2 Ft., no reaction with HCl, 1' layer of sand with clay, 5Y 5/1 gray						
-59.7	11.6		At El. -59.2 Ft., little fine-grained sand-sized shell, weak reaction with HCl, 5Y 6/1 gray						
-59.8	11.7		CLAY, fat, few fine-grained sand-sized quartz, no reaction with HCl, 5G 4/1 dark greenish gray (CH)						
			No Recovery						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																					
PROJECT Jacksonville Harbor O&M			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																					
LOCATION COORDINATES X = 530,528 Y = 2,204,435			ELEVATION TOP OF BORING -48.1 Ft.																								
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
-66.3	18.2						-66.3																				
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/0.8</td> <td>SP-SM*</td> </tr> <tr> <td>2</td> <td>2.0/2.3</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>3.0/3.3</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> <tr> <td>3</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/0.8	SP-SM*	2	2.0/2.3	SM*	3	3.0/3.3	SM*	1	Percent Visual Shell	2	Percent Visual Shell	3	Percent Visual Shell				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	0.5/0.8	SP-SM*																									
2	2.0/2.3	SM*																									
3	3.0/3.3	SM*																									
1	Percent Visual Shell																										
2	Percent Visual Shell																										
3	Percent Visual Shell																										

Boring Designation VB-MPB2014M-3

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Jacksonville Harbor O&M				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPB2014M-3		LOCATION COORDINATES X = 529,679 Y = 2,204,476		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ		CONTRACTOR FILE NO. 6738-15-5400		11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 0	
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 11-03-14		COMPLETED 11-03-14	
8. TOTAL DEPTH OF BORING 17.3 Ft.				16. ELEVATION TOP OF BORING -48.3 Ft.		17. TOTAL RECOVERY FOR BORING 84 %			
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-48.3	0.0		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, trace shell, weak reaction with HCl, 5Y 5/2 olive gray (SP-SM) From El. -49.1 to -49.5 Ft., occasional silty fine sand seams, N 4/ dark gray At El. -50.4 Ft., 1" silt seam From El. -51.3 to -53.3 Ft., trace organic matter At El. -52.8 Ft., occasional silty fine sand seams, N 3/ very dark gray	1450	1		-48.3		
							-49.3		
					2		-51.3		
					3		-53.3		
-53.9	5.6		SILT, inorganic-H, few fine-grained sand-sized quartz, no reaction with HCl, N 2.5/ black (MH)						
-54.8	6.5								
-55.4	7.1		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, no reaction with HCl, 5Y 5/2 olive gray (SP-SM) From El. -55.2 to -58.1 Ft., clay seams SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace clay, no reaction with HCl, 5Y 6/1 gray (SP) From El. -58.8 to -59.0 Ft., clay seams From El. -60.3 to -60.5 Ft., clay seams From El. -61.3 to -61.4 Ft., clay seams						
-62.8	14.5		No Recovery						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																					
PROJECT Jacksonville Harbor O&M			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																					
LOCATION COORDINATES X = 529,679 Y = 2,204,476			ELEVATION TOP OF BORING -48.3 Ft.																								
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
-65.6	17.3						-65.6																				
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/1.3</td> <td>SP-SM*</td> </tr> <tr> <td>2</td> <td>3.0/3.3</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>5.0/5.3</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> <tr> <td>3</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.3	SP-SM*	2	3.0/3.3	SP-SM*	3	5.0/5.3	SP-SM*	1	Percent Visual Shell	2	Percent Visual Shell	3	Percent Visual Shell				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	1.0/1.3	SP-SM*																									
2	3.0/3.3	SP-SM*																									
3	5.0/5.3	SP-SM*																									
1	Percent Visual Shell																										
2	Percent Visual Shell																										
3	Percent Visual Shell																										

Boring Designation VB-MPB2014M-4

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 1 SHEETS		
1. PROJECT Jacksonville Harbor O&M				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPB2014M-4		LOCATION COORDINATES X = 529,109 Y = 2,204,021		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ		CONTRACTOR FILE NO. 6738-15-5400		11. MANUFACTURER'S DESIGNATION OF DRILL			☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 0		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 0	
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 10-31-14		COMPLETED 10-31-14	
8. TOTAL DEPTH OF BORING 7.5 Ft.				16. ELEVATION TOP OF BORING -46.8 Ft.					
				17. TOTAL RECOVERY FOR BORING 83 %					
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-46.8	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, no reaction with HCl, N 2.5/ black (MH) At El. -47.7 Ft., some fine-grained sand-sized quartz, weak reaction with HCl, 5Y 4/2 olive gray At El. -47.8 Ft., few fine-grained sand-sized quartz, no reaction with HCl, N 3/ very dark gray	83			-46.8		
-52.0	5.2		CLAY, fat, little fine-grained sand-sized quartz, no reaction with HCl, N 3/ very dark gray (CH)						
-52.9	6.1								
-53.0	6.2		SILT, organic-L, no reaction with HCl, N 3/ very dark gray (OL) No Recovery						
-54.3	7.5		At El. -54.3 Ft. Tight, hard clay/silt at base. BORING TERMINATED IN REFUSAL NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.				-54.3		

Boring Designation VB-MPB2014M-5

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 1 SHEETS	
1. PROJECT Jacksonville Harbor O&M				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPB2014M-5		LOCATION COORDINATES X = 533,297 Y = 2,205,009		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ		CONTRACTOR FILE NO. 6738-15-5400		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 3	UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER				
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 11-02-14	COMPLETED 11-02-14	
8. TOTAL DEPTH OF BORING 6.8 Ft.				16. ELEVATION TOP OF BORING		-48.3 Ft.		
				17. TOTAL RECOVERY FOR BORING		78 %		
				18. SIGNATURE AND TITLE OF INSPECTOR				

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RCD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE																		
-48.3	0.0						-48.3																				
			SAND, poorly-graded, mostly sand to gravel-sized shell, some fine to medium-grained sand-sized quartz, trace silt, weak reaction with HCl, 5Y 6/2 light olive gray (SP)	530			-49.3																				
					1																						
-50.9	2.6						-51.3																				
			SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, trace shell, weak reaction with HCl, 5Y 4/2 olive gray (SP-SM)		2		-52.8																				
			At El. -52.8 Ft., discontinue shell		3																						
-53.6	5.3		No Recovery																								
-55.1	6.8						-55.1																				
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> <tr> <td>1</td> <td>1.0/1.3</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>3.0/3.3</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>4.5/4.8</td> <td>SP-SM*</td> </tr> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table style="width:100%; border-collapse: collapse;"> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> <tr> <td>3</td> <td>Percent Visual Shell</td> </tr> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/1.3	SP*	2	3.0/3.3	SP-SM*	3	4.5/4.8	SP-SM*	1	Percent Visual Shell	2	Percent Visual Shell	3	Percent Visual Shell				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	1.0/1.3	SP*																									
2	3.0/3.3	SP-SM*																									
3	4.5/4.8	SP-SM*																									
1	Percent Visual Shell																										
2	Percent Visual Shell																										
3	Percent Visual Shell																										

Boring Designation VB-MP13-1

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MP13-1				LOCATION COORDINATES X = 525,631 Y = 2,203,593		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAJ				CONTRACTOR FILE NO. 6738-13-5275		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 2
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		15. DATE BORING		STARTED 06-18-13
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -37.8 Ft.		COMPLETED 06-18-13		
8. TOTAL DEPTH OF BORING 12.0 Ft.				17. TOTAL RECOVERY FOR BORING N/A		18. SIGNATURE AND TITLE OF INSPECTOR		

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-37.8	0.0		SILT, inorganic-H, trace quartz, weak reaction with HCl, moist, N 4/ dark gray (MH)				-37.8		
					1		-40.8		
					2		-43.8		
			At El. -45.7 Ft., few quartz, trace shell						
			At El. -46.5 Ft., trace quartz, discontinue shell						
-47.2	9.4								
-47.6	9.8		SAND, poorly-graded, mostly fine to coarse-grained sand-sized quartz, little fine-grained sand-sized quartz, trace silt, trace limestone, strong reaction with HCl, moist, 10Y 7/1 light greenish gray (SP)		3		-47.8		
-47.8	10.0		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, little fine to coarse-grained sand-sized shell, 2.5Y 6/1 gray (SP)						
-49.8	12.0		SAND, poorly-graded with silt, mostly fine to coarse-grained sand-sized shell, some fine-grained sand-sized quartz, few silt, strong reaction with HCl, moist, 2.5Y 6/1 gray (SP-SM)				-49.8		
			At El. -48.5 Ft., 1.5" limestone piece				Abbreviations:		
			NOTES:						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																					
PROJECT Mayport Vibracoring 2013			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																					
LOCATION COORDINATES X = 525,631 Y = 2,203,593			ELEVATION TOP OF BORING -37.8 Ft.																								
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
			1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>3.0/3.2</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>6.0/6.2</td> <td>MH*</td> </tr> <tr> <td>3</td> <td>10.0/10.2</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> <tr> <td>3</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	3.0/3.2	MH*	2	6.0/6.2	MH*	3	10.0/10.2	SP-SM*	1	Percent Visual Shell	2	Percent Visual Shell	3	Percent Visual Shell						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	3.0/3.2	MH*																									
2	6.0/6.2	MH*																									
3	10.0/10.2	SP-SM*																									
1	Percent Visual Shell																										
2	Percent Visual Shell																										
3	Percent Visual Shell																										

Boring Designation VB-MP13-2

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS											
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks														
2. BORING DESIGNATION VB-MP13-2				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW										
3. DRILLING AGENCY Corps of Engineers - CESAJ				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER												
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0										
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0												
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER														
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-18-13		COMPLETED 06-18-13										
8. TOTAL DEPTH OF BORING 9.0 Ft.				16. ELEVATION TOP OF BORING		-44.9 Ft.												
				17. TOTAL RECOVERY FOR BORING		N/A												
				18. SIGNATURE AND TITLE OF INSPECTOR														
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE									
-44.9	0.0						-44.9											
-45.9	1.0		SILT, inorganic-H, trace fine-grained sand-sized quartz, weak reaction with HCl, moist, N 4/ dark gray (MH)		1		-45.4											
-46.9	2.0		SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, few sand to gravel-sized shell, trace silt, weak reaction with HCl, moist, 10YR 5/1 gray (SP)		2		-46.9											
-48.6	3.7		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few coarse-grained sand-sized shell, few silt, weak reaction with HCl, moist, 10YR 5/1 gray (SP-SM)		3		-48.9											
-49.7	4.8		SILT, inorganic-H, trace fine-grained sand-sized quartz, weak reaction with HCl, moist, N 4/ dark gray (MH)		4		-50.9											
-49.8	4.9		SAND, poorly-graded with silt, some sand to gravel-sized limestone, little fine-grained sand-sized quartz, few fine to coarse-grained sand-sized shell, few silt, strong reaction with HCl, moist, 2.5Y 8/2 pale yellow (SP-SM)															
-50.2	5.3		LIMESTONE, soft, slightly weathered, medium grained, intensely fractured, little fine-grained sand-sized quartz, trace silt, 2.5Y 7/4 pale yellow															
-50.9	6.0		SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, trace shell, weak reaction with HCl, moist, 2.5Y 5/4 light olive brown (SP-SM)															
-53.9	9.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, no reaction with HCl, moist, 2.5Y 4/3 olive brown (SM)				-53.9											
			Abbreviations: NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/0.7</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>2.0/2.2</td> <td>SP-SM*</td> </tr> </tbody> </table>							SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/0.7	MH*	2	2.0/2.2	SP-SM*
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																
1	0.5/0.7	MH*																
2	2.0/2.2	SP-SM*																

DRILLING LOG (Cont. Sheet)				INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS	
PROJECT Mayport Vibracoring 2013				COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
LOCATION COORDINATES X = 526,278 Y = 2,203,678				ELEVATION TOP OF BORING -44.9 Ft.					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			3 4.0/4.2 MH*						
			4 6.0/6.2 SM*						
			*Lab visual classification based on gradation curve						
			4. Additional Laboratory Testing						
			1 Percent Visual Shell						
			2 Percent Visual Shell						
			3 Percent Visual Shell						
			4 Percent Visual Shell						

Boring Designation VB-MP13-3

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MP13-3				LOCATION COORDINATES X = 526,398 Y = 2,204,433		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAJ				CONTRACTOR FILE NO. 6738-13-5275		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 0
6. THICKNESS OF OVERBURDEN N/A				7. DEPTH DRILLED INTO ROCK N/A		14. ELEVATION GROUND WATER		15. DATE BORING 06-18-13
8. TOTAL DEPTH OF BORING 8.2 Ft.				16. ELEVATION TOP OF BORING -44.4 Ft.		17. TOTAL RECOVERY FOR BORING N/A		18. SIGNATURE AND TITLE OF INSPECTOR

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE									
-44.4	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, no reaction with HCl, moist, N 4/ dark gray (MH)				-44.4											
					1		-47.4											
					2		-50.4											
-52.6	8.2		From El. -52.1 to -52.3 Ft., little silt, fine to medium sand seam, 2.5Y 3/2 very dark grayish brown				-52.6											
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">SAMPLE ID</td> <td style="text-align: center;">SAMPLE DEPTH</td> <td style="text-align: center;">LABORATORY CLASSIFICATION</td> </tr> <tr> <td style="text-align: center;">1</td> <td style="text-align: center;">3.0/3.2</td> <td style="text-align: center;">MH*</td> </tr> <tr> <td style="text-align: center;">2</td> <td style="text-align: center;">6.0/6.2</td> <td style="text-align: center;">MH*</td> </tr> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	3.0/3.2	MH*	2	6.0/6.2	MH*				Abbreviations:		
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																
1	3.0/3.2	MH*																
2	6.0/6.2	MH*																

Boring Designation VB-MP13-3

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS		
PROJECT Mayport Vibracoring 2013			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW			
LOCATION COORDINATES X = 526,398 Y = 2,204,433			ELEVATION TOP OF BORING -44.4 Ft.						
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			4. Additional Laboratory Testing 1 Percent Visual Shell 2 Percent Visual Shell						

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Boring Designation VB-MP13-4

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP13-4				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ				11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-18-13		COMPLETED 06-18-13	
8. TOTAL DEPTH OF BORING 10.1 Ft.				16. ELEVATION TOP OF BORING -50.5 Ft.					
				17. TOTAL RECOVERY FOR BORING N/A					
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-50.5	0.0		SILT, inorganic-H, few fine-grained sand-sized quartz, weak reaction with HCl, moist, N 4/ dark gray (MH)				-50.5		
			At El. -52.5 Ft., some fine-grained sand-sized quartz		1		-52.5		
-54.7	4.2		At El. -54.4 Ft., little fine-grained sand-sized quartz, trace shell, 2.5Y 3/1 very dark gray SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, trace silt, trace shell, weak reaction with HCl, moist, 2.5Y 3/1 very dark gray (SP)		2		-55.5		
			At El. -58.5 Ft., 2.5Y 3/2 very dark grayish brown						
-60.6	10.1		At El. -60.3 Ft., some fine to medium-grained sand-sized quartz, little fine-grained sand-sized limestone, little sand to gravel-sized shell, strong reaction with HCl, 5Y 6/3 pale olive				-60.6		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results				Abbreviations:		
			SAMPLE SAMPLE LABORATORY						

Boring Designation VB-MP13-4

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS															
			PROJECT Mayport Vibracoring 2013		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW														
LOCATION COORDINATES X = 528,806 Y = 2,203,966			ELEVATION TOP OF BORING -50.5 Ft.																			
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE													
			<table style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 10%;">ID</th> <th style="width: 10%;">DEPTH</th> <th style="width: 80%;">CLASSIFICATION</th> </tr> <tr> <td>1</td> <td>2.0/2.2</td> <td>MH*</td> </tr> <tr> <td>2</td> <td>5.0/5.2</td> <td>SP*</td> </tr> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%;">1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> </table>	ID	DEPTH	CLASSIFICATION	1	2.0/2.2	MH*	2	5.0/5.2	SP*	1	Percent Visual Shell	2	Percent Visual Shell						
ID	DEPTH	CLASSIFICATION																				
1	2.0/2.2	MH*																				
2	5.0/5.2	SP*																				
1	Percent Visual Shell																					
2	Percent Visual Shell																					

Boring Designation VB-MP13-5

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP13-5		LOCATION COORDINATES X = 535,015 Y = 2,205,088		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ		CONTRACTOR FILE NO. 6738-13-5275		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		0			
6. THICKNESS OF OVERBURDEN		N/A		14. ELEVATION GROUND WATER					
7. DEPTH DRILLED INTO ROCK		N/A		15. DATE BORING		STARTED 06-18-13		COMPLETED 06-18-13	
8. TOTAL DEPTH OF BORING 10.0 Ft.				16. ELEVATION TOP OF BORING		-49.6 Ft.			
				17. TOTAL RECOVERY FOR BORING		N/A			
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-49.6	0.0						-49.6		
-50.6	1.0		SAND, poorly-graded, mostly shell, strong reaction with HCl, moist, 2.5Y 5/3 light olive brown (SP)						
			SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, some sand to gravel-sized shell, trace silt, 2.5Y 5/2 grayish brown (SP)		1		-51.6		
-53.8	4.2								
			SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, few medium-grained sand-sized shell, weak reaction with HCl, moist, N 5/ gray (SP-SM)		2		-54.6		
			At El. -57.0 Ft., trace limestone						
-59.6	10.0						-59.6		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results SAMPLE ID SAMPLE DEPTH LABORATORY CLASSIFICATION ----- 1 2.0/2.2 SP*				Abbreviations:		

Boring Designation VB-MP13-5

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS		
PROJECT Mayport Vibracoring 2013			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW			
LOCATION COORDINATES X = 535,015 Y = 2,205,088			ELEVATION TOP OF BORING -49.6 Ft.						
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			2 5.0/5.2 SP-SM* *Lab visual classification based on gradation curve 4. Additional Laboratory Testing 1 Percent Visual Shell 2 Percent Visual Shell						

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Boring Designation VB-MP13-6

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP13-6				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ				LOCATION COORDINATES X = 534,612 Y = 2,205,642		11. MANUFACTURER'S DESIGNATION OF DRILL ☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES 2		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING			
6. THICKNESS OF OVERBURDEN N/A				13. TOTAL NUMBER CORE BOXES 0		14. ELEVATION GROUND WATER			
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING 06-18-13		STARTED 06-18-13		COMPLETED 06-18-13	
8. TOTAL DEPTH OF BORING 8.0 Ft.				16. ELEVATION TOP OF BORING -49.1 Ft.		17. TOTAL RECOVERY FOR BORING N/A			
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-49.1	0.0		SAND, poorly-graded, mostly fine to coarse-grained sand-sized quartz, little sand to gravel-sized shell up to 1/2", trace silt, strong reaction with HCl, moist, 10YR 6/1 gray (SP)				-49.1		
			At El. -51.7 Ft., some shell At El. -52.1 Ft., little shell		1		-51.1		
-53.0	3.9		At El. -52.7 Ft., some shell						
-53.2	4.1		SAND, silty, some fine to coarse-grained sand-sized quartz, little silt, little sand to gravel-sized shell up to 1/2", strong reaction with HCl, moist, 5Y 5/1 gray (SM)		2		-54.1		
			CLAY, fat, few fine-grained sand-sized quartz, no reaction with HCl, moist, 10Y 4/1 dark greenish gray (CH)						
-57.1	8.0		At El. -54.1 Ft., trace fine-grained sand-sized quartz				-57.1		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results SAMPLE ID SAMPLE DEPTH LABORATORY CLASSIFICATION ----- 1 2.0/2.2 SP* 2 5.0/5.2 CH* *Lab visual classification based on gradation curve 4. Additional Laboratory Testing				Abbreviations:		

Boring Designation VB-MP13-6

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS		
PROJECT Mayport Vibracoring 2013			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW			
LOCATION COORDINATES X = 534,612 Y = 2,205,642			ELEVATION TOP OF BORING -49.1 Ft.						
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			1 Percent Visual Shell 2 Percent Visual Shell						

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Boring Designation VB-MP13-7

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP13-7		LOCATION COORDINATES X = 542,969 Y = 2,204,844		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ		CONTRACTOR FILE NO. 6738-13-5275		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING			
6. THICKNESS OF OVERBURDEN N/A				13. TOTAL NUMBER CORE BOXES 0		14. ELEVATION GROUND WATER			
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-18-13		COMPLETED 06-18-13	
8. TOTAL DEPTH OF BORING 11.8 Ft.				16. ELEVATION TOP OF BORING -49.4 Ft.		17. TOTAL RECOVERY FOR BORING N/A			
				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-49.4	0.0						-49.4		
-49.6	0.2		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, weak reaction with HCl, moist, 5Y 7/3 pale yellow (SP)		1		-49.9		
-50.5	1.1		SAND, silty, mostly fine-grained sand-sized quartz, little silt, weak reaction with HCl, moist, 5Y 5/2 olive gray (SM)						
-50.8	1.4		At El. -50.4 Ft., some silt						
			SILT, inorganic-H, little fine-grained sand-sized quartz, weak reaction with HCl, moist, 5Y 4/2 olive gray (MH)				-52.4		
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, weak reaction with HCl, moist, 5Y 7/2 light gray (SP)		2				
			At El. -52.4 Ft., little fine-grained sand-sized shell						
-54.6	5.2		SAND, silty, mostly fine-grained sand-sized quartz, little silt, no reaction with HCl, moist, 2.5" silt seam, 5Y 5/2 olive gray (SM)						
			At El. -55.6 Ft., some silt, 3.0" silt seam						
-57.4	8.0		SILT, inorganic-H, little fine-grained sand-sized quartz, weak reaction with HCl, moist, 5Y 3/2 dark olive gray (MH)						
-57.6	8.2		SAND, silty, mostly fine-grained sand-sized quartz, little silt, weak reaction with HCl, moist, 5Y 4/1 dark gray (SM)						
			At El. -59.2 Ft., some silt, trace shell, 10Y 4/1 dark greenish gray						
-61.2	11.8						-61.2		
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System.				Abbreviations:					

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																
PROJECT Mayport Vibracoring 2013			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																
LOCATION COORDINATES X = 542,969 Y = 2,204,844			ELEVATION TOP OF BORING -49.4 Ft.																			
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE													
			3. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/0.7</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>3.0/3.2</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve 4. Additional Laboratory Testing <table border="1"> <tbody> <tr> <td>1</td> <td>Percent Visual Shell</td> </tr> <tr> <td>2</td> <td>Percent Visual Shell</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/0.7	SM*	2	3.0/3.2	SP*	1	Percent Visual Shell	2	Percent Visual Shell						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																				
1	0.5/0.7	SM*																				
2	3.0/3.2	SP*																				
1	Percent Visual Shell																					
2	Percent Visual Shell																					

Boring Designation VB-MP13-8

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MP13-8				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Corps of Engineers - CESAJ				LOCATION COORDINATES X = 545,468 Y = 2,204,540		11. MANUFACTURER'S DESIGNATION OF DRILL ☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 0		14. ELEVATION GROUND WATER			
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 06-18-13		COMPLETED 06-18-13	
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -50.1 Ft.		17. TOTAL RECOVERY FOR BORING N/A			
8. TOTAL DEPTH OF BORING 10.6 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR					
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-50.1	0.0		SILT, inorganic-H, little fine-grained sand-sized quartz, weak reaction with HCl, moist, 5Y 4/3 olive (MH) At El. -50.3 Ft., little silt, 0.5" sand seam At El. -50.8 Ft., 1.0" sand seam SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace shell, weak reaction with HCl, moist, 3.0" sand seam, 5Y 4/3 olive (SM) At El. -52.3 Ft., 5Y 4/2 olive gray At El. -53.1 Ft., some silt, few medium-grained sand-sized shell, 5Y 4/1 dark gray				-50.1		
-51.1	1.0						-51.1		
				1			-53.1		
				2					
-55.0	4.9		CLAY, fat, few fine-grained sand-sized quartz, weak reaction with HCl, moist, 10Y 4/1 dark greenish gray (CH)						
-60.7	10.6						-60.7		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results SAMPLE ID SAMPLE DEPTH LABORATORY CLASSIFICATION				Abbreviations:		

Boring Designation VB-MP13-8

DRILLING LOG (Cont. Sheet)			INSTALLATION				SHEET 2		
			Jacksonville District				OF 2 SHEETS		
PROJECT			COORDINATE SYSTEM/DATUM		HORIZONTAL	VERTICAL			
Mayport Vibracoring 2013			State Plane, FLE (U.S. Ft.)		NAD83	MLLW			
LOCATION COORDINATES			ELEVATION TOP OF BORING						
X = 545,468 Y = 2,204,540			-50.1 Ft.						
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			1 1.0/1.2 SM* 2 3.0/3.2 SM* *Lab visual classification based on gradation curve 4. Additional Laboratory Testing 1 Percent Visual Shell 2 Percent Visual Shell						

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Boring Designation VB-MP13-9

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT Mayport Vibracoring 2013 Logging and Lab Analysis				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MP13-9				LOCATION COORDINATES X = 548,323 Y = 2,203,510		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Corps of Engineers - CESAJ				CONTRACTOR FILE NO. 6738-13-5275		11. MANUFACTURER'S DESIGNATION OF DRILL		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 0
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		15. DATE BORING		STARTED 06-18-13
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -49.8 Ft.		COMPLETED 06-18-13		
8. TOTAL DEPTH OF BORING 11.0 Ft.				17. TOTAL RECOVERY FOR BORING N/A		18. SIGNATURE AND TITLE OF INSPECTOR		

ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/1 FT.	N-VALUE
-49.8	0.0						-49.8		
-50.2	0.4		SAND, silty, mostly fine-grained sand-sized quartz, some silt, weak reaction with HCl, moist, 5Y 5/2 olive gray (SM)				-50.8		
			SILT, inorganic-L, some fine-grained sand-sized quartz, trace shell, weak reaction with HCl, moist, N 4/ dark gray (ML)		1				
			At El. -51.9 Ft., 1.5" sand seam, 5Y 5/2 olive gray						
-52.8	3.0		At El. -52.7 Ft., 5" sand seam		2		-52.8		
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace shell, weak reaction with HCl, moist, N 4/ dark gray (SM)						
			At El. -57.4 Ft., some silt, trace shell, 10Y 4/1 dark greenish gray						
-60.8	11.0						-60.8		
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Laboratory Testing Results SAMPLE SAMPLE LABORATORY							Abbreviations:		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS				
PROJECT Mayport Vibracoring 2013			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW				
LOCATION COORDINATES X = 548,323 Y = 2,203,510			ELEVATION TOP OF BORING -49.8 Ft.							
ELEV.	DEPTH	LEGEND	CLASSIFICATION OF MATERIALS		% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
			ID	DEPTH	CLASSIFICATION					
			1	1.0/1.2	ML*					
			2	3.0/3.2 /1.0	SM* *					
			*Lab visual classification based on gradation curve							
			4. Additional Laboratory Testing							
			1 Percent Visual Shell							
			2 Percent Visual Shell							

Boring Designation VB-MPH08-1

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-1		LOCATION COORDINATES X = 525,630 Y = 2,204,943		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 4	UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		1		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		Tidal		
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-10-08	COMPLETED 06-10-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-45.0 Ft.		
				17. TOTAL RECOVERY FOR BORING		100 %		
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-45.0	0.0		SAND, silty, mostly medium-grained sand-sized quartz, little silt, few angular sand to gravel-sized shell up to 1", wet, 2.5Y 5/2 grayish brown (SM) At El. -47.0 Ft., discontinue shell, 2.5Y 4/2 dark grayish brown	100			-45.0		
				100	1		-46.0 Vibracore		
							-46.5 Vibracore		
				100			Vibracore		
							-49.0		
				100	2		-49.5 Vibracore		
-51.0	6.0		SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, few silt, wet, 2.5Y 3/2 very dark grayish brown (SP-SM)	100			Vibracore		
							-52.0		
				100	3		-52.5 Vibracore		
							Vibracore		
							-56.5		
				100	4		-57.0 Vibracore		
				100			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																		
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																		
LOCATION COORDINATES X = 525,630 Y = 2,204,943			ELEVATION TOP OF BORING -45.0 Ft.																					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE															
-62.0	17.0																							
-63.0	18.0		SAND, clayey, mostly medium-grained sand-sized quartz, wet, 2.5Y 4/1 dark gray (SC)	100			Vibracore																	
-65.0	20.0		SILT, inorganic-L, low plasticity, firm, some fine-grained sand-sized quartz, trace clay, wet, 2.5Y 6/2 light brownish gray (ML)																					
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>7.0/20.0</td> <td>SP*</td> </tr> <tr> <td>4</td> <td>11.5/20.0</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	4.0/20.0	SM*	3	7.0/20.0	SP*	4	11.5/20.0	SM*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																						
1	1.0/20.0	SM*																						
2	4.0/20.0	SM*																						
3	7.0/20.0	SP*																						
4	11.5/20.0	SM*																						

Boring Designation VB-MPH08-2

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-2		LOCATION COORDINATES X = 525,443 Y = 2,204,136		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES 5		DISTURBED 5		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal		
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING 06-10-08		STARTED 06-10-08		COMPLETED 06-10-08
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -44.2 Ft.		17. TOTAL RECOVERY FOR BORING 100 %		
8. TOTAL DEPTH OF BORING 16.8 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

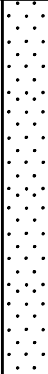
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-44.2	0.0						-44.2		
			SILT, inorganic-L, low plasticity, very soft, some fine-grained sand-sized quartz, wet, 2.5Y 2.5/1 black (ML)	100			Vibracore		
				100	1		-45.7		
							-46.2		
-47.2	3.0		At El. -47.0 Ft., trace angular sand to gravel-sized shell up to 2"	100			Vibracore		
			SAND, silty, mostly medium-grained sand-sized quartz, little silt, trace angular sand to gravel-sized shell up to 1/4", wet, 2.5Y 6/2 light brownish gray (SM)	100	2		-48.2		
				100			-48.7		
				100			Vibracore		
				100	3		-50.2		
							-50.7		
-51.2	7.0			100			Vibracore		
				100	4		-53.2		
							-53.7		
				100			Vibracore		
				100	5		-56.2		
							-56.7		
				100			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																												
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																												
LOCATION COORDINATES X = 525,443 Y = 2,204,136			ELEVATION TOP OF BORING -44.2 Ft.																															
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																									
-61.0	16.8			100			Vibracore																											
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.5/16.8</td> <td>ML</td> </tr> <tr> <td>2</td> <td>4.0/16.8</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>6.0/16.8</td> <td>SM*</td> </tr> <tr> <td>4</td> <td>9.0/16.8</td> <td>SP-SM*</td> </tr> <tr> <td>5</td> <td>12.0/16.8</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>1.5</td> <td>Atterberg (PI)</td> <td>50</td> </tr> <tr> <td>1.5</td> <td>Specific Gravity</td> <td>2.35</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.5/16.8	ML	2	4.0/16.8	SM*	3	6.0/16.8	SM*	4	9.0/16.8	SP-SM*	5	12.0/16.8	SP-SM*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	1.5	Atterberg (PI)	50	1.5	Specific Gravity	2.35					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																																
1	1.5/16.8	ML																																
2	4.0/16.8	SM*																																
3	6.0/16.8	SM*																																
4	9.0/16.8	SP-SM*																																
5	12.0/16.8	SP-SM*																																
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																																
1.5	Atterberg (PI)	50																																
1.5	Specific Gravity	2.35																																

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-3		LOCATION COORDINATES X = 525,105 Y = 2,203,200		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW		
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5	UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal			
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 06-10-08	COMPLETED 06-10-08		
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -42.4 Ft.		17. TOTAL RECOVERY FOR BORING 95 %			
8. TOTAL DEPTH OF BORING 19.7 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-42.4	0.0						-42.4		
-43.4	1.0		SILT, inorganic-L, low plasticity, very soft, little fine-grained sand-sized quartz, wet, 2.5Y 2.5/1 black (ML)	100			-42.9 Vibracore		
		100		1	-43.4 Vibracore				
			SHELL, mostly angular sand to gravel-sized shell up to 2", little fine-grained sand-sized quartz, strong reaction with HCl, wet, 2.5Y 6/1 gray	100			Vibracore		
						-44.4			
				2	-44.9 Vibracore				
					Vibracore				
								-49.4	
	3	-49.9 Vibracore							
					Vibracore				
-51.4	9.0			100			-51.4		
-53.4	11.0		SAND, silty, mostly medium-grained sand-sized quartz, wet, 2.5Y 5/2 grayish brown (SM)	100	4		-51.9 Vibracore		
							Vibracore		
		SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, wet, 2.5Y 5/2 grayish brown (SP-SM)	100			-55.4			
			100	5	-55.9 Vibracore				
			85		Vibracore				

Boring Designation VB-MPH08-3

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																					
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																					
LOCATION COORDINATES X = 525,105 Y = 2,203,200			ELEVATION TOP OF BORING -42.4 Ft.																								
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
-61.2	18.8	NR		85			Vibracore																				
-62.1	19.7					-62.1																					
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/19.7</td> <td>ML*</td> </tr> <tr> <td>2</td> <td>2.0/19.7</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>7.0/19.7</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>9.0/19.7</td> <td>SM*</td> </tr> <tr> <td>5</td> <td>13.0/19.7</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/19.7	ML*	2	2.0/19.7	SP-SM*	3	7.0/19.7	SP-SM*	4	9.0/19.7	SM*	5	13.0/19.7	SP-SM*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	0.5/19.7	ML*																									
2	2.0/19.7	SP-SM*																									
3	7.0/19.7	SP-SM*																									
4	9.0/19.7	SM*																									
5	13.0/19.7	SP-SM*																									

DRILLING LOG				DIVISION South Atlantic		INSTALLATION Jacksonville District				SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin						9. SIZE AND TYPE OF BIT See Remarks						
						10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW		
2. BORING DESIGNATION VB-MPH08-6			LOCATION COORDINATES X = 526,513 Y = 2,204,686			11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit				☐ AUTO HAMMER ☐ MANUAL HAMMER		
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.				CONTRACTOR FILE NO. 2007D16		12. TOTAL SAMPLES 3		DISTURBED 3		UNDISTURBED (UD) 0		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.						13. TOTAL NUMBER CORE BOXES 1						
						14. ELEVATION GROUND WATER Tidal						
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED			DEG. FROM VERTICAL		BEARING		15. DATE BORING 06-10-08		STARTED 06-10-08		COMPLETED 06-10-08	
6. THICKNESS OF OVERBURDEN N/A				7. DEPTH DRILLED INTO ROCK N/A		16. ELEVATION TOP OF BORING -45.0 Ft.						
8. TOTAL DEPTH OF BORING 20.0 Ft.						17. TOTAL RECOVERY FOR BORING 100 %						
						18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist						
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS		% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS		BLOWS/ 1 FT.	N-VALUE	
-45.0	0.0							-45.0				
			CLAY, fat, high plasticity, very soft, wet, 5Y 3/2 dark olive gray (CH)		100			Vibracore				
					100	1	-48.0 -48.5					
					100		Vibracore					
-51.0	6.0		SAND, silty, mostly fine to medium-grained sand-sized quartz, few silt, wet, 5Y 4/2 olive gray (SM)		100	2		-51.0 -51.5				
					100		Vibracore					
					100		Vibracore					
-55.0	10.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 4/2 olive gray (SP)		100	3		-56.0 -56.5				
					100		Vibracore					
					100		Vibracore					

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																							
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																							
LOCATION COORDINATES X = 526,513 Y = 2,204,686			ELEVATION TOP OF BORING -45.0 Ft.																										
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																				
-63.0	18.0	[Symbol]		100			Vibracore																						
-65.0	20.0	[Symbol]	SAND, silty, mostly fine-grained sand-sized quartz, little silt, wet, 5Y 7/2 light gray (SM) At El. -64.0 Ft., few coarse-grained sand-sized quartz				-65.0																						
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">SAMPLE ID</th> <th style="text-align: center;">SAMPLE DEPTH</th> <th style="text-align: center;">LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">1</td> <td style="text-align: center;">3.0/20.0</td> <td style="text-align: center;">CH</td> </tr> <tr> <td style="text-align: center;">2</td> <td style="text-align: center;">6.0/20.0</td> <td style="text-align: center;">SM*</td> </tr> <tr> <td style="text-align: center;">3</td> <td style="text-align: center;">11.0/20.0</td> <td style="text-align: center;">SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">SAMPLE DEPTH</th> <th style="text-align: center;">LABORATORY SOIL TESTING</th> <th style="text-align: center;">RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">3.0</td> <td style="text-align: center;">Atterberg (PI)</td> <td style="text-align: center;">44</td> </tr> <tr> <td style="text-align: center;">3.0</td> <td style="text-align: center;">Specific Gravity</td> <td style="text-align: center;">2.76</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	3.0/20.0	CH	2	6.0/20.0	SM*	3	11.0/20.0	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	3.0	Atterberg (PI)	44	3.0	Specific Gravity	2.76					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																											
1	3.0/20.0	CH																											
2	6.0/20.0	SM*																											
3	11.0/20.0	SP*																											
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																											
3.0	Atterberg (PI)	44																											
3.0	Specific Gravity	2.76																											

Boring Designation VB-MPH08-8

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-8		LOCATION COORDINATES X = 526,009 Y = 2,202,959		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1					
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-10-08		COMPLETED 06-10-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -38.9 Ft.					
				17. TOTAL RECOVERY FOR BORING 80 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-38.9	0.0		CLAY, fat, high plasticity, very soft, wet, 5Y 3/2 dark olive gray (CH)				-38.9		
				100			Vibracore		
-42.9	4.0			100	1		-42.4 -42.9 Vibracore		
-43.4	4.5		CLAY, lean, medium plasticity, soft, some fine-grained sand-sized quartz, wet, 5Y 5/3 olive (CL) CLAY, fat, high plasticity, very soft, wet, 5Y 3/2 dark olive gray (CH)	100			Vibracore		
							-45.9 -46.4 Vibracore		
				100	2		Vibracore		
-48.4	9.5			100			Vibracore		
				100	3		-48.9 -49.4 Vibracore		
			SAND, poorly-graded with silt, mostly medium to coarse-grained sand-sized quartz, some angular sand to gravel-sized shell up to 1", wet, 2.5Y 8/4 pale yellow (SP-SM)				Vibracore		
				100			Vibracore		
				100	4		-51.9 -52.4 Vibracore		
				38			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																	
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																	
LOCATION COORDINATES X = 526,009 Y = 2,202,959			ELEVATION TOP OF BORING -38.9 Ft.																				
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE														
-54.9	16.0	NO RECOVERY		38			Vibracore																
-58.9	20.0										-58.9												
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>3.5/20.0</td> <td>CH*</td> </tr> <tr> <td>2</td> <td>7.0/20.0</td> <td>CH*</td> </tr> <tr> <td>3</td> <td>10.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>13.0/20.0</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	3.5/20.0	CH*	2	7.0/20.0	CH*	3	10.0/20.0	SP-SM*	4	13.0/20.0	SP-SM*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																					
1	3.5/20.0	CH*																					
2	7.0/20.0	CH*																					
3	10.0/20.0	SP-SM*																					
4	13.0/20.0	SP-SM*																					

Boring Designation VB-MPH08-13

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-13		LOCATION COORDINATES X = 527,020 Y = 2,202,950		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 4	UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal		
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 06-10-08	COMPLETED 06-10-08	
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -44.7 Ft.		17. TOTAL RECOVERY FOR BORING 100 %		
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-44.7	0.0						-44.7		
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, few angular sand to gravel-sized shell up to 1/4", wet, 2.5Y 7/3 pale yellow (SM)	100			Vibracore		
-46.2	1.5			100	1		-45.7 -46.2		
			SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, few silt, wet, 2.5Y 3/2 very dark grayish brown (SP-SM)	100			Vibracore		
				100	2		-48.7 -49.2		
-50.7	6.0			100			Vibracore		
			SAND, poorly-graded, mostly medium-grained sand-sized quartz, trace silt, wet, 2.5Y 3/2 very dark grayish brown (SP)	100	3		-51.7 -52.2		
				100			Vibracore		
				100	4		-56.7 -57.2		
				100			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																	
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																	
LOCATION COORDINATES X = 527,020 Y = 2,202,950			ELEVATION TOP OF BORING -44.7 Ft.																				
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE														
-62.7	18.0			100			Vibracore																
-64.7	20.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace angular sand to gravel-sized shell up to 1/4", wet, 5Y 5/2 olive gray (SM)				-64.7																
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>7.0/20.0</td> <td>SP*</td> </tr> <tr> <td>4</td> <td>12.0/20.0</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	4.0/20.0	SP-SM*	3	7.0/20.0	SP*	4	12.0/20.0	SP*					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																					
1	1.0/20.0	SM*																					
2	4.0/20.0	SP-SM*																					
3	7.0/20.0	SP*																					
4	12.0/20.0	SP*																					

Boring Designation VB-MPH08-14

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Turning Basin				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-14				10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW		
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.				CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5	UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL	BEARING	13. TOTAL NUMBER CORE BOXES 1			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING STARTED 06-10-08 COMPLETED 06-10-08					
8. TOTAL DEPTH OF BORING 19.0 Ft.				16. ELEVATION TOP OF BORING -42.5 Ft.					
				17. TOTAL RECOVERY FOR BORING 100 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-42.5	0.0		CLAY, fat, high plasticity, very soft, wet, 5Y 3/2 dark olive gray (CH)	100			-42.5		0
							Vibracore		
							-44.5		
				100	1		-45.0	Vibracore	
-45.5	3.0		SHELL, mostly angular sand to gravel-sized shell up to 2", some medium-grained sand-sized quartz, trace silt, wet, 2.5Y 8/2 pale yellow	100					
							Vibracore		
							-47.5		
				100	2		-48.0	Vibracore	
-49.5	7.0		SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, few silt, wet, 2.5Y 6/6 olive yellow (SP-SM)	100					
							Vibracore		
							-50.5		
-51.0	8.5		SAND, poorly-graded, mostly medium-grained sand-sized quartz, trace silt, wet, 2.5Y 3/2 very dark grayish brown (SP)	100	3		-51.0	Vibracore	
							Vibracore		
							-52.5		
				100	4		-53.0	Vibracore	
				100					
							Vibracore		
				100	5		-55.5	Vibracore	
							-56.0	Vibracore	
				100				Vibracore	

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																					
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																					
LOCATION COORDINATES X = 527,840 Y = 2,203,920			ELEVATION TOP OF BORING -42.5 Ft.																								
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
-60.5	18.0			100			Vibracore																				
-61.5	19.0		SAND, poorly-graded with silt, mostly fine to coarse-grained sand-sized quartz, little silt, wet, 5Y 7/2 light gray (SP-SM)				-61.5																				
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.0/19.0</td> <td>CH*</td> </tr> <tr> <td>2</td> <td>5.0/19.0</td> <td>SP*</td> </tr> <tr> <td>3</td> <td>8.0/19.0</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>10.0/19.0</td> <td>SP*</td> </tr> <tr> <td>5</td> <td>13.0/19.0</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/19.0	CH*	2	5.0/19.0	SP*	3	8.0/19.0	SP-SM*	4	10.0/19.0	SP*	5	13.0/19.0	SP*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	2.0/19.0	CH*																									
2	5.0/19.0	SP*																									
3	8.0/19.0	SP-SM*																									
4	10.0/19.0	SP*																									
5	13.0/19.0	SP*																									

Boring Designation VB-MPH08-15

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-15		LOCATION COORDINATES X = 528,386 Y = 2,203,815		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES 5		DISTURBED 5		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal		
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING 06-10-08		STARTED 06-10-08		COMPLETED 06-10-08
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -43.0 Ft.		17. TOTAL RECOVERY FOR BORING 98 %		
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.0	0.0		CLAY, fat, high plasticity, very soft, wet, 5Y 3/2 dark olive gray (CH)	100			-43.0		
				100	1		-45.0		
							-45.5		
-46.5	3.5		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace angular sand to gravel-sized shell up to 1/2", wet, 5Y 7/3 pale yellow (SM)	100			Vibracore		
-47.5	4.5		CLAY, fat, high plasticity, soft, wet, 5Y 4/1 dark gray (CH)						
-48.0	5.0		SHELL, mostly angular sand to gravel-sized shell up to 2", some fine-grained sand-sized quartz, wet, 2.5Y 7/3 pale yellow	100	2		-49.0		
							-49.5		
-50.0	7.0		SAND, poorly-graded, mostly medium-grained sand-sized quartz, trace silt, wet, 2.5Y 6/3 light yellowish brown (SP)	100			Vibracore		
				100	3		-50.7		
							-51.2		
			At El. -51.5 Ft., 5Y 4/2 olive gray	100			Vibracore		
				100	4		-52.5		
							-53.0		
				100			Vibracore		
							-56.0		
				100	5		-56.5		
				94			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																													
			PROJECT NAVSTA Mayport CVN Dredging		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																												
LOCATION COORDINATES X = 528,386 Y = 2,203,815			ELEVATION TOP OF BORING -43.0 Ft.																																	
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																											
-60.0	17.0	•••••	SAND, poorly-graded with clay, low plasticity, mostly fine-grained sand-sized quartz, little silt, few clay, trace coarse-grained sand-sized quartz, wet, 5Y 8/1 white (SP-SC)	94			Vibracore																													
-62.6	19.6	•••••																																		
-63.0	20.0	NO REC																																		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>2.0/20.0</td><td>CH</td></tr> <tr><td>2</td><td>6.0/20.0</td><td>SM*</td></tr> <tr><td>3</td><td>7.7/20.0</td><td>SP*</td></tr> <tr><td>4</td><td>9.5/20.0</td><td>SP*</td></tr> <tr><td>5</td><td>13.0/20.0</td><td>SP*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve <table style="width:100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY SOIL TESTING</th> <th style="text-align: left;">RESULT UNIT</th> </tr> </thead> <tbody> <tr><td>2.0</td><td>Atterberg (PI)</td><td>43</td></tr> <tr><td>2.0</td><td>Specific Gravity</td><td>2.16</td></tr> </tbody> </table>							SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/20.0	CH	2	6.0/20.0	SM*	3	7.7/20.0	SP*	4	9.5/20.0	SP*	5	13.0/20.0	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	2.0	Atterberg (PI)	43	2.0	Specific Gravity	2.16
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																																		
1	2.0/20.0	CH																																		
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5	13.0/20.0	SP*																																		
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																																		
2.0	Atterberg (PI)	43																																		
2.0	Specific Gravity	2.16																																		

Boring Designation VB-MPH08-16

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-16		LOCATION COORDINATES X = 528,809 Y = 2,204,311		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1					
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-10-08		COMPLETED 06-10-08	
8. TOTAL DEPTH OF BORING 16.3 Ft.				16. ELEVATION TOP OF BORING -45.4 Ft.					
				17. TOTAL RECOVERY FOR BORING 87 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-45.4	0.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, trace clay, wet, 5Y 4/3 olive (SM)	100			-45.4		0
				100	1	-45.9	Vibracore		
				100		-46.4	Vibracore		
				100		-46.9	Vibracore		
-47.4	2.0			100	2	-47.4	Vibracore		
-48.4	3.0		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, wet, 5Y 6/2 light olive gray (SP-SM)	100					5
				100	3	-49.9	Vibracore		
				100		-50.4	Vibracore		
				100			Vibracore		
-53.9	8.5			100	4	-53.4	Vibracore		
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, wet, 5Y 8/1 white (SP) At El. -55.4 Ft., 5Y 8/2 pale yellow	100					10
				100	5	-55.4	Vibracore		
				100		-55.9	Vibracore		
				64			Vibracore		
-59.6	14.2								
		NR							15

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																										
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																										
LOCATION COORDINATES X = 528,809 Y = 2,204,311			ELEVATION TOP OF BORING -45.4 Ft.																													
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																							
-61.7	16.3	NR		64			Vibracore																									
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr><td>1</td><td>0.5/16.3</td><td>SM*</td></tr> <tr><td>2</td><td>1.5/16.3</td><td>SM*</td></tr> <tr><td>3</td><td>4.5/16.3</td><td>SP*</td></tr> <tr><td>4</td><td>8.0/16.3</td><td>SP*</td></tr> <tr><td>5</td><td>10.0/16.3</td><td>SP*</td></tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>4.5</td> <td>Specific Gravity</td> <td>2.67</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/16.3	SM*	2	1.5/16.3	SM*	3	4.5/16.3	SP*	4	8.0/16.3	SP*	5	10.0/16.3	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	4.5	Specific Gravity	2.67						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																														
1	0.5/16.3	SM*																														
2	1.5/16.3	SM*																														
3	4.5/16.3	SP*																														
4	8.0/16.3	SP*																														
5	10.0/16.3	SP*																														
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																														
4.5	Specific Gravity	2.67																														

Boring Designation VB-MPH08-18

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-18		LOCATION COORDINATES X = 529,795 Y = 2,204,614		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1				
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal				
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-05-08		COMPLETED 06-05-08
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -44.9 Ft.				
				17. TOTAL RECOVERY FOR BORING 98 %				
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE	
-44.9	0.0						-44.9			
			SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, discontinuous silt, wet, 5Y 8/1 white (SP-SM)	100			Vibracore			
								-46.4		
				100	1			-46.9		
				100				Vibracore		
-47.9	3.0							-48.4		
				100	2			-48.9		
				100				Vibracore		
								-51.4		
		100	3			-51.9				
		100				Vibracore				
						-52.9				
		100	4			-53.4				
		100				Vibracore				
						-55.4				
		100	5			-55.9				
				96			Vibracore			

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																													
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW																													
LOCATION COORDINATES X = 529,795 Y = 2,204,614			ELEVATION TOP OF BORING -44.9 Ft.																																	
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																											
-64.5	19.6	•••••		96			Vibracore																													
-64.9	20.0	NO REC					-64.9																													
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">SAMPLE ID</th> <th style="text-align: center;">SAMPLE DEPTH</th> <th style="text-align: center;">LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">1</td> <td style="text-align: center;">1.5/20.0</td> <td style="text-align: center;">SP-SM*</td> </tr> <tr> <td style="text-align: center;">2</td> <td style="text-align: center;">3.5/20.0</td> <td style="text-align: center;">SP*</td> </tr> <tr> <td style="text-align: center;">3</td> <td style="text-align: center;">6.5/20.0</td> <td style="text-align: center;">SP*</td> </tr> <tr> <td style="text-align: center;">4</td> <td style="text-align: center;">8.0/20.0</td> <td style="text-align: center;">SP*</td> </tr> <tr> <td style="text-align: center;">5</td> <td style="text-align: center;">10.5/20.0</td> <td style="text-align: center;">SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">SAMPLE DEPTH</th> <th style="text-align: center;">LABORATORY SOIL TESTING</th> <th style="text-align: center;">RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">3.5</td> <td style="text-align: center;">Specific Gravity</td> <td style="text-align: center;">2.60</td> </tr> <tr> <td style="text-align: center;">8.0</td> <td style="text-align: center;">Specific Gravity</td> <td style="text-align: center;">2.64</td> </tr> </tbody> </table>							SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.5/20.0	SP-SM*	2	3.5/20.0	SP*	3	6.5/20.0	SP*	4	8.0/20.0	SP*	5	10.5/20.0	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	3.5	Specific Gravity	2.60	8.0	Specific Gravity	2.64
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																																		
1	1.5/20.0	SP-SM*																																		
2	3.5/20.0	SP*																																		
3	6.5/20.0	SP*																																		
4	8.0/20.0	SP*																																		
5	10.5/20.0	SP*																																		
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																																		
3.5	Specific Gravity	2.60																																		
8.0	Specific Gravity	2.64																																		

Boring Designation VB-MPH08-19

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-19		LOCATION COORDINATES X = 530,353 Y = 2,204,305		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5	UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		1		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		Tidal		
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-05-08	COMPLETED 06-05-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-42.6 Ft.		
				17. TOTAL RECOVERY FOR BORING		88 %		
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-42.6	0.0						-42.6		
-43.4	0.8		CLAY, lean, low plasticity, very soft, some silt, wet, 5Y 4/2 olive gray (CL)	100			-43.1 Vibracore		
				100	1		-43.4 Vibracore		
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, wet, 5Y 8/1 white (SP)						
				100			Vibracore		
				100	2		-46.1 -46.6 Vibracore		
				100			Vibracore		
				100	3		-48.6 -49.1 Vibracore		
				100			Vibracore		
				100	4		-52.1 -52.6 Vibracore		
				100			Vibracore		
				100	5		-55.1 -55.6 Vibracore		
				66			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																													
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																													
LOCATION COORDINATES X = 530,353 Y = 2,204,305			ELEVATION TOP OF BORING -42.6 Ft.																																
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																										
-60.2	17.6			66			Vibracore																												
-62.6	20.0	NR					-62.6																												
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/20.0</td> <td>CL*</td> </tr> <tr> <td>2</td> <td>3.5/20.0</td> <td>SP*</td> </tr> <tr> <td>3</td> <td>6.0/20.0</td> <td>SP*</td> </tr> <tr> <td>4</td> <td>9.5/20.0</td> <td>SP*</td> </tr> <tr> <td>5</td> <td>12.5/20.0</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>0.5</td> <td>Specific Gravity</td> <td>2.61</td> </tr> <tr> <td>6.0</td> <td>Specific Gravity</td> <td>2.48</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/20.0	CL*	2	3.5/20.0	SP*	3	6.0/20.0	SP*	4	9.5/20.0	SP*	5	12.5/20.0	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	0.5	Specific Gravity	2.61	6.0	Specific Gravity	2.48					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																																	
1	0.5/20.0	CL*																																	
2	3.5/20.0	SP*																																	
3	6.0/20.0	SP*																																	
4	9.5/20.0	SP*																																	
5	12.5/20.0	SP*																																	
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																																	
0.5	Specific Gravity	2.61																																	
6.0	Specific Gravity	2.48																																	

Boring Designation VB-MPH08-20

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-20				LOCATION COORDINATES X = 530,731 Y = 2,204,819		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.				CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 1
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal		15. DATE BORING 06-08-08		STARTED 06-08-08
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -43.7 Ft.		17. TOTAL RECOVERY FOR BORING 100 %		COMPLETED 06-08-08
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE	
-43.7	0.0		CLAY, fat, high plasticity, very soft, wet, 5Y 3/2 dark olive gray (CH)				-43.7			
				100			Vibracore			
				100	1		-46.2 -46.7	Vibracore		
				100			Vibracore			
				100	2		-51.2 -51.7	Vibracore		
				100			Vibracore			
				100			Vibracore			
				100			Vibracore			
				100			Vibracore			
				100			Vibracore			

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																							
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																							
LOCATION COORDINATES X = 530,731 Y = 2,204,819			ELEVATION TOP OF BORING -43.7 Ft.																										
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																				
-63.7	20.0			100			Vibracore																						
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.5/20.0</td> <td>CH*</td> </tr> <tr> <td>2</td> <td>7.5/20.0</td> <td>CH</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT</th> <th>UNIT</th> </tr> </thead> <tbody> <tr> <td>7.5</td> <td>Atterberg (PI)</td> <td>29</td> <td></td> </tr> <tr> <td>7.5</td> <td>Specific Gravity</td> <td>2.44</td> <td></td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.5/20.0	CH*	2	7.5/20.0	CH	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT	7.5	Atterberg (PI)	29		7.5	Specific Gravity	2.44						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																											
1	2.5/20.0	CH*																											
2	7.5/20.0	CH																											
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT																										
7.5	Atterberg (PI)	29																											
7.5	Specific Gravity	2.44																											

Boring Designation VB-MPH08-22

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-22		LOCATION COORDINATES X = 531,676 Y = 2,205,049		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1				
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal				
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-04-08		COMPLETED 06-04-08
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -49.6 Ft.				
				17. TOTAL RECOVERY FOR BORING 100 %				
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-49.6	0.0		SAND, silty, mostly fine-grained sand-sized quartz, trace silt, trace clay, wet, 2.5Y 8/1 white (SM)				-49.6		
				100			Vibracore		
				100	1		-50.6 -51.1	Vibracore	
				100			Vibracore		
				100	2		-52.6 -53.1	Vibracore	
				100			Vibracore		
				100	3		-56.6 -57.1	Vibracore	
				100			Vibracore		
				100			Vibracore		
				100			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																				
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																				
LOCATION COORDINATES X = 531,676 Y = 2,205,049			ELEVATION TOP OF BORING -49.6 Ft.																							
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																	
-69.6	20.0		At El. -67.6 Ft., 2.5Y 6/2 light brownish gray	100			Vibracore																			
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>3.0/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>7.0/20.0</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>1.0</td> <td>Specific Gravity</td> <td>2.62</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	3.0/20.0	SM*	3	7.0/20.0	SM*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	1.0	Specific Gravity	2.62					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																								
1	1.0/20.0	SM*																								
2	3.0/20.0	SM*																								
3	7.0/20.0	SM*																								
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																								
1.0	Specific Gravity	2.62																								

Boring Designation VB-MPH08-24

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-24		LOCATION COORDINATES X = 532,983 Y = 2,204,998		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES 3		DISTURBED 0		UNDISTURBED (UD)	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal			
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING 06-08-08		STARTED 06-08-08		COMPLETED 06-08-08	
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -49.9 Ft.		17. TOTAL RECOVERY FOR BORING 94 %			
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-49.9	0.0		SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, few silt, trace angular shell up to 2", wet, 5Y 3/2 dark olive gray (SP-SM)	100			-49.9		
				100	1		-51.9		
							-52.4		
				100					
-53.9	4.0		SAND, silty, mostly medium-grained sand-sized quartz, little silt, trace angular shell up to 2", wet, 5Y 3/2 dark olive gray (SM)	100	2		-53.9		
							-54.4		
				100					
-55.9	6.0		SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, few silt, trace angular shell up to 2", wet, 5Y 3/2 dark olive gray (SP-SM)	100	3		-55.9		
							-56.4		
			At El. -59.9 Ft., mostly medium to coarse-grained sand-sized quartz, few angular shell up to 1", wet, 5Y 3/2 dark olive gray	91					
-61.9	12.0		CLAY, lean, medium plasticity, firm, some silt, wet, 5Y 6/1 gray (CL)						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																				
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																				
LOCATION COORDINATES X = 532,983 Y = 2,204,998			ELEVATION TOP OF BORING -49.9 Ft.																							
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																	
-68.7	18.8			91			Vibracore																			
-69.9	20.0	NR					-69.9																			
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>6.0/20.0</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>4.0</td> <td>Specific Gravity</td> <td>2.50</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/20.0	SP-SM*	2	4.0/20.0	SM*	3	6.0/20.0	SP-SM*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	4.0	Specific Gravity	2.50						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																								
1	2.0/20.0	SP-SM*																								
2	4.0/20.0	SM*																								
3	6.0/20.0	SP-SM*																								
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																								
4.0	Specific Gravity	2.50																								

Boring Designation VB-MPH08-25

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Mayport Entrance Channel				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-25		LOCATION COORDINATES X = 533,761 Y = 2,205,012		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1					
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-09-08		COMPLETED 06-09-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -46.3 Ft.					
				17. TOTAL RECOVERY FOR BORING 86 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-46.3	0.0		SHELL, mostly angular sand to gravel-sized shell up to 1", some medium-grained sand-sized quartz, wet, 5Y 8/2 pale yellow	100			-46.3		
				100	1		-47.8		
							-48.3		
-49.3	3.0		SAND, poorly-graded, mostly medium-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	100			Vibracore		
				100	2		-51.5		
							-52.0		
				100			Vibracore		
-53.3	7.0		SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, wet, 5Y 4/2 olive gray (SP-SM)	100	3		-53.3		
							-53.8		
				77			Vibracore		

Boring Designation VB-MPH08-25

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																				
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																				
LOCATION COORDINATES X = 533,761 Y = 2,205,012			ELEVATION TOP OF BORING -46.3 Ft.																							
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																	
-63.4	17.1	NO RECOVERY		77			Vibracore																			
-66.3	20.0										-66.3															
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.5/20.0</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>5.2/20.0</td> <td>SP*</td> </tr> <tr> <td>3</td> <td>7.0/20.0</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>1.5</td> <td>Specific Gravity</td> <td>2.68</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.5/20.0	SP*	2	5.2/20.0	SP*	3	7.0/20.0	SP-SM*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	1.5	Specific Gravity	2.68						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																								
1	1.5/20.0	SP*																								
2	5.2/20.0	SP*																								
3	7.0/20.0	SP-SM*																								
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																								
1.5	Specific Gravity	2.68																								

DRILLING LOG				DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3						9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-26			LOCATION COORDINATES X = 532,651 Y = 2,205,401			10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.				CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit				☐ AUTO HAMMER ☐ MANUAL HAMMER	
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.						12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED						DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 1	
6. THICKNESS OF OVERBURDEN N/A						14. ELEVATION GROUND WATER Tidal		15. DATE BORING		STARTED 06-09-08 COMPLETED 06-09-08	
7. DEPTH DRILLED INTO ROCK N/A						16. ELEVATION TOP OF BORING -47.6 Ft.		17. TOTAL RECOVERY FOR BORING 100 %		18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist	
8. TOTAL DEPTH OF BORING 20.0 Ft.											
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE		
-47.6	0.0						-47.6				
			SAND, poorly-graded with silt, mostly medium-grained sand-sized quartz, few silt, trace clay, wet, 5Y 4/2 olive gray (SP-SM)	100			Vibracore				
				100	1		-49.6 -50.1 Vibracore				
				100			Vibracore				
				100	2		-51.6 -52.1 Vibracore				
				100			Vibracore				
				100	3		-52.8 -53.3 Vibracore				
				100			Vibracore				
-58.6	11.0		CLAY, lean, low plasticity, firm, some silt, wet, 2.5Y 7/1 light gray (CL)								

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS														
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW														
LOCATION COORDINATES X = 532,651 Y = 2,205,401			ELEVATION TOP OF BORING -47.6 Ft.																	
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE											
-67.6	20.0			100			Vibracore													
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>5.2/20.0</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/20.0	SP-SM*	2	4.0/20.0	SP-SM*	3	5.2/20.0	SP-SM*					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																		
1	2.0/20.0	SP-SM*																		
2	4.0/20.0	SP-SM*																		
3	5.2/20.0	SP-SM*																		

Boring Designation VB-MPH08-38

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-38		LOCATION COORDINATES X = 535,770 Y = 2,205,586		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1				
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal				
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-12-08		COMPLETED 06-12-08
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -48.9 Ft.				
				17. TOTAL RECOVERY FOR BORING 99 %				
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-48.9	0.0						-48.9		
			SAND, silty, mostly fine to medium-grained sand-sized quartz, few angular sand to gravel-sized shell up to 1", few silt, wet, 5Y 7/1 light gray (SM)	100			Vibracore		
-50.2	1.3			100	1		-49.9 -50.2 Vibracore		
			SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, trace silt, wet, 5Y 4/3 olive (SP-SM)	100			Vibracore		
				100	2		-51.9 -52.4 Vibracore		
				100			Vibracore		
				100	3		-54.9 -55.4 Vibracore		
				99			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																					
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																					
LOCATION COORDINATES X = 535,770 Y = 2,205,586			ELEVATION TOP OF BORING -48.9 Ft.																								
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																		
-68.7	19.8	NO REC		99			Vibracore																				
-68.9	20.0																										
		NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>3.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>6.0/20.0</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>3.0</td> <td>Specific Gravity</td> <td>2.42</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	3.0/20.0	SP-SM*	3	6.0/20.0	SP-SM*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	3.0	Specific Gravity	2.42					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																									
1	1.0/20.0	SM*																									
2	3.0/20.0	SP-SM*																									
3	6.0/20.0	SP-SM*																									
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																									
3.0	Specific Gravity	2.42																									

Boring Designation VB-MPH08-40

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS			
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks						
2. BORING DESIGNATION VB-MPH08-40		LOCATION COORDINATES X = 537,692 Y = 2,204,768		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW		
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER				
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES 2		DISTURBED 2		UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal				
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING 06-12-08		STARTED 06-12-08		COMPLETED 06-12-08		
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -52.4 Ft.		17. TOTAL RECOVERY FOR BORING 89 %				
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist						
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE	
-52.4	0.0		CLAY, fat, high plasticity, soft, trace angular sand to gravel-sized shell up to 1/4", wet, 5Y 3/1 very dark gray (CH) At El. -52.9 Ft., discontinue shell	100			-52.4		0	
							Vibracore			
				100	1		-53.4			
							-53.9			Vibracore
				100						Vibracore
				100	2		-55.4			
							-55.9			
				86			Vibracore		5	
									10	
-62.9	10.5		SAND, poorly-graded, mostly fine-grained sand-sized quartz, trace silt, wet, 5Y 8/1 white (SP)							

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																							
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																							
LOCATION COORDINATES X = 537,692 Y = 2,204,768			ELEVATION TOP OF BORING -52.4 Ft.																										
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																				
-70.1	17.7			86			Vibracore																						
-72.4	20.0	NR					-72.4																						
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>CH</td> </tr> <tr> <td>2</td> <td>3.0/20.0</td> <td>CH*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT</th> <th>UNIT</th> </tr> </thead> <tbody> <tr> <td>1.0</td> <td>Atterberg (PI)</td> <td>38</td> <td></td> </tr> <tr> <td>1.0</td> <td>Specific Gravity</td> <td>2.30</td> <td></td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	CH	2	3.0/20.0	CH*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT	1.0	Atterberg (PI)	38		1.0	Specific Gravity	2.30						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																											
1	1.0/20.0	CH																											
2	3.0/20.0	CH*																											
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT																										
1.0	Atterberg (PI)	38																											
1.0	Specific Gravity	2.30																											

Boring Designation VB-MPH08-41

DRILLING LOG		DIVISION South Atlantic	INSTALLATION Jacksonville District	SHEET 1 OF 2 SHEETS
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3			9. SIZE AND TYPE OF BIT See Remarks	
2. BORING DESIGNATION VB-MPH08-41		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		VERTICAL MLLW
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.		12. TOTAL SAMPLES 3		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED		13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal
6. THICKNESS OF OVERBURDEN N/A		15. DATE BORING 06-07-08		STARTED 06-07-08
7. DEPTH DRILLED INTO ROCK N/A		16. ELEVATION TOP OF BORING -47.8 Ft.		COMPLETED 06-07-08
8. TOTAL DEPTH OF BORING 20.0 Ft.		17. TOTAL RECOVERY FOR BORING 89 %		18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.8	0.0		CLAY, fat, high plasticity, soft, wet, 5Y 3/1 very dark gray (CH)	100			-47.8		
				100	1		-48.3 Vibracore		
							-48.8 Vibracore		
			At El. -49.8 Ft., 5Y 5/2 olive gray	100			Vibracore		
-51.3	3.5		SAND, silty, mostly fine-grained sand-sized quartz, some silt, wet, 5Y 8/1 white (SM)	100	2		-51.8		
							-52.3 Vibracore		
				100			Vibracore		
-55.8	8.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	100	3		-56.3		
							-56.8 Vibracore		
			At El. -57.8 Ft., 5YR 7/2 pinkish gray				Vibracore		
-59.8	12.0		CLAY, fat, high plasticity, soft, wet, 5Y 3/1 very dark gray (CH)	80			Vibracore		
-60.8	13.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)						
-62.3	14.5								
-62.8	15.0		CLAY, fat, high plasticity, soft, wet,						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																							
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																							
LOCATION COORDINATES X = 539,213 Y = 2,205,270			ELEVATION TOP OF BORING -47.8 Ft.																										
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																				
-65.6	17.8		5Y 3/1 very dark gray (CH) SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	80			Vibracore																						
-67.8	20.0	NIR					-67.8																						
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/20.0</td> <td>CH</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>8.5/20.0</td> <td>SP*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>0.5</td> <td>Atterberg (PI)</td> <td>28</td> </tr> <tr> <td>0.5</td> <td>Specific Gravity</td> <td>2.18</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/20.0	CH	2	4.0/20.0	SM*	3	8.5/20.0	SP*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	0.5	Atterberg (PI)	28	0.5	Specific Gravity	2.18					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																											
1	0.5/20.0	CH																											
2	4.0/20.0	SM*																											
3	8.5/20.0	SP*																											
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																											
0.5	Atterberg (PI)	28																											
0.5	Specific Gravity	2.18																											

Boring Designation VB-MPH08-45

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-45		LOCATION COORDINATES X = 540,244 Y = 2,205,165		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW		
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 4	UNDISTURBED (UD) 0		
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal			
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 06-07-08	COMPLETED 06-07-08		
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING -43.7 Ft.		17. TOTAL RECOVERY FOR BORING 95 %			
8. TOTAL DEPTH OF BORING 20.0 Ft.				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.7	0.0		CLAY, fat, high plasticity, firm, wet, 5Y 4/1 dark gray (CH)				-43.7		
-44.7	1.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, wet, 5Y 5/2 olive gray (SM)	100			Vibracore		
				100	1		-45.2		
							-45.7		
-46.2	2.5		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP-SM)	100			Vibracore		
				100	2		-47.7		
							-48.2		
				100			Vibracore		
				100	3		-51.7		
							-52.2		
			At El. -53.7 Ft., 2.5Y 6/3 light yellowish brown	100			Vibracore		
-56.2	12.5		SAND, silty, mostly fine-grained sand-sized quartz, some silt, wet, 2.5YR 6/3 light reddish brown (SM)	100	4		-56.2		
							-56.7		
				84			Vibracore		

Boring Designation VB-MPH08-45

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																	
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																	
LOCATION COORDINATES X = 540,244 Y = 2,205,165			ELEVATION TOP OF BORING -43.7 Ft.																				
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE														
-62.6	18.9	NR		84			Vibracore																
-63.7	20.0					-63.7																	
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.5/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>8.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>12.5/20.0</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.5/20.0	SM*	2	4.0/20.0	SP-SM*	3	8.0/20.0	SP-SM*	4	12.5/20.0	SM*					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																					
1	1.5/20.0	SM*																					
2	4.0/20.0	SP-SM*																					
3	8.0/20.0	SP-SM*																					
4	12.5/20.0	SM*																					

Boring Designation VB-MPH08-49

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-49		LOCATION COORDINATES X = 541,181 Y = 2,205,080		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		14. ELEVATION GROUND WATER		Tidal	
6. THICKNESS OF OVERBURDEN N/A				15. DATE BORING		STARTED 06-07-08		COMPLETED 06-07-08	
7. DEPTH DRILLED INTO ROCK N/A				16. ELEVATION TOP OF BORING		-37.8 Ft.			
8. TOTAL DEPTH OF BORING 20.0 Ft.				17. TOTAL RECOVERY FOR BORING		98 %			
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-37.8	0.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	100			-37.8		
				100	1		-38.3 Vibracore		
							-38.8 Vibracore		
				100			Vibracore		
							-41.8		
				100	2		-42.3 Vibracore		
-43.3	5.5		SAND, silty, mostly fine-grained sand-sized quartz, little silt, wet, 5Y 3/2 dark olive gray (SM)	100			Vibracore		
							-44.8		
			At El. -45.3 Ft., 5Y 4/3 olive	100	3		-45.3 Vibracore		
				100			Vibracore		
							-48.8		
			At El. -48.3 Ft., 10Y 4/1 dark greenish gray	100	4		-49.3 Vibracore		
-50.8	13.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	100			Vibracore		
-52.3	14.5								
-52.5	14.7		CLAY, fat, high plasticity, soft, wet,						

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																										
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																										
LOCATION COORDINATES X = 541,181 Y = 2,205,080			ELEVATION TOP OF BORING -37.8 Ft.																													
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																							
-54.8	17.0		10Y 4/1 dark greenish gray (CH) SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	100			Vibracore																									
-55.2	17.4		SAND, silty, mostly fine-grained sand-sized quartz, wet, 10Y 4/1 dark greenish gray (SM)	100	5		Vibracore																									
-55.8	18.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP) CLAY, fat, high plasticity, soft, wet, 10Y 4/1 dark greenish gray (CH) SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	80			Vibracore																									
-56.1	18.3																															
-57.3	19.5																															
-57.8	20.0	NR																														
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/20.0</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SP*</td> </tr> <tr> <td>3</td> <td>7.0/20.0</td> <td>SM*</td> </tr> <tr> <td>4</td> <td>11.0/20.0</td> <td>SM*</td> </tr> <tr> <td>5</td> <td>17.0/20.0</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>0.5</td> <td>Specific Gravity</td> <td>2.65</td> </tr> </tbody> </table>			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/20.0	SP*	2	4.0/20.0	SP*	3	7.0/20.0	SM*	4	11.0/20.0	SM*	5	17.0/20.0	SM*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	0.5	Specific Gravity	2.65						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																														
1	0.5/20.0	SP*																														
2	4.0/20.0	SP*																														
3	7.0/20.0	SM*																														
4	11.0/20.0	SM*																														
5	17.0/20.0	SM*																														
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																														
0.5	Specific Gravity	2.65																														

Boring Designation VB-MPH08-51

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-51		LOCATION COORDINATES X = 541,612 Y = 2,204,235		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 4		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1				
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal				
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-07-08		COMPLETED 06-07-08
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -44.0 Ft.				
				17. TOTAL RECOVERY FOR BORING 95 %				
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE	
-44.0	0.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, few angular shell up to 1/2", wet, 5Y 5/1 gray (SM)				-44.0			
				100			Vibracore			
				100	1		-45.0 -45.5	Vibracore		
				100			Vibracore			
				100	2		-48.0 -48.5	Vibracore		
-49.5	5.5		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, wet, 5Y 8/1 white (SP-SM)							
				100			Vibracore			
				100	3		-52.5 -53.0	Vibracore		
				100			Vibracore			
-56.0	12.0		SILT, inorganic-L, low plasticity, soft, wet, 5Y 7/2 light gray (ML)				-56.0			
				100	4		-56.5	Vibracore		
				87			Vibracore			

DRILLING LOG (Cont. Sheet)				INSTALLATION Jacksonville District				SHEET 2 OF 2 SHEETS																					
				PROJECT NAVSTA Mayport CVN Dredging		COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW																			
LOCATION COORDINATES X = 541,612 Y = 2,204,235				ELEVATION TOP OF BORING -44.0 Ft.																									
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																				
-63.0	19.0	NR	At El. -62.0 Ft., 5Y 7/1 light gray	87			Vibracore																						
-64.0	20.0						-64.0																						
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table style="width:100%; border-collapse: collapse;"> <tr> <th style="text-align: left;">SAMPLE ID</th> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY CLASSIFICATION</th> </tr> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>4.0/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>8.5/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>4</td> <td>12.0/20.0</td> <td>ML*</td> </tr> </table> *Lab visual classification based on gradation curve <table style="width:100%; border-collapse: collapse;"> <tr> <th style="text-align: left;">SAMPLE DEPTH</th> <th style="text-align: left;">LABORATORY SOIL TESTING</th> <th style="text-align: left;">RESULT UNIT</th> </tr> <tr> <td>8.5</td> <td>Specific Gravity</td> <td>2.55</td> </tr> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	4.0/20.0	SM*	3	8.5/20.0	SP-SM*	4	12.0/20.0	ML*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	8.5	Specific Gravity	2.55					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																											
1	1.0/20.0	SM*																											
2	4.0/20.0	SM*																											
3	8.5/20.0	SP-SM*																											
4	12.0/20.0	ML*																											
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																											
8.5	Specific Gravity	2.55																											

Boring Designation VB-MPH08-53

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-53		LOCATION COORDINATES X = 542,205 Y = 2,204,909		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 5		UNDISTURBED (UD) 0
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		1		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		Tidal		
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-06-08		COMPLETED 06-06-08
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-43.0 Ft.		
				17. TOTAL RECOVERY FOR BORING		81 %		
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-43.0	0.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 7/1 light gray (SP)	100			-43.0		
				100	1		-45.0		
				100			-45.5		
				100					
-47.8	4.8		SAND, silty, mostly fine-grained sand-sized quartz, little silt, wet, 5Y 5/2 olive gray (SM)	100	2		-48.0		
				100			-48.5		
				100					
-51.0	8.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 5/2 olive gray (SP)	100	3		-51.0		
				100			-51.5		
				100					
-53.5	10.5		SAND, silty, mostly fine-grained sand-sized quartz, little silt, wet, 5Y 5/2 olive gray (SM)	100	4		-53.5		
				100			-54.0		
				100					
-55.0	12.0		SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP-SM)	100	5		-55.5		
				100			-56.0		
				44					

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																				
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																				
LOCATION COORDINATES X = 542,205 Y = 2,204,909			ELEVATION TOP OF BORING -43.0 Ft.																							
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																	
-59.1	16.1	NO RECOVERY	At El. -58.0 Ft., trace silt	44			Vibracore																			
-63.0	20.0																									
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.0/20.0</td> <td>SP*</td> </tr> <tr> <td>2</td> <td>5.0/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>8.0/20.0</td> <td>SP*</td> </tr> <tr> <td>4</td> <td>10.5/20.0</td> <td>SM*</td> </tr> <tr> <td>5</td> <td>12.5/20.0</td> <td>SP-SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/20.0	SP*	2	5.0/20.0	SM*	3	8.0/20.0	SP*	4	10.5/20.0	SM*	5	12.5/20.0	SP-SM*					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																								
1	2.0/20.0	SP*																								
2	5.0/20.0	SM*																								
3	8.0/20.0	SP*																								
4	10.5/20.0	SM*																								
5	12.5/20.0	SP-SM*																								

DRILLING LOG						DIVISION South Atlantic			INSTALLATION Jacksonville District				SHEET 1 OF 2 SHEETS			
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3									9. SIZE AND TYPE OF BIT See Remarks							
2. BORING DESIGNATION VB-MPH08-57						LOCATION COORDINATES X = 543,175 Y = 2,204,834			10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)				HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.						CONTRACTOR FILE NO. 2007D16			11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit				☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.									12. TOTAL SAMPLES				DISTURBED 4		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED									DEG. FROM VERTICAL		BEARING		13. TOTAL NUMBER CORE BOXES 1			
									14. ELEVATION GROUND WATER Tidal							
6. THICKNESS OF OVERBURDEN N/A									15. DATE BORING				STARTED 06-06-08		COMPLETED 06-06-08	
7. DEPTH DRILLED INTO ROCK N/A									16. ELEVATION TOP OF BORING -40.8 Ft.							
8. TOTAL DEPTH OF BORING 20.0 Ft.									17. TOTAL RECOVERY FOR BORING 98 %							
									18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist							
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE							
-40.8	0.0						-40.8									
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 7/1 light gray (SP)	100			Vibracore									
				100	1		-42.8 -43.3	Vibracore								
-44.8	4.0															
			SAND, silty, mostly fine-grained sand-sized quartz, some silt, trace angular sand to gravel-sized shell up to 1/2", wet, 5Y 4/2 olive gray (SM)	100			Vibracore									
								-47.8 -48.3	Vibracore							
					100			Vibracore								
					100	3		-52.3 -52.8	Vibracore							
				100			Vibracore									
				100	4		-54.8 -55.3	Vibracore								
				91			Vibracore									

Boring Designation VB-MPH08-57

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																	
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																	
LOCATION COORDINATES X = 543,175 Y = 2,204,834			ELEVATION TOP OF BORING -40.8 Ft.																				
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE														
-58.8	18.0		SAND, poorly-graded, mostly fine-grained sand-sized quartz, wet, 5Y 8/1 white (SP)	91			Vibracore																
-60.3	19.5																						
-60.8	20.0																						
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results																							
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SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																					
1	2.0/20.0	SP*																					
2	7.0/20.0	SM*																					
4	14.0/20.0	SM*																					
3	11.5/20.0	SM*																					
*Lab visual classification based on gradation curve																							

Boring Designation VB-MPH08-69

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-69		LOCATION COORDINATES X = 546,162 Y = 2,204,453		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1					
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-05-08		COMPLETED 06-05-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -44.7 Ft.					
				17. TOTAL RECOVERY FOR BORING 98 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-44.7	0.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, trace clay, wet, 5Y 6/2 light olive gray (SM)	100			-44.7		
				100	1		Vibracore		
							-45.7		
							-46.2		
-46.7	2.0		CLAY, fat, high plasticity, soft, wet, 5Y 2.5/1 black (CH)	100			Vibracore		
-50.2	5.5		CLAY, lean, low plasticity, firm, some silt, few fine-grained sand-sized quartz, trace angular sand to gravel-sized shell up to 1/2", wet, 5Y 5/1 gray (CL)	100	2		-51.2		
							-51.7		
				100			Vibracore		
-54.7	10.0		SAND, silty, mostly fine-grained sand-sized quartz, little silt, wet, 5Y 5/1 gray (SM)	100	3		-55.2		
							-55.7		
				96			Vibracore		
			At El. -58.7 Ft., 5Y 6/1 gray						

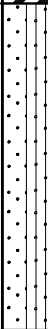
DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS														
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW														
LOCATION COORDINATES X = 546,162 Y = 2,204,453			ELEVATION TOP OF BORING -44.7 Ft.																	
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE											
-64.3	19.6	NO REC		96			Vibracore													
-64.7	20.0																			
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>6.5/20.0</td> <td>CL*</td> </tr> <tr> <td>3</td> <td>10.5/20.0</td> <td>SM*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	6.5/20.0	CL*	3	10.5/20.0	SM*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																		
1	1.0/20.0	SM*																		
2	6.5/20.0	CL*																		
3	10.5/20.0	SM*																		

Boring Designation VB-MPH08-71

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-71		LOCATION COORDINATES X = 546,576 Y = 2,203,739		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 3	UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		1		
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		Tidal		
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-05-08	COMPLETED 06-05-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING		-46.8 Ft.		
				17. TOTAL RECOVERY FOR BORING		100 %		
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-46.8	0.0						-46.8		
			CLAY, fat, high plasticity, soft, wet, 5Y 2.5/1 black (CH)	100			Vibracore		
	100			1	-48.3				
					-48.8				
-49.3	2.5		SAND, silty, mostly quartz, little silt, trace angular sand to gravel-sized shell up to 1/2", wet, 5Y 4/1 dark gray (SM)	100			Vibracore		
				100	2	-52.3			
					-52.8				
-54.8	8.0		CLAY, lean, low plasticity, soft, wet, 5Y 4/1 dark gray (CL)	100			Vibracore		
						-55.3			
-55.8	9.0			100	3		Vibracore		
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, few angular sand to gravel-sized shell up to 1/2", wet, 5Y 8/1 white (SP)	100			Vibracore		

DRILLING LOG (Cont. Sheet)				INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS														
PROJECT NAVSTA Mayport CVN Dredging				COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW														
LOCATION COORDINATES X = 546,576 Y = 2,203,739				ELEVATION TOP OF BORING -46.8 Ft.																	
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	ROD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE												
-66.8	20.0		At El. -61.8 Ft., trace angular sand to gravel-sized shell up to 1/4"	100			Vibracore														
			At El. -65.8 Ft., little angular sand to gravel-sized shell up to 1-1/2", oyster shell fragments																		
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.5/20.0</td> <td>CH*</td> </tr> <tr> <td>2</td> <td>5.5/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>8.5/20.0</td> <td>CL*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.5/20.0	CH*	2	5.5/20.0	SM*	3	8.5/20.0	CL*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																			
1	1.5/20.0	CH*																			
2	5.5/20.0	SM*																			
3	8.5/20.0	CL*																			

DRILLING LOG			DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS	
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3					9. SIZE AND TYPE OF BIT See Remarks				
2. BORING DESIGNATION VB-MPH08-73			LOCATION COORDINATES X = 547,216 Y = 2,204,329		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.			CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER		
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.					12. TOTAL SAMPLES		DISTURBED 3	UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED					13. TOTAL NUMBER CORE BOXES 1		14. ELEVATION GROUND WATER Tidal		
6. THICKNESS OF OVERBURDEN N/A					15. DATE BORING		STARTED 06-04-08	COMPLETED 06-04-08	
7. DEPTH DRILLED INTO ROCK N/A					16. ELEVATION TOP OF BORING -47.0 Ft.		17. TOTAL RECOVERY FOR BORING 98 %		
8. TOTAL DEPTH OF BORING 20.0 Ft.					18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist				
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-47.0	0.0						-47.0		
			CLAY, fat, high plasticity, soft, wet, 5Y 2.5/1 black (CH)	100			Vibracore		
				100	1		-49.0		
							-49.5	Vibracore	
			SAND, poorly-graded with silt, mostly fine-grained sand-sized quartz, few silt, trace angular shell up to 1/4", wet, 5Y 3/2 dark olive gray (SP-SM)	86			Vibracore		
-51.8	4.8						-53.0		
				100	2		-53.5	Vibracore	
			CLAY, fat, high plasticity, soft, trace angular sand to gravel-sized shell up to 1/4", wet, 5Y 2.5/1 black (CH)	100			Vibracore		
-56.0	9.0						-56.0		
				100	3		-56.5	Vibracore	
				100			Vibracore		

Boring Designation VB-MPH08-73

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS																							
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW																							
LOCATION COORDINATES X = 547,216 Y = 2,204,329			ELEVATION TOP OF BORING -47.0 Ft.																										
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE																				
-67.0	20.0			100			Vibracore																						
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>2.0/20.0</td> <td>CH</td> </tr> <tr> <td>2</td> <td>6.0/20.0</td> <td>SP-SM*</td> </tr> <tr> <td>3</td> <td>9.0/20.0</td> <td>CH*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve <table border="1"> <thead> <tr> <th>SAMPLE DEPTH</th> <th>LABORATORY SOIL TESTING</th> <th>RESULT UNIT</th> </tr> </thead> <tbody> <tr> <td>2.0</td> <td>Atterberg (PI)</td> <td>24</td> </tr> <tr> <td>2.0</td> <td>Specific Gravity</td> <td>2.52</td> </tr> </tbody> </table>	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	2.0/20.0	CH	2	6.0/20.0	SP-SM*	3	9.0/20.0	CH*	SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT	2.0	Atterberg (PI)	24	2.0	Specific Gravity	2.52					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																											
1	2.0/20.0	CH																											
2	6.0/20.0	SP-SM*																											
3	9.0/20.0	CH*																											
SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT UNIT																											
2.0	Atterberg (PI)	24																											
2.0	Specific Gravity	2.52																											

Boring Designation VB-MPH08-81

DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-81		LOCATION COORDINATES X = 549,173 Y = 2,204,040		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES 3		DISTURBED 3		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES 1					
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER Tidal					
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING 06-02-08		STARTED 06-02-08		COMPLETED 06-02-08	
8. TOTAL DEPTH OF BORING 20.0 Ft.				16. ELEVATION TOP OF BORING -51.2 Ft.					
				17. TOTAL RECOVERY FOR BORING 98 %					
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-51.2	0.0		SAND, silty, mostly fine-grained sand-sized quartz, trace angular sand to gravel-sized shell up to 1/4", wet, 5Y 3/2 dark olive gray (SM)	100			-51.2		
				100	1		Vibracore		
				100			-52.2		
				100			-52.7		
				100	2		Vibracore		
				100			-53.7		
				100			-54.2		
				100			Vibracore		
-55.7	4.5		CLAY, fat, high plasticity, soft, wet, 5Y 3/1 very dark gray (CH)	100	3		-55.7		
				100			-56.2		
				97			Vibracore		

DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS														
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW														
LOCATION COORDINATES X = 549,173 Y = 2,204,040			ELEVATION TOP OF BORING -51.2 Ft.																	
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE											
-70.8	19.6			97			Vibracore													
-71.2	20.0									NO REC			-71.2							
NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1.0/20.0</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>2.5/20.0</td> <td>SM*</td> </tr> <tr> <td>3</td> <td>4.5/20.0</td> <td>CH*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve			SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	1.0/20.0	SM*	2	2.5/20.0	SM*	3	4.5/20.0	CH*						
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION																		
1	1.0/20.0	SM*																		
2	2.5/20.0	SM*																		
3	4.5/20.0	CH*																		

Boring Designation VB-MPH08-83

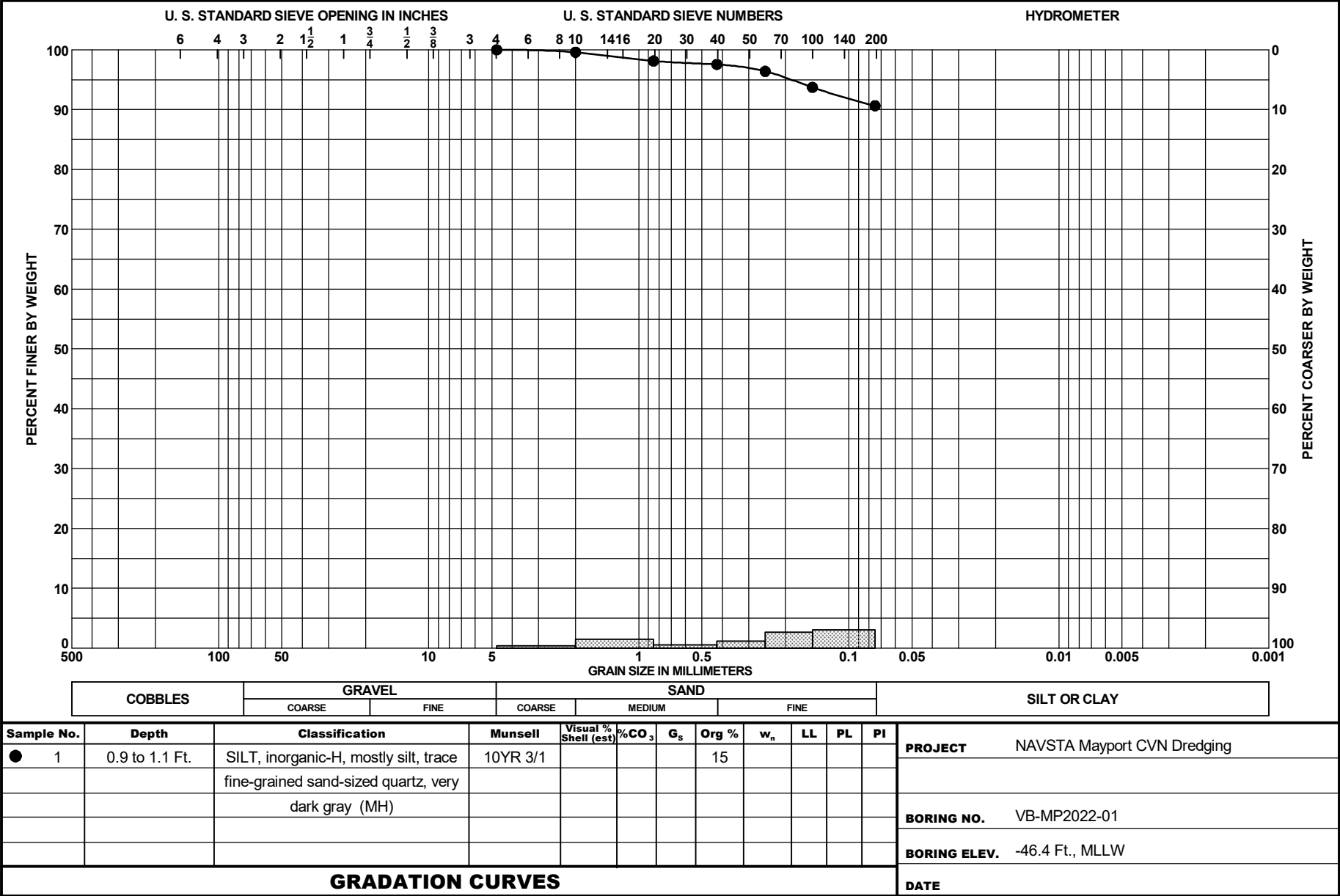
DRILLING LOG		DIVISION South Atlantic		INSTALLATION Jacksonville District			SHEET 1 OF 2 SHEETS		
1. PROJECT NAVSTA Mayport CVN Dredging Jacksonville Harbor Bar Cut-3				9. SIZE AND TYPE OF BIT See Remarks					
2. BORING DESIGNATION VB-MPH08-83		LOCATION COORDINATES X = 549,557 Y = 2,203,268		10. COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83		VERTICAL MLLW	
3. DRILLING AGENCY Challenge Engineering & Testing, Inc.		CONTRACTOR FILE NO. 2007D16		11. MANUFACTURER'S DESIGNATION OF DRILL Alpine 271 Vibracore Unit		☐ AUTO HAMMER ☐ MANUAL HAMMER			
4. NAME OF DRILLER Alpine Ocean Seismic, Inc.				12. TOTAL SAMPLES		DISTURBED 2		UNDISTURBED (UD) 0	
5. DIRECTION OF BORING ☒ VERTICAL ☐ INCLINED				13. TOTAL NUMBER CORE BOXES		1			
6. THICKNESS OF OVERBURDEN N/A				14. ELEVATION GROUND WATER		Tidal			
7. DEPTH DRILLED INTO ROCK N/A				15. DATE BORING		STARTED 06-02-08		COMPLETED 06-02-08	
8. TOTAL DEPTH OF BORING 17.2 Ft.				16. ELEVATION TOP OF BORING		-52.6 Ft.			
				17. TOTAL RECOVERY FOR BORING		88 %			
				18. SIGNATURE AND TITLE OF INSPECTOR Bob Momberger, Geologist					
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-52.6	0.0		SAND, silty, mostly fine-grained sand-sized quartz, some silt, wet, 5Y 3/1 very dark gray (SM)	100			-52.6		0
				100	1		-53.1	Vibracore	
				100			-53.6	Vibracore	
-54.1	1.5			100			-54.1	Vibracore	
			CLAY, lean, medium plasticity, soft, little silt, trace angular medium to coarse-grained sand-sized shell up to 1/8", wet, 5Y 3/1 very dark gray (CL)	100	2		-54.6	Vibracore	5
			At El. -55.6 Ft., low plasticity, firm, some silt, few fine-grained sand-sized quartz, trace angular sand to gravel-sized shell up to 1/4", 5Y 5/2 olive gray						10
-60.6	8.0		CLAY, fat, high plasticity, firm, wet, 5Y 2.5/1 black (CH)	87			Vibracore		15

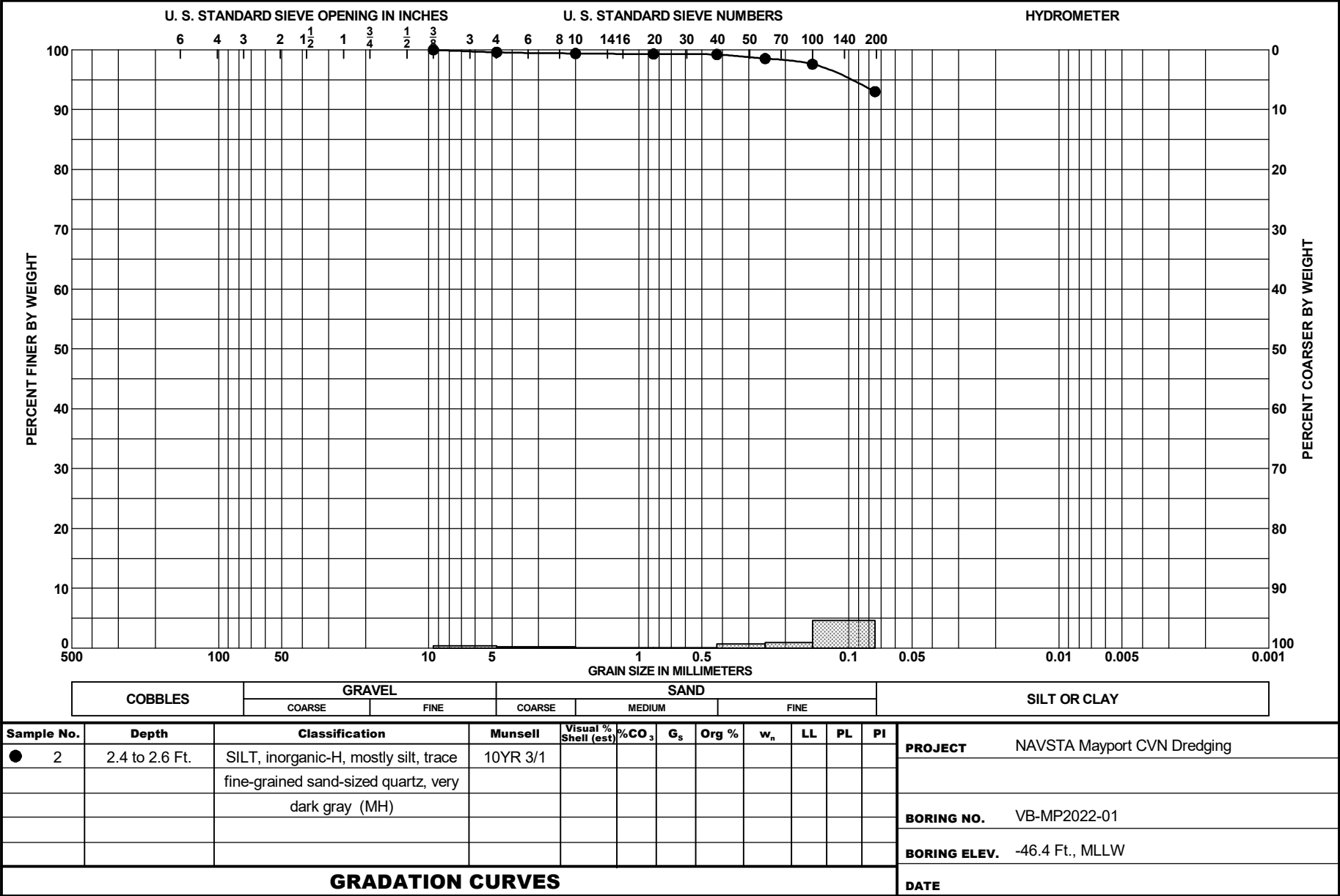
Boring Designation VB-MPH08-83

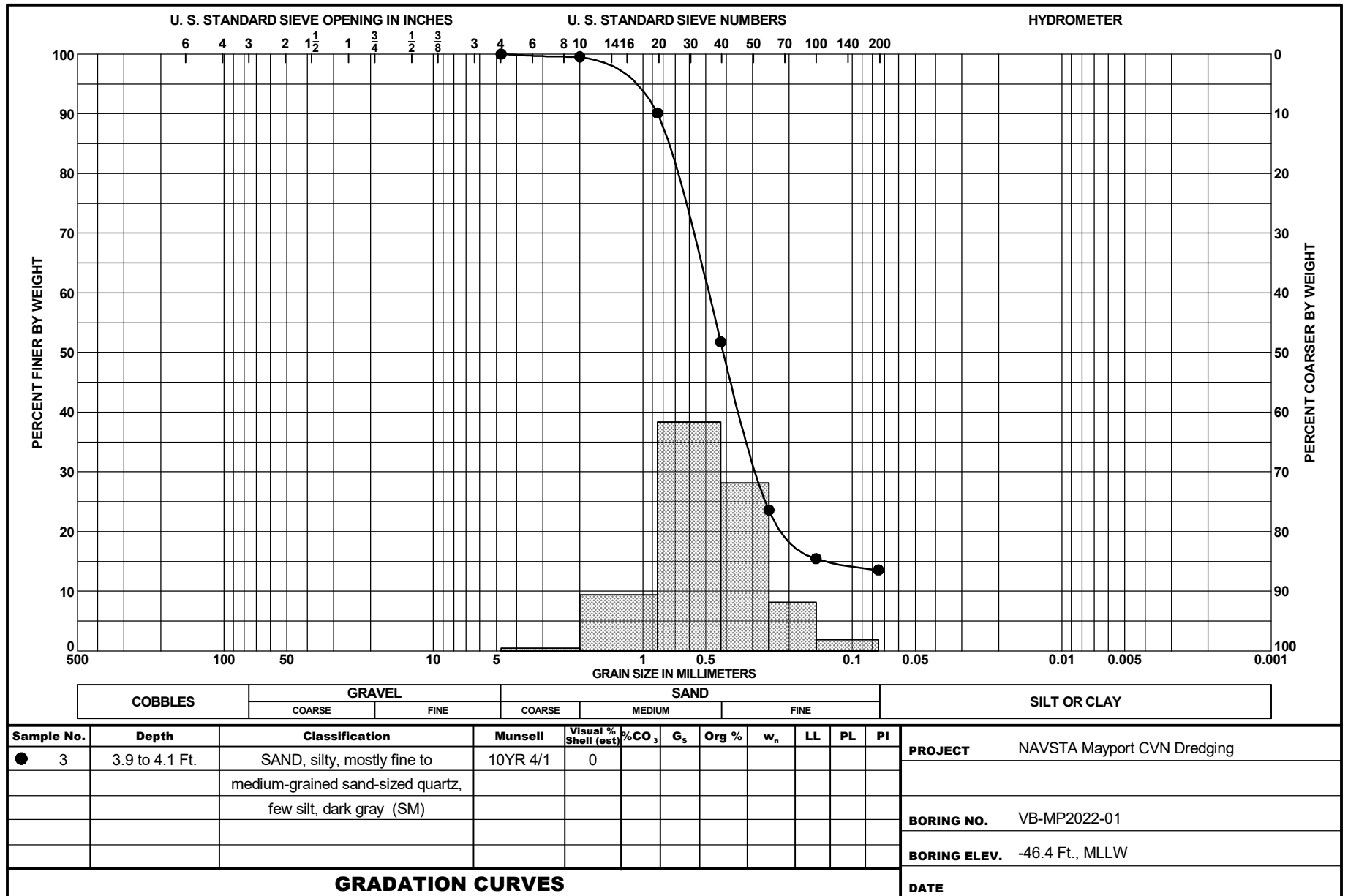
DRILLING LOG (Cont. Sheet)			INSTALLATION Jacksonville District			SHEET 2 OF 2 SHEETS											
PROJECT NAVSTA Mayport CVN Dredging			COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)		HORIZONTAL NAD83	VERTICAL MLLW											
LOCATION COORDINATES X = 549,557 Y = 2,203,268			ELEVATION TOP OF BORING -52.6 Ft.														
ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE								
-67.8	15.2	NR		87			Vibracore										
-69.8	17.2						-69.8										
			NOTES: 1. USACE Jacksonville is the custodian for these original files. 2. Soils are field visually classified in accordance with the Unified Soils Classification System. 3. Vibracore Borings Were Sampled With An Alpine 271 Pneumatic Powered Unit Using A 3 7/8" Lexan Liner To Termination Depth Specified. 4. Laboratory Testing Results <table border="1"> <thead> <tr> <th>SAMPLE ID</th> <th>SAMPLE DEPTH</th> <th>LABORATORY CLASSIFICATION</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.5/17.2</td> <td>SM*</td> </tr> <tr> <td>2</td> <td>1.5/17.2</td> <td>CL*</td> </tr> </tbody> </table> *Lab visual classification based on gradation curve	SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION	1	0.5/17.2	SM*	2	1.5/17.2	CL*					
SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION															
1	0.5/17.2	SM*															
2	1.5/17.2	CL*															

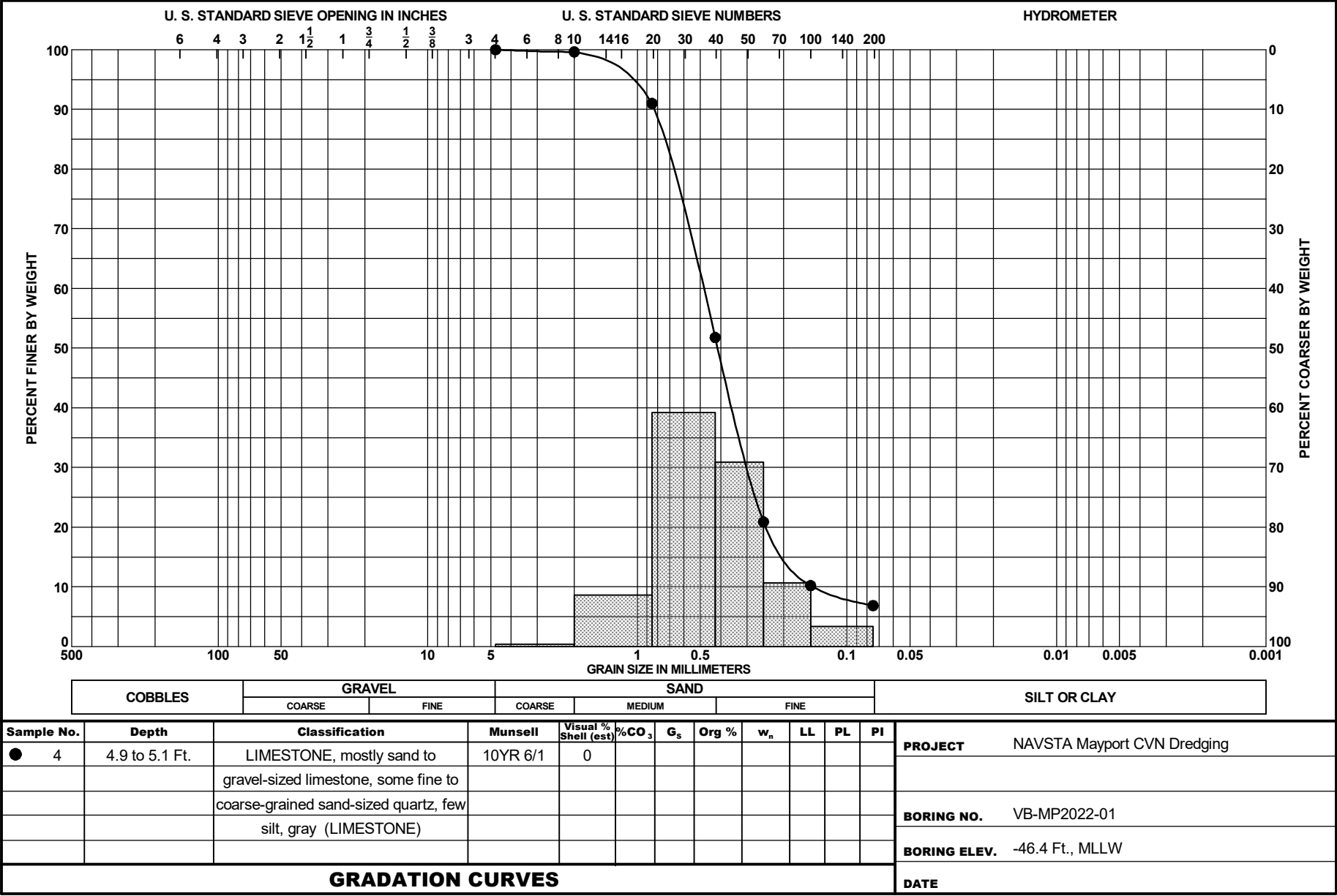
4.6 LABORATORY TESTING DATA

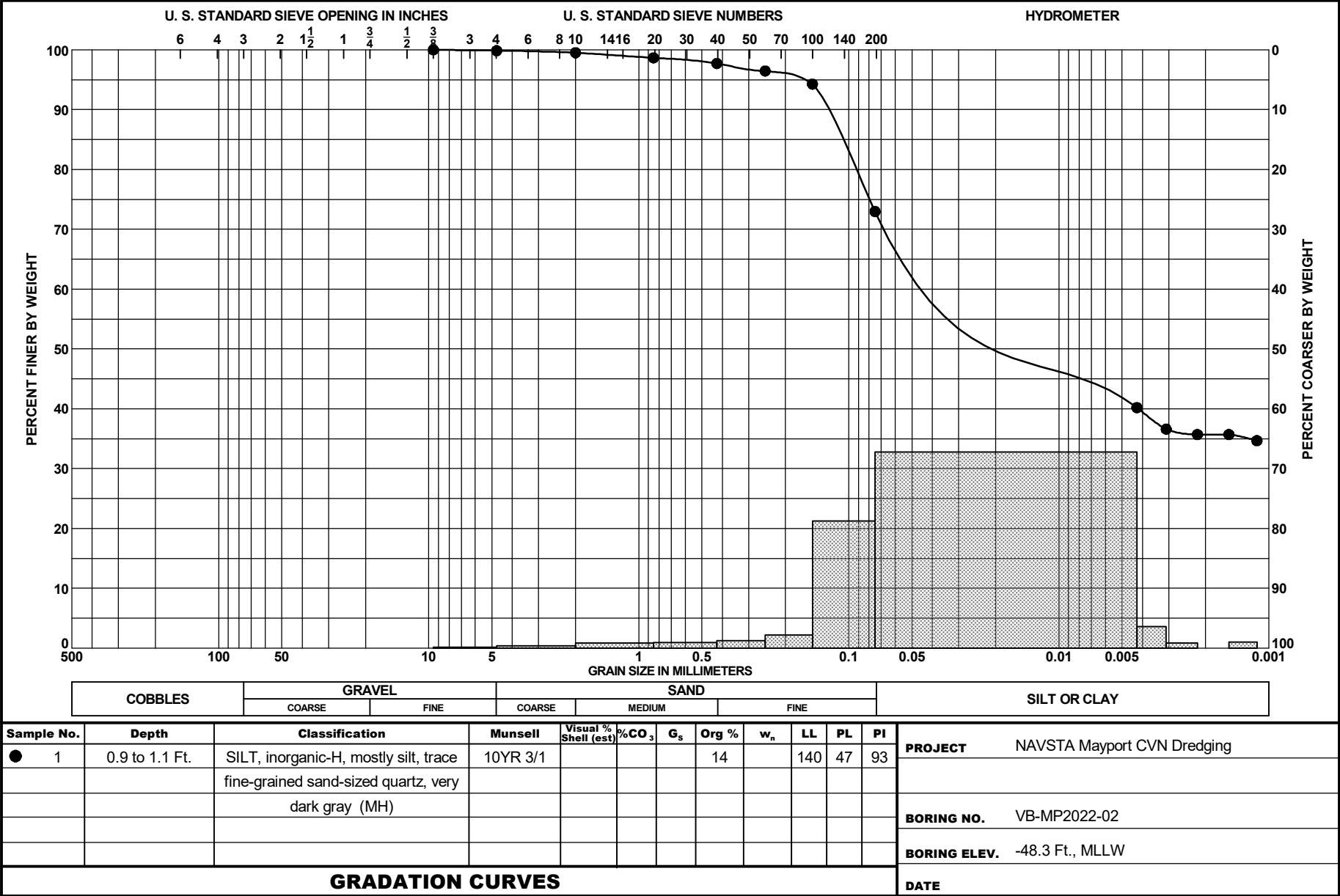
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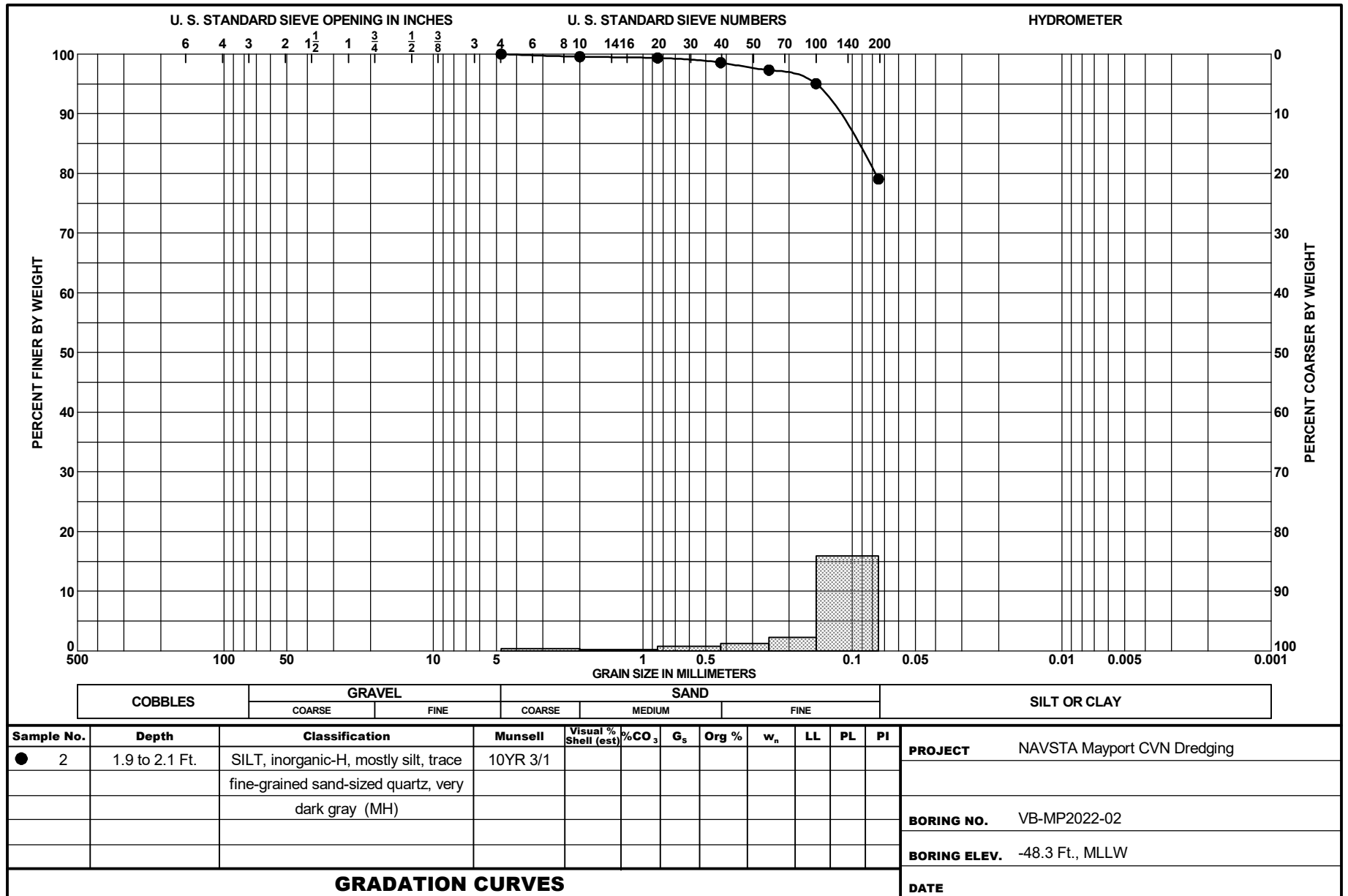


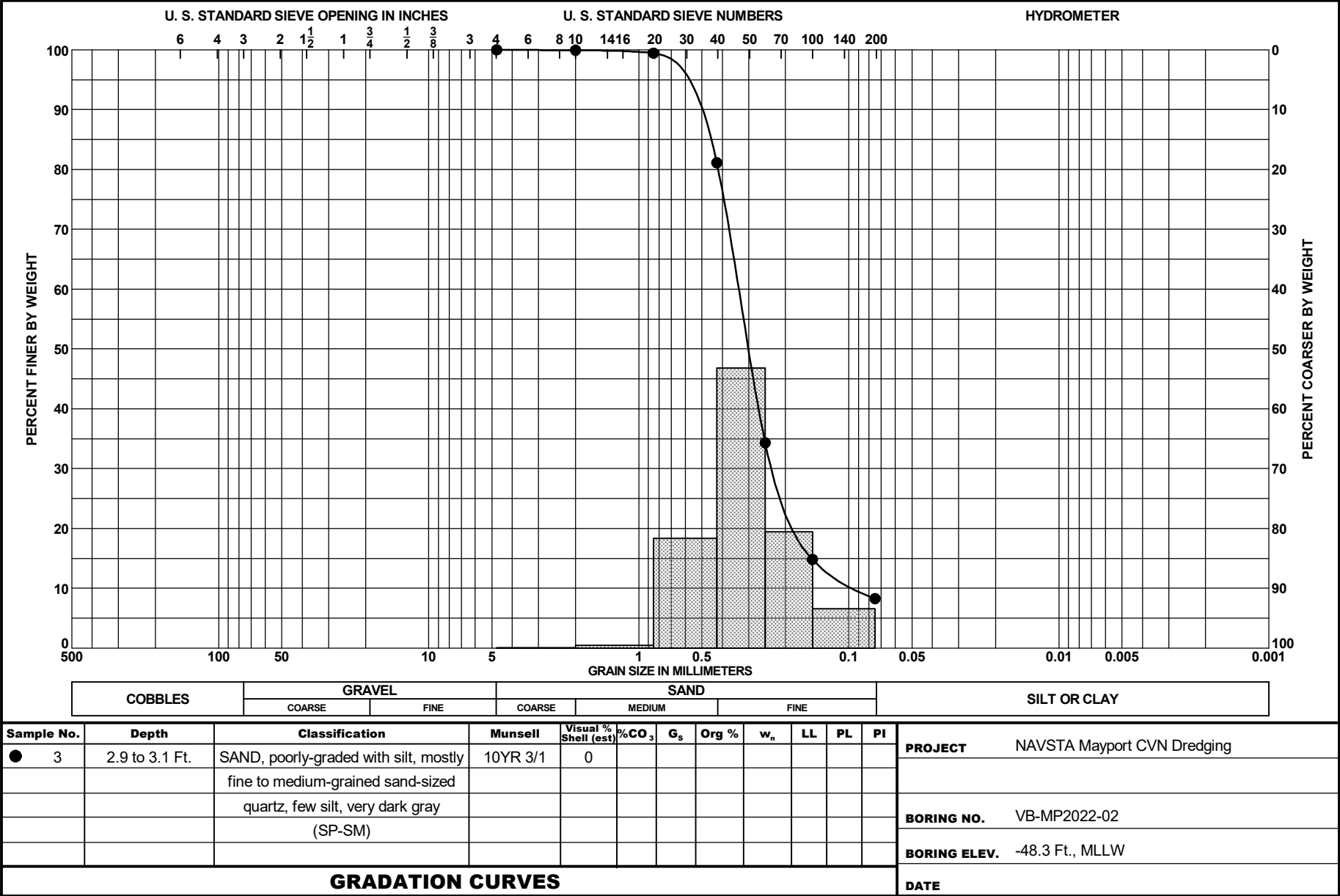


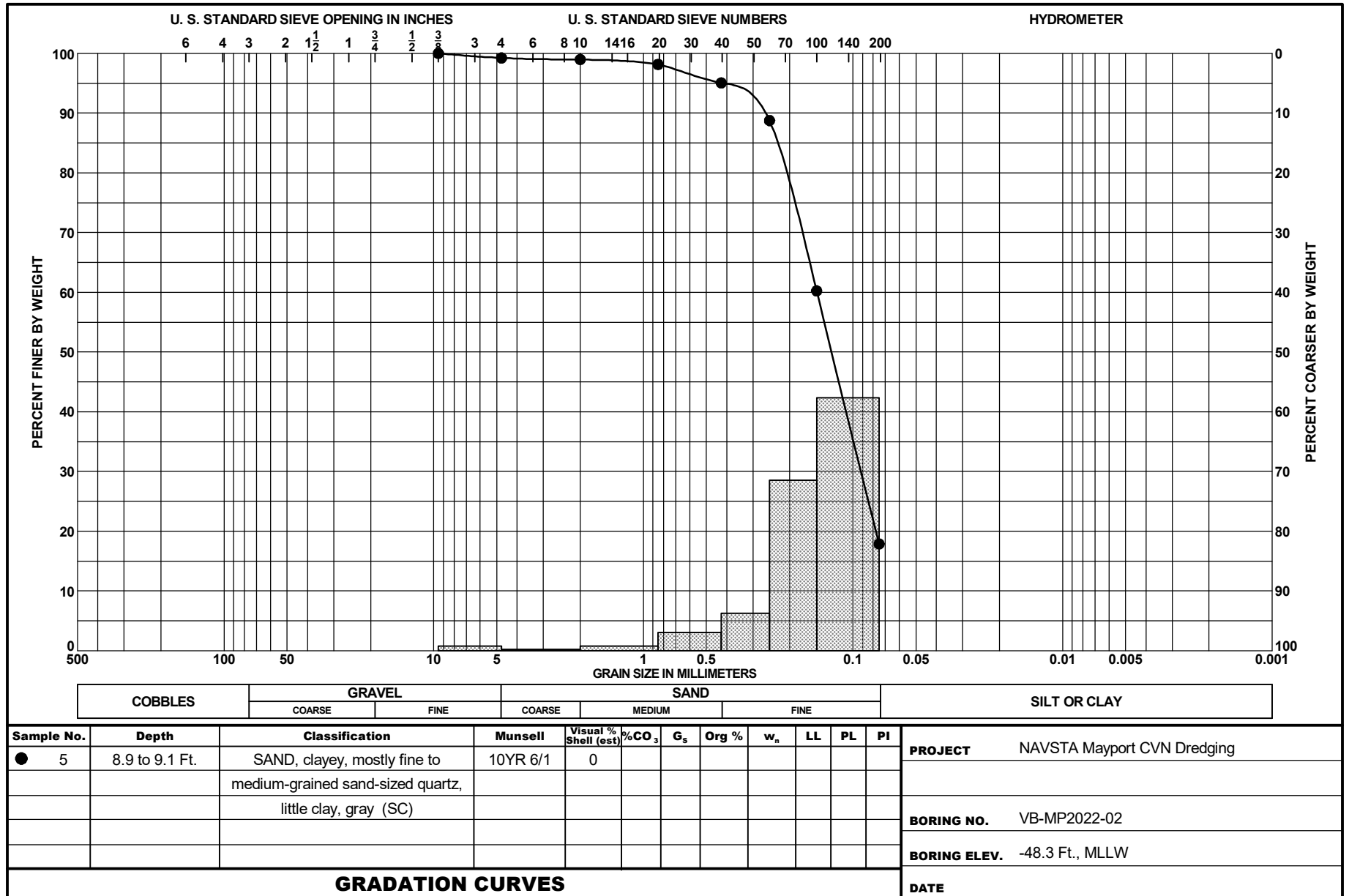


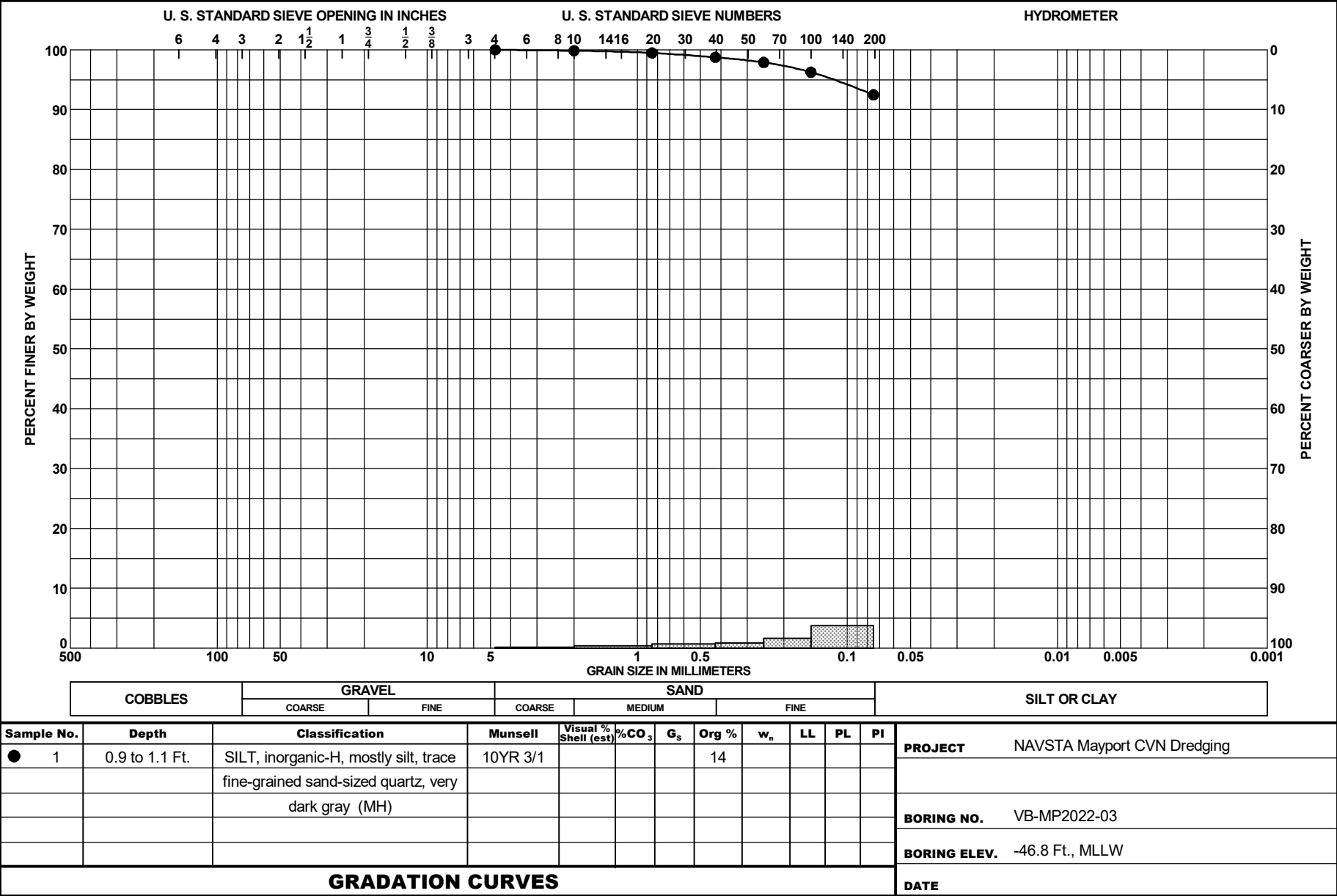


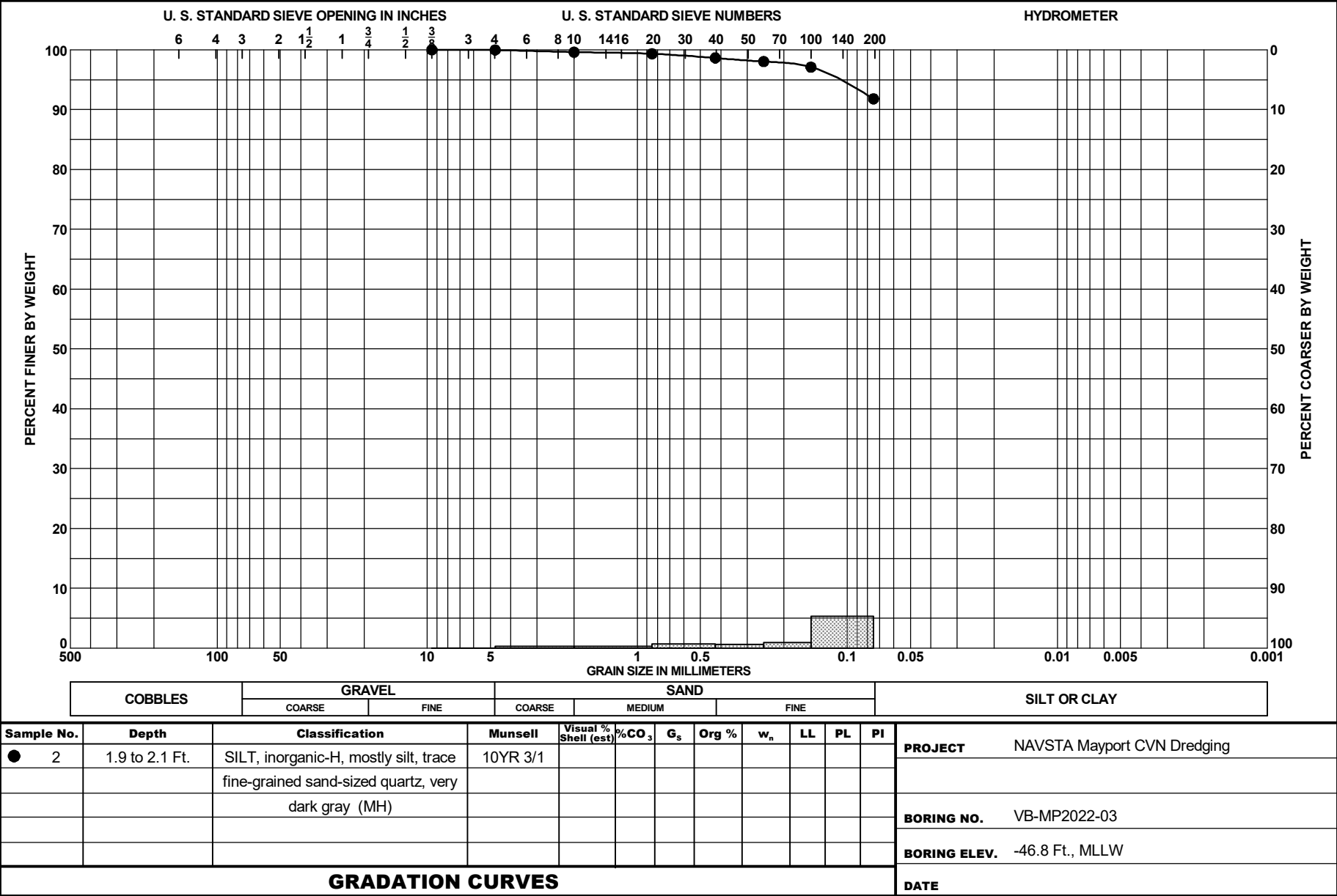


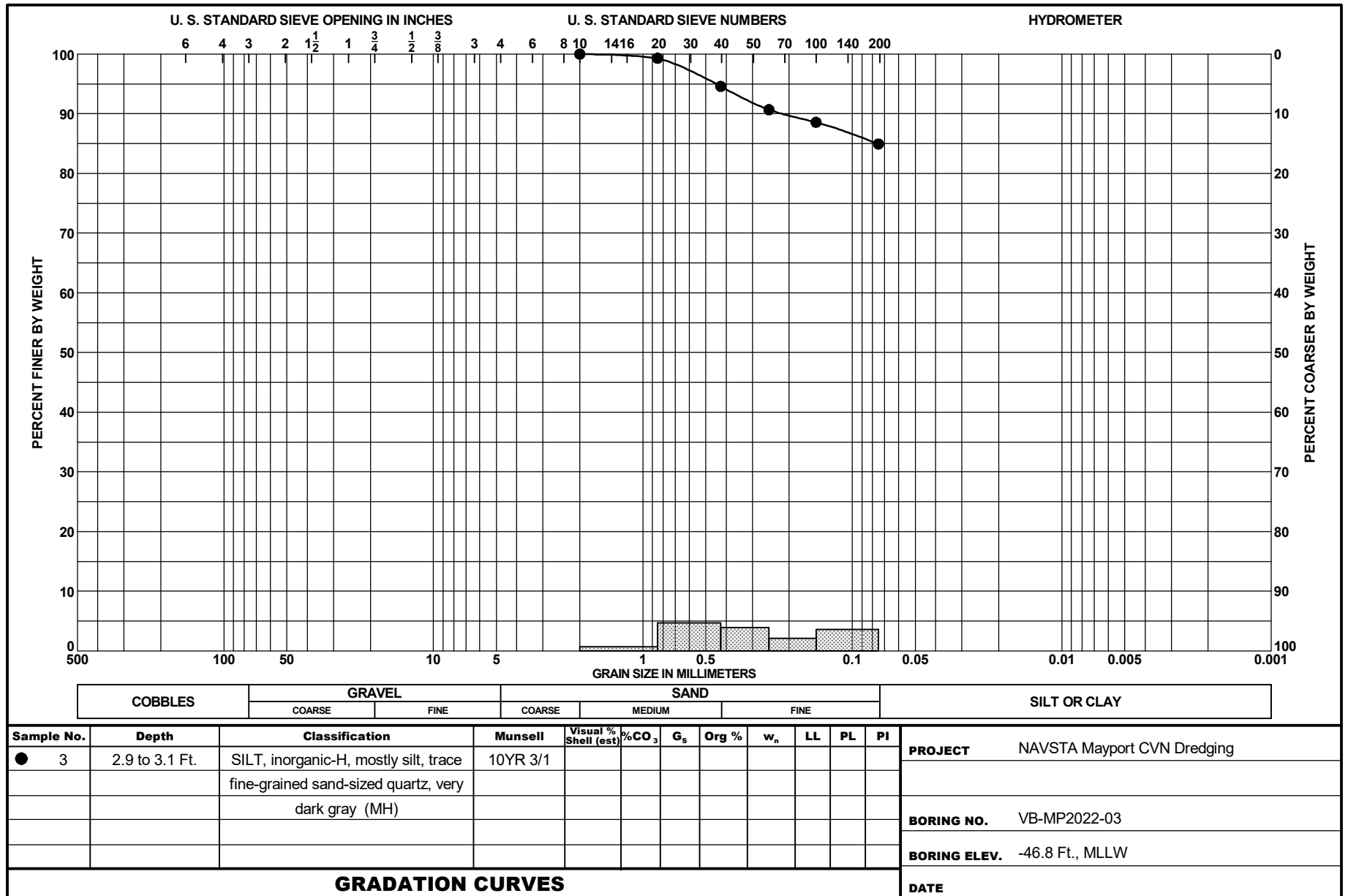




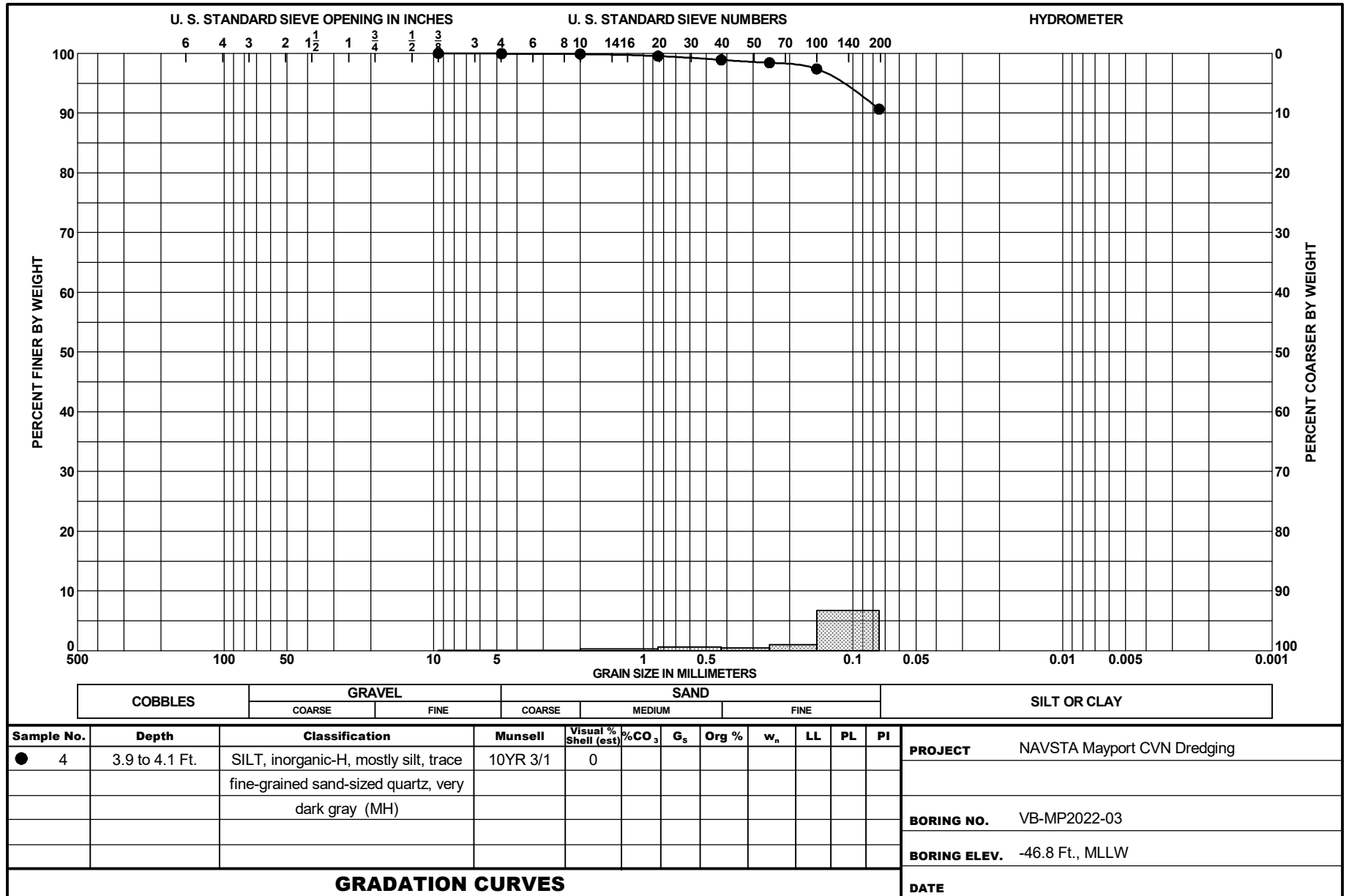


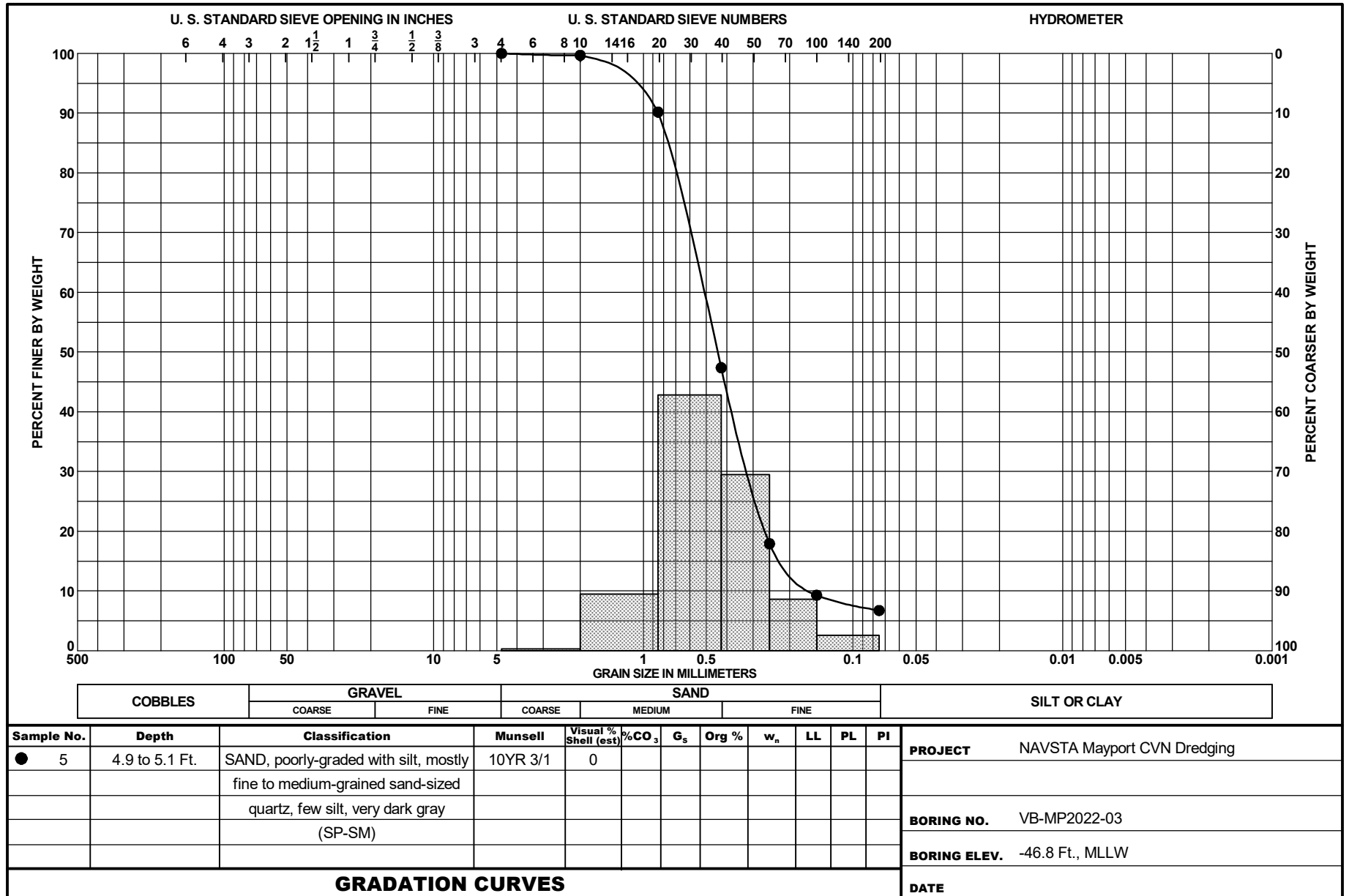


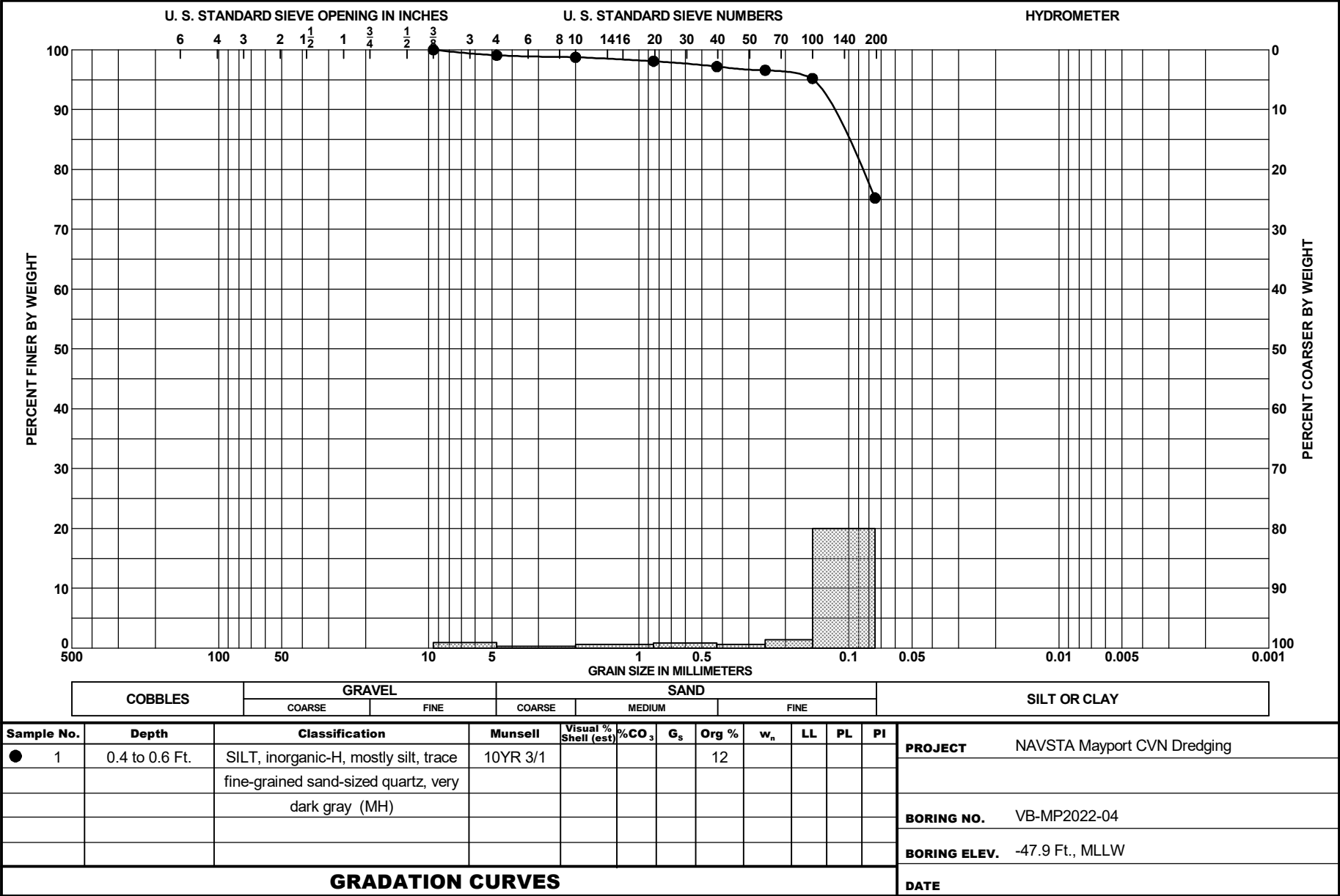


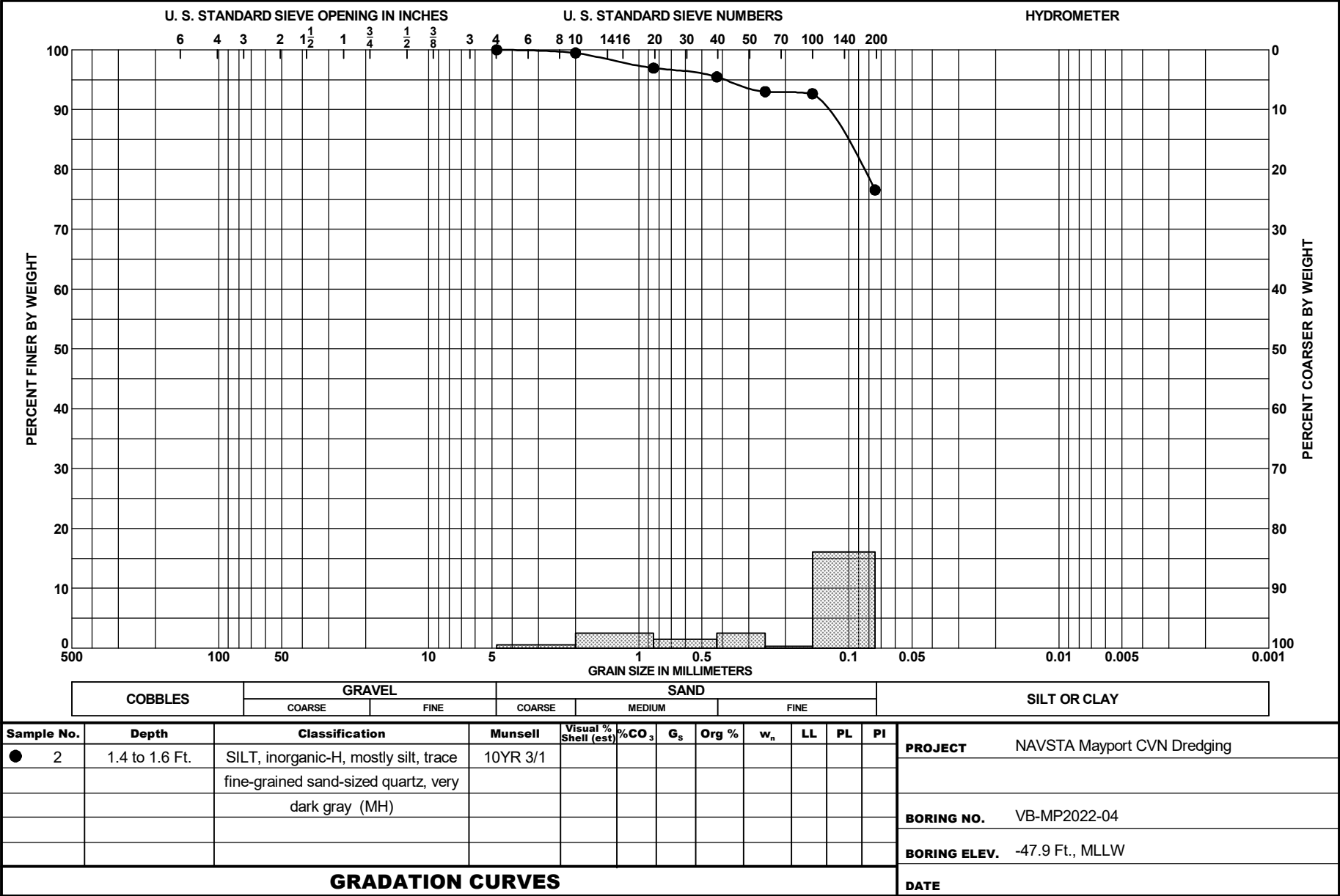


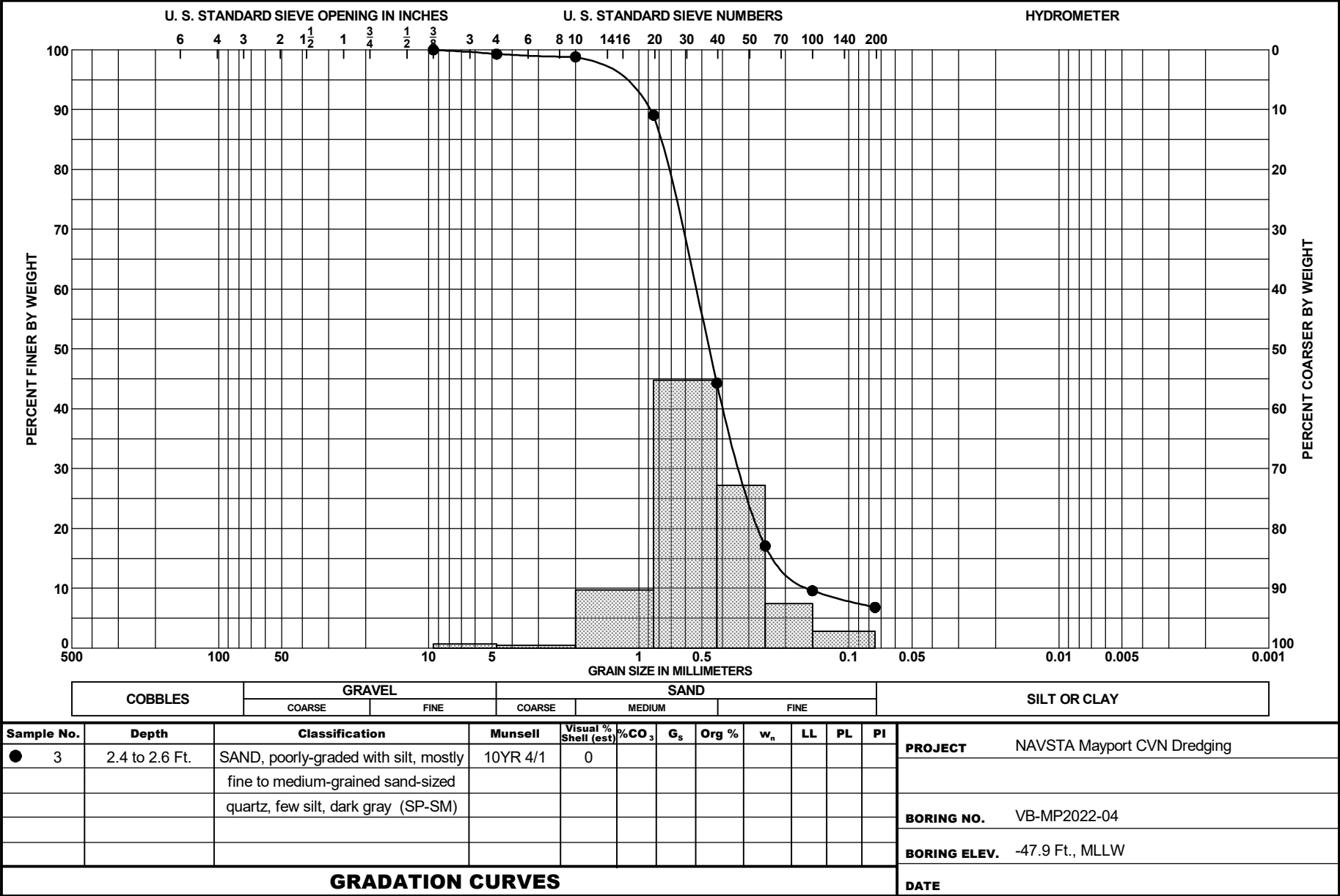
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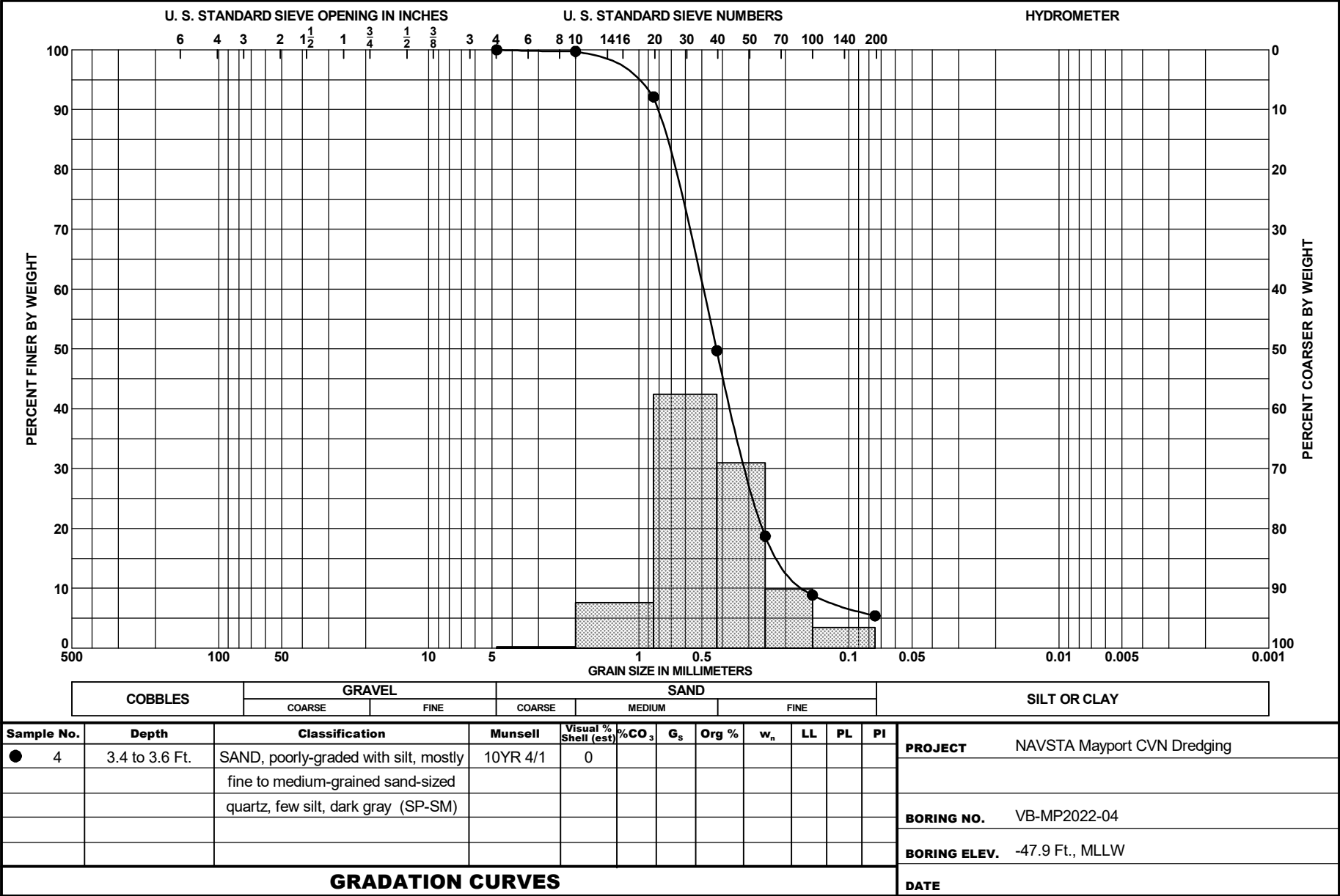


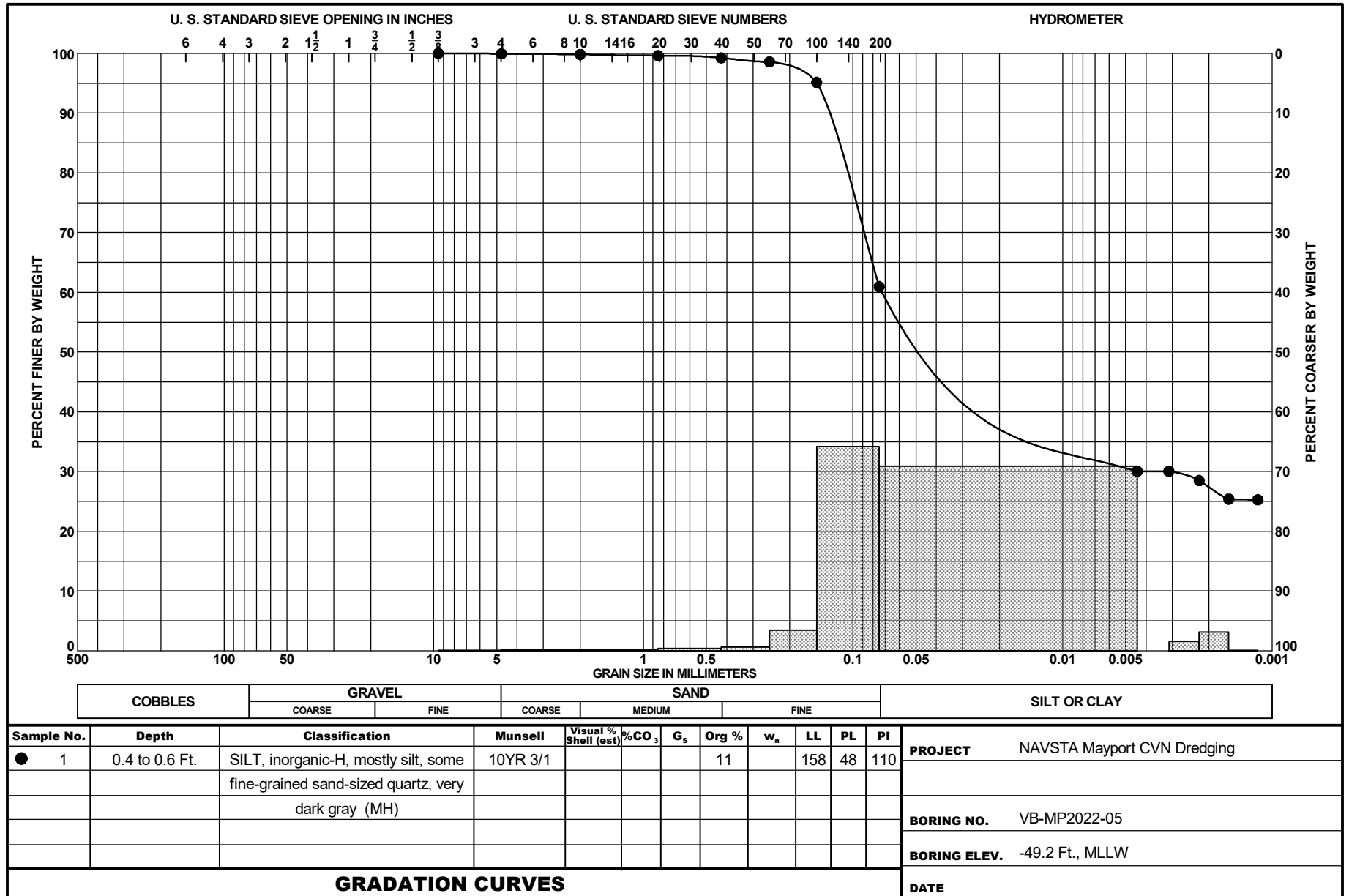


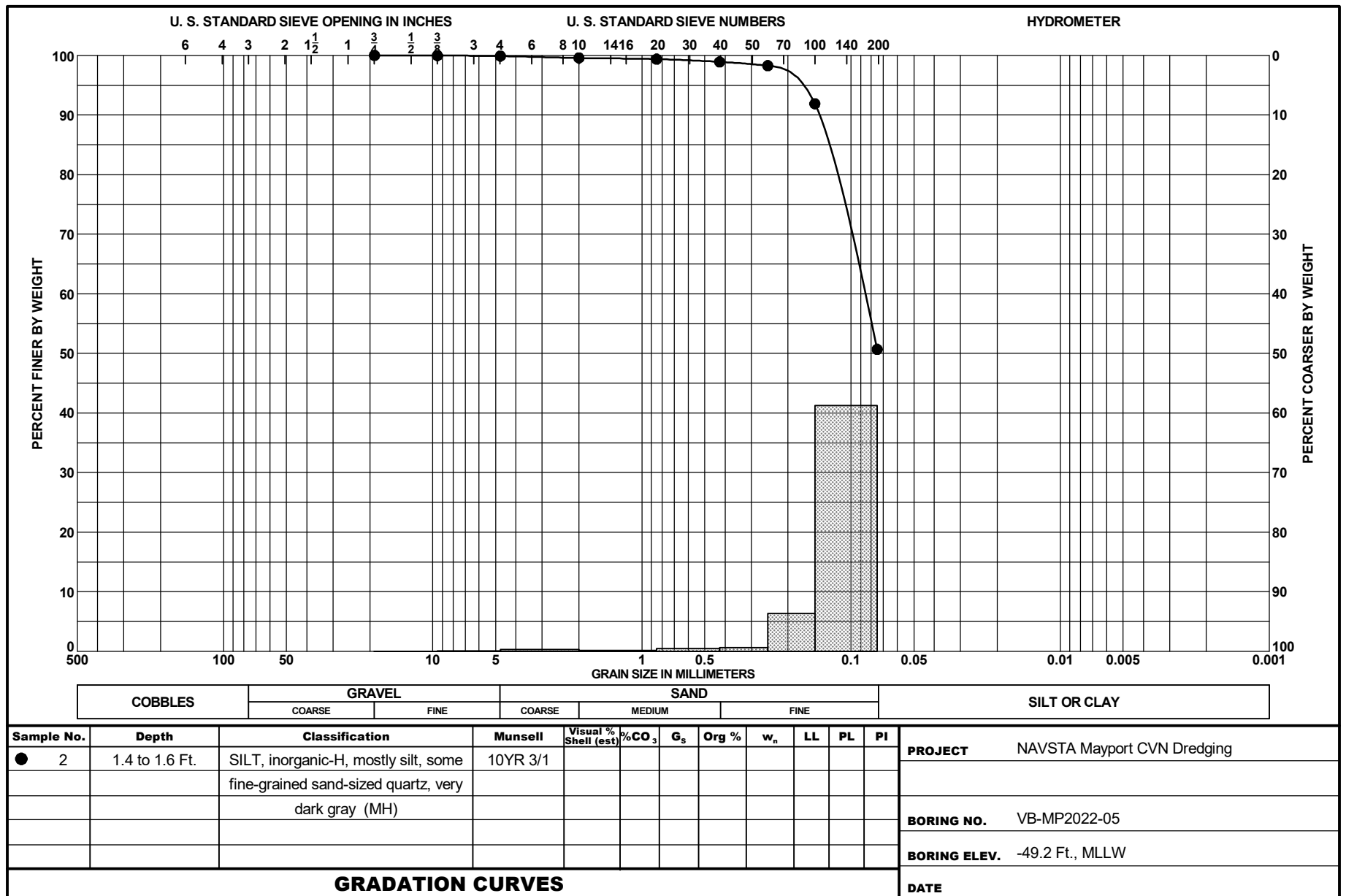


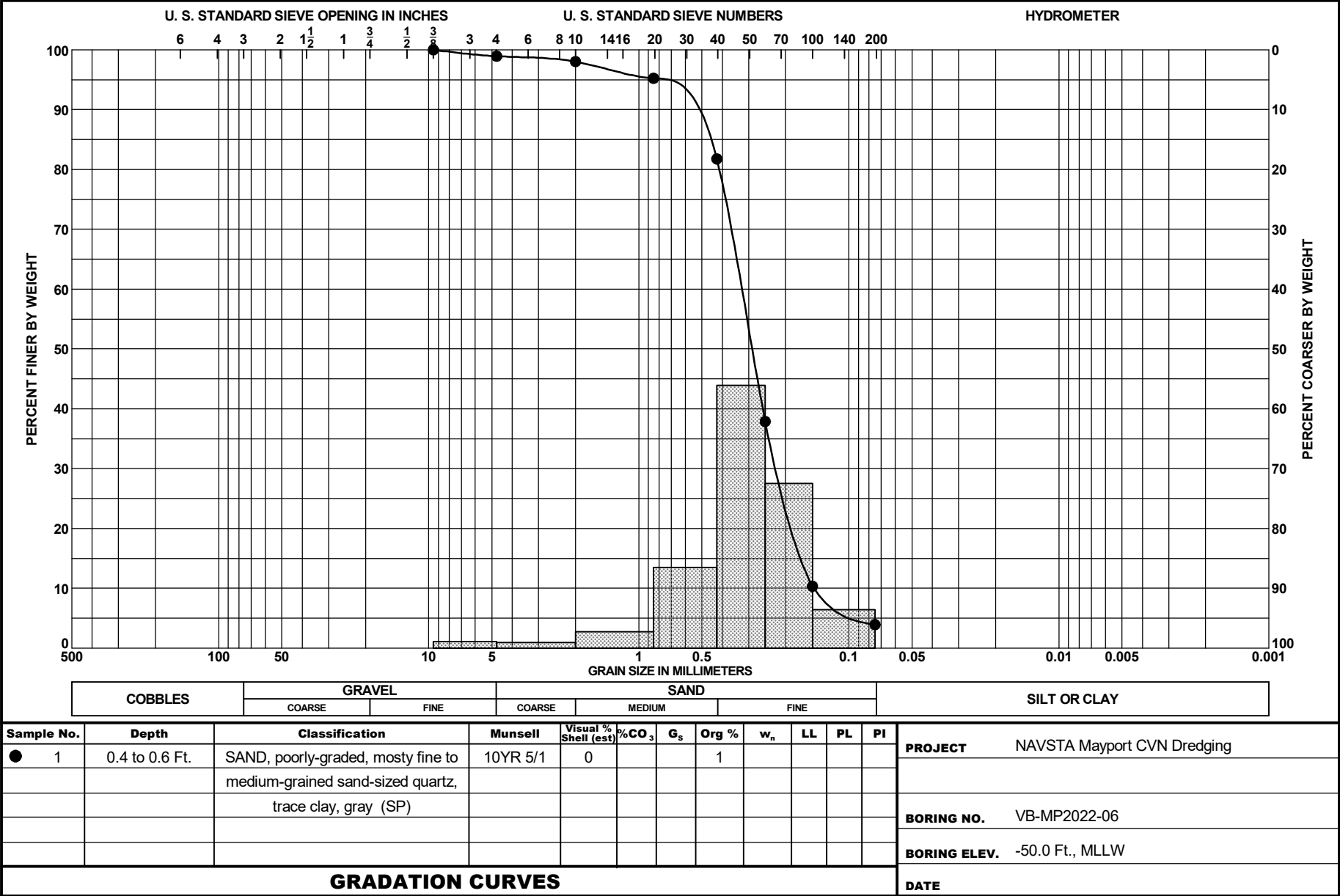


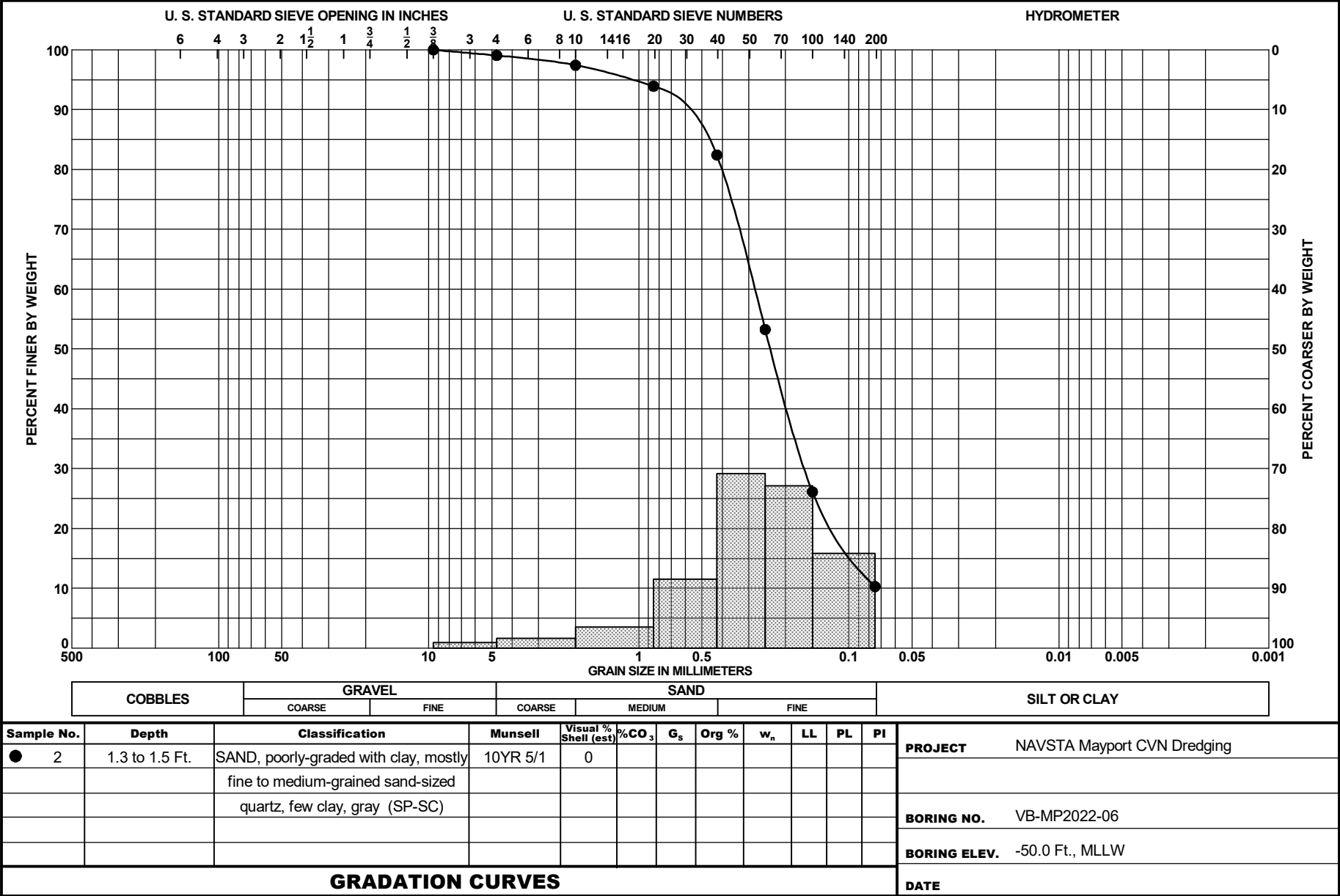


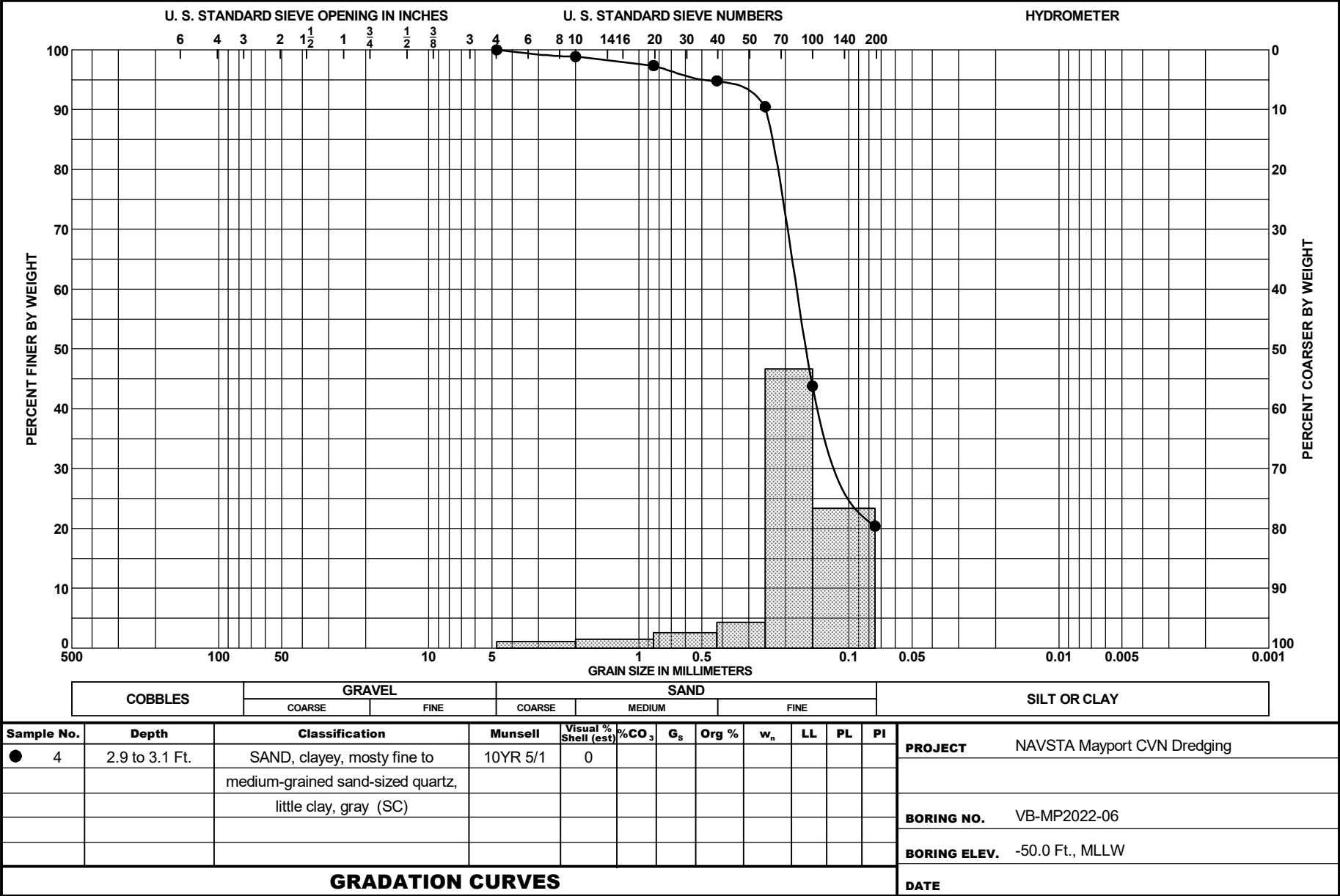


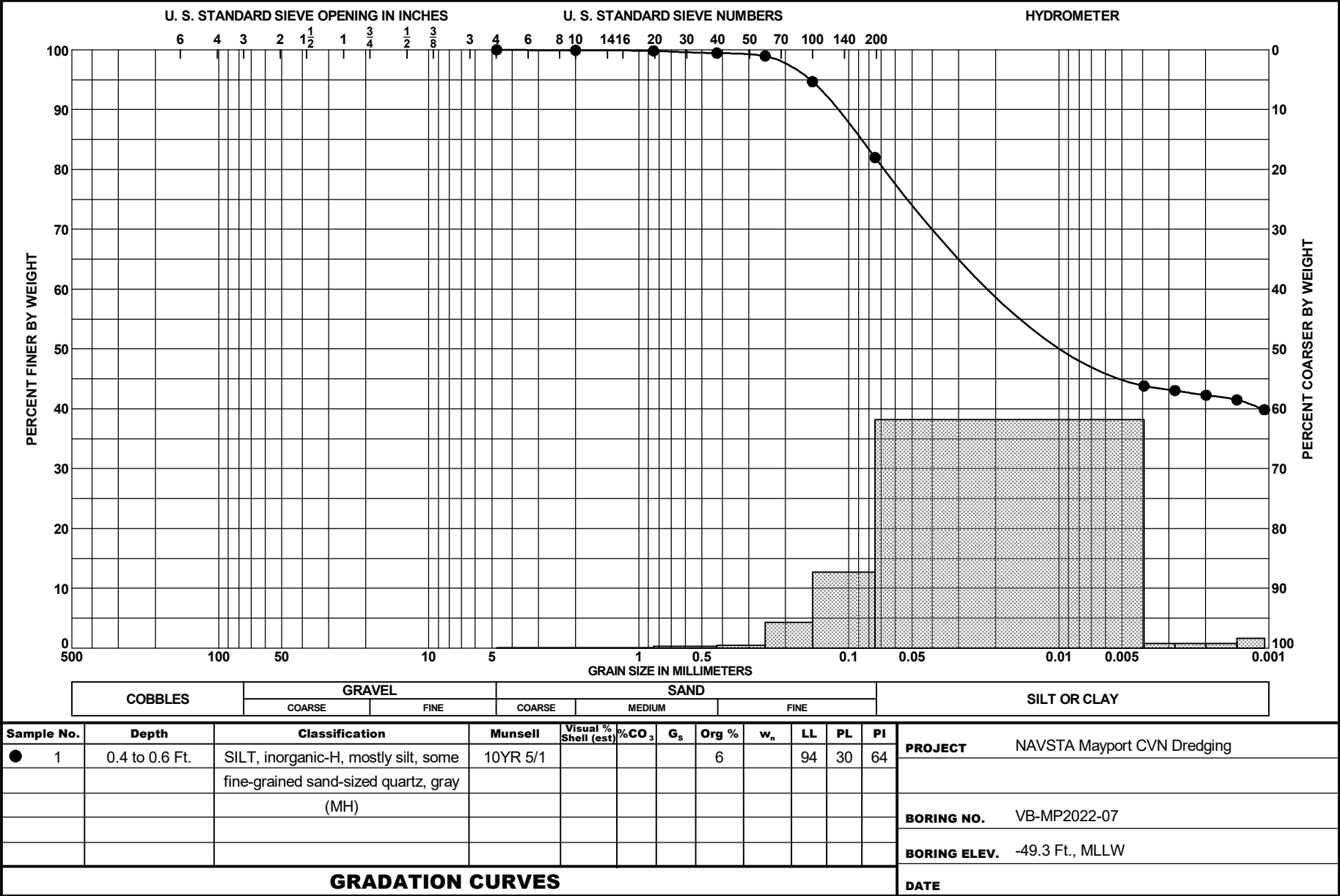


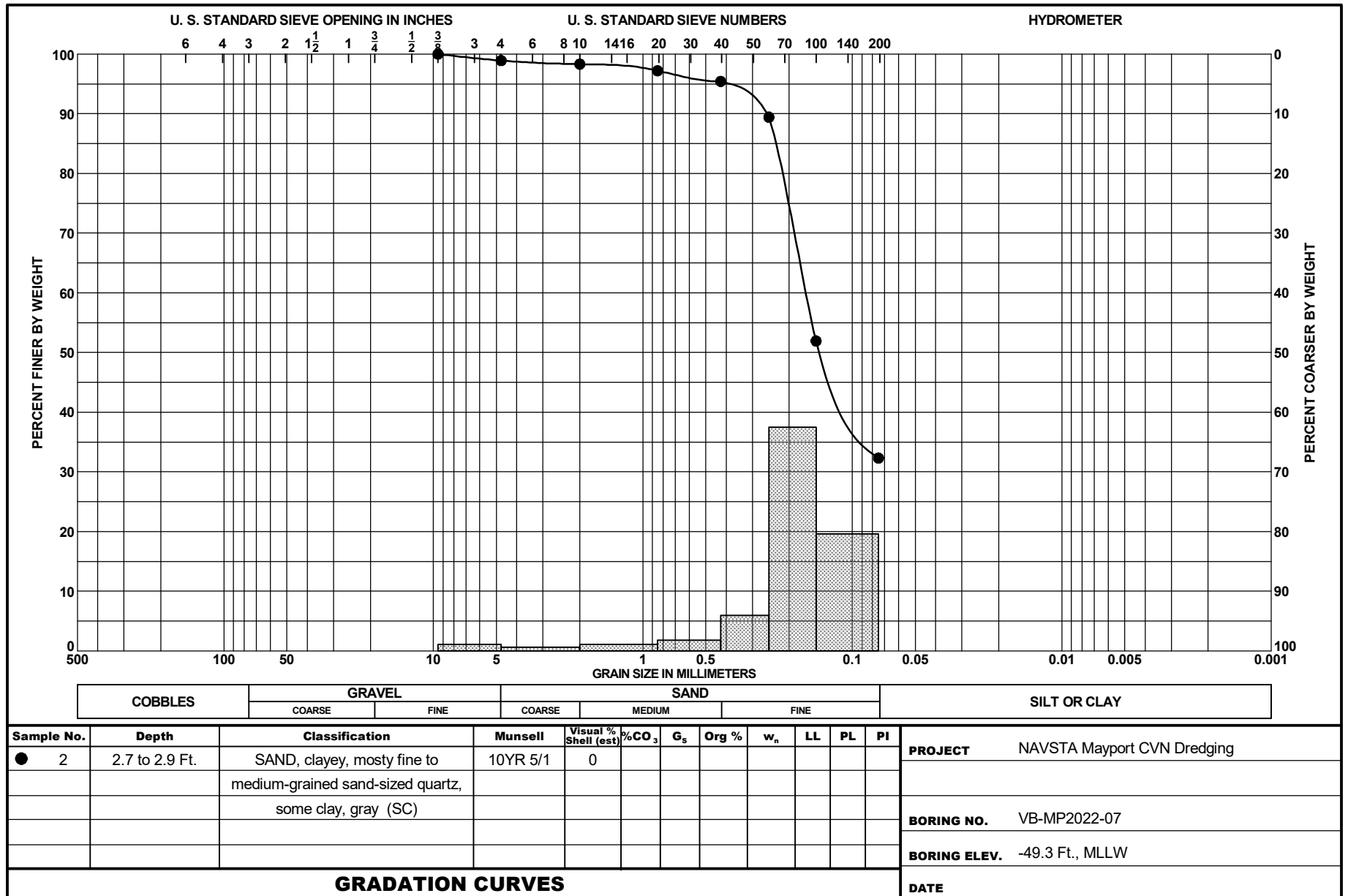


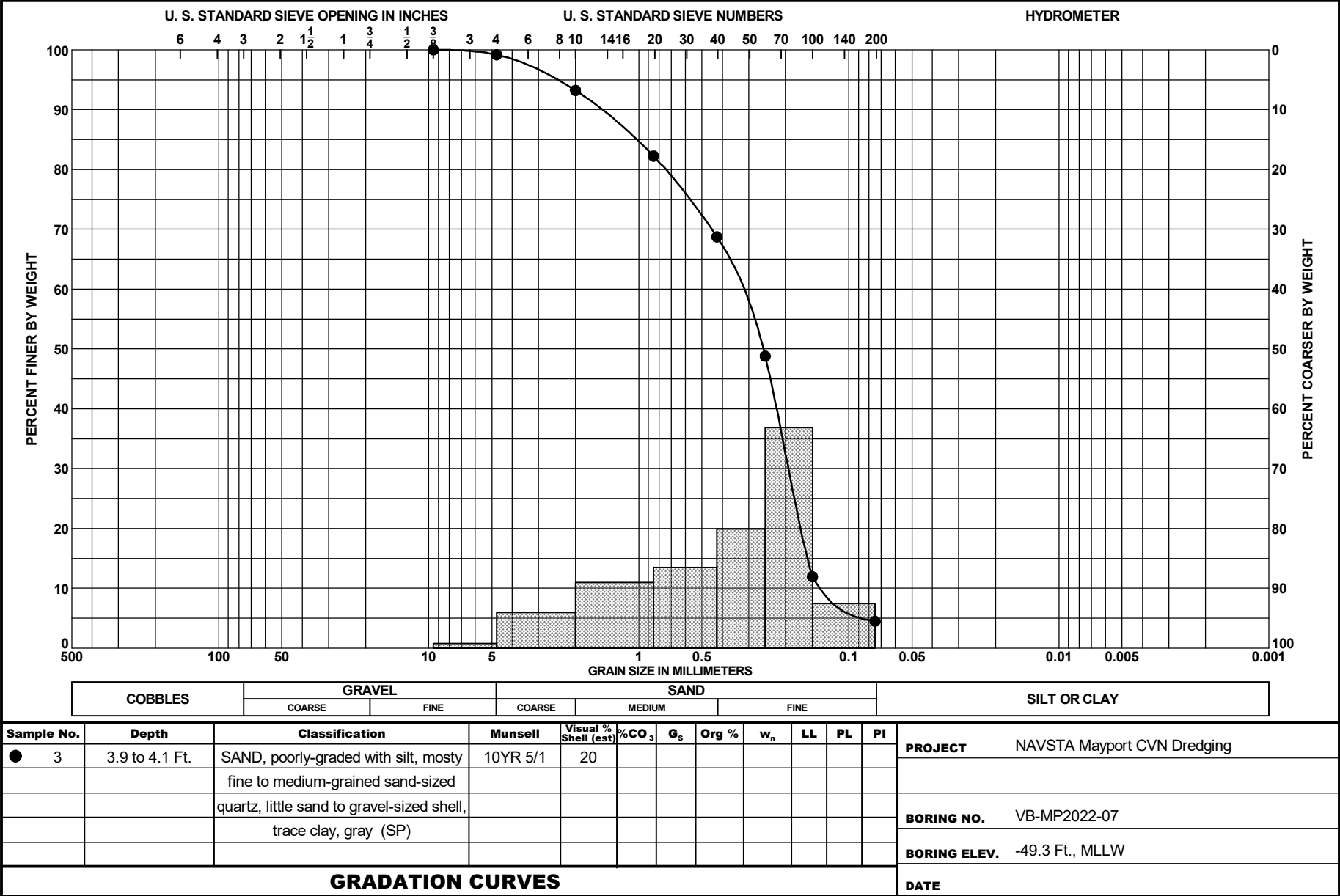


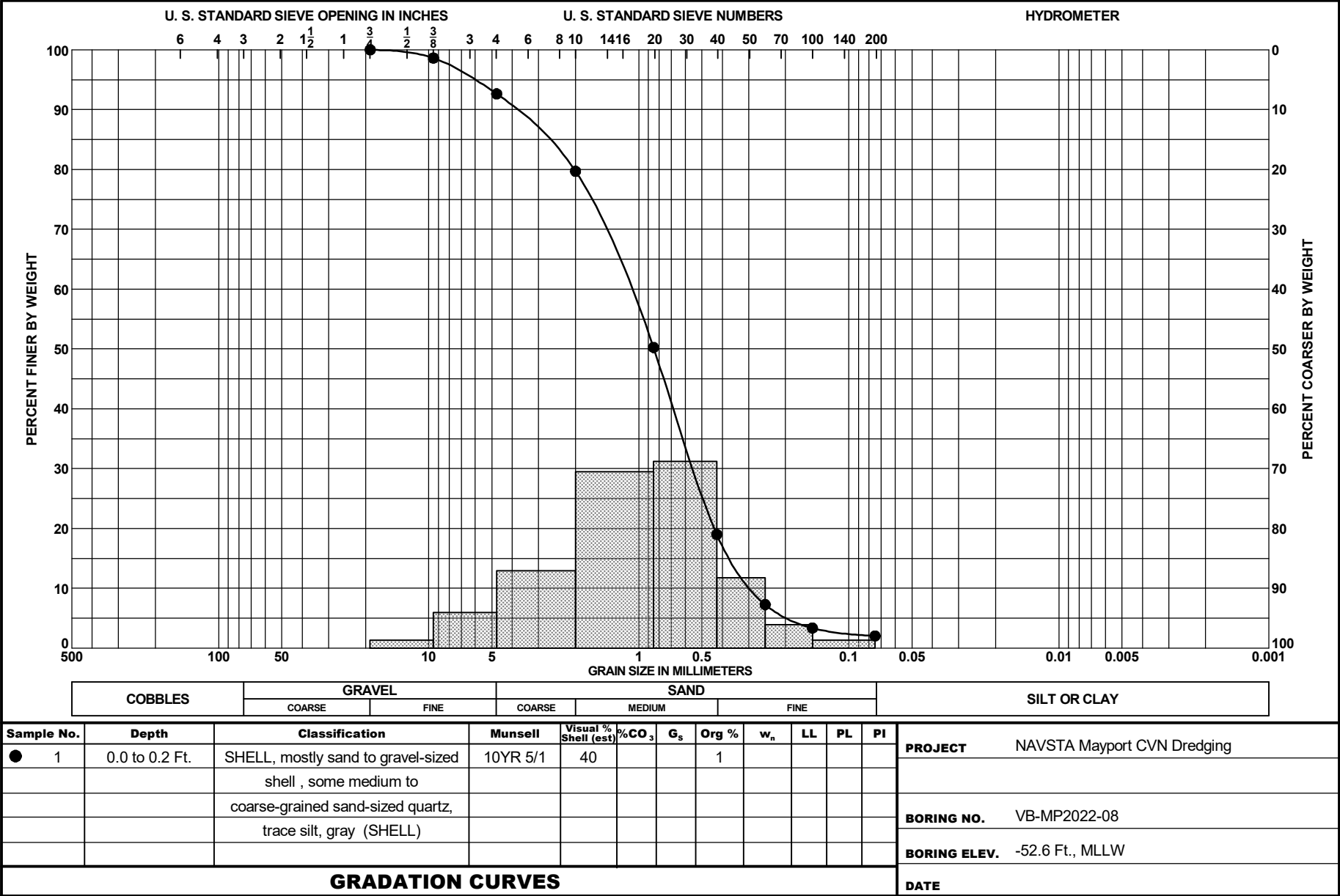


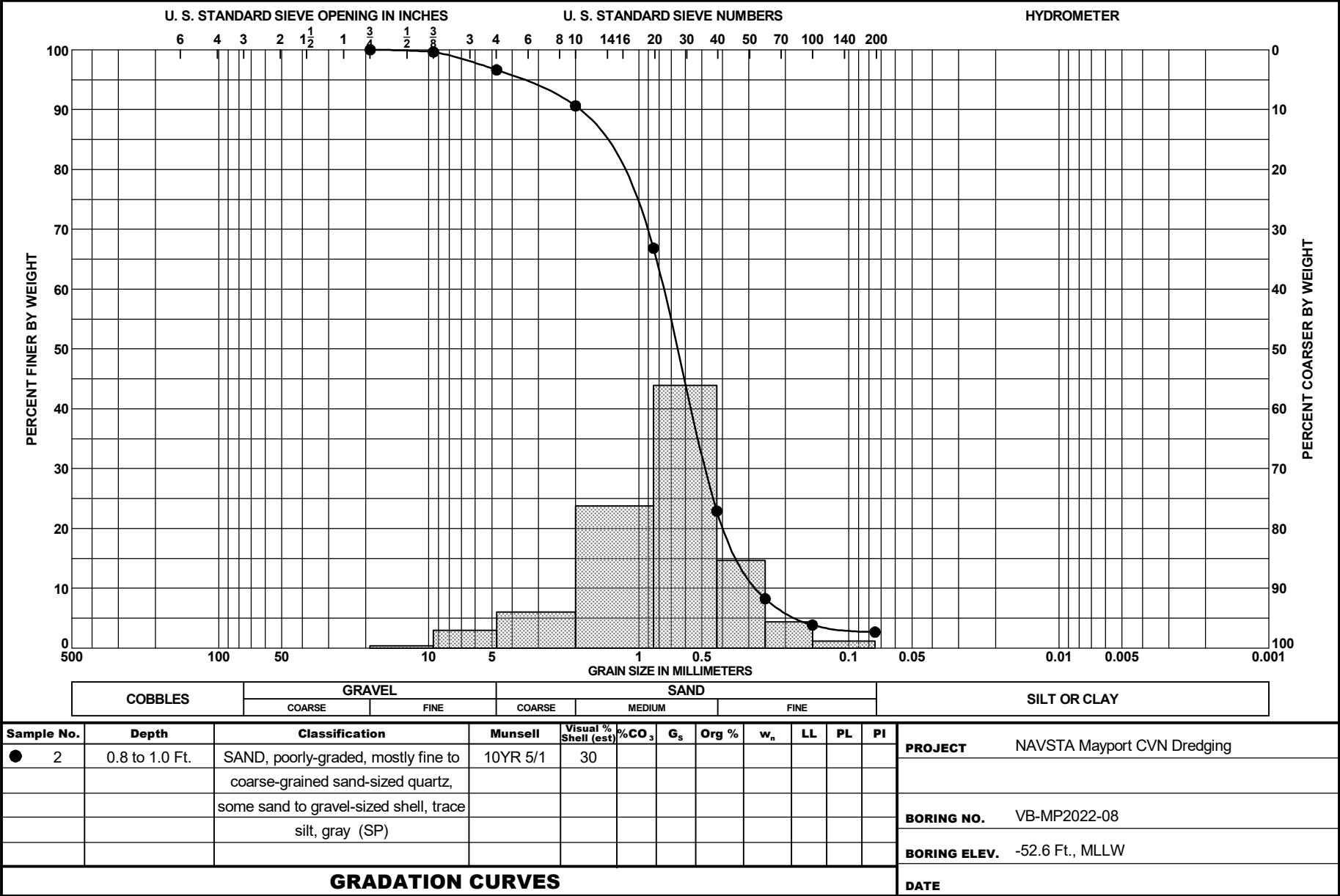


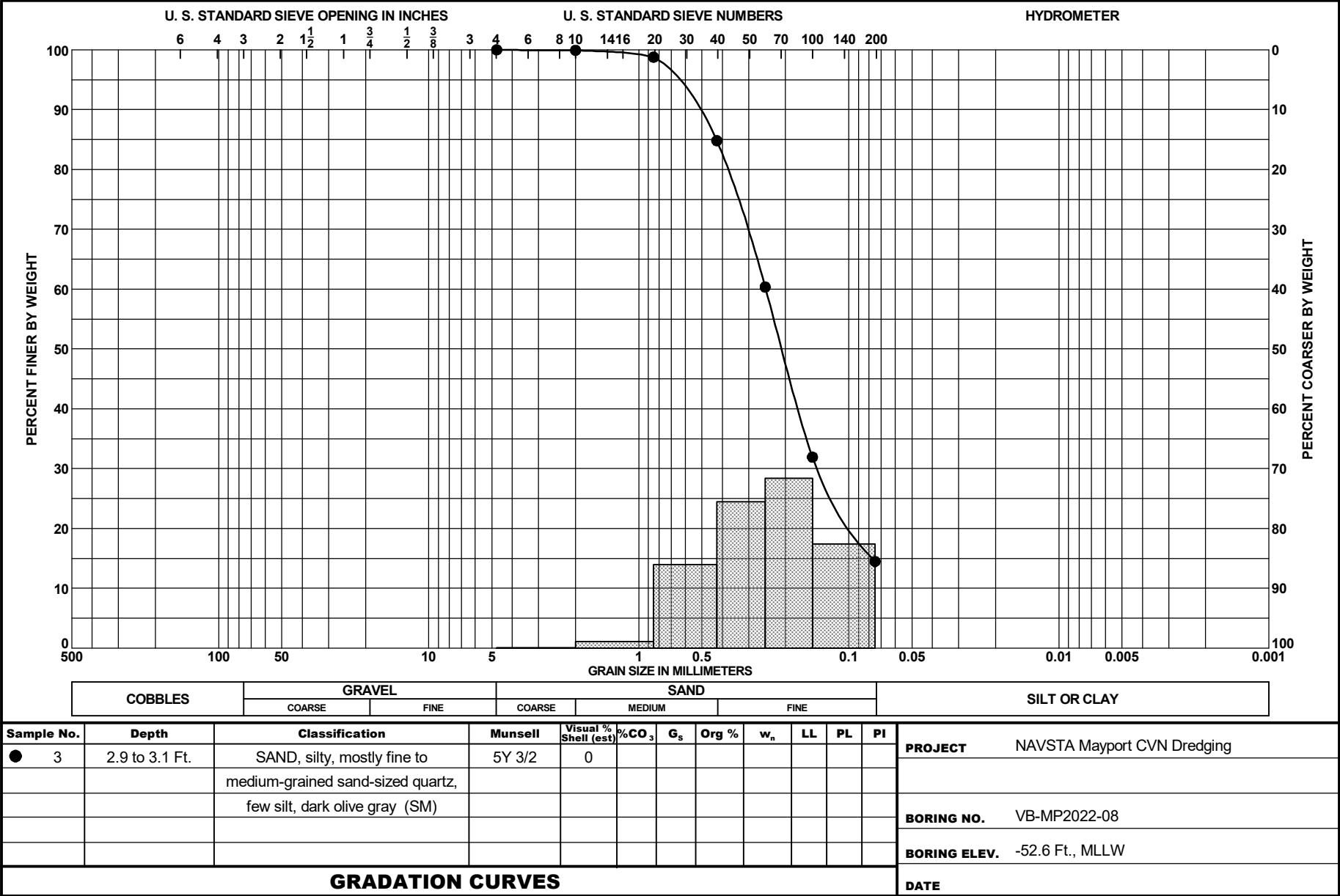


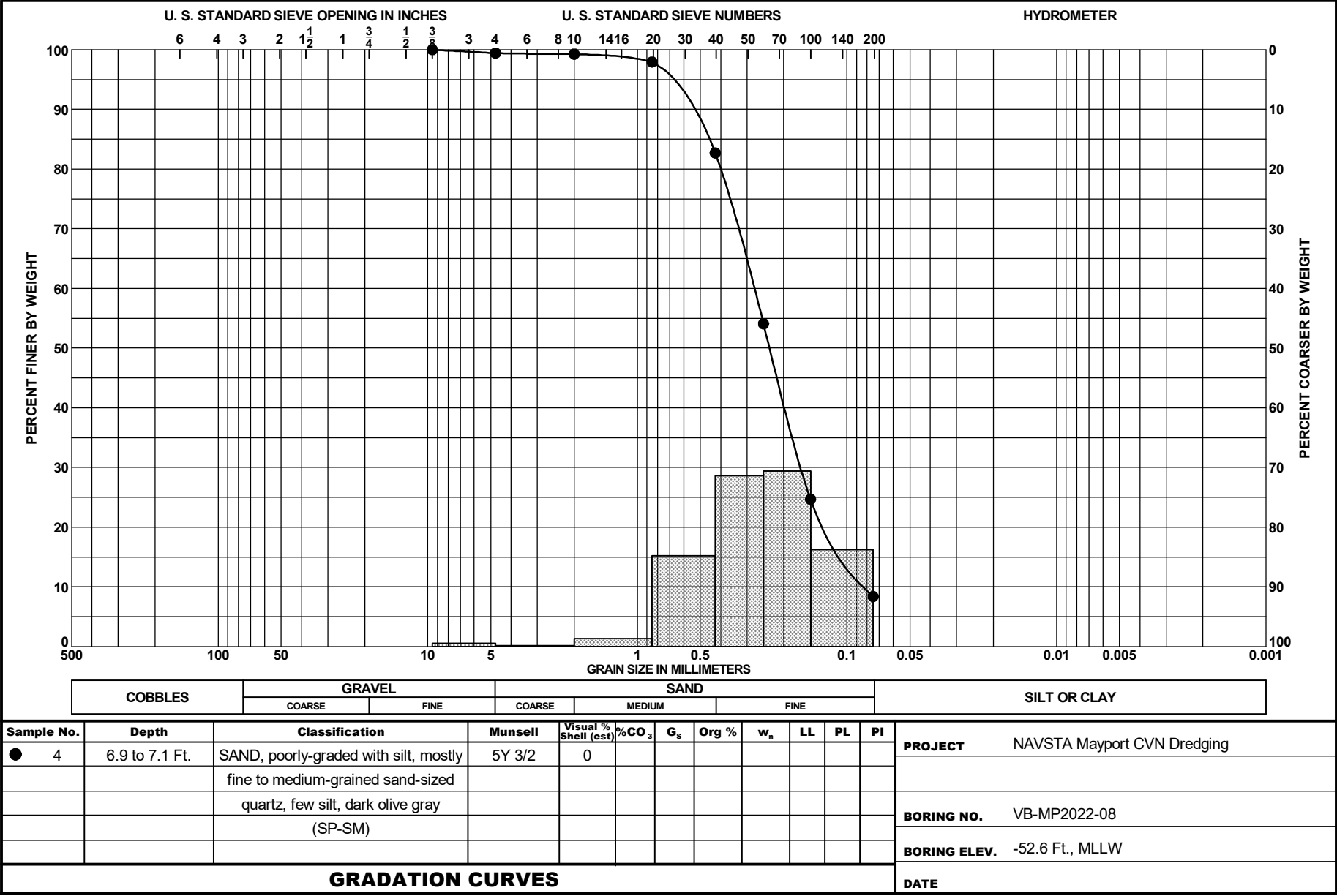


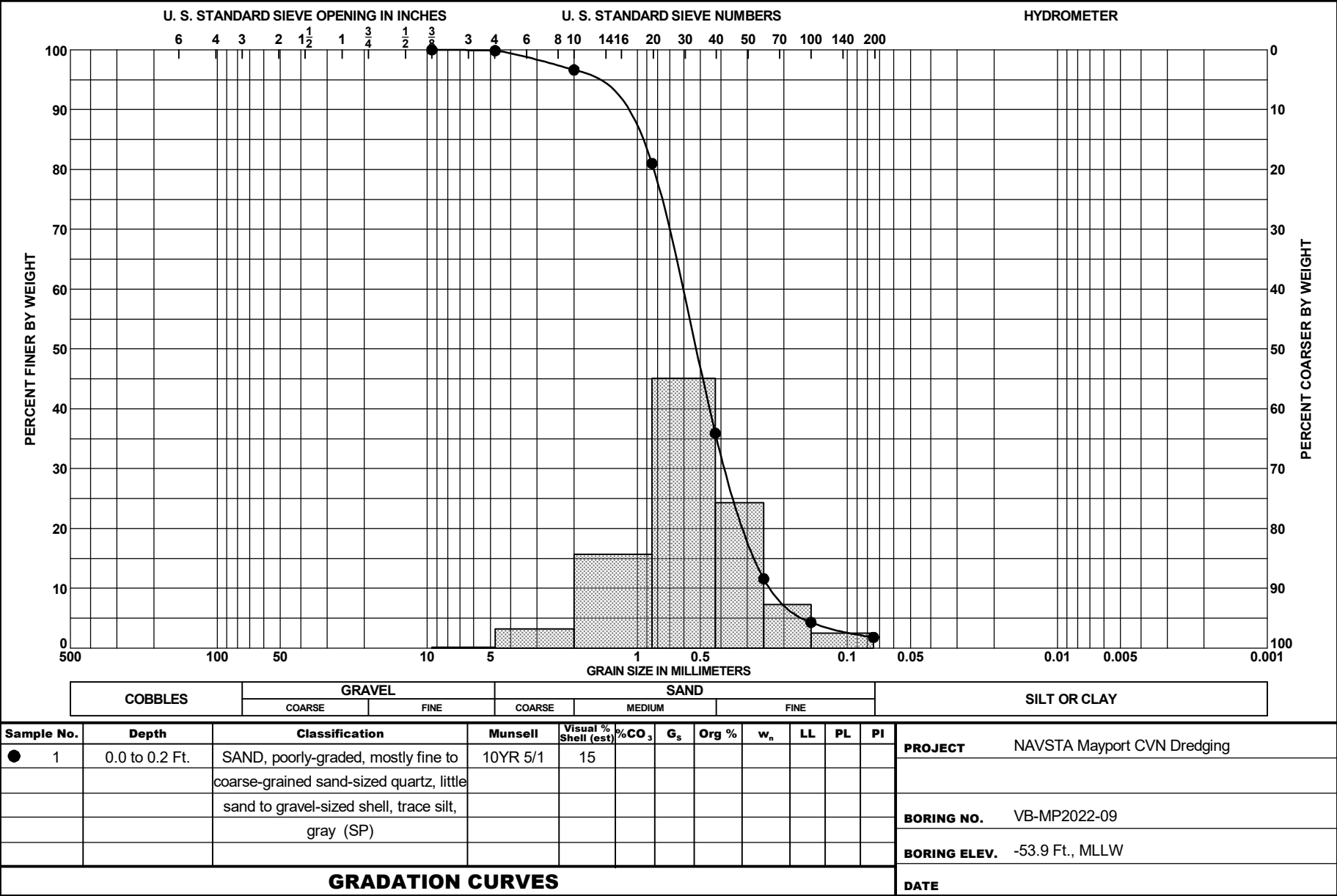


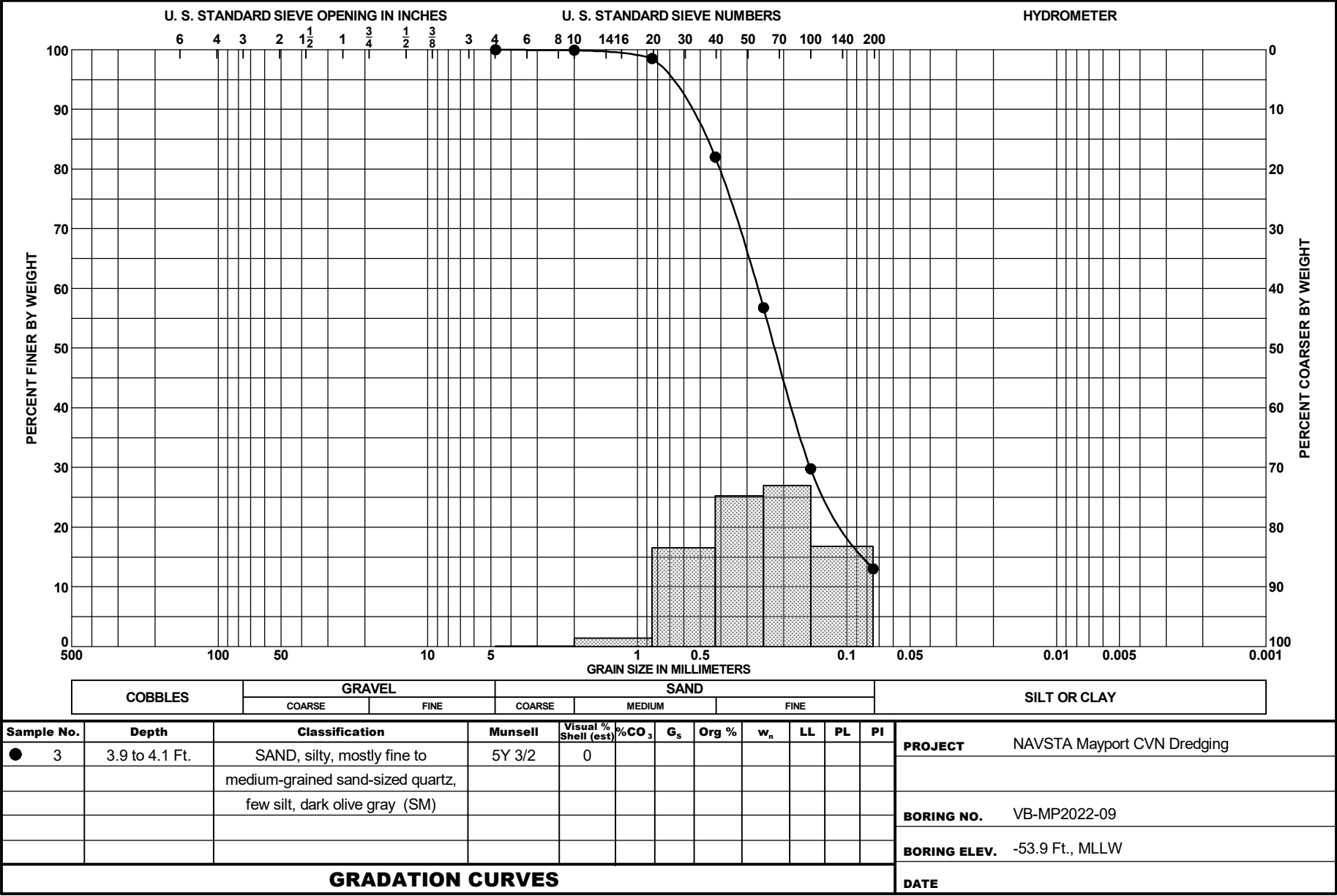


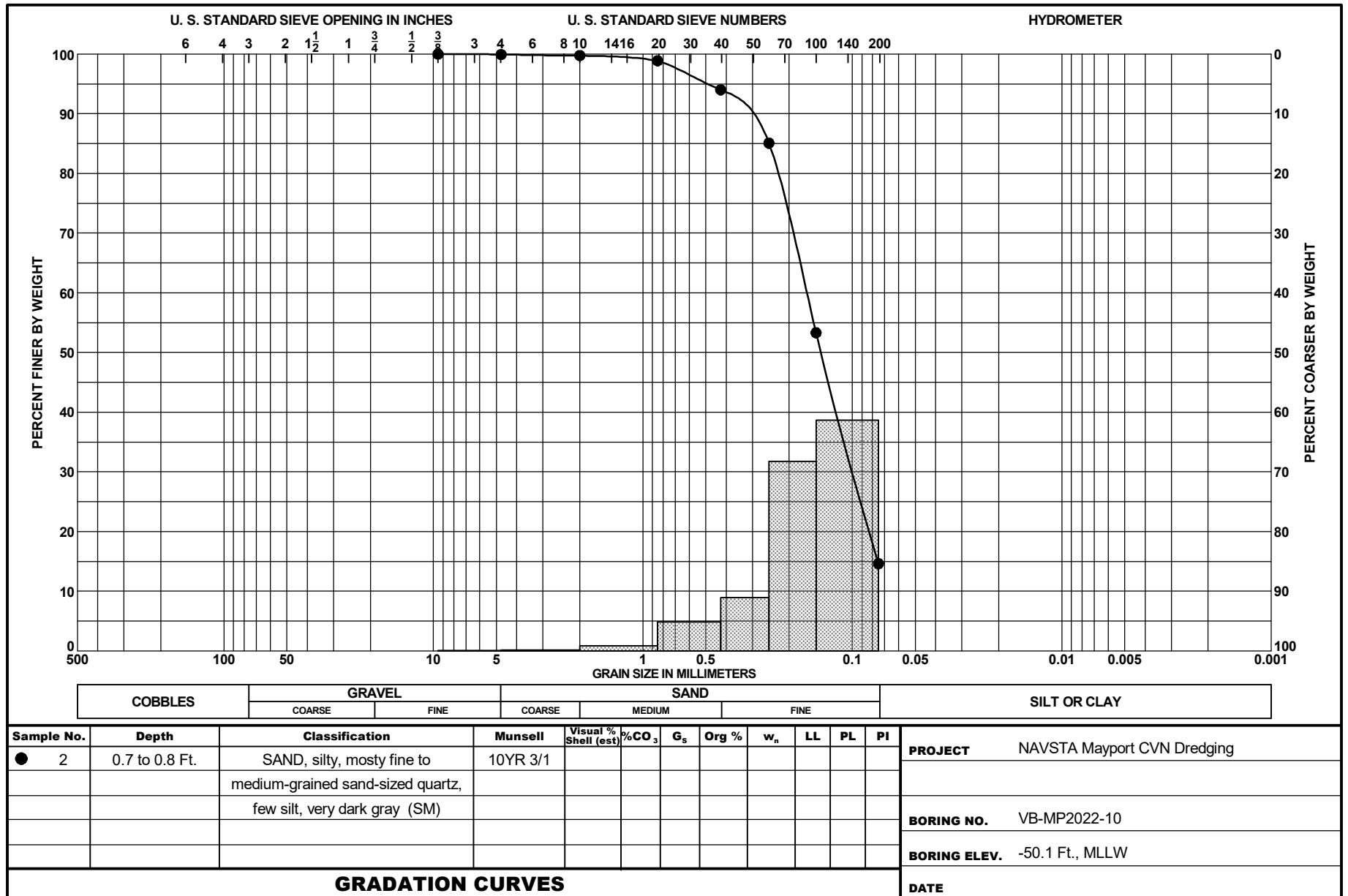


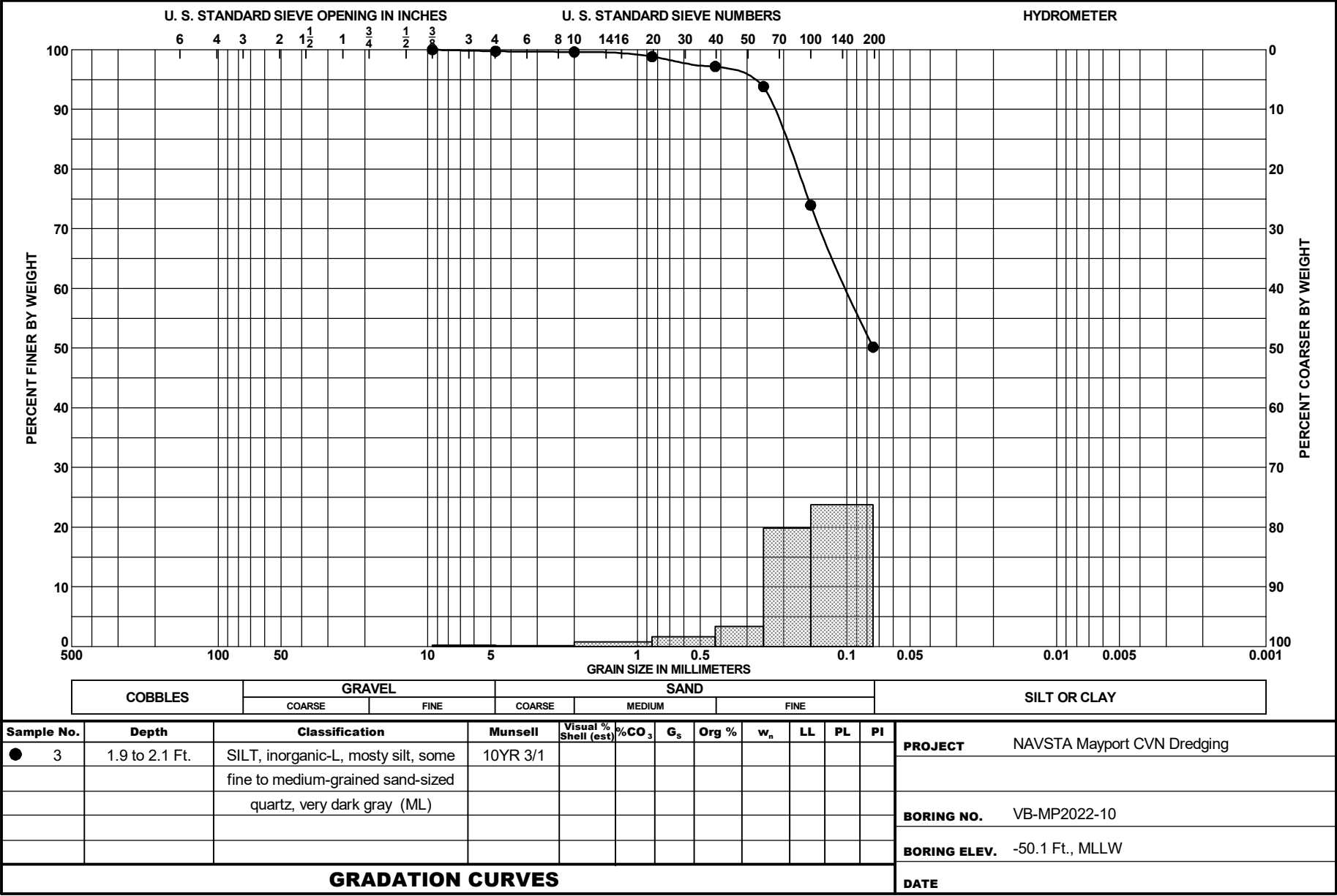


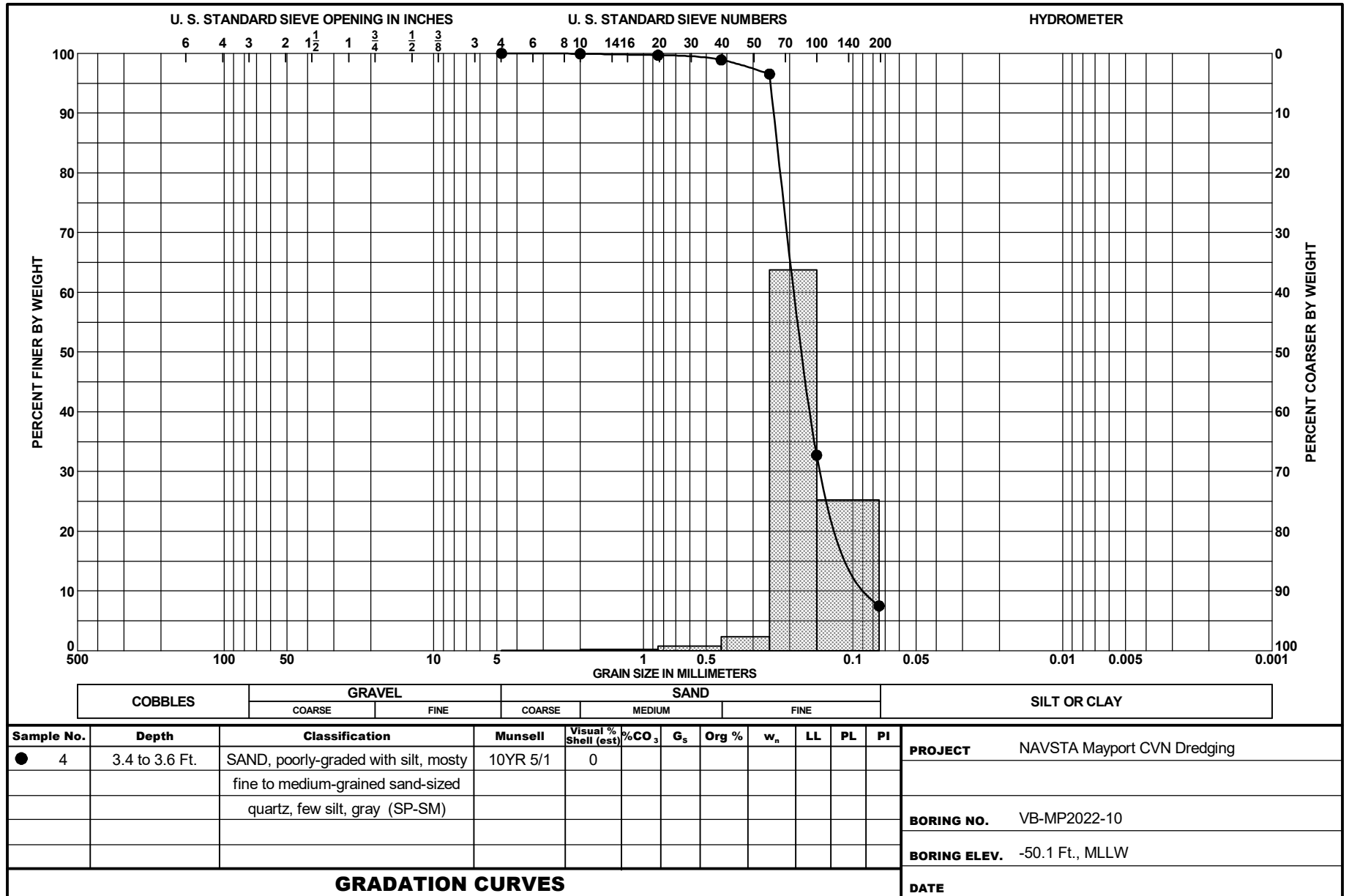


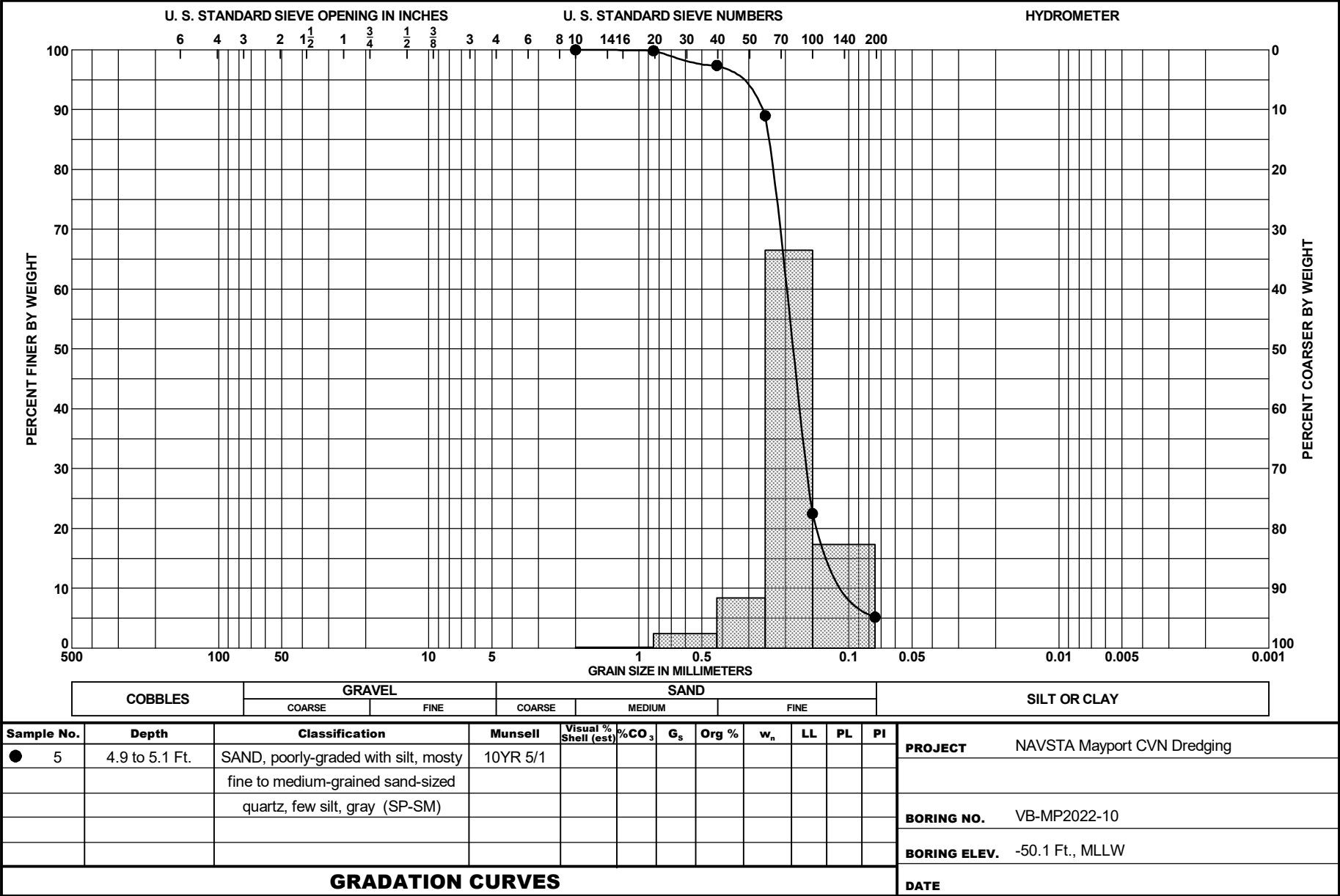


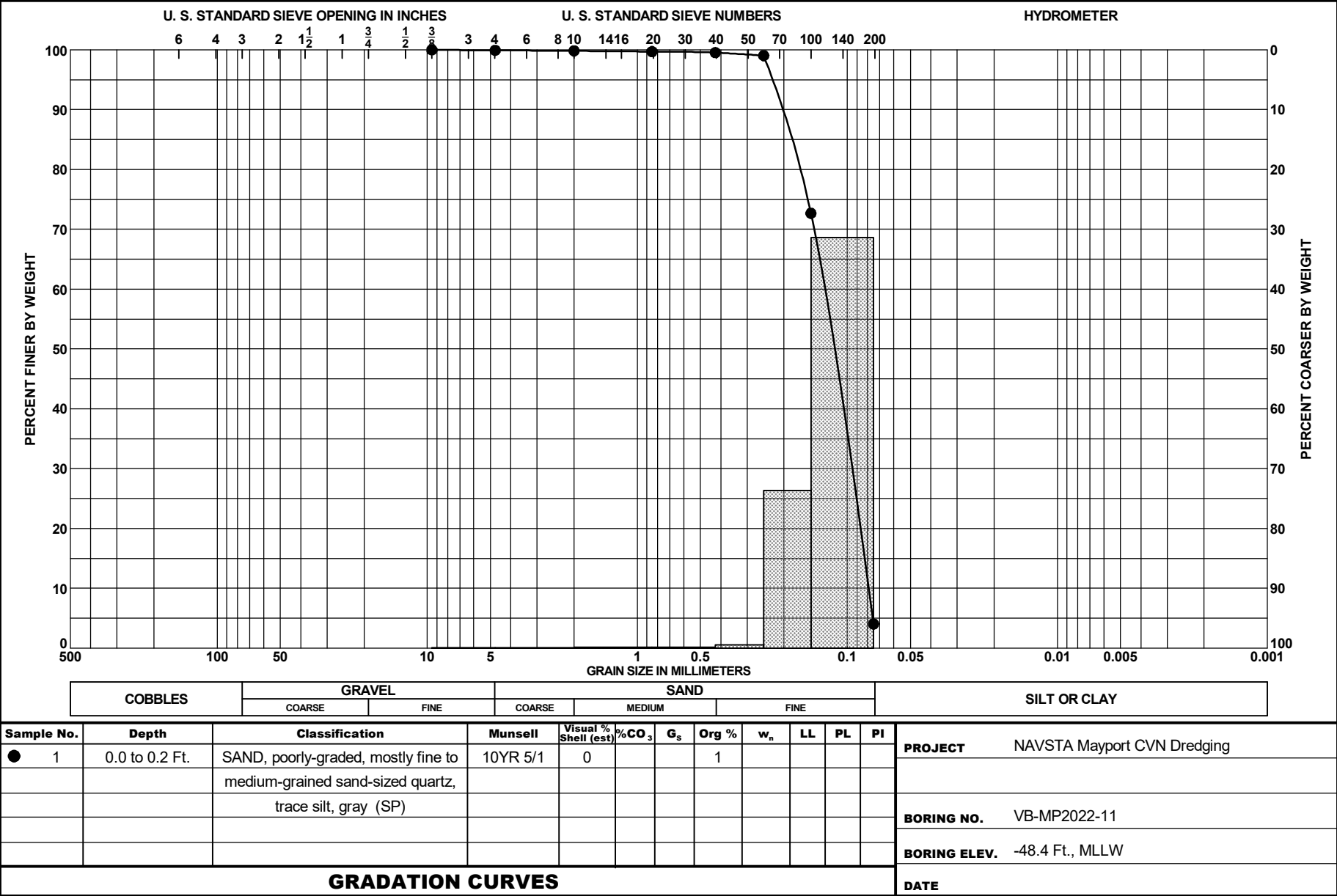


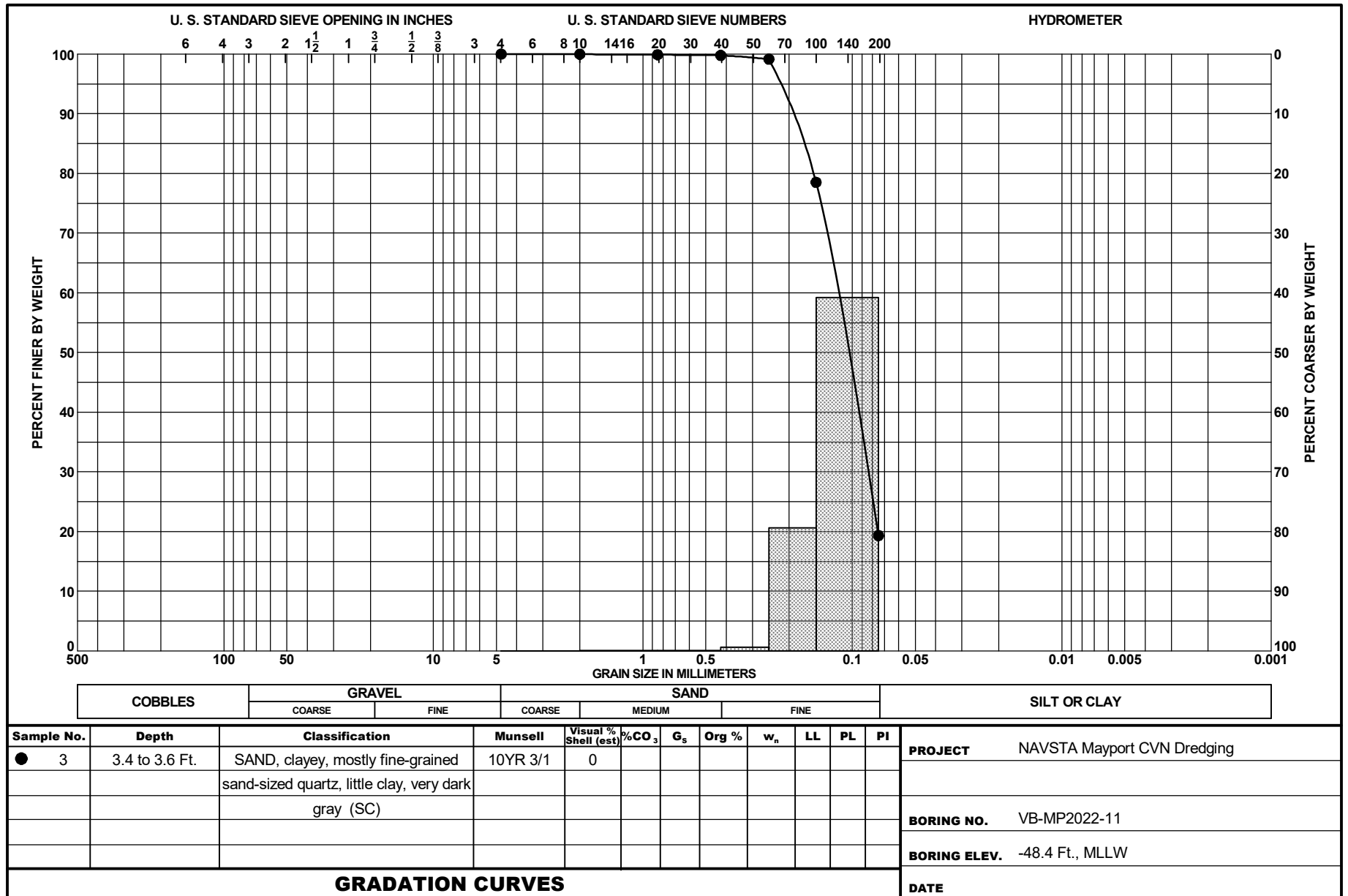


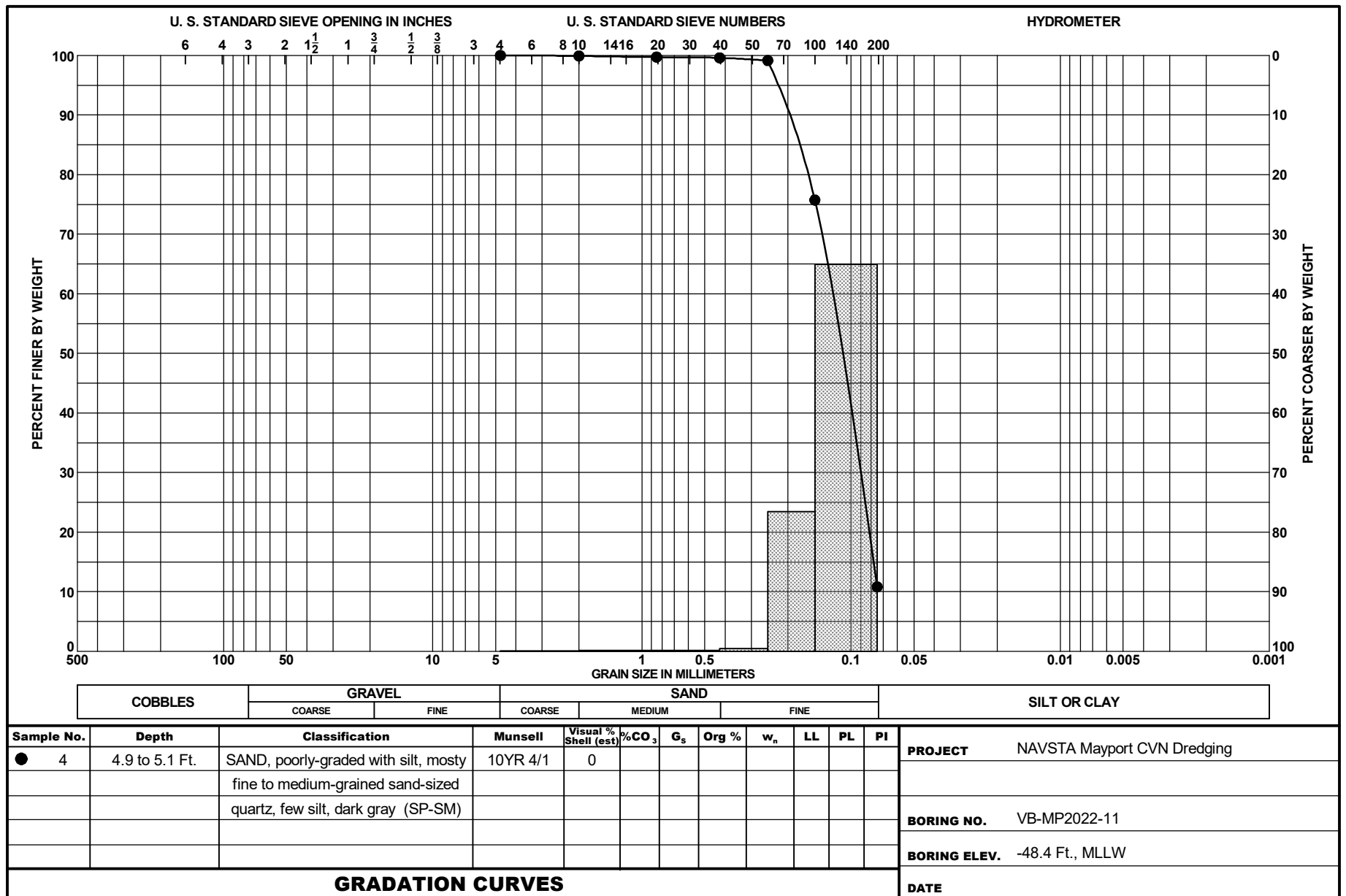


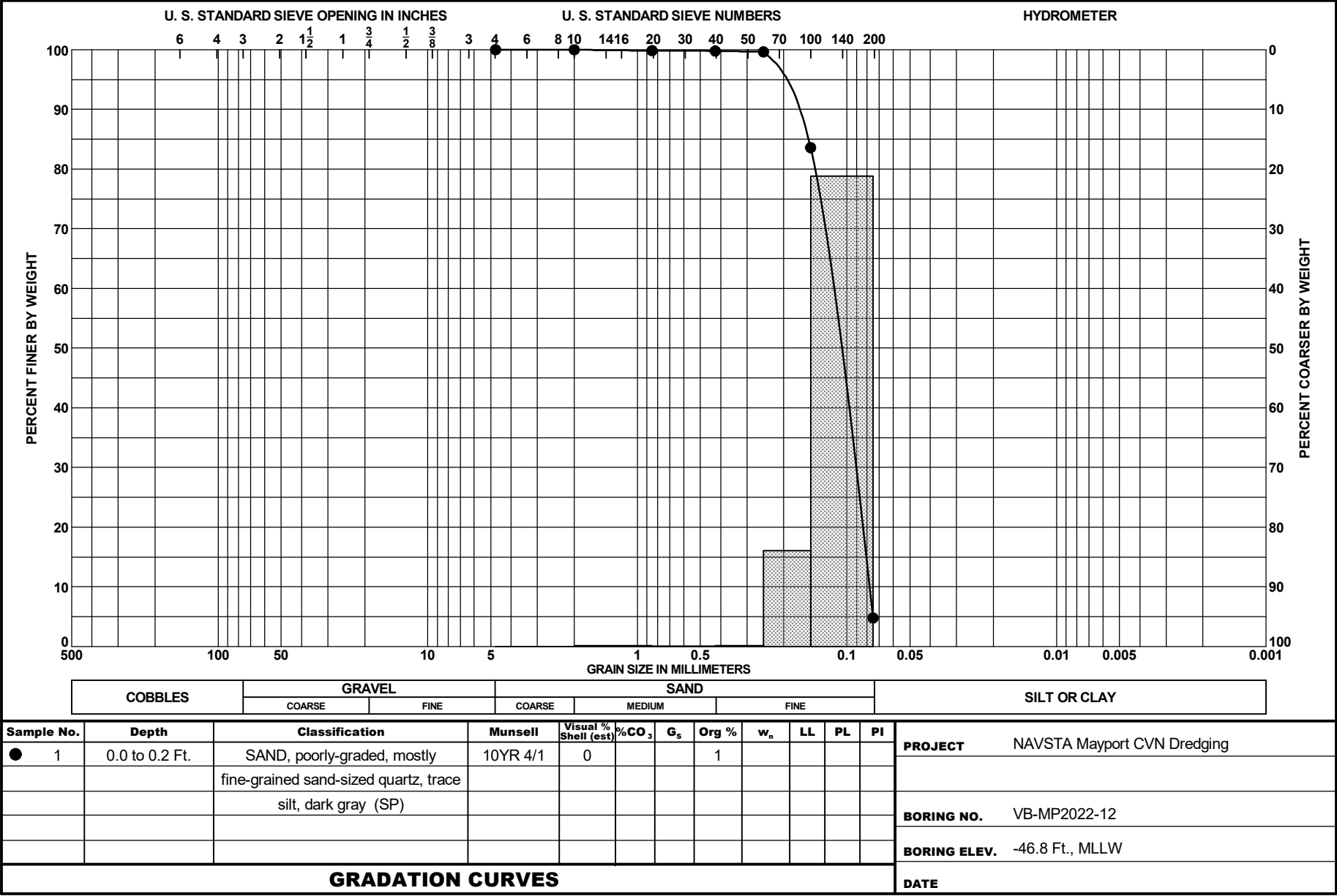


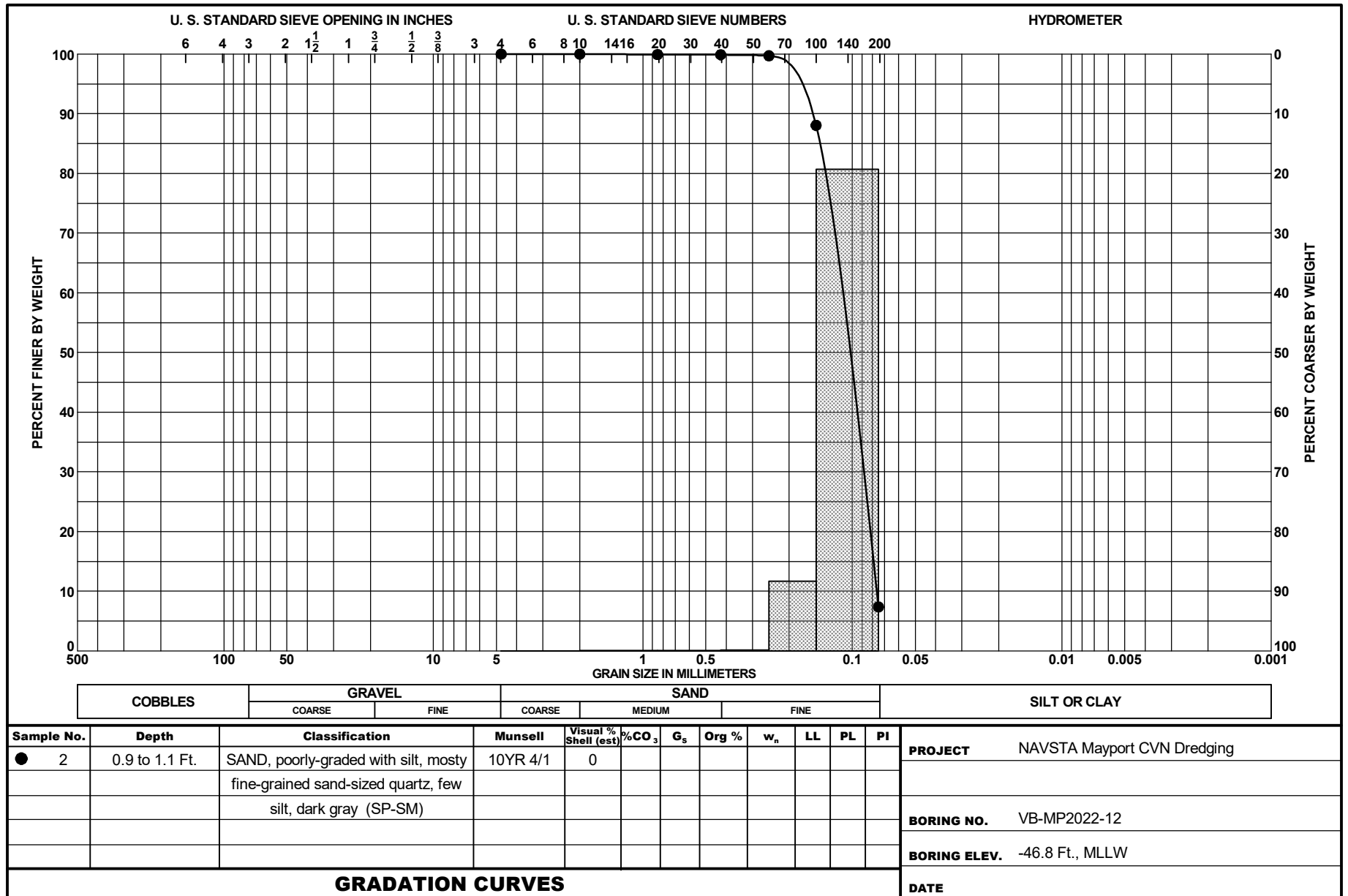


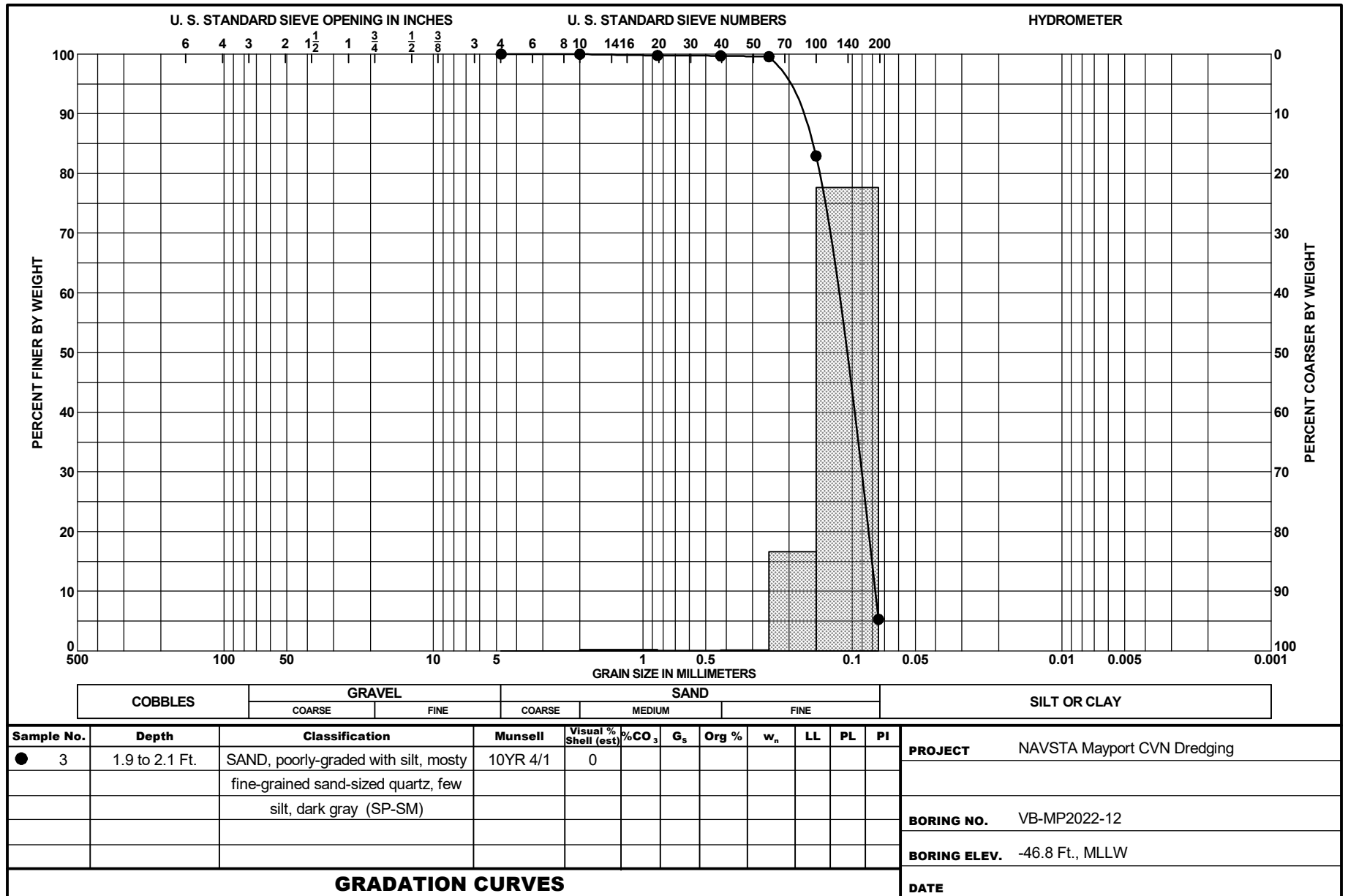


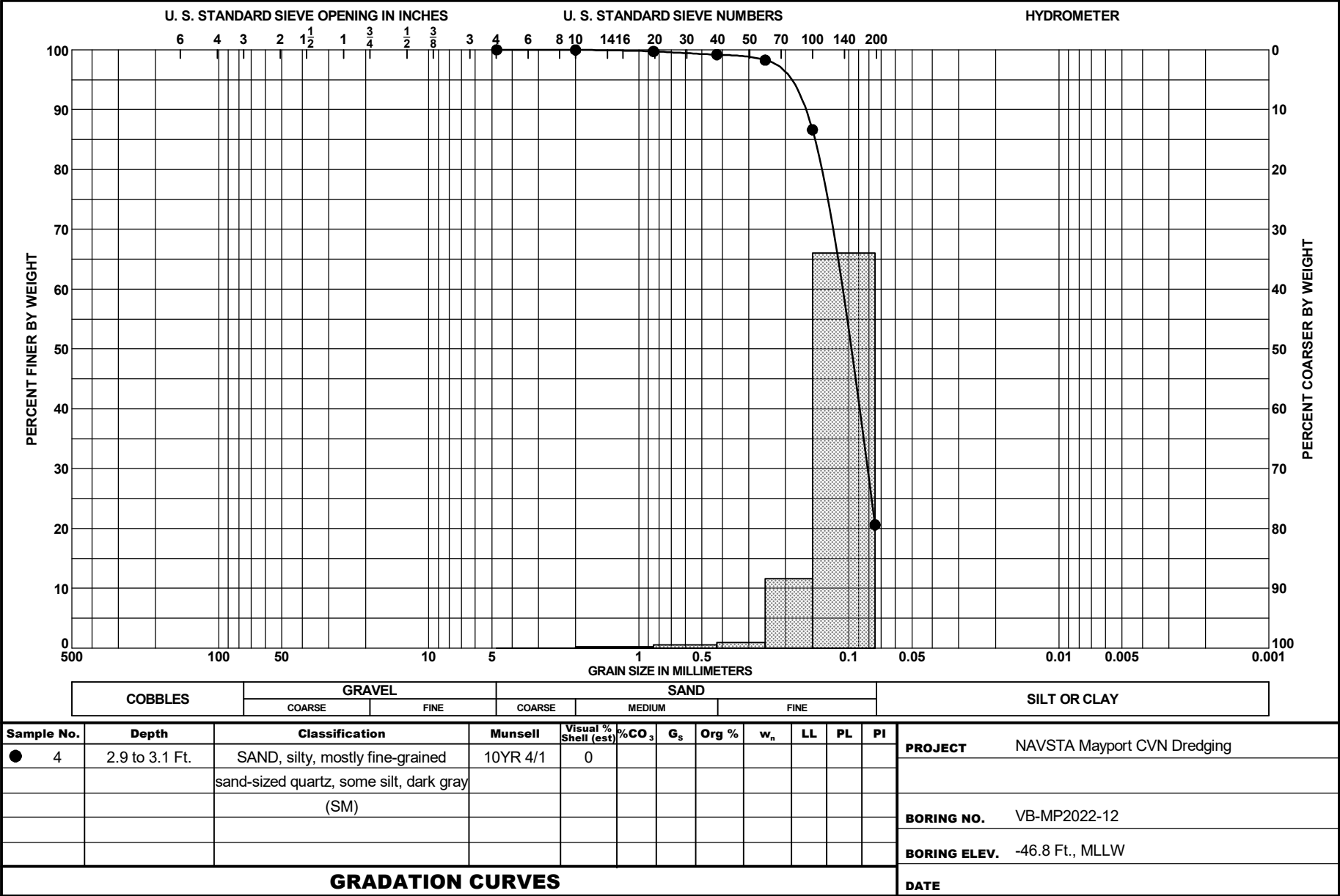


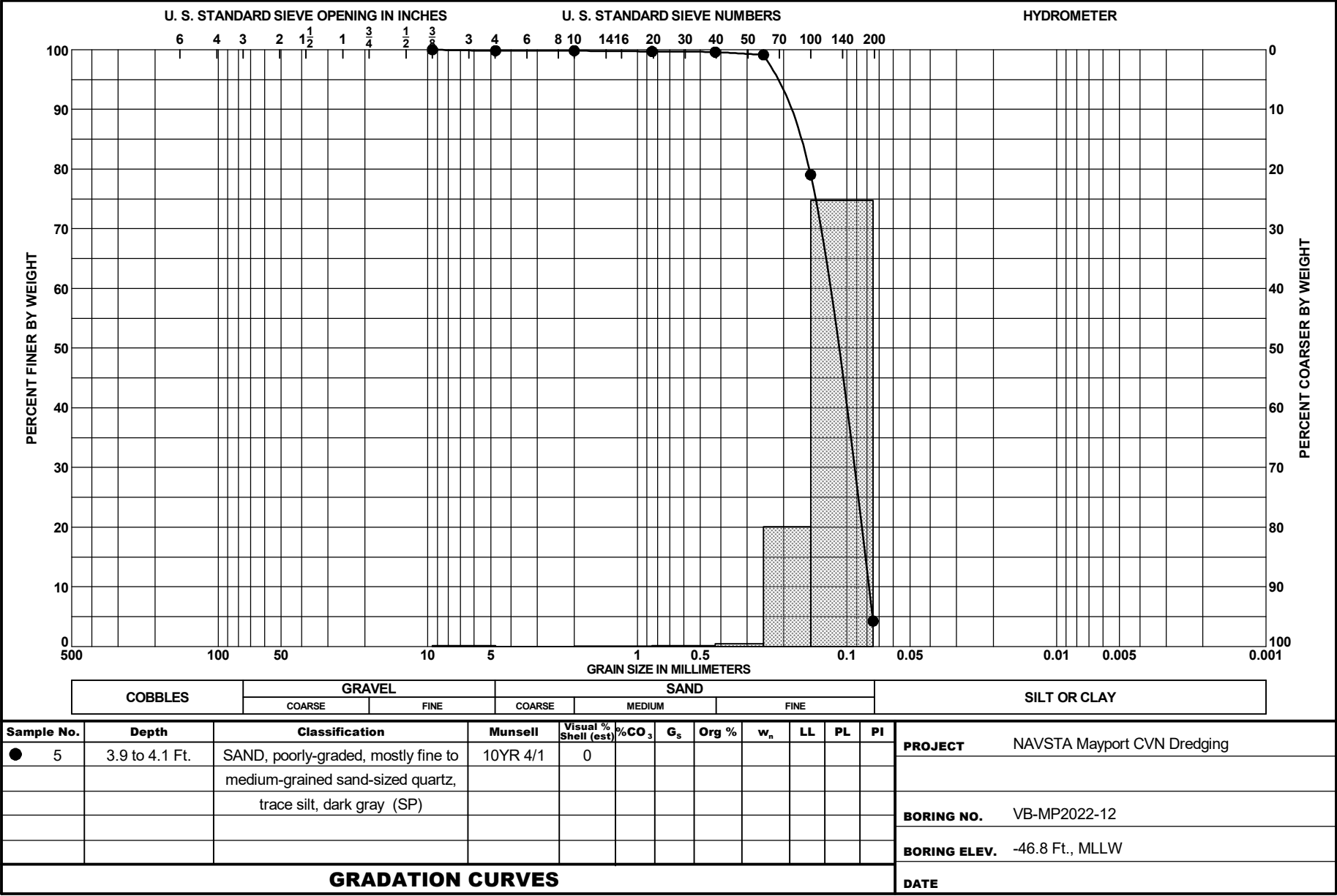


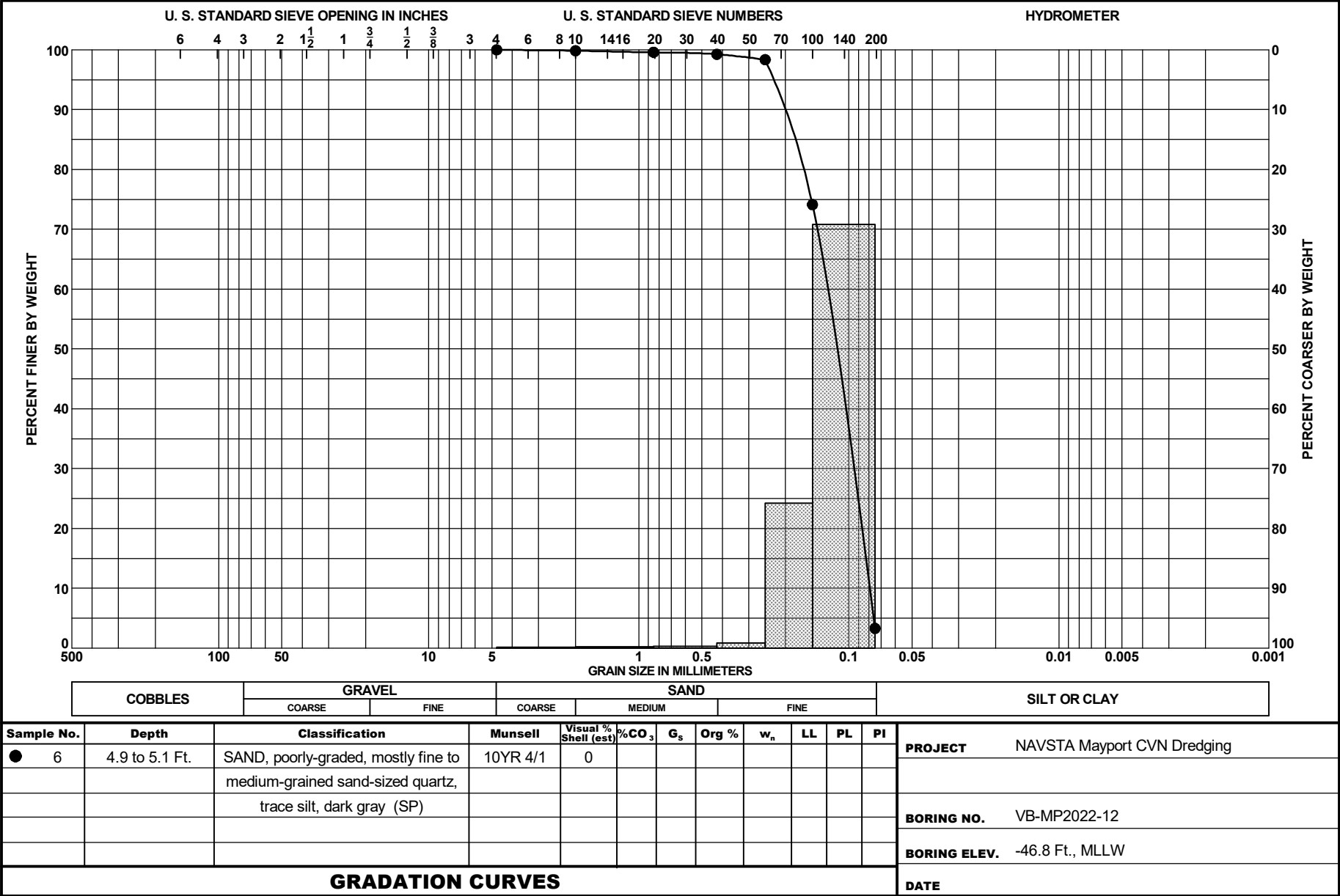


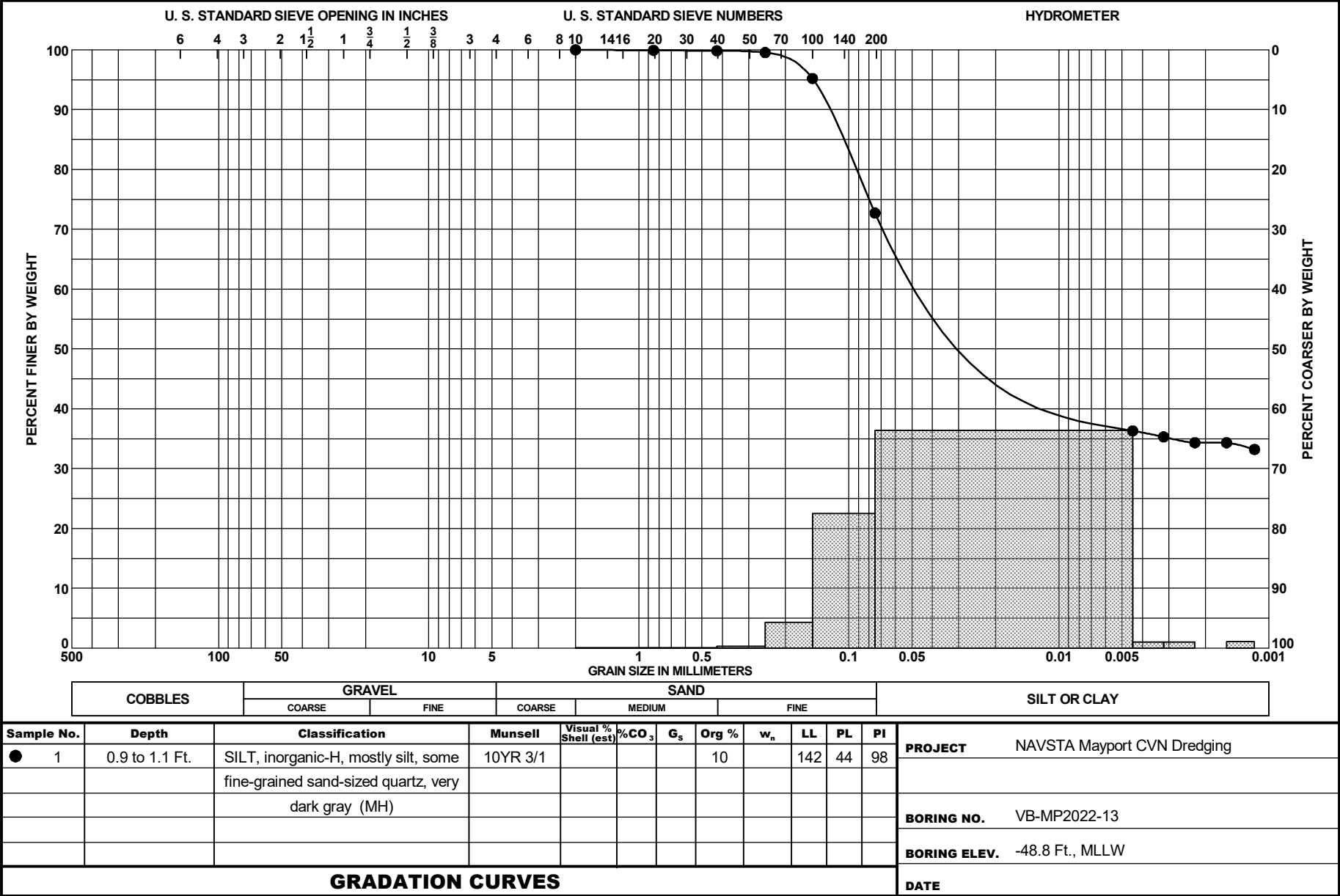


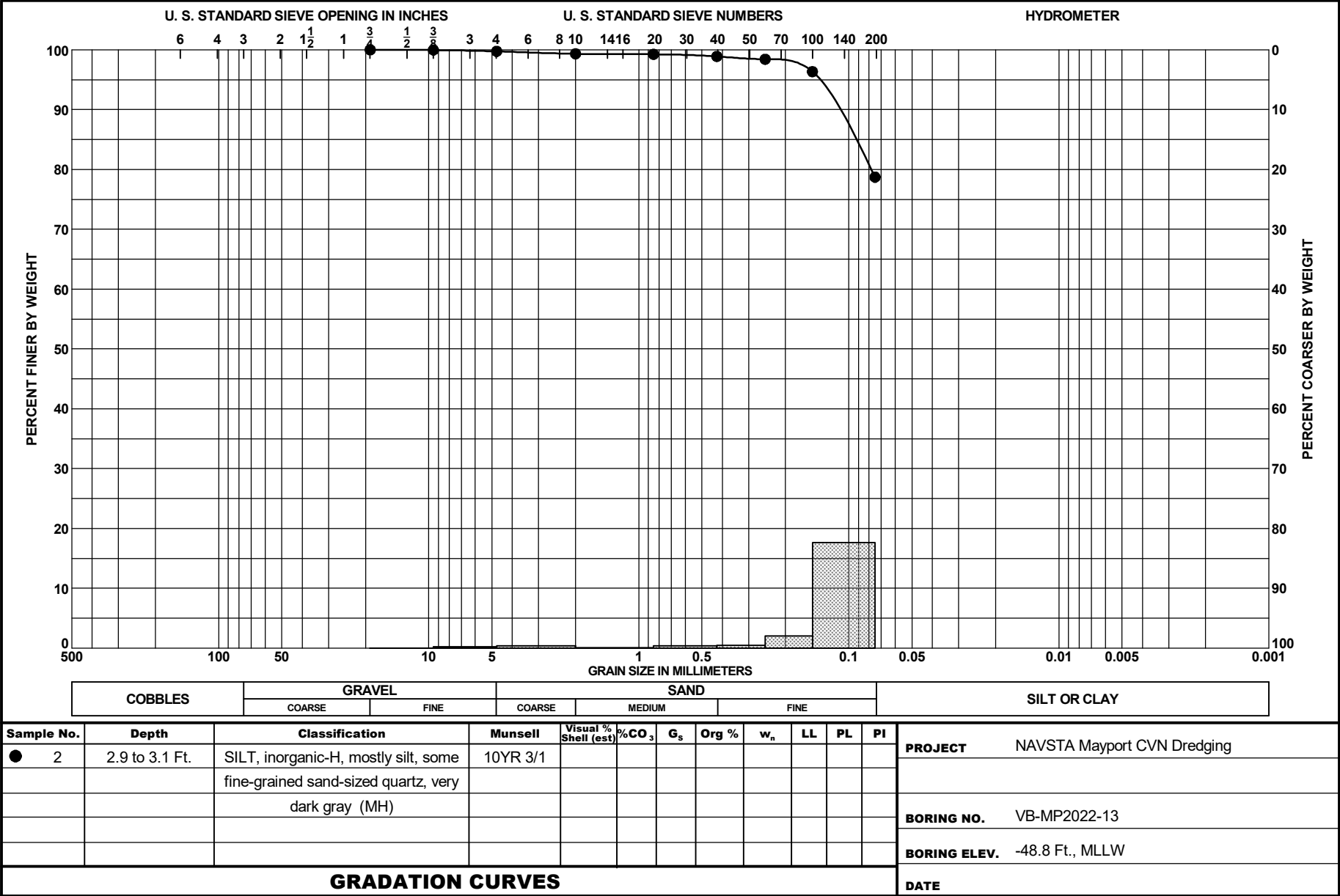


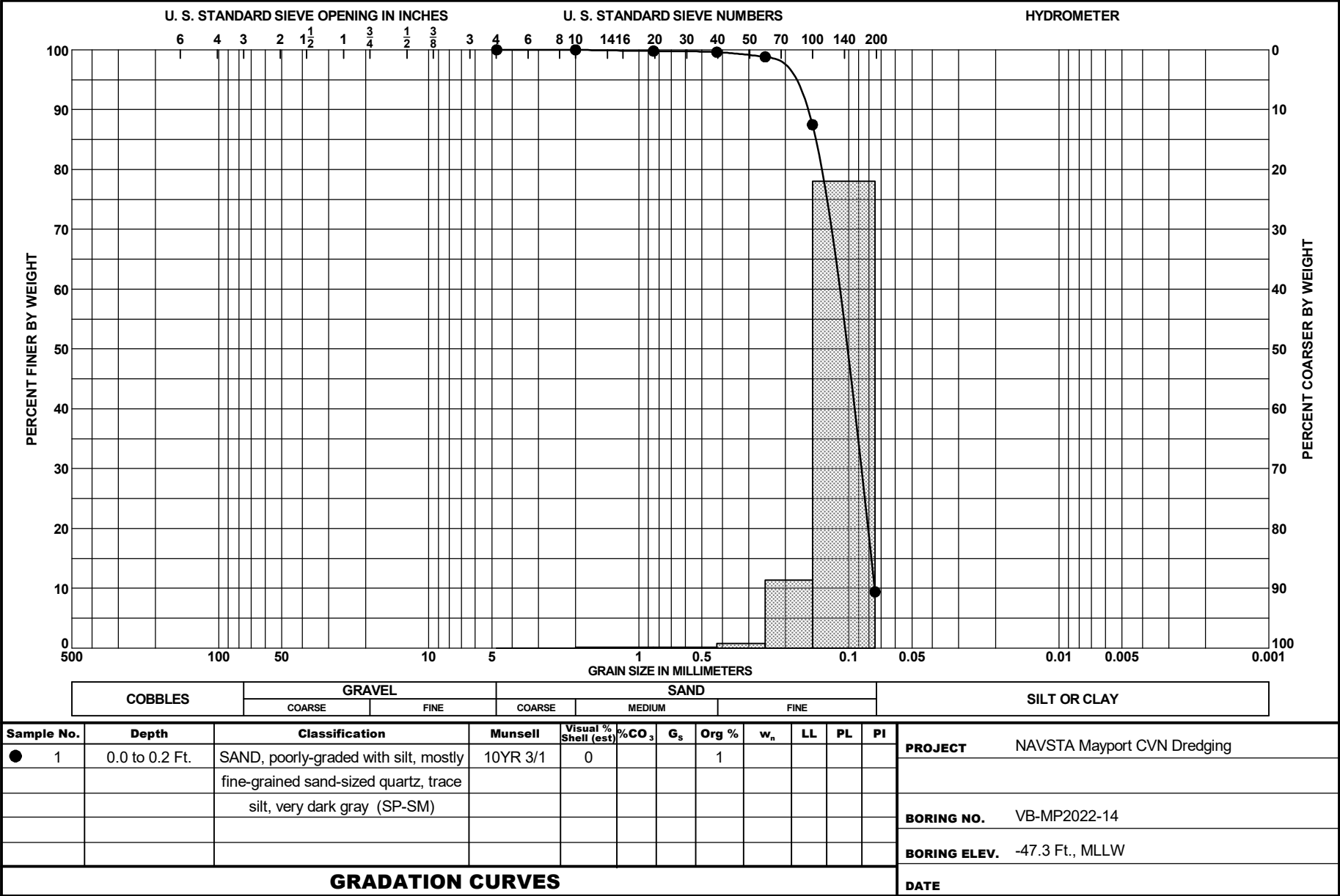


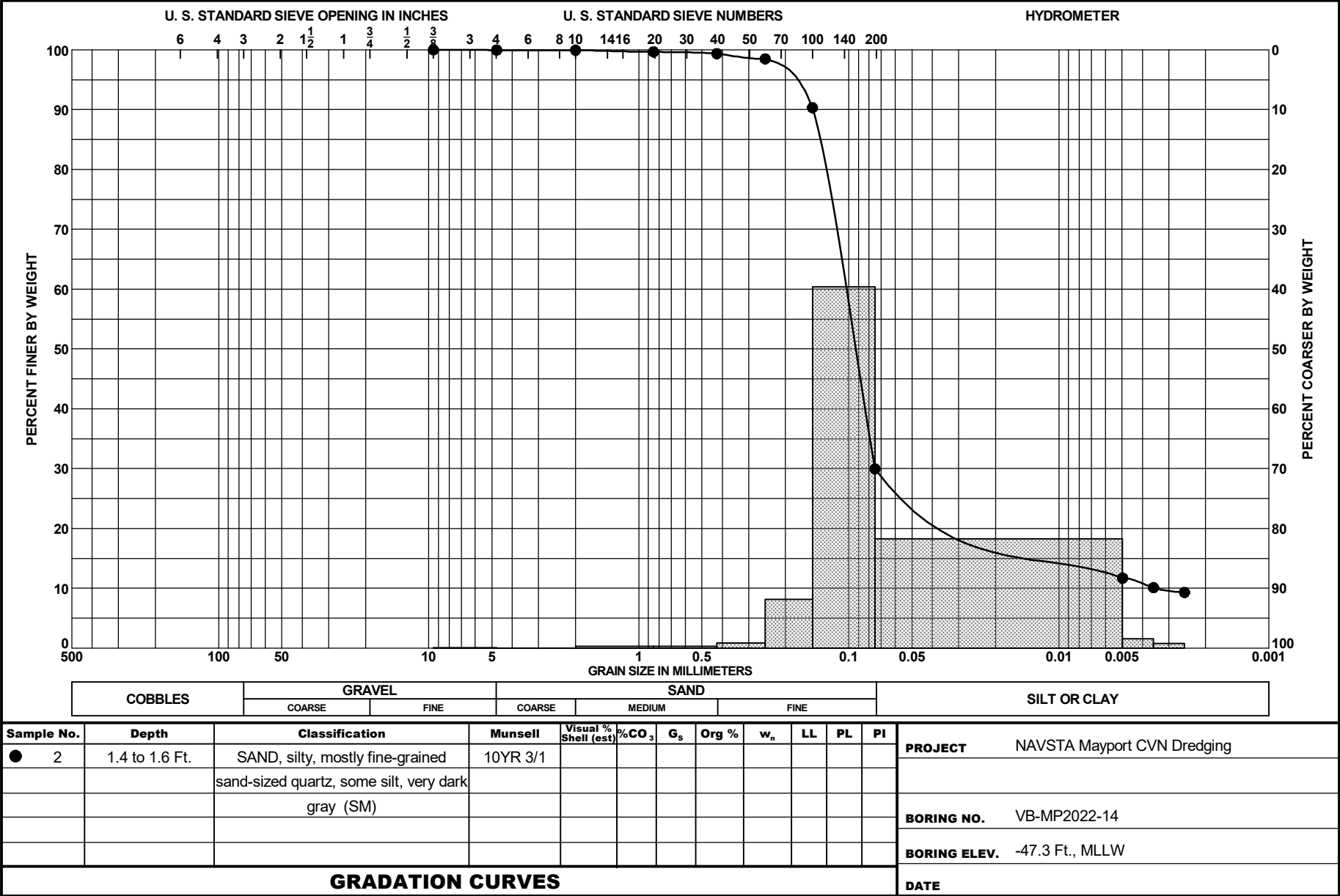


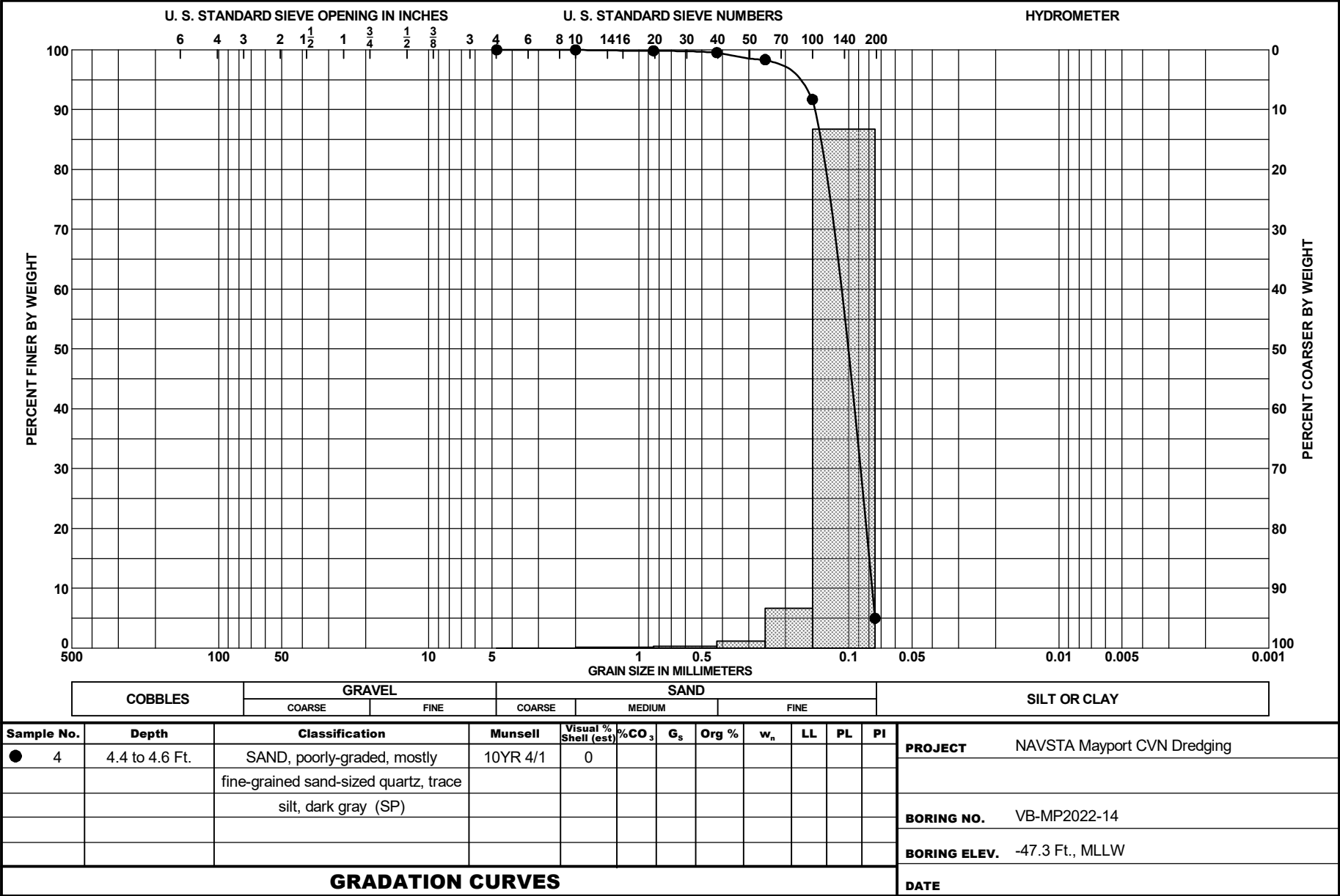


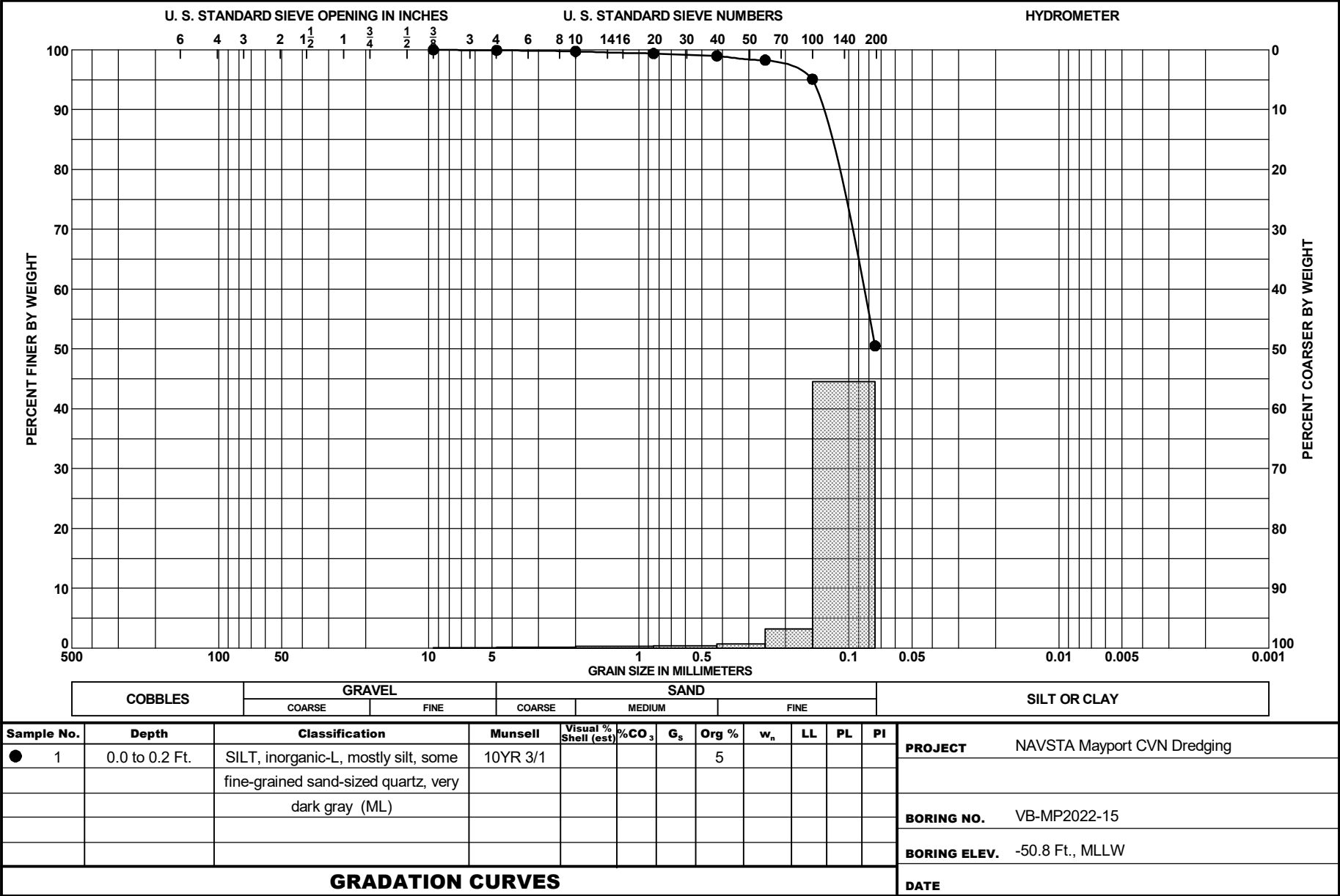


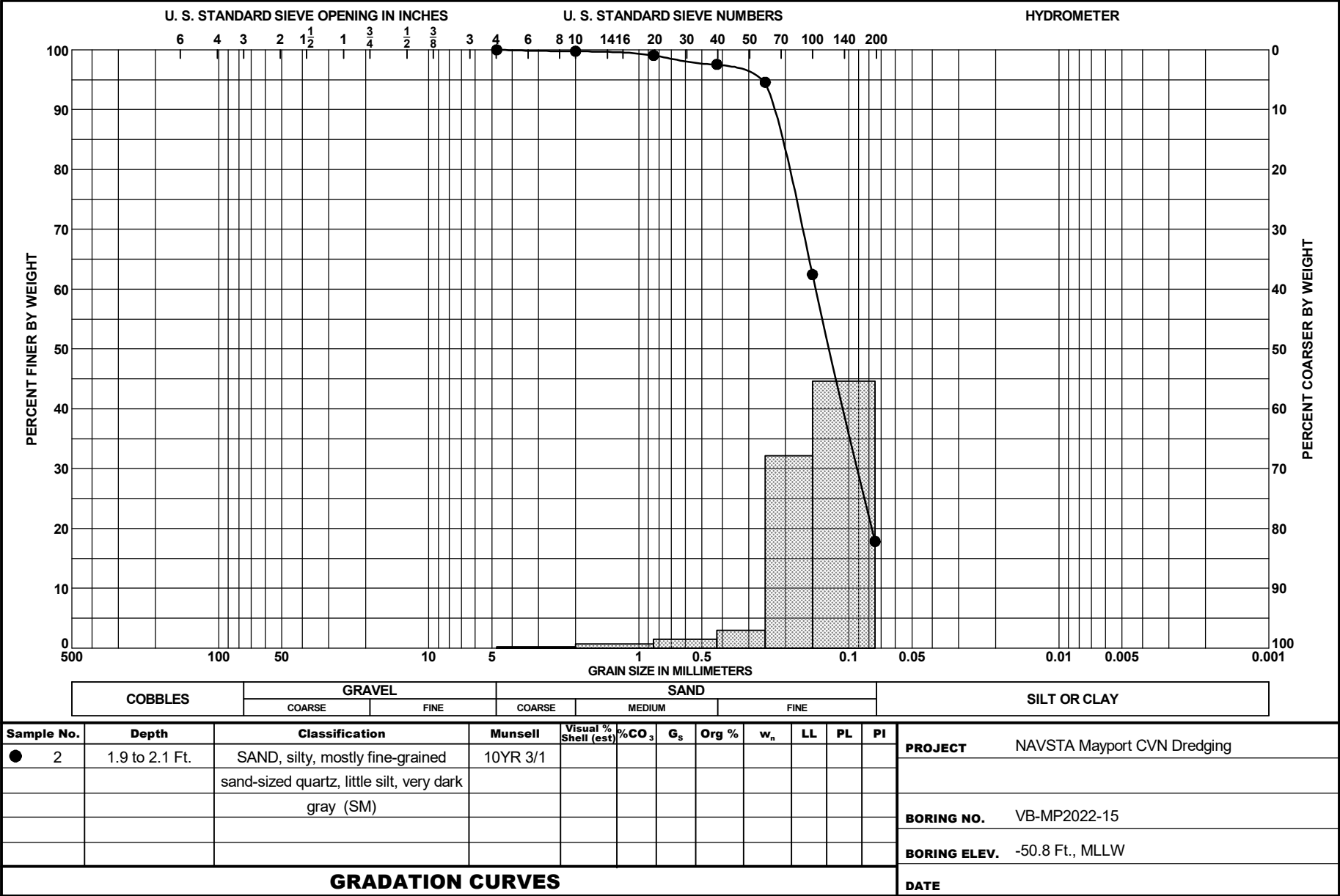




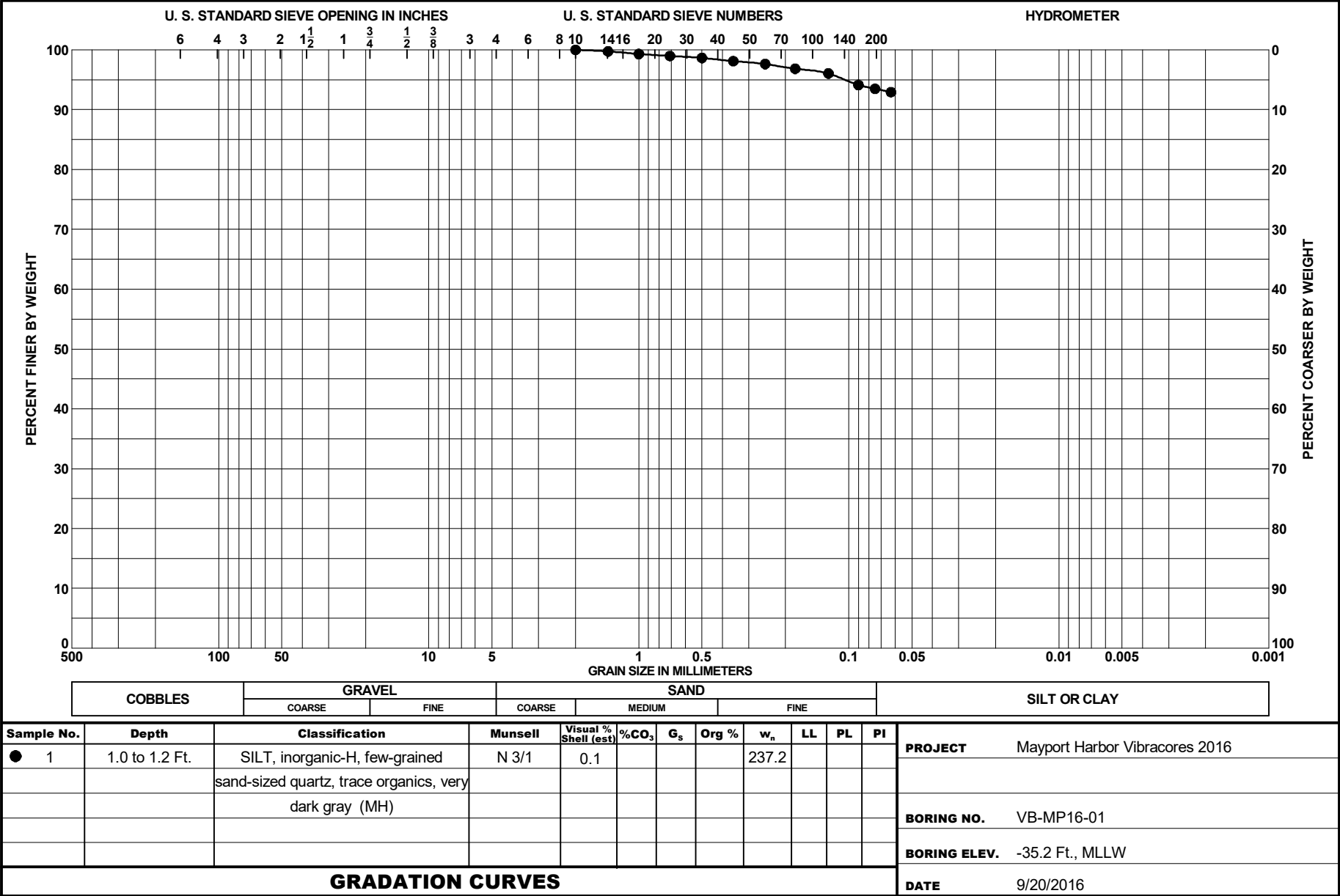




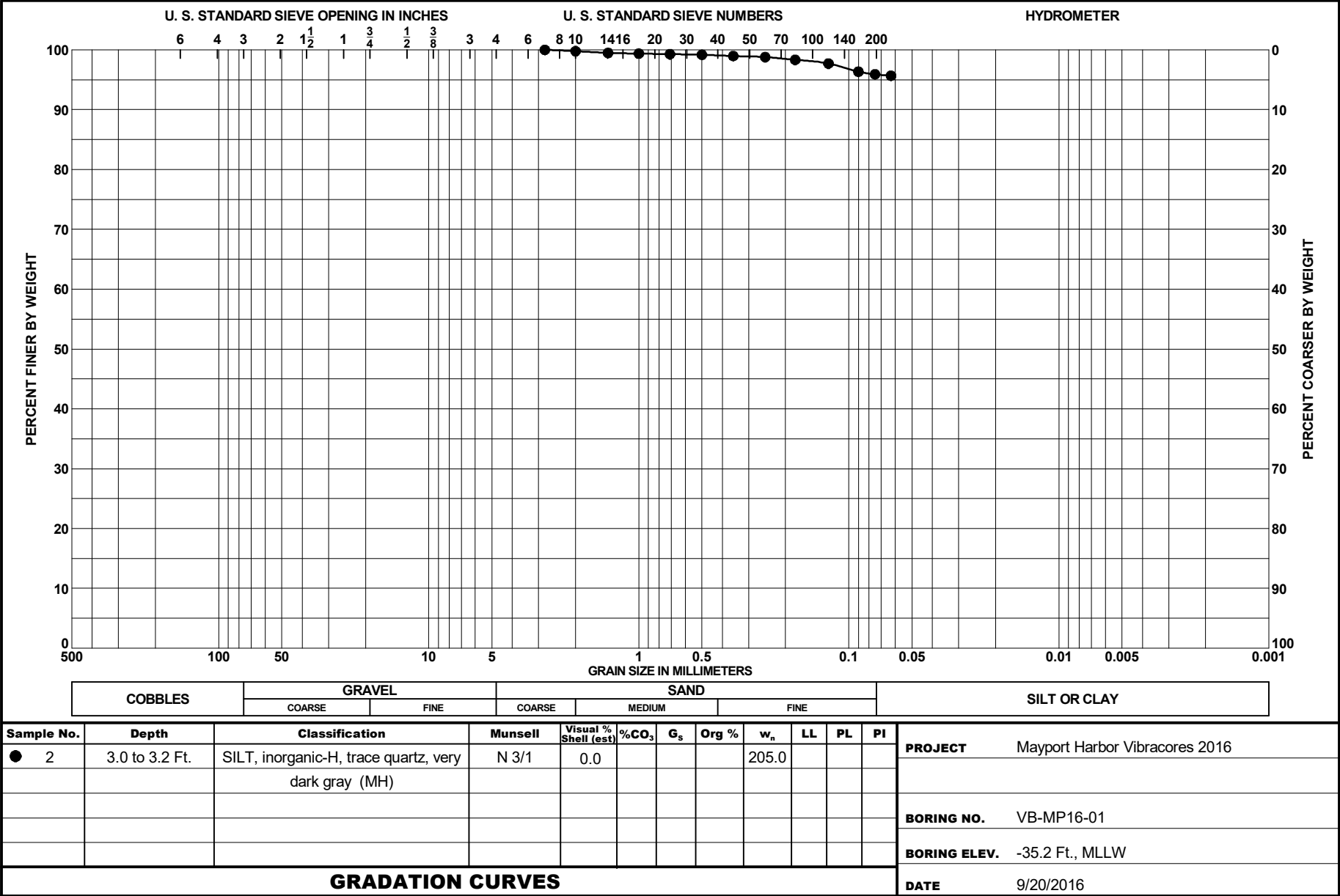




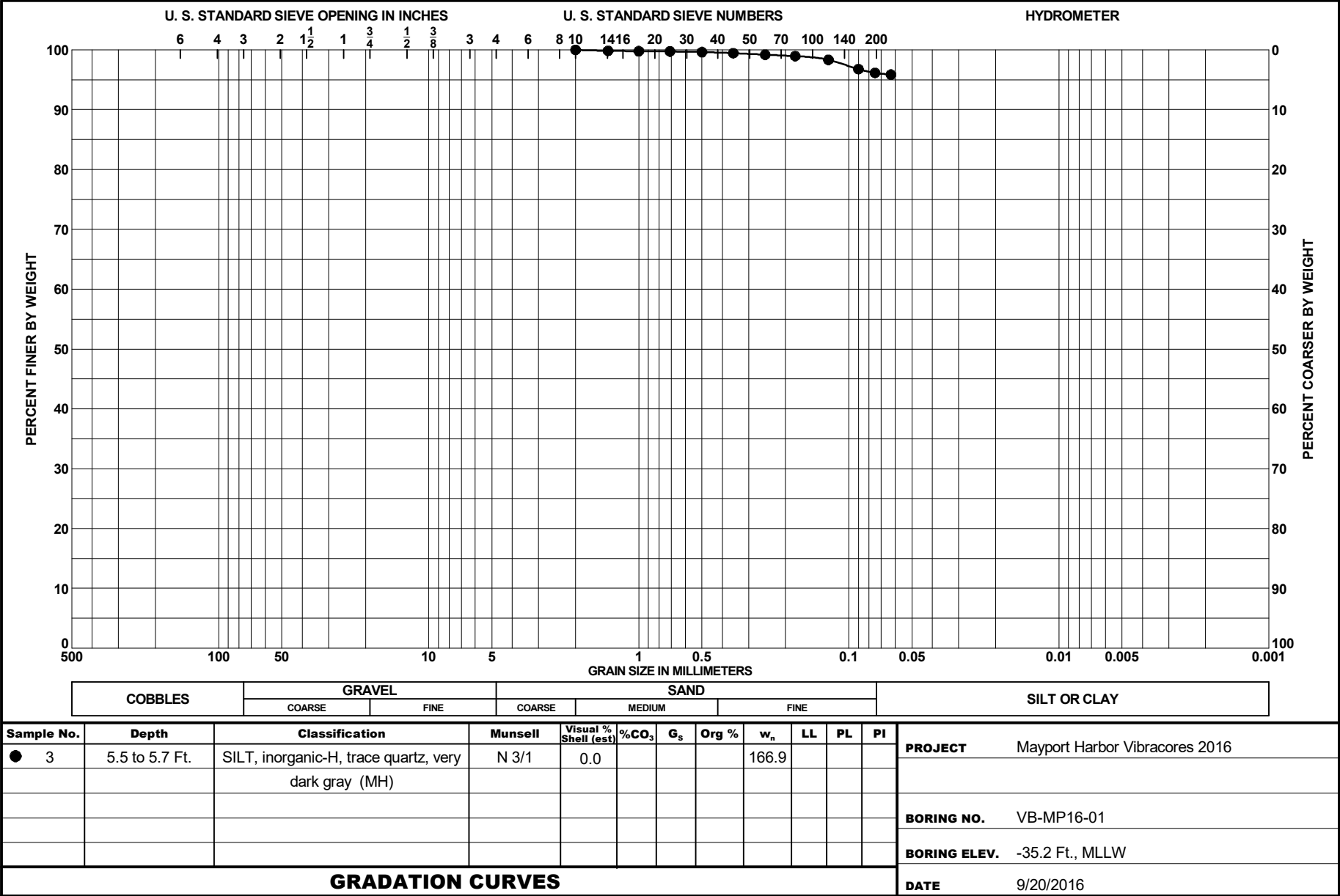
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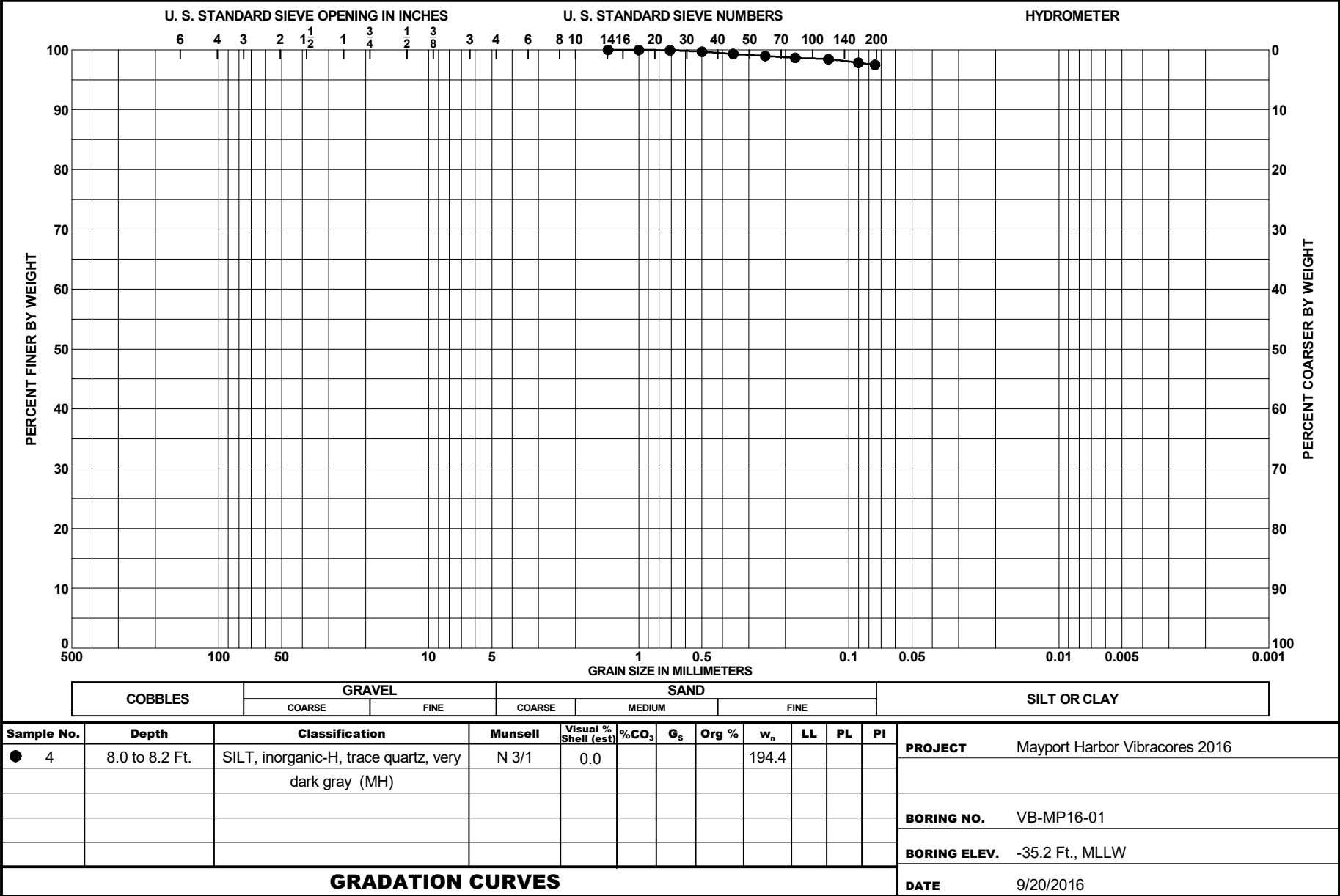
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00320-3



00320-4

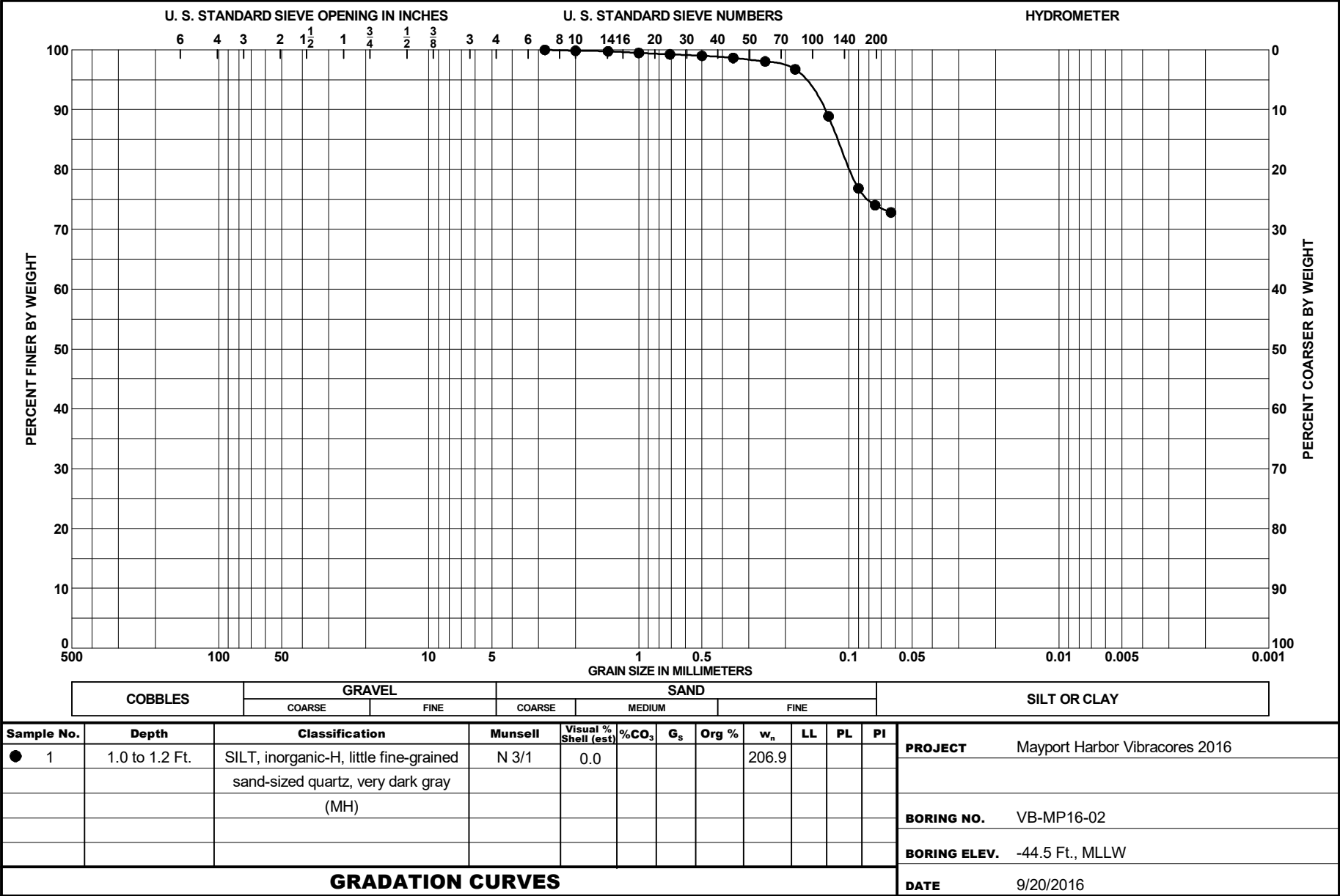


Sample No.	Depth	Classification	Munsell	Visual Shell % (est)	%CO ₃	G _s	Org %	w _n	LL	PL	PI		
● 4	8.0 to 8.2 Ft.	SILT, inorganic-H, trace quartz, very dark gray (MH)	N 3/1	0.0				194.4				PROJECT	Mayport Harbor Vibracores 2016
												BORING NO.	VB-MP16-01
												BORING ELEV.	-35.2 Ft., MLLW
GRADATION CURVES												DATE	9/20/2016



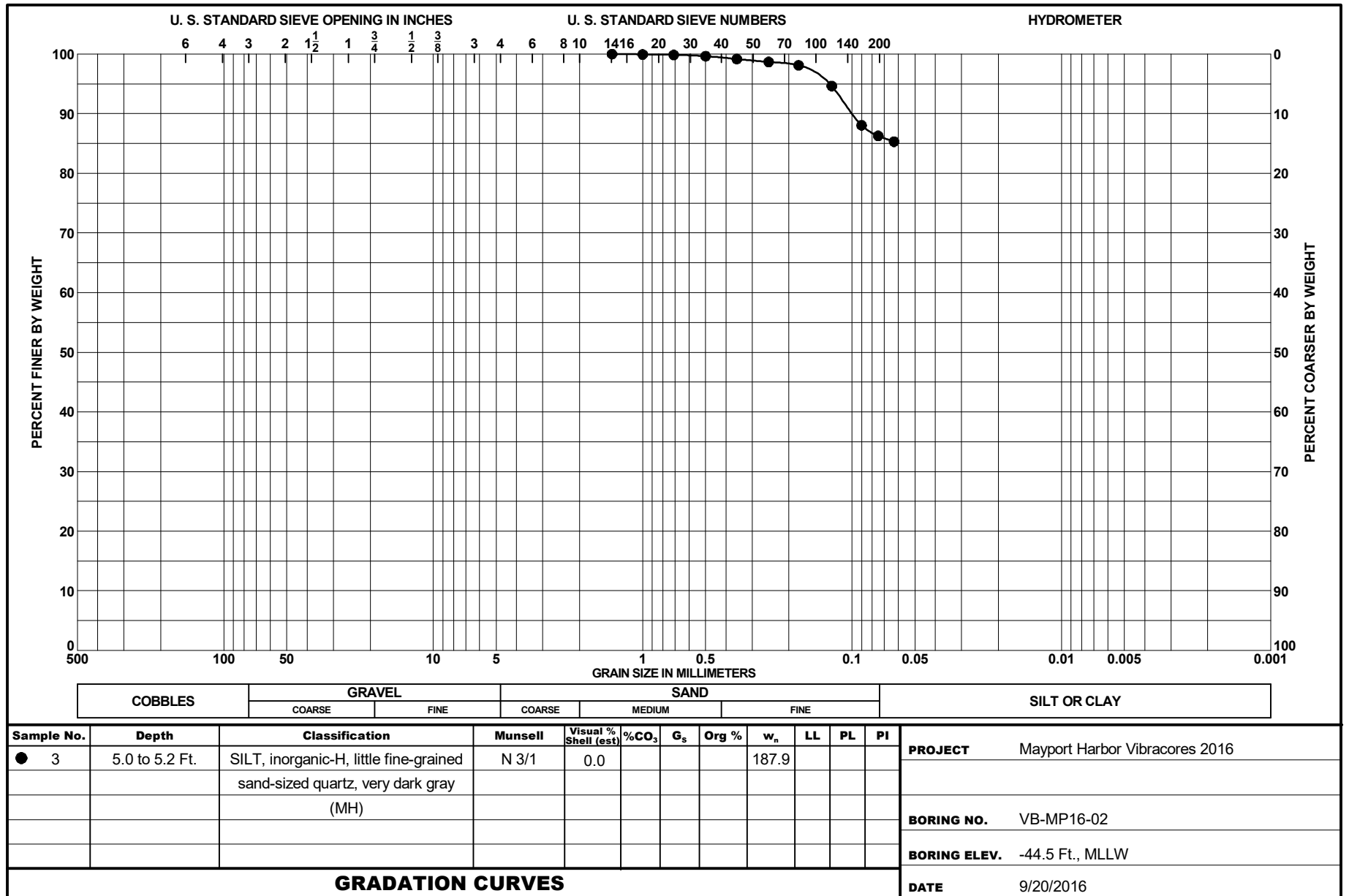


00320-7



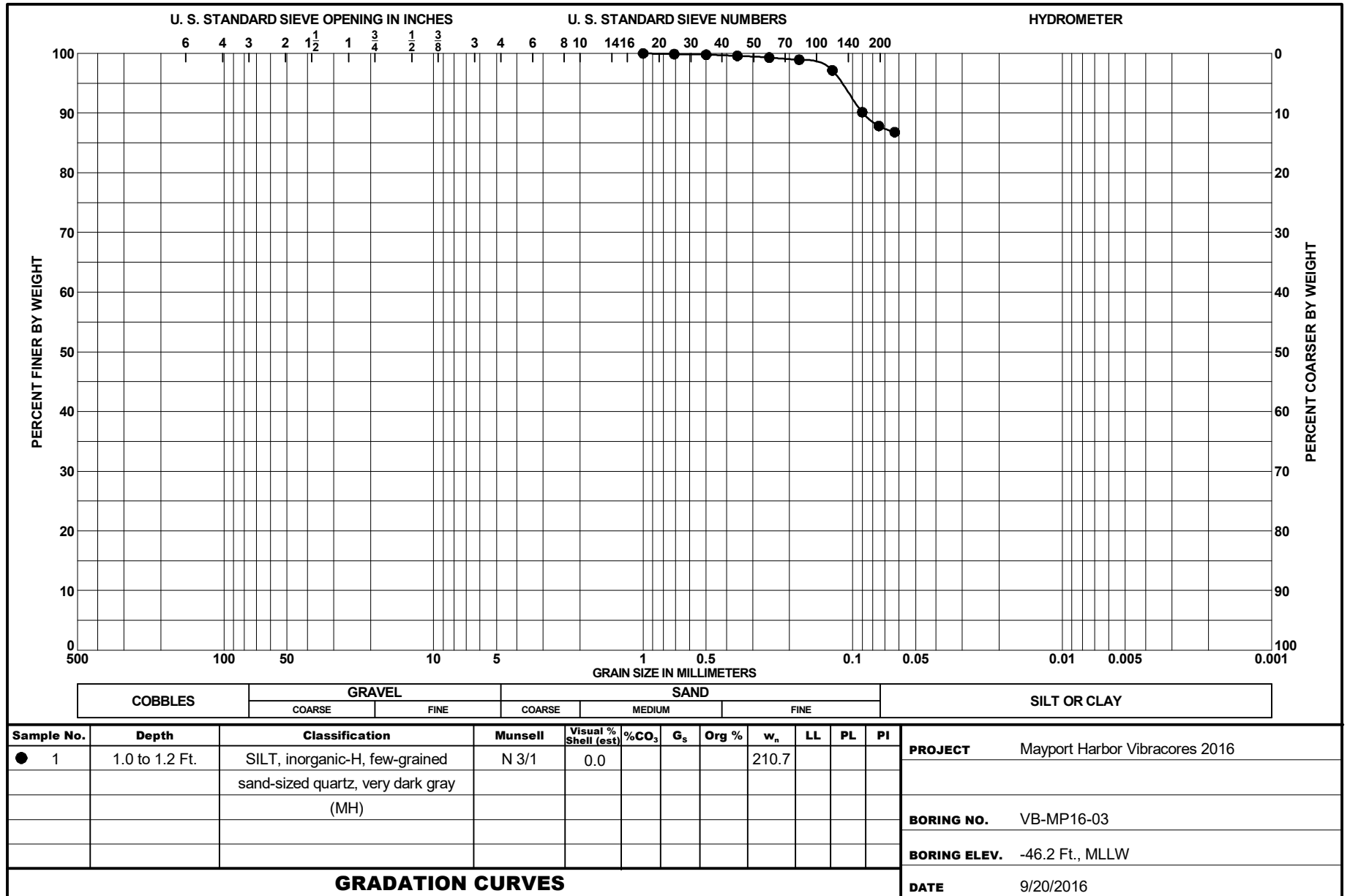


00320-9

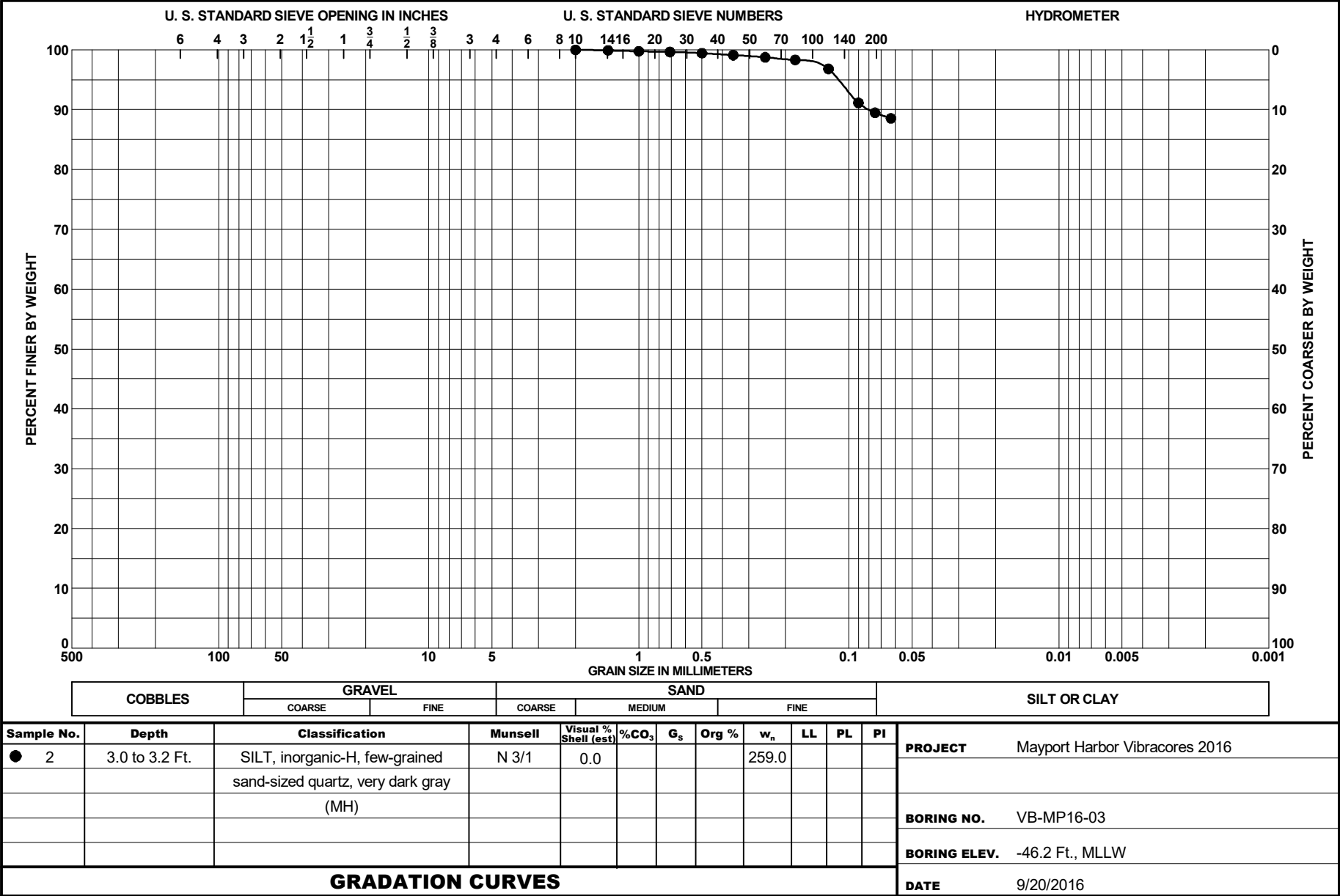




00320-11



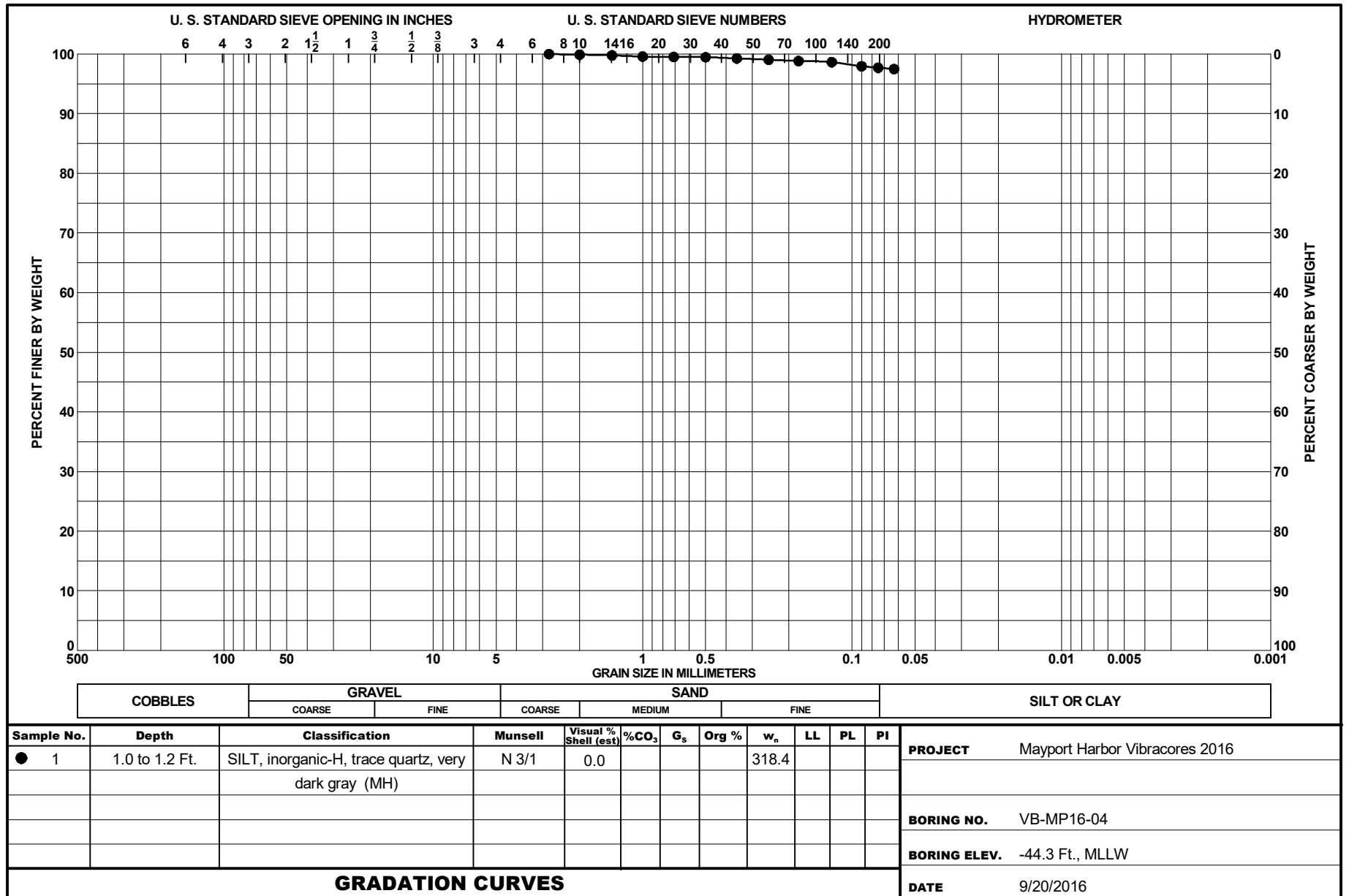
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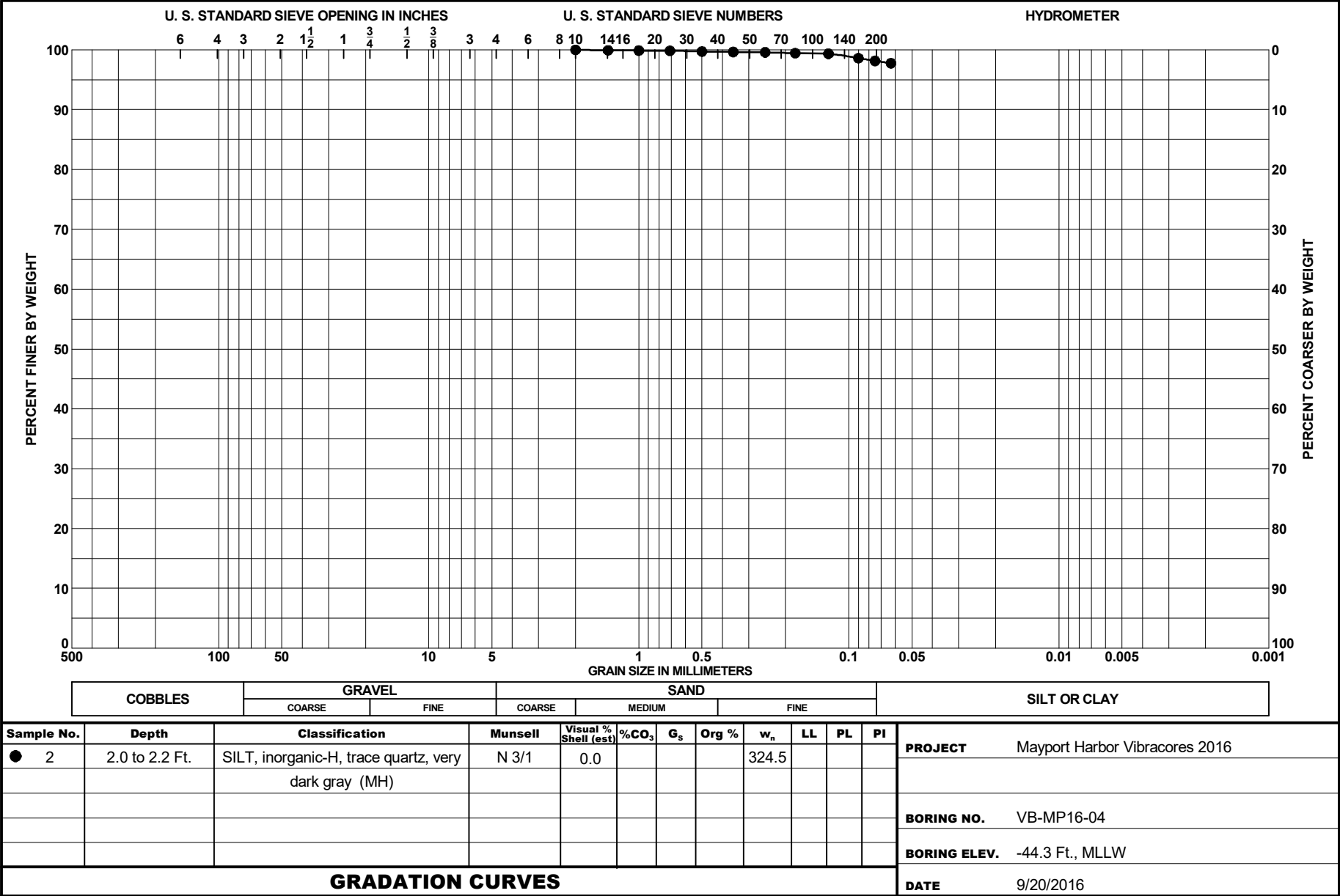




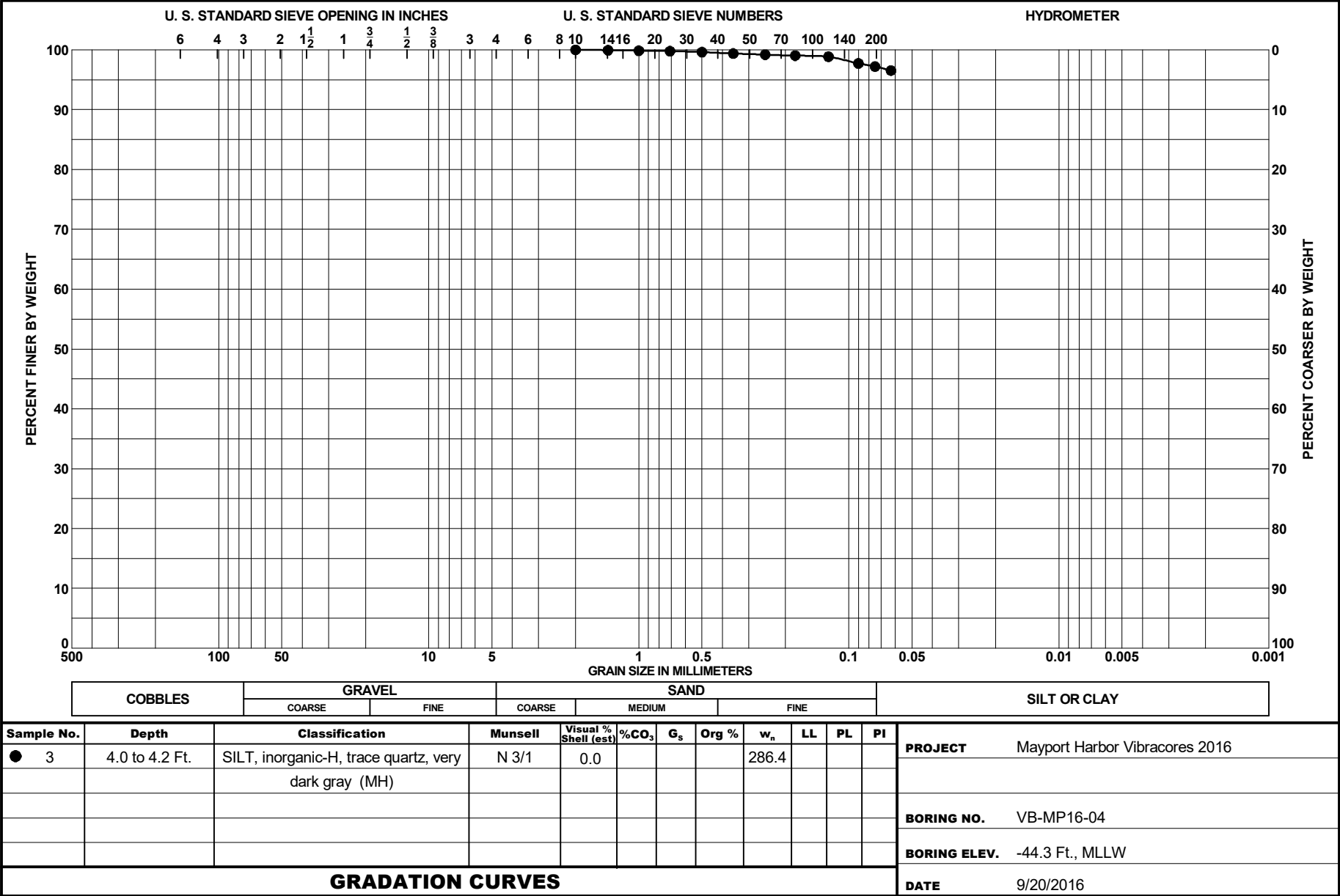
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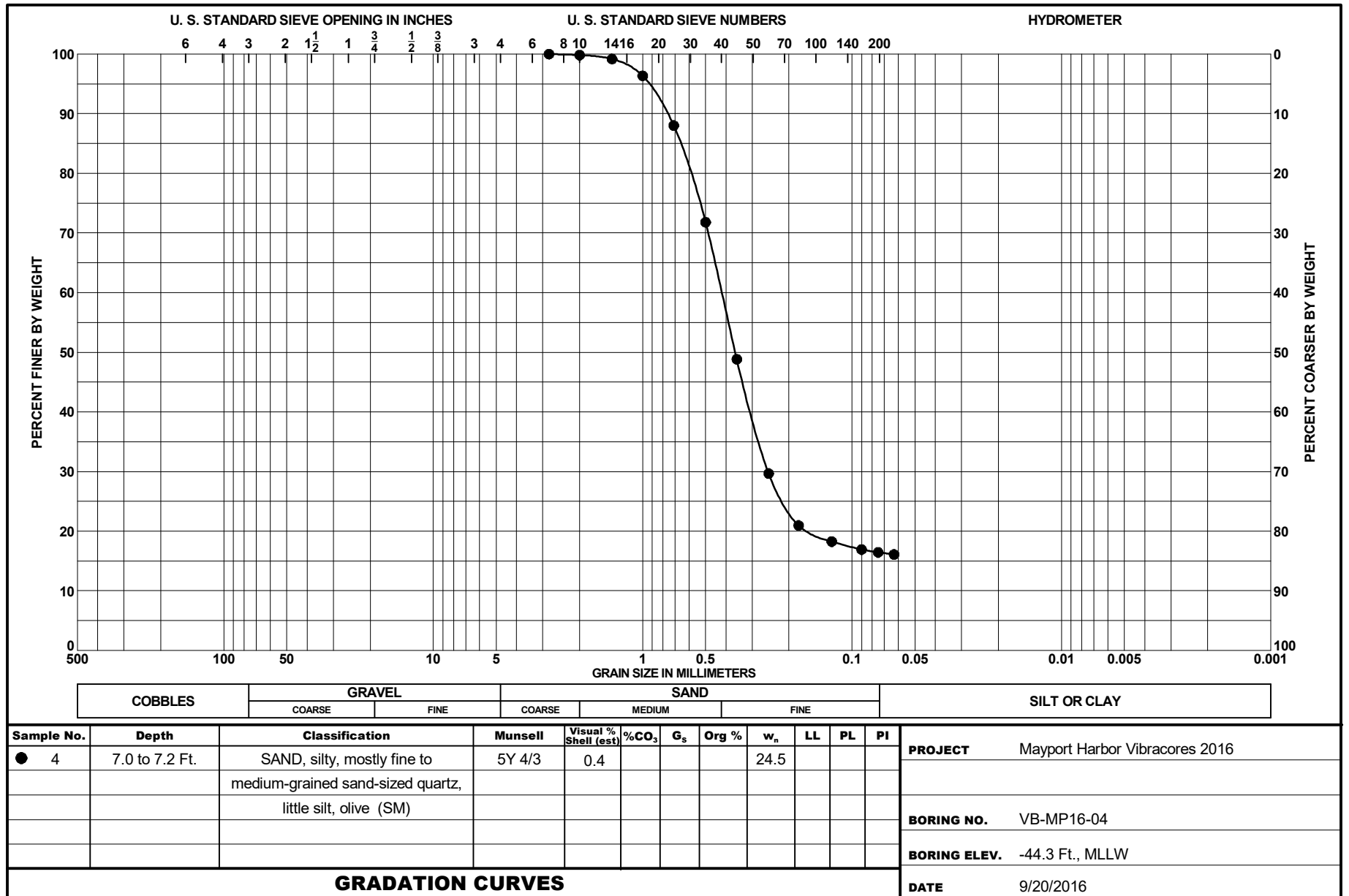
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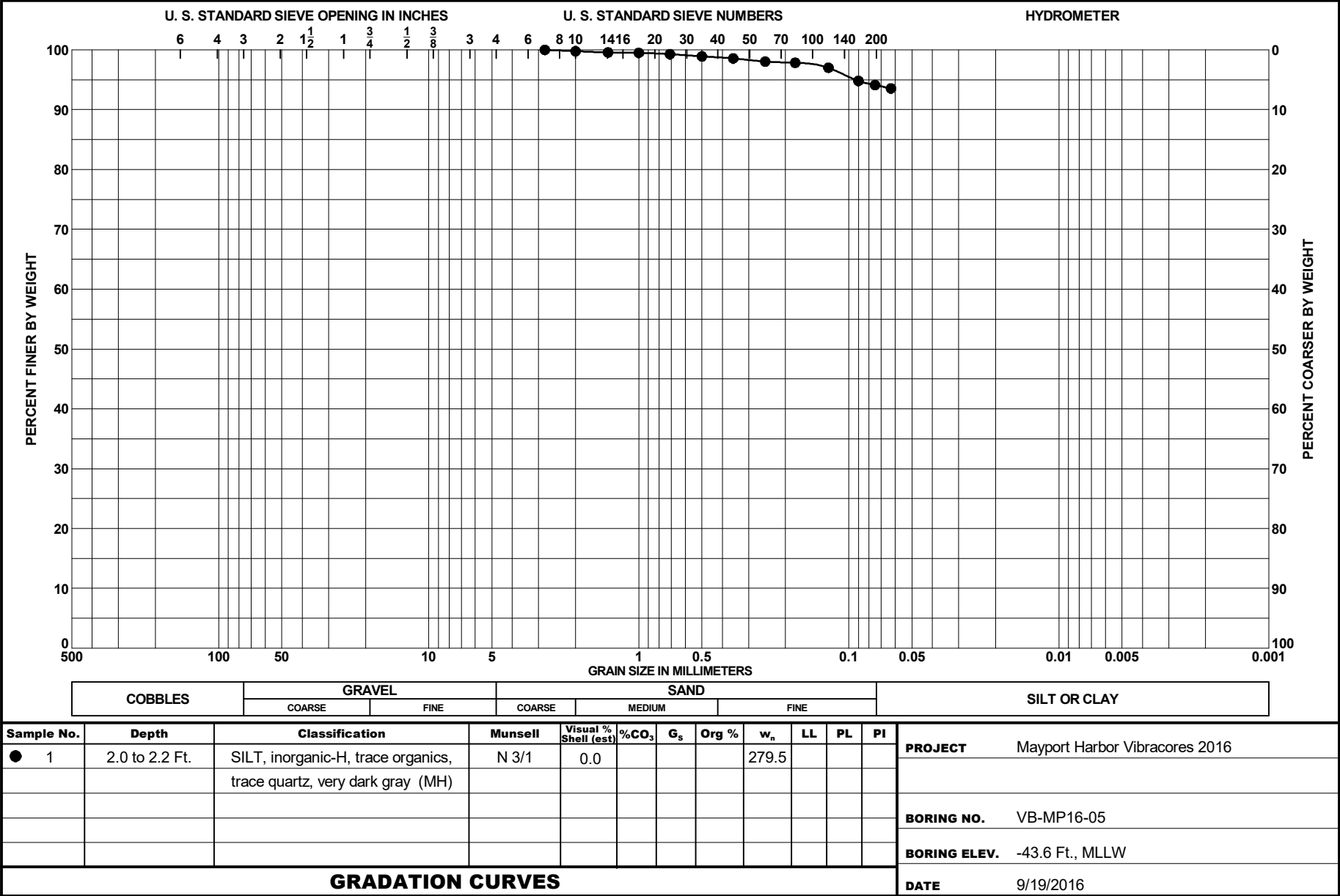
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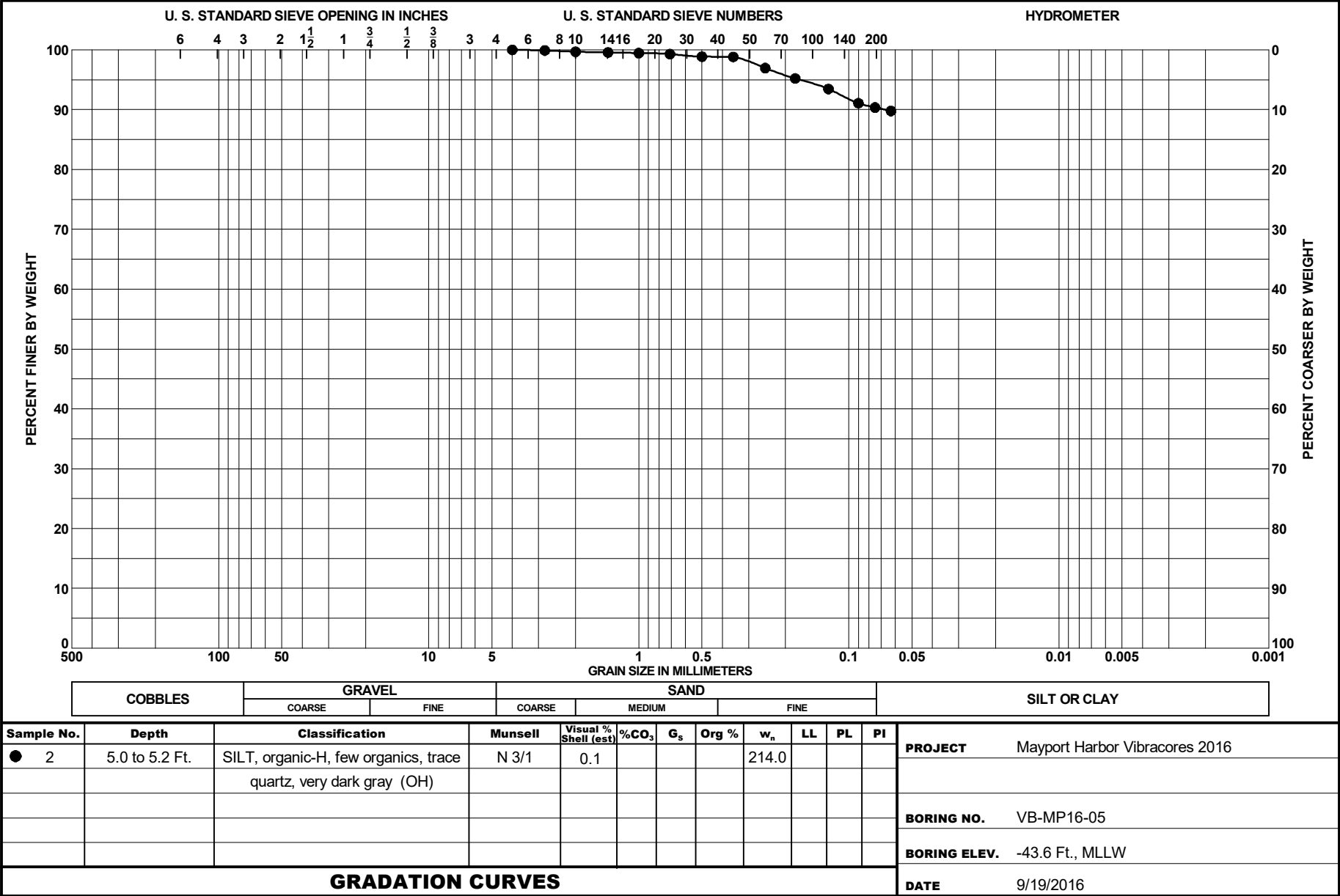
00320-18



00320-19



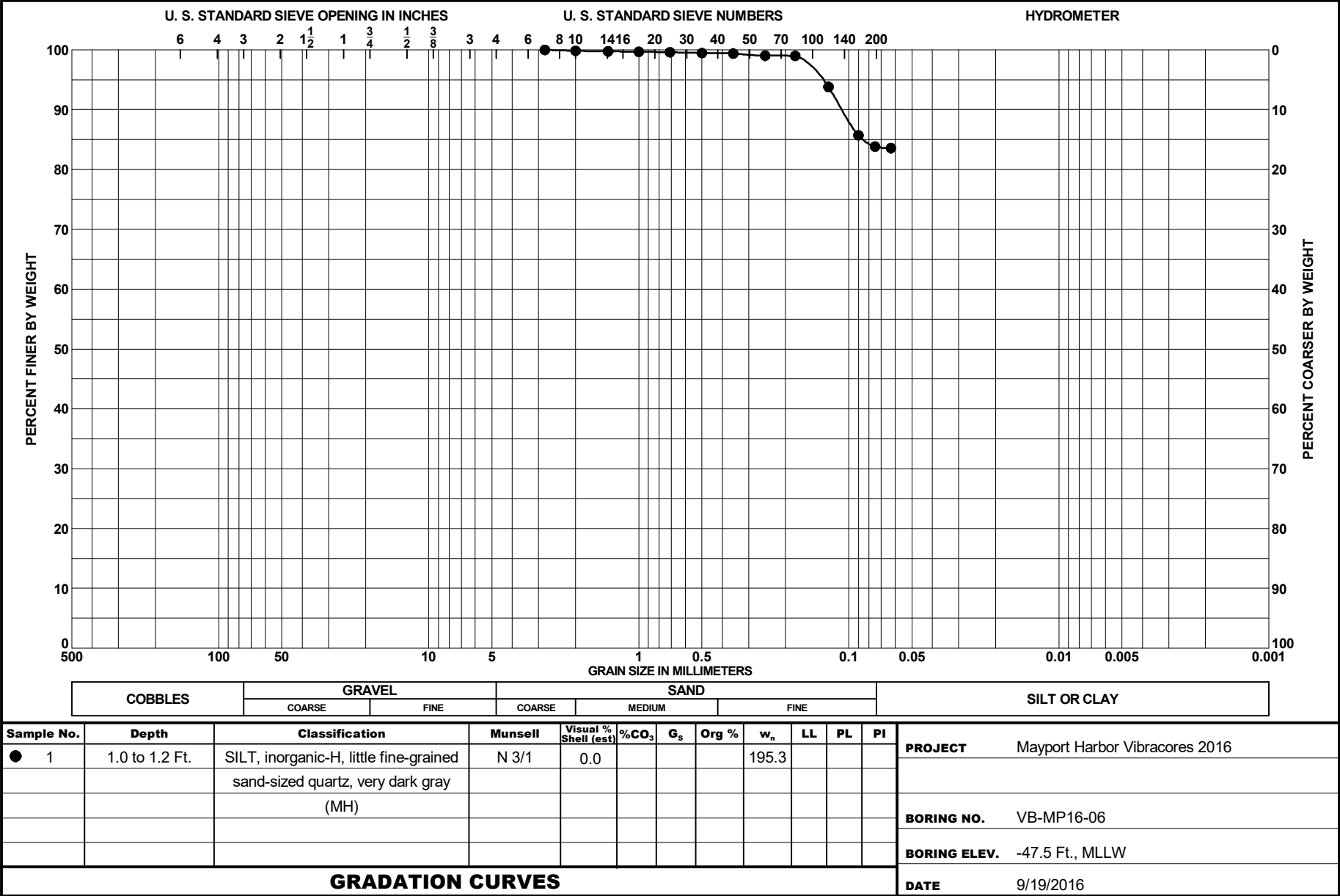
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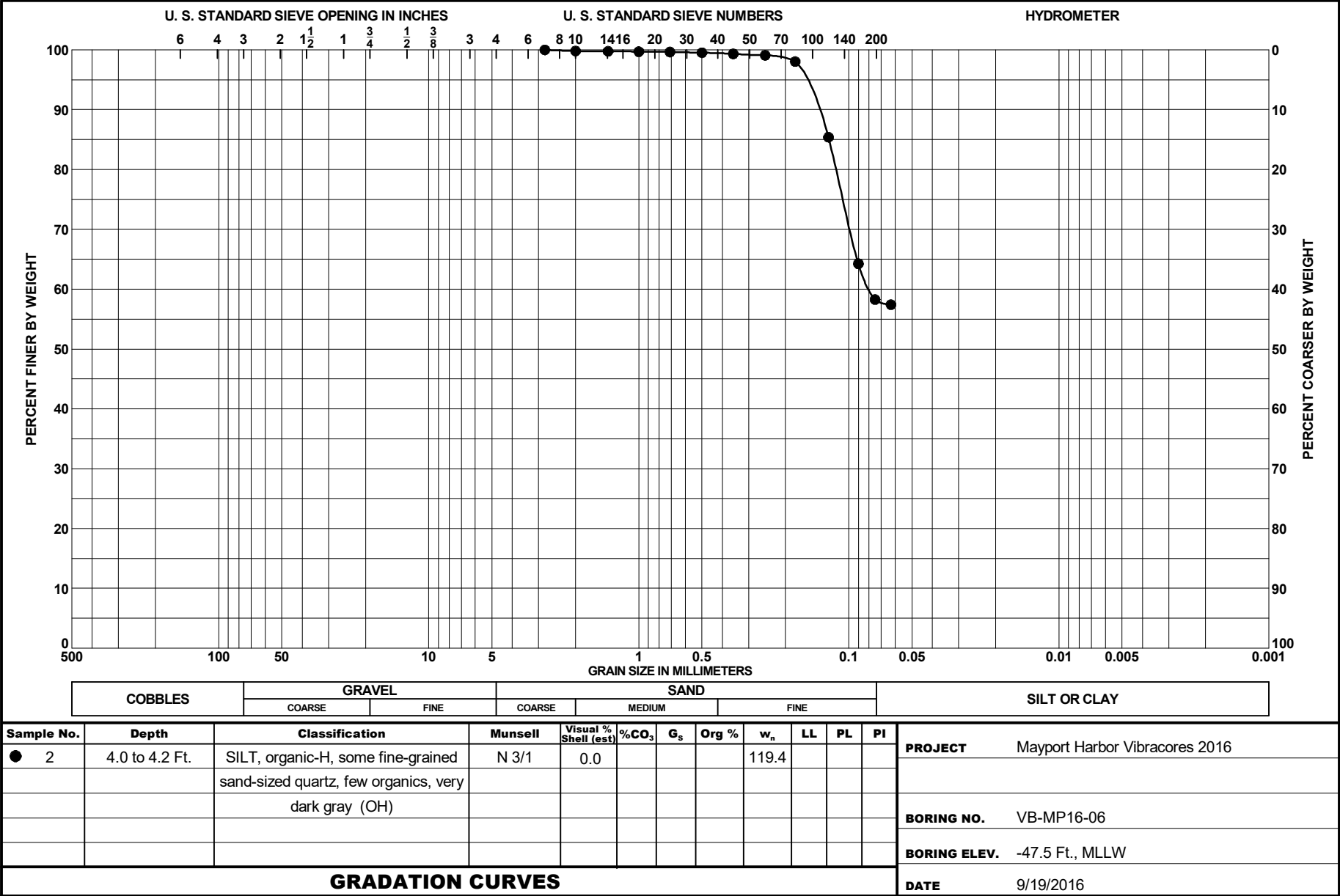




00320-23

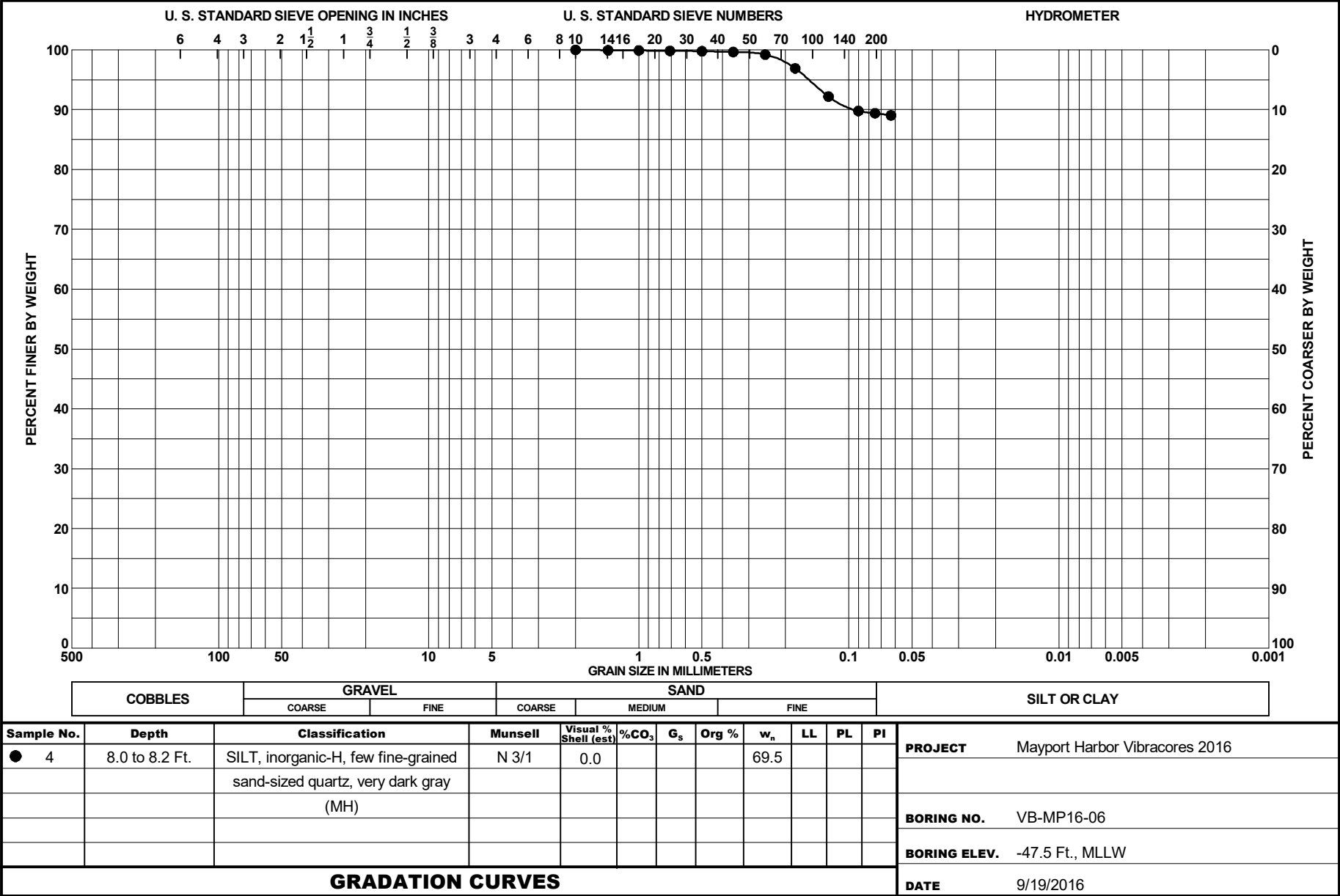


00320-24





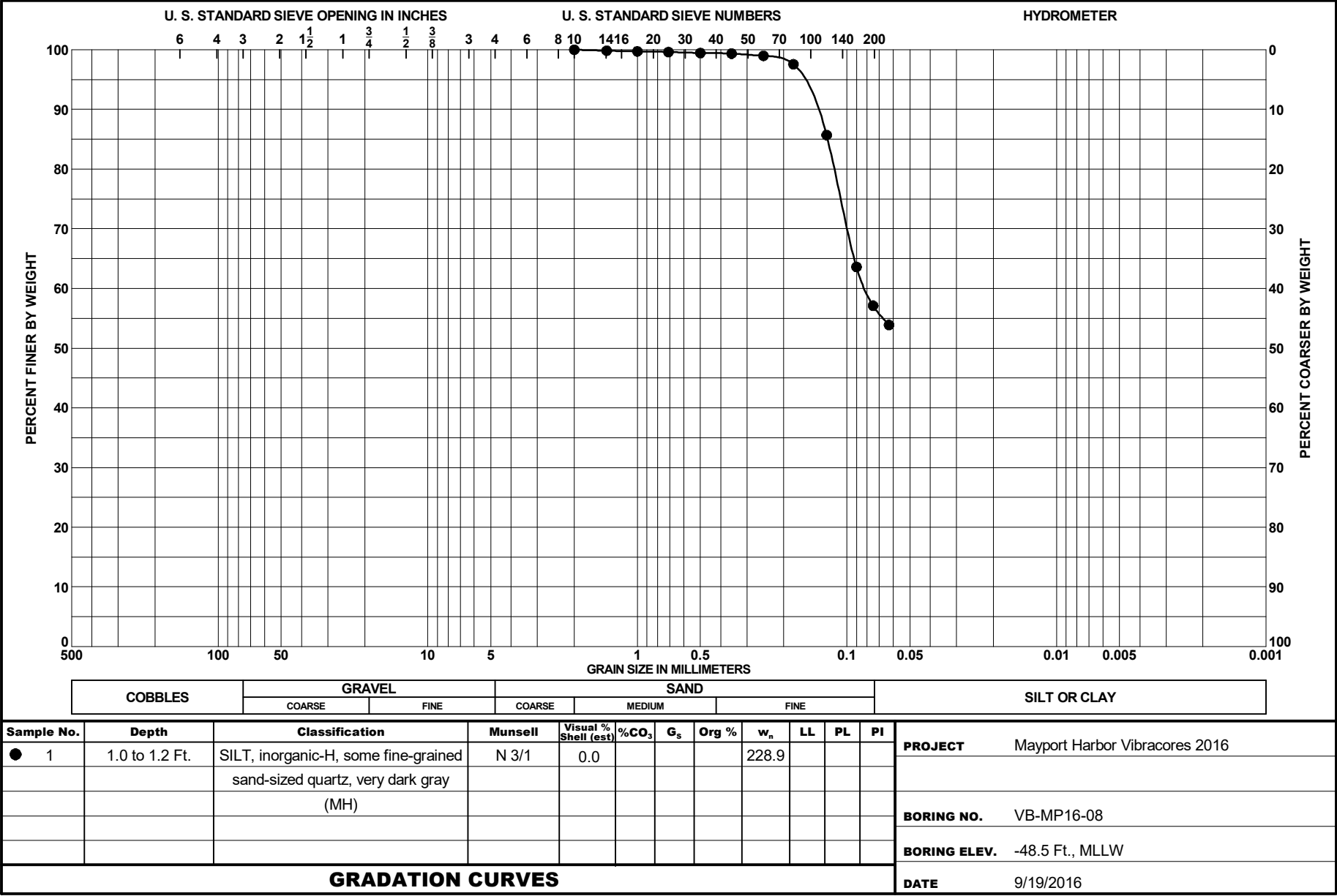
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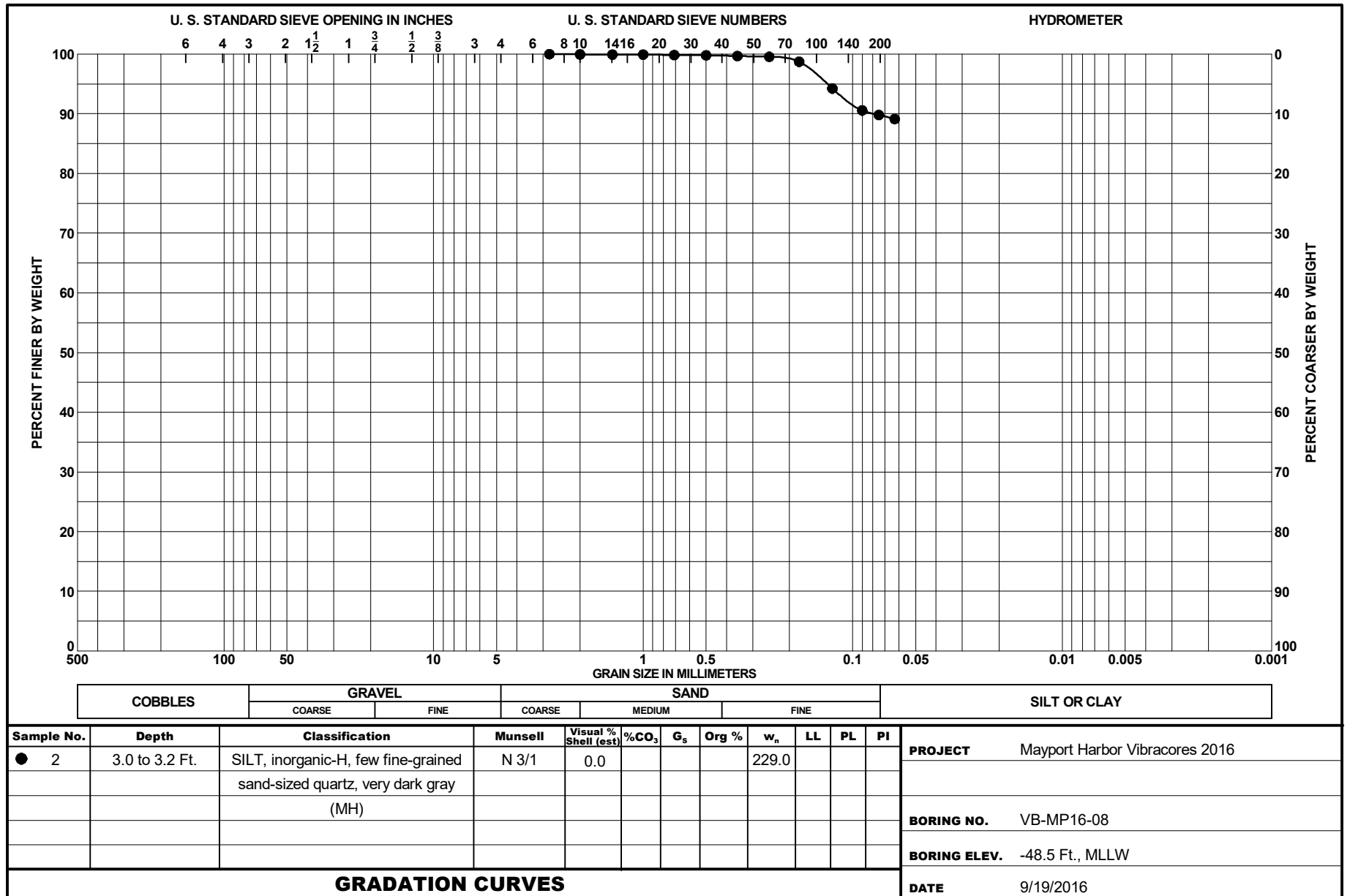




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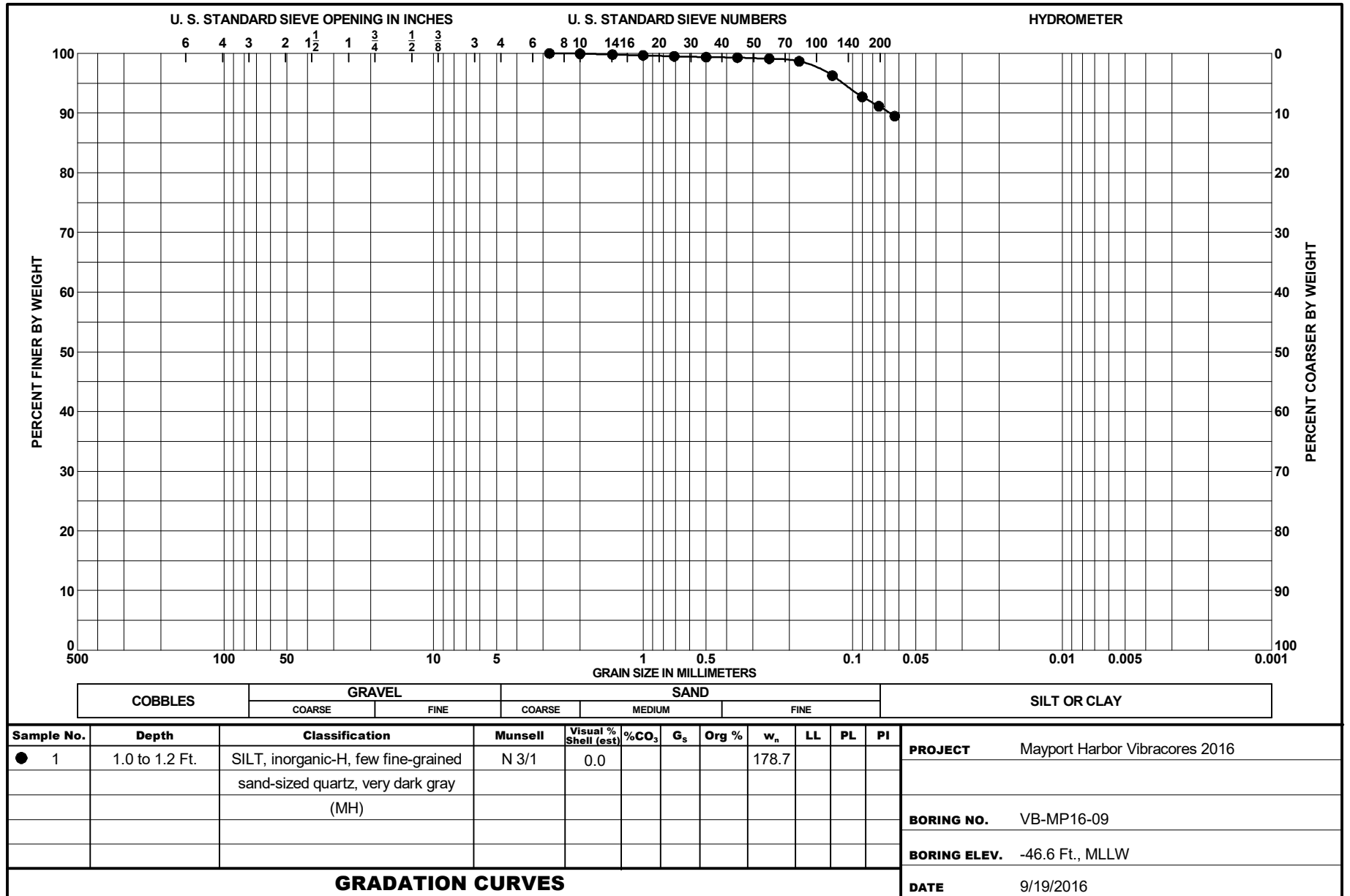
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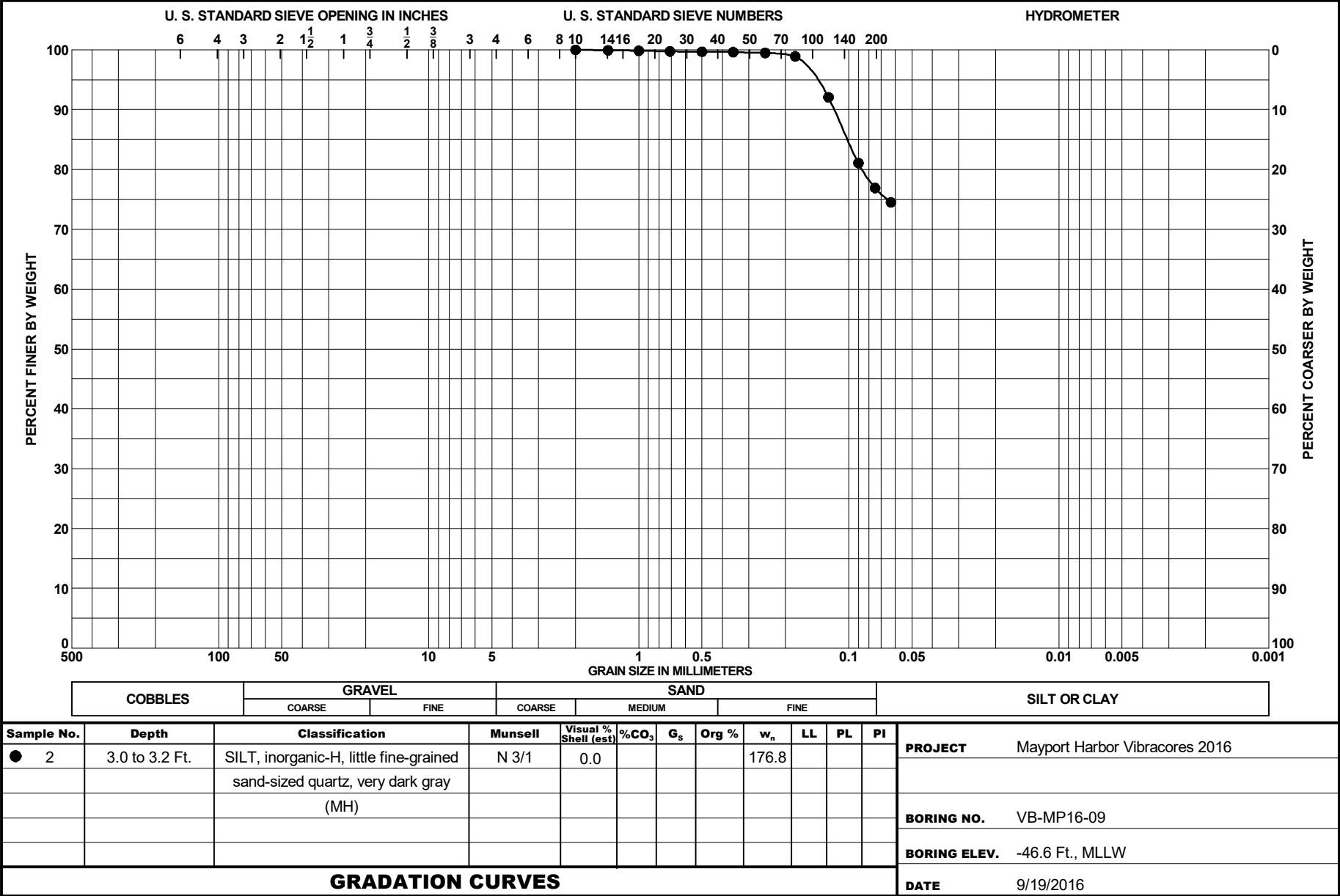




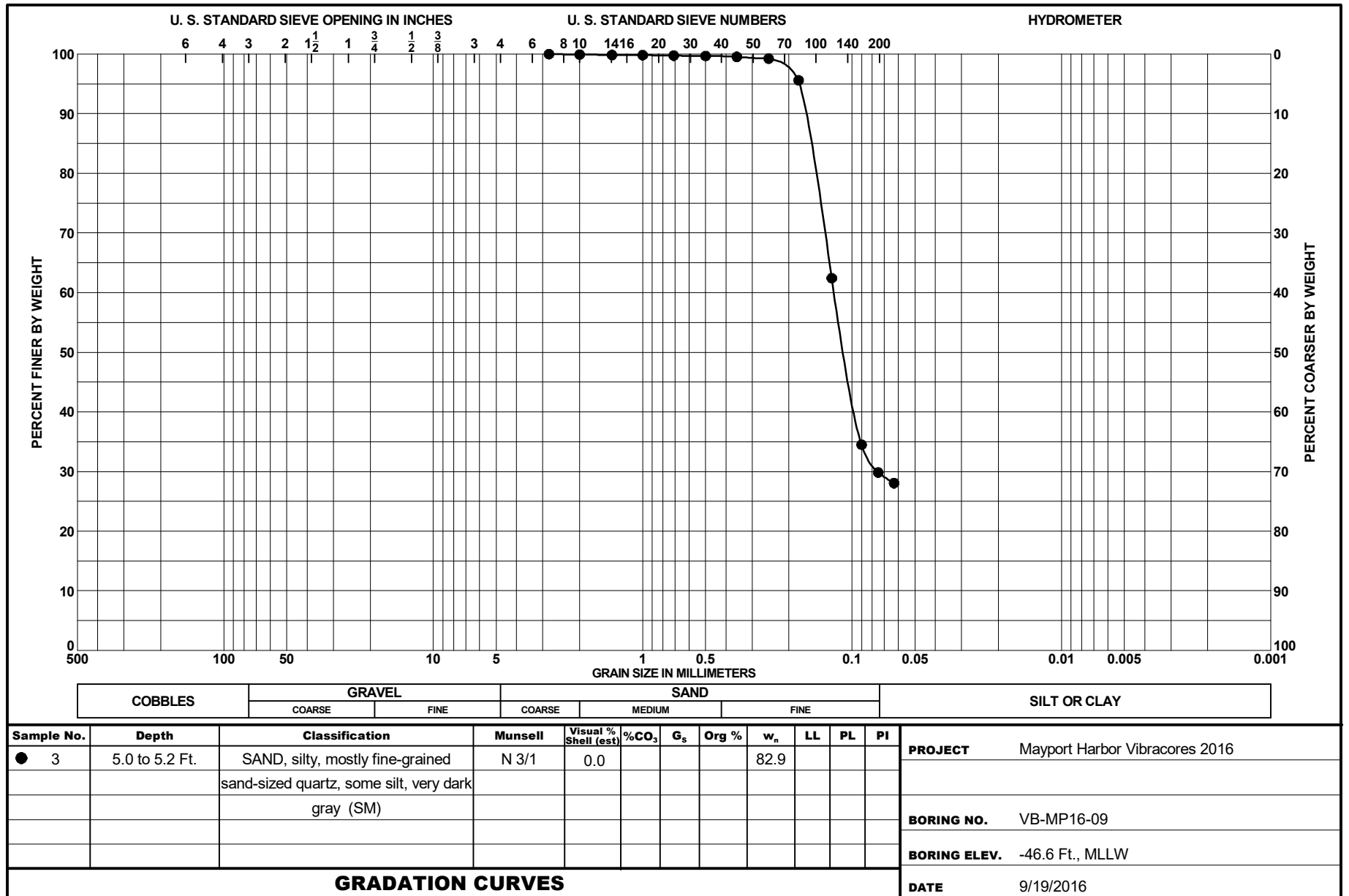
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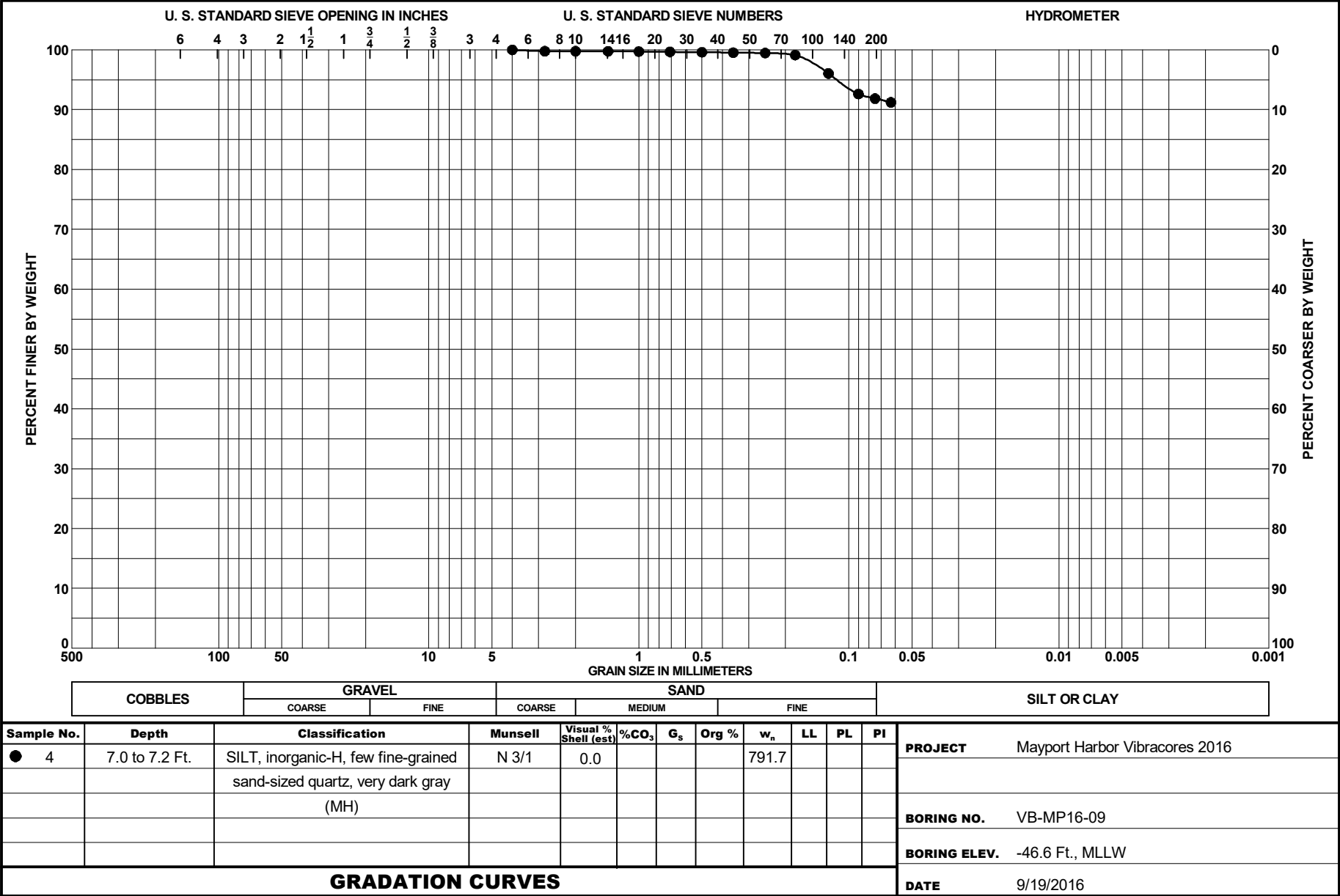
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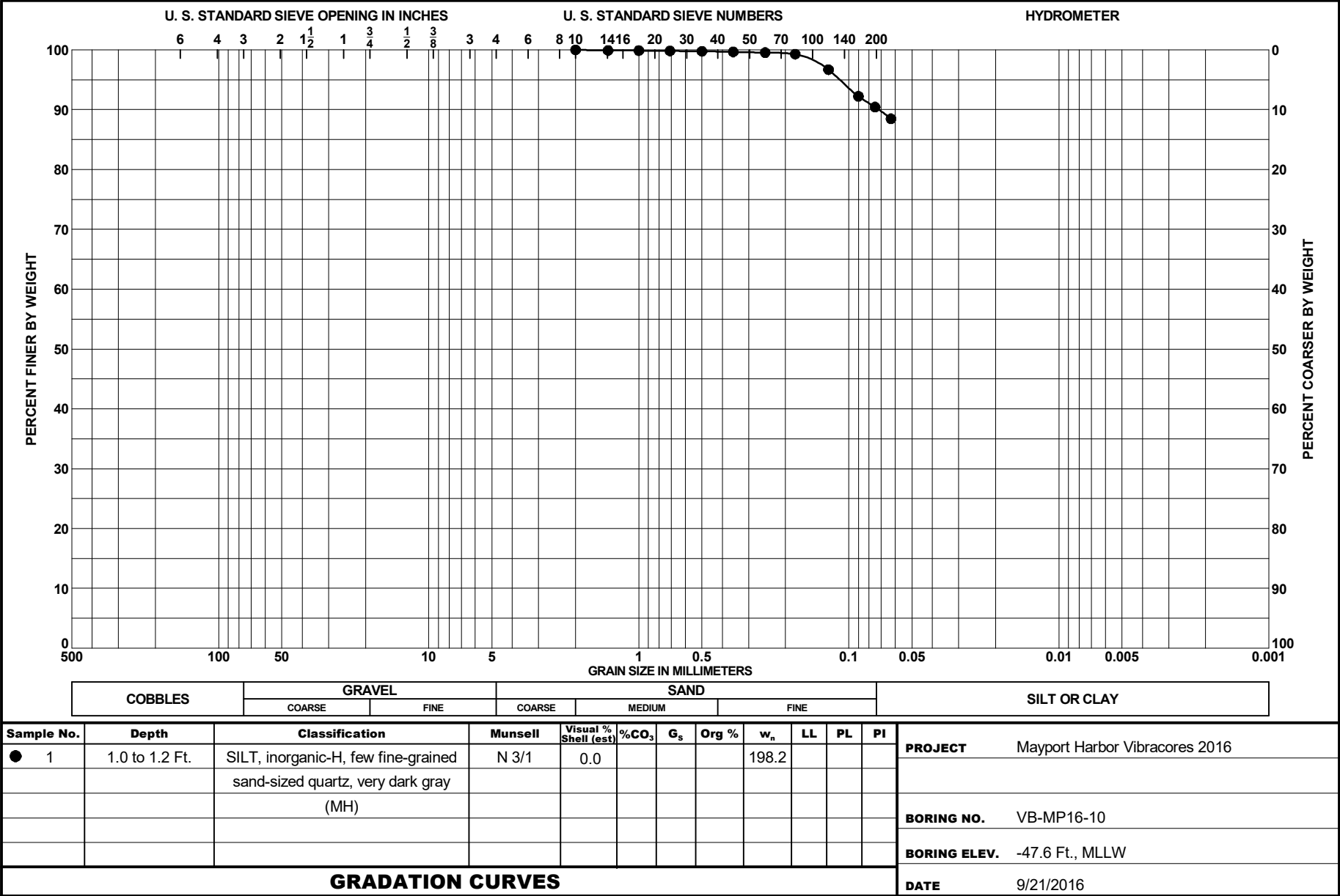
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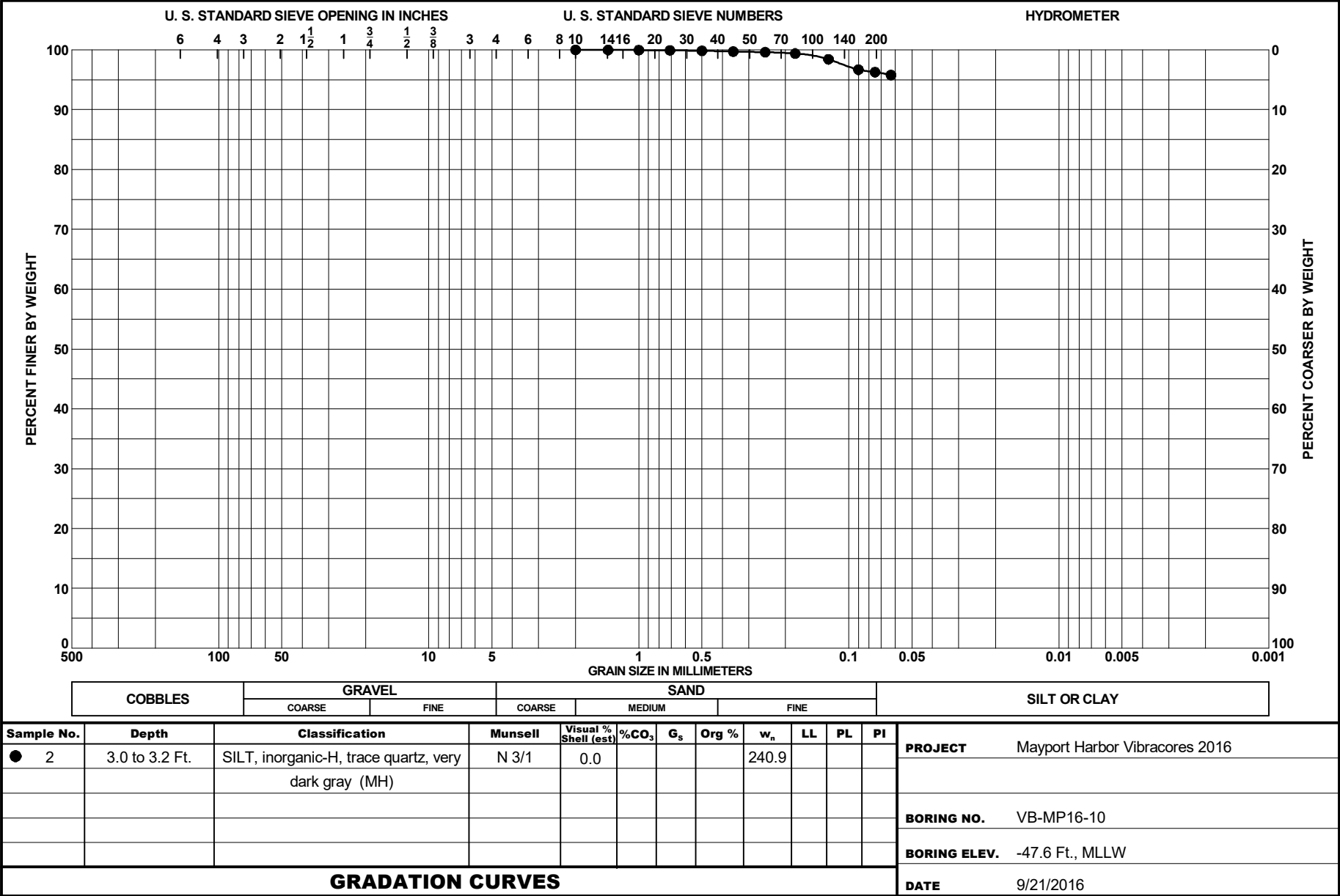
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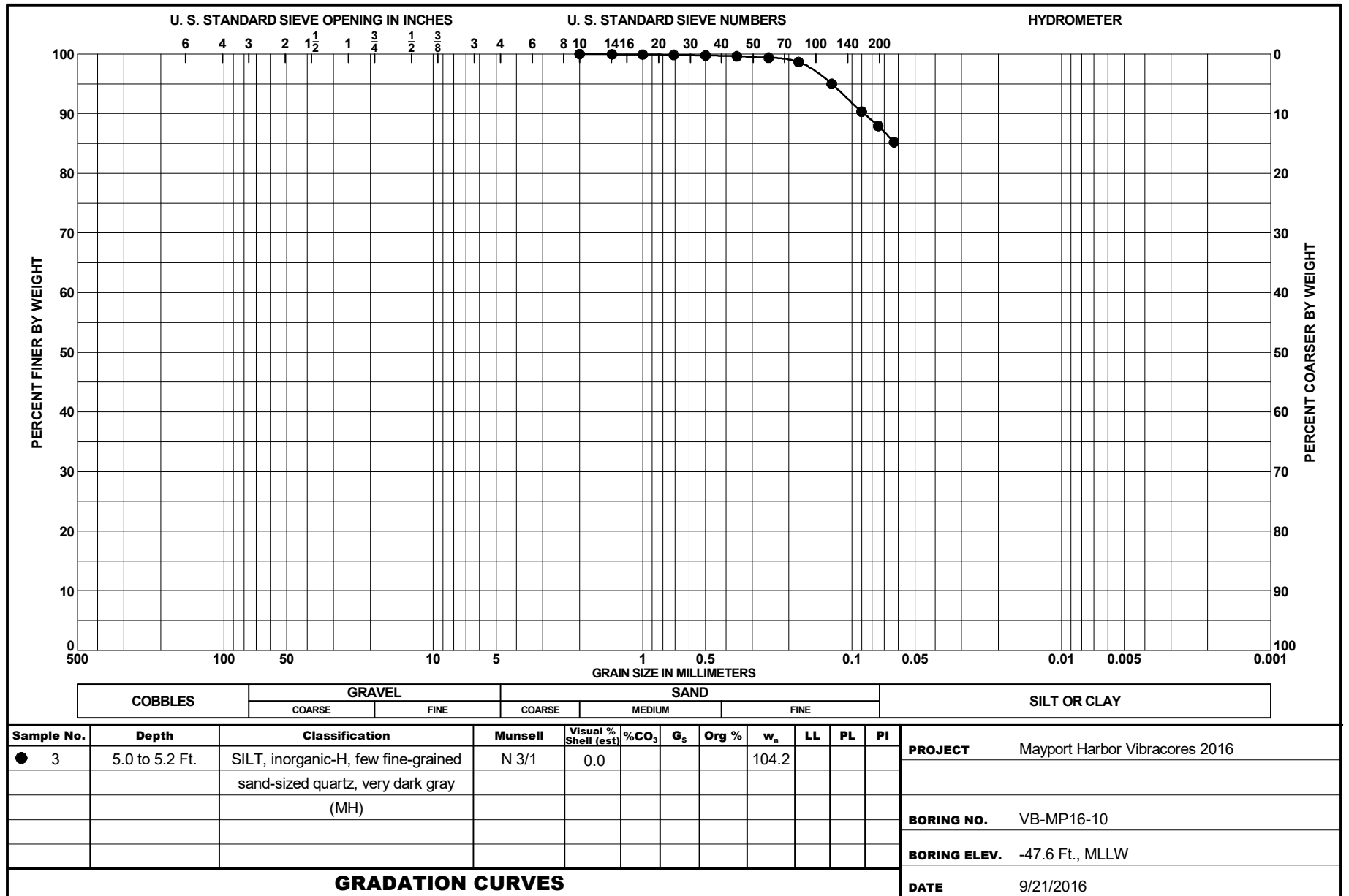
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00320-38

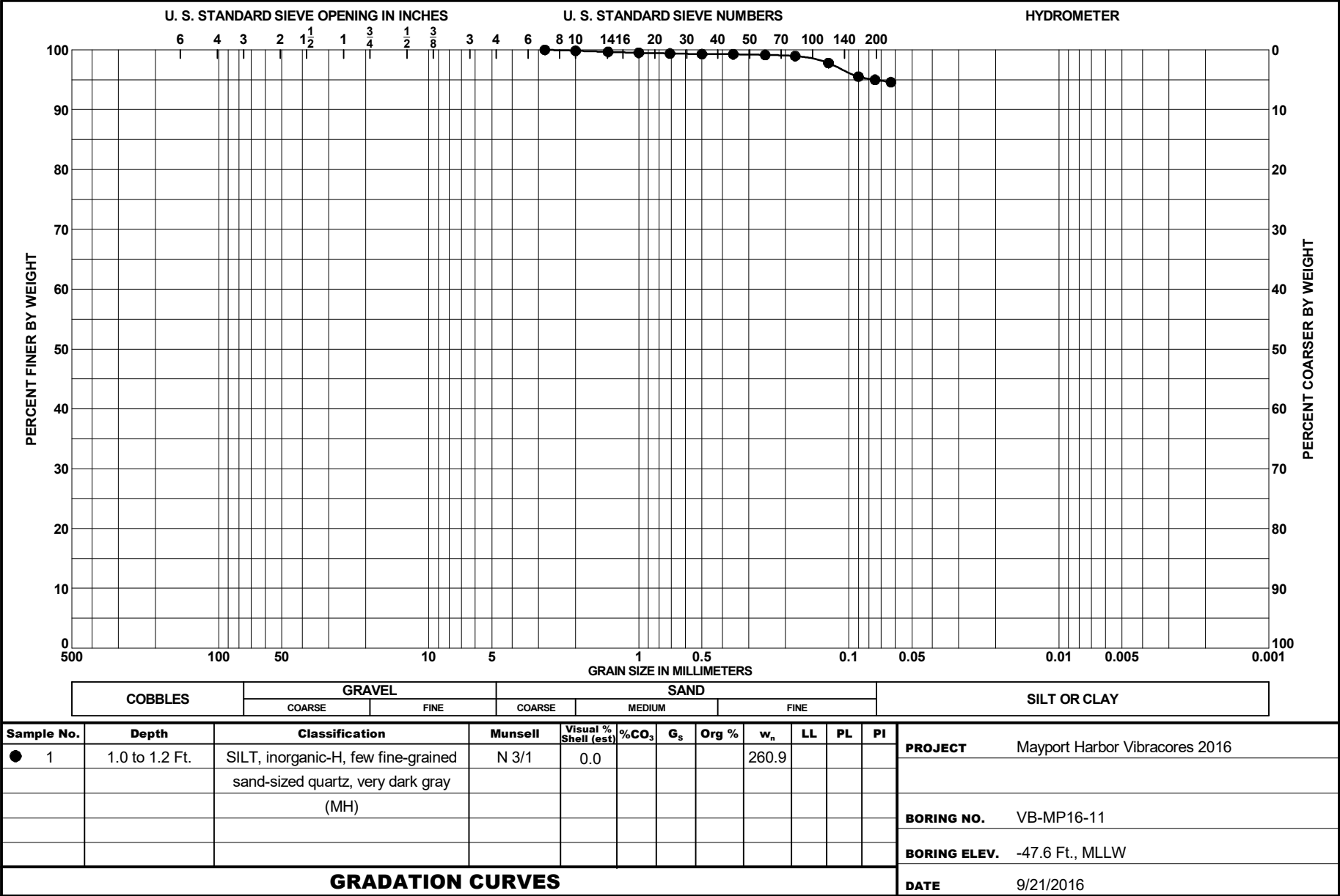


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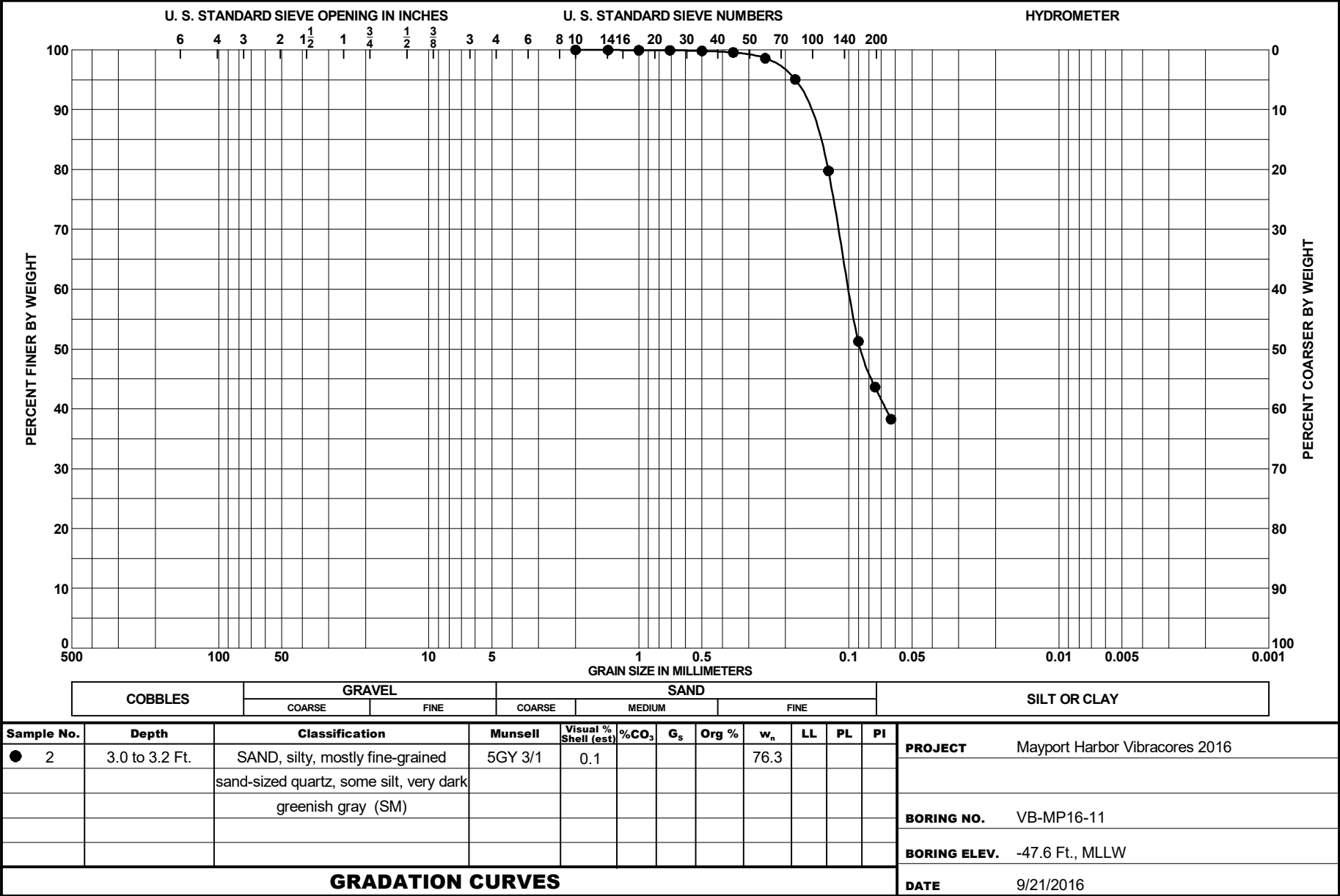




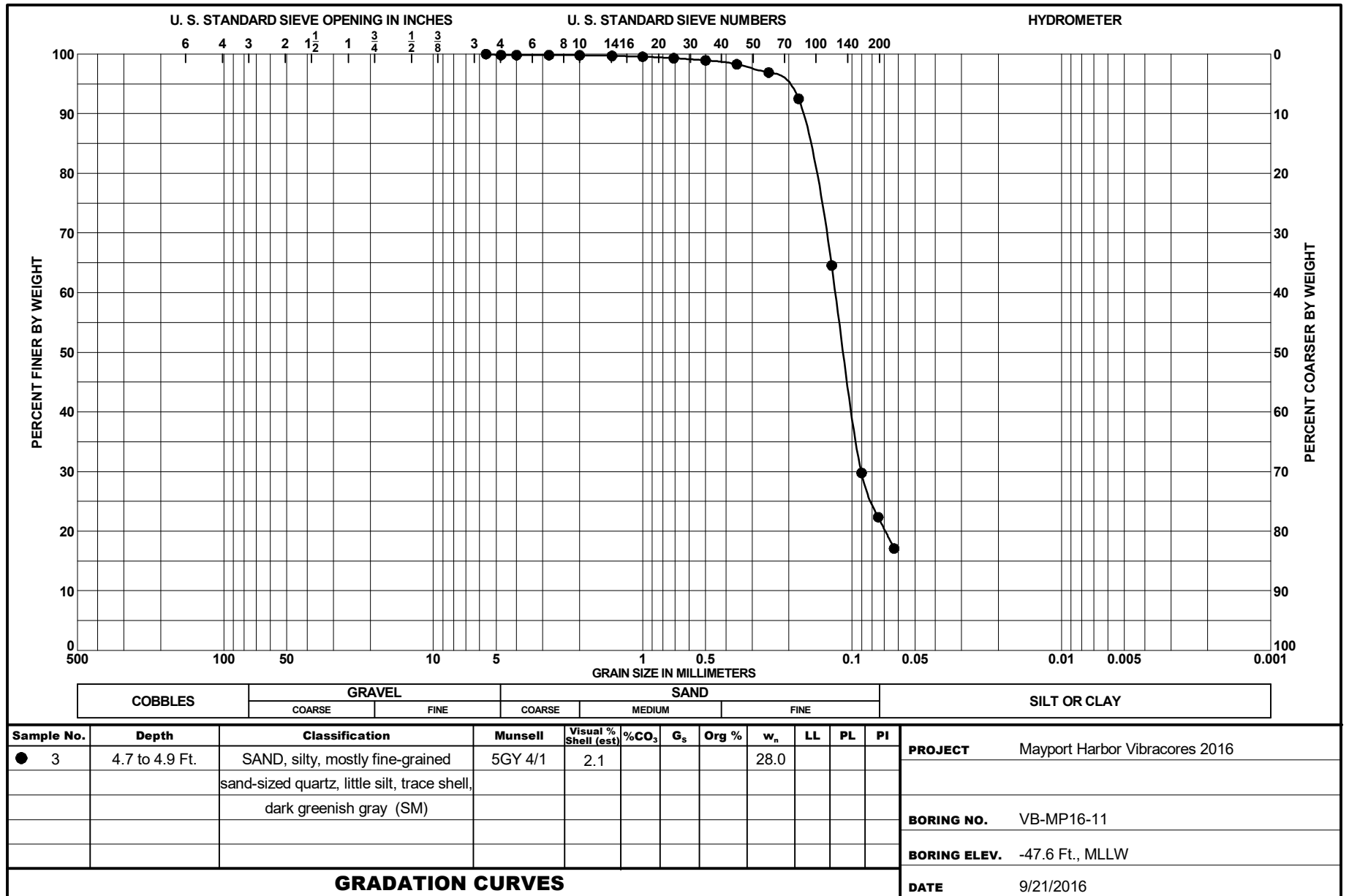
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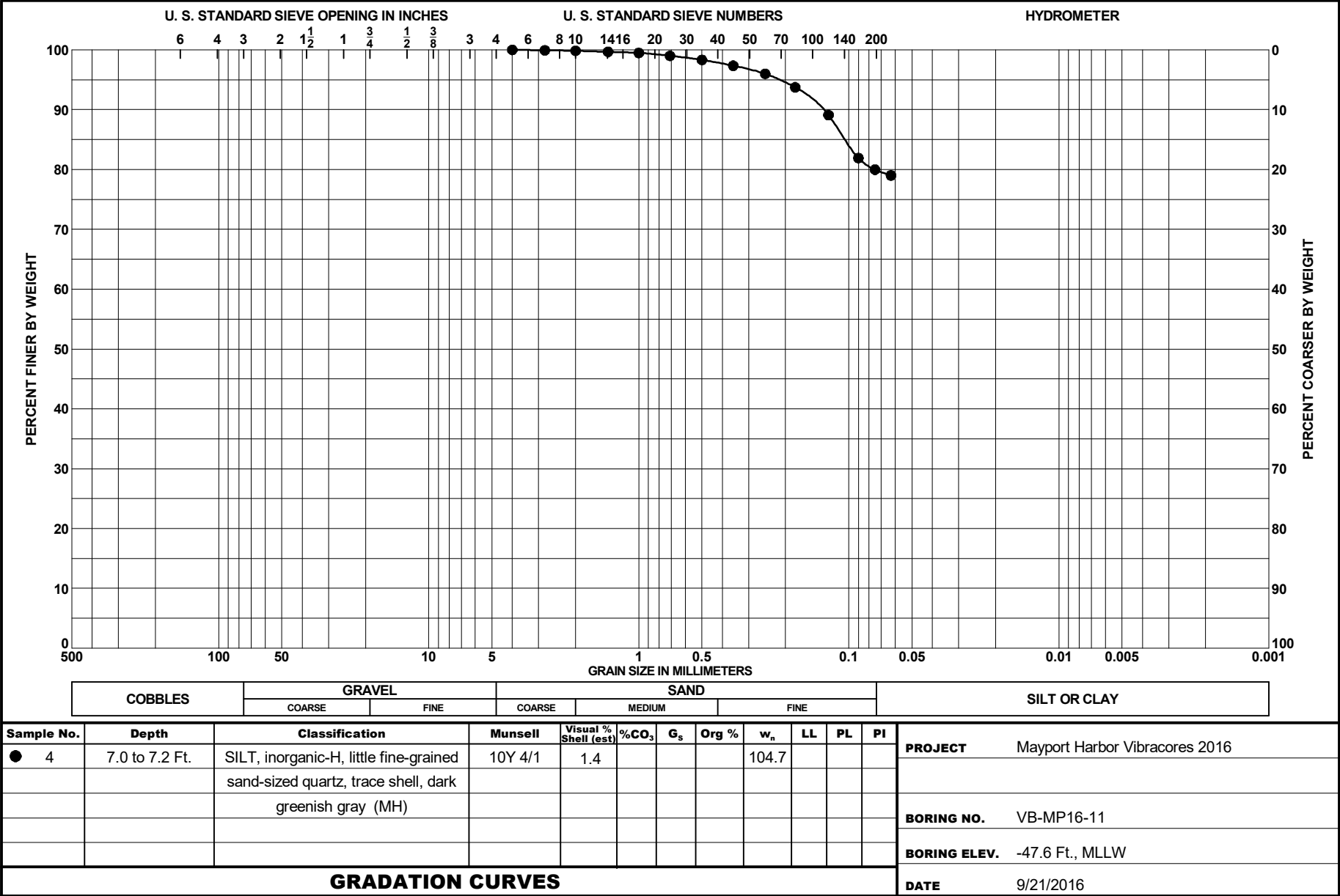
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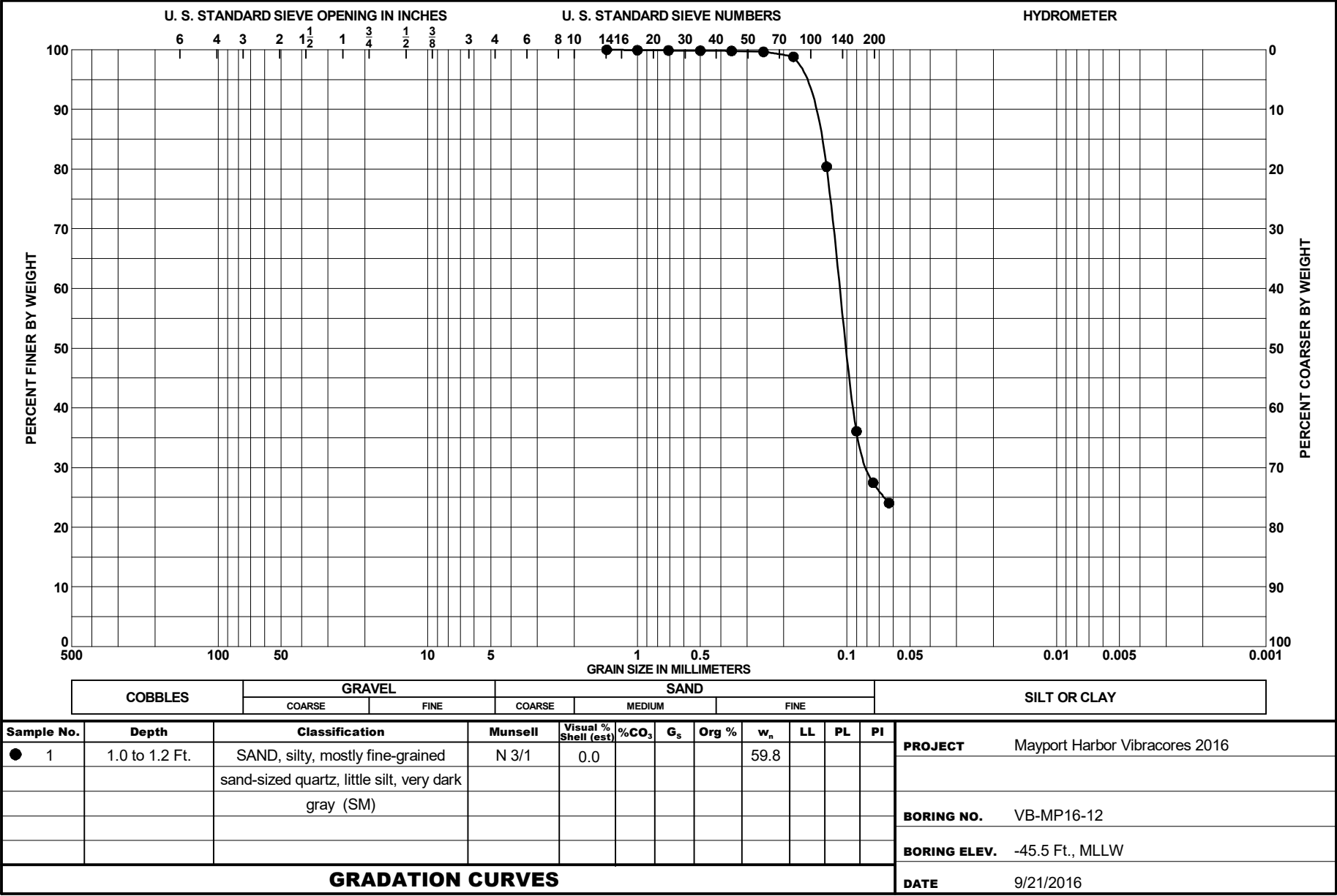
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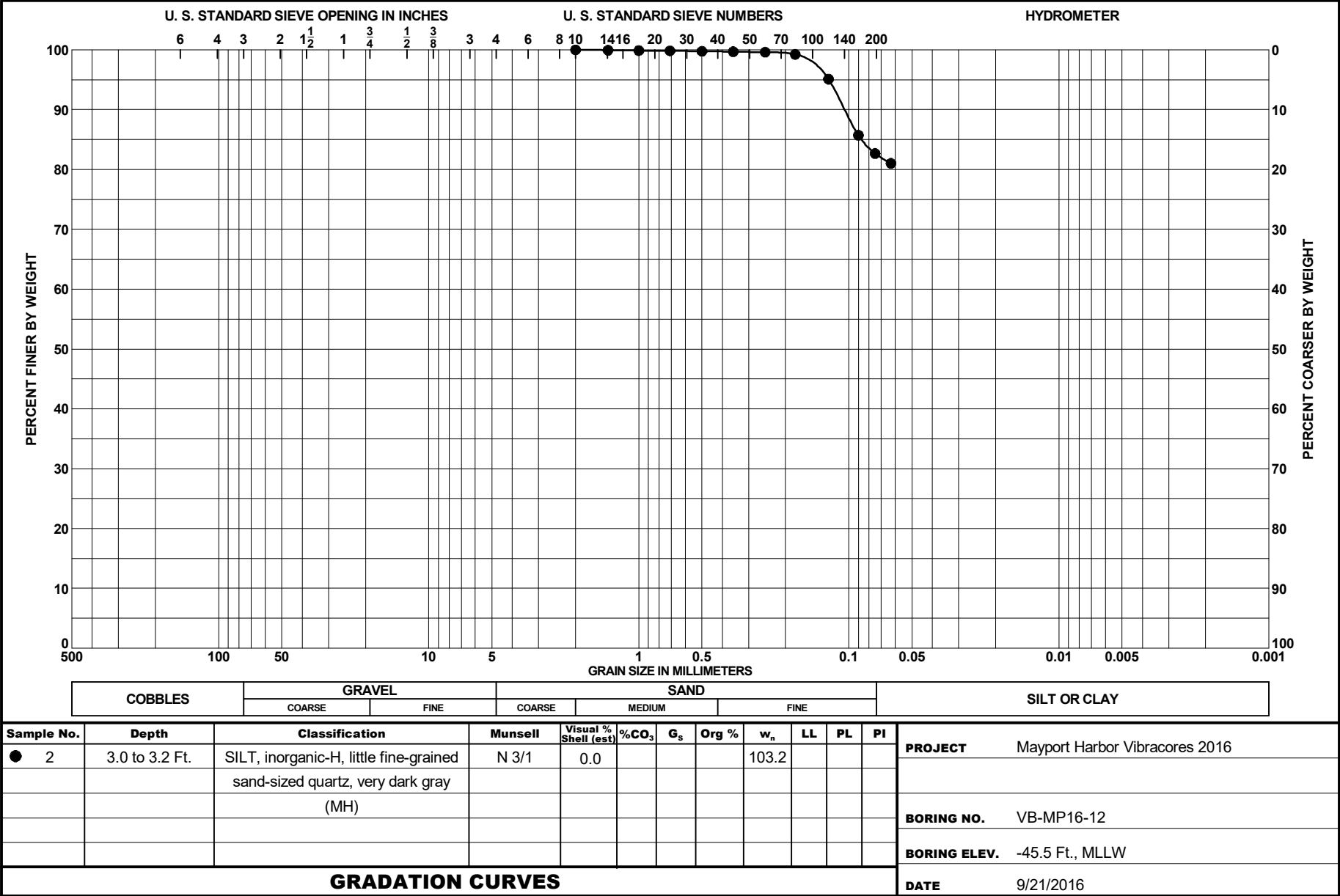
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00320-45

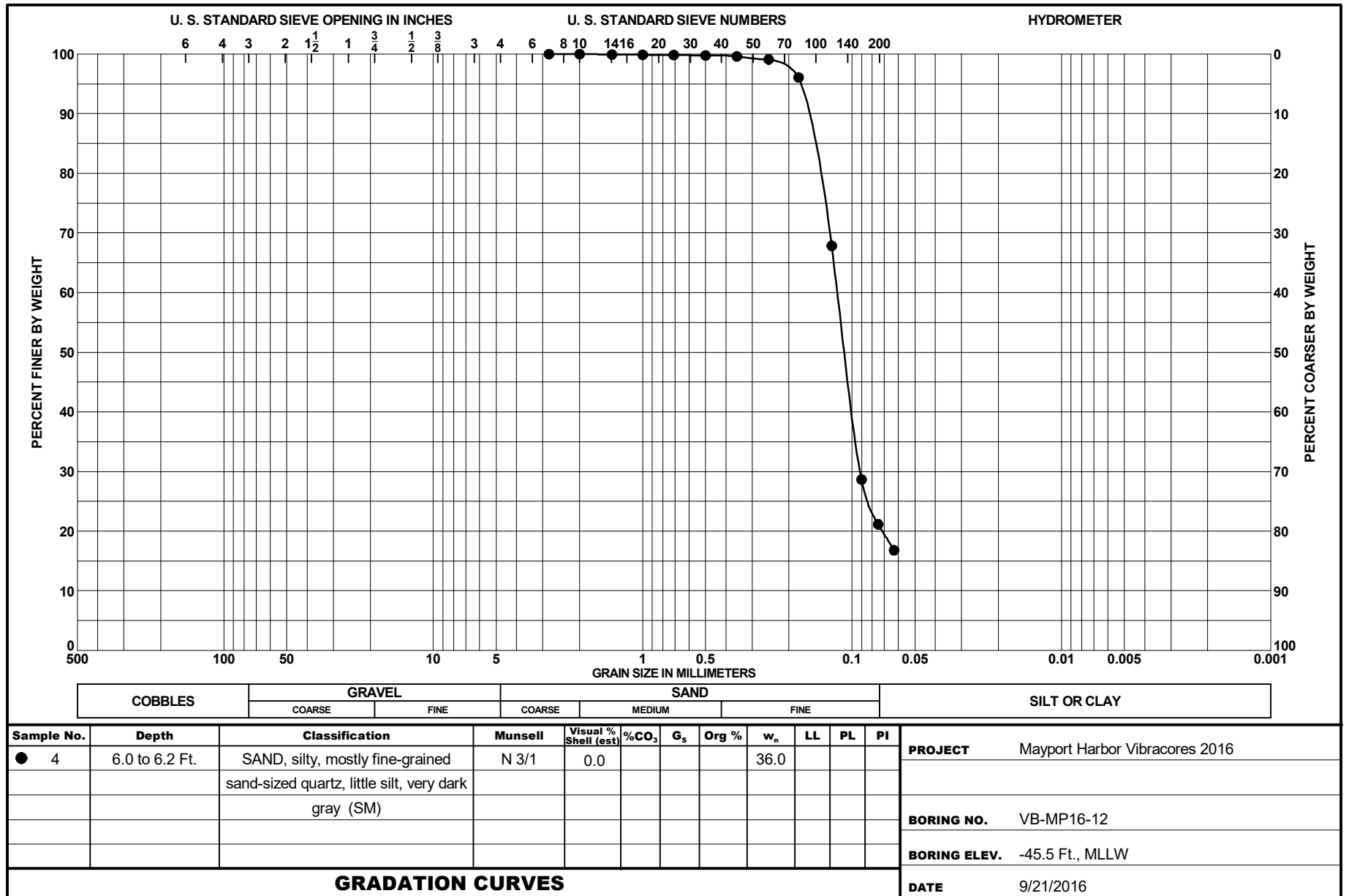


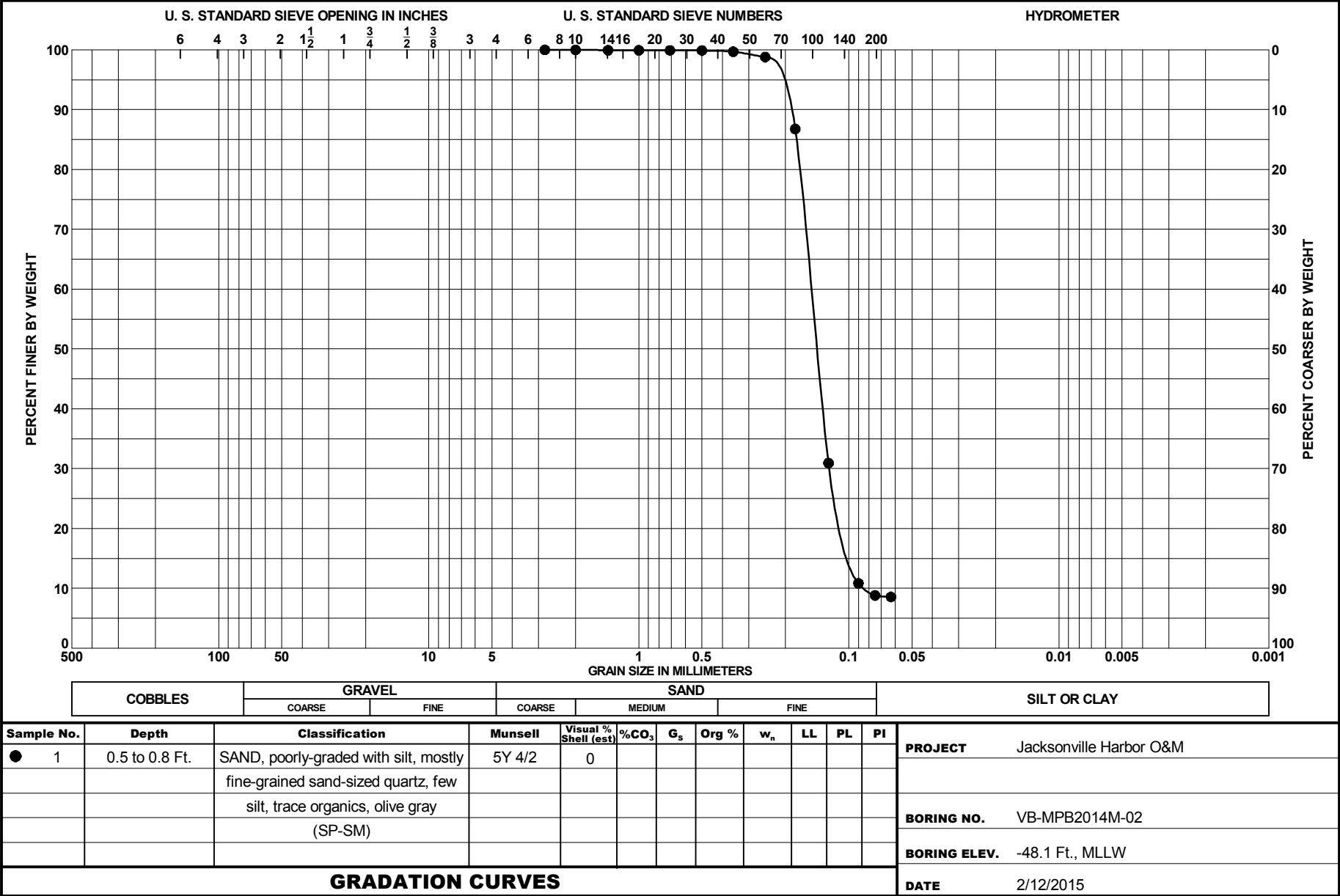
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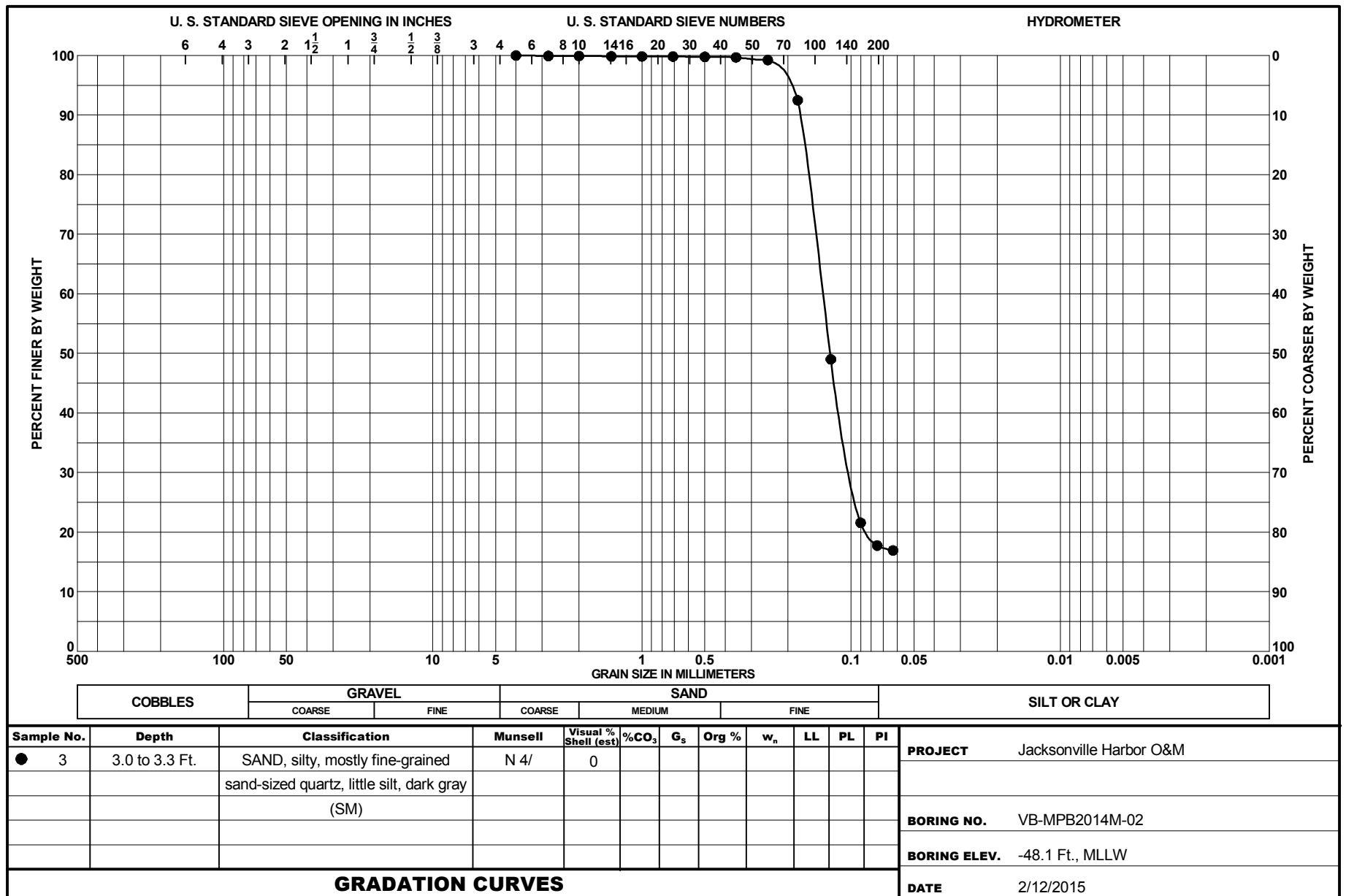


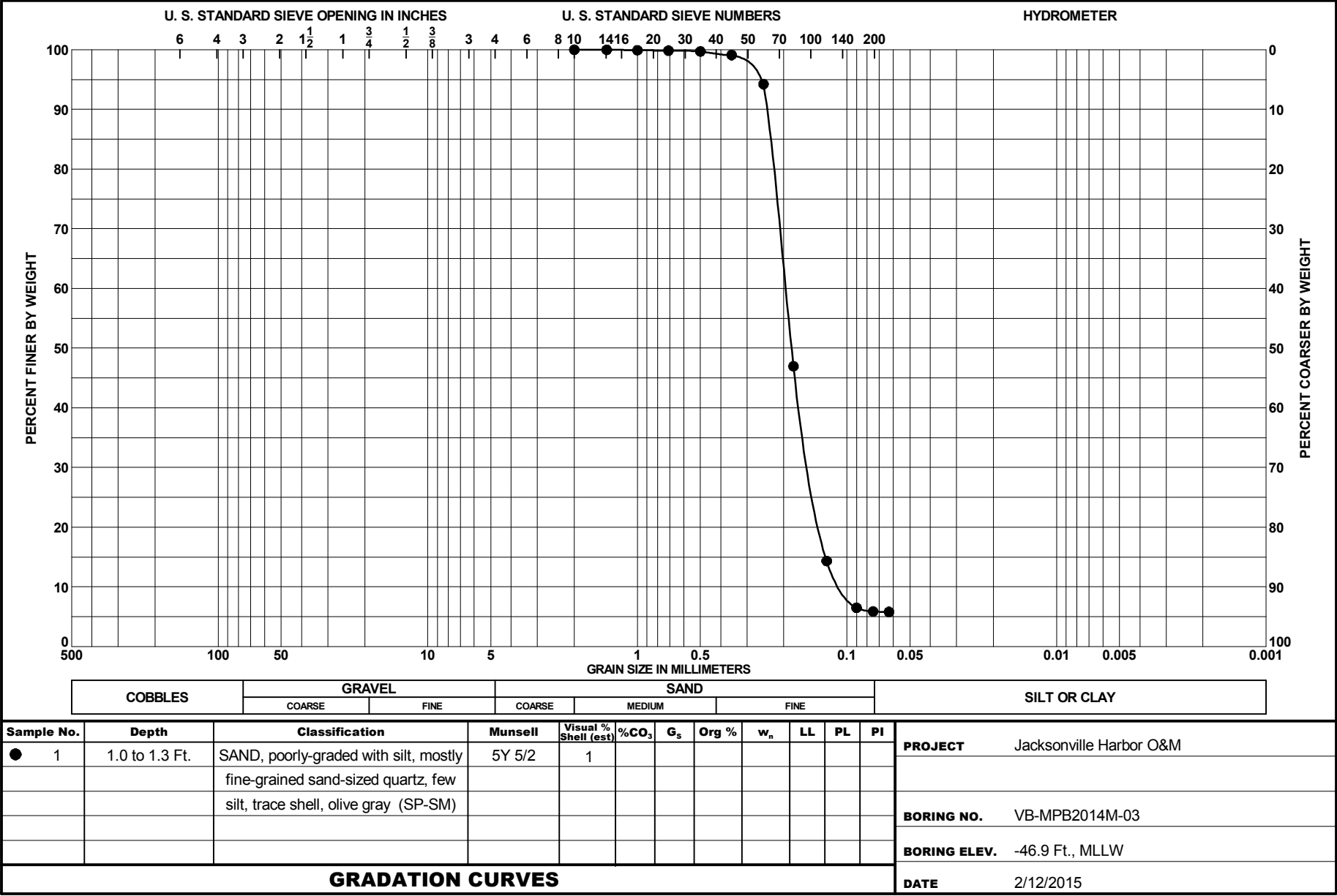


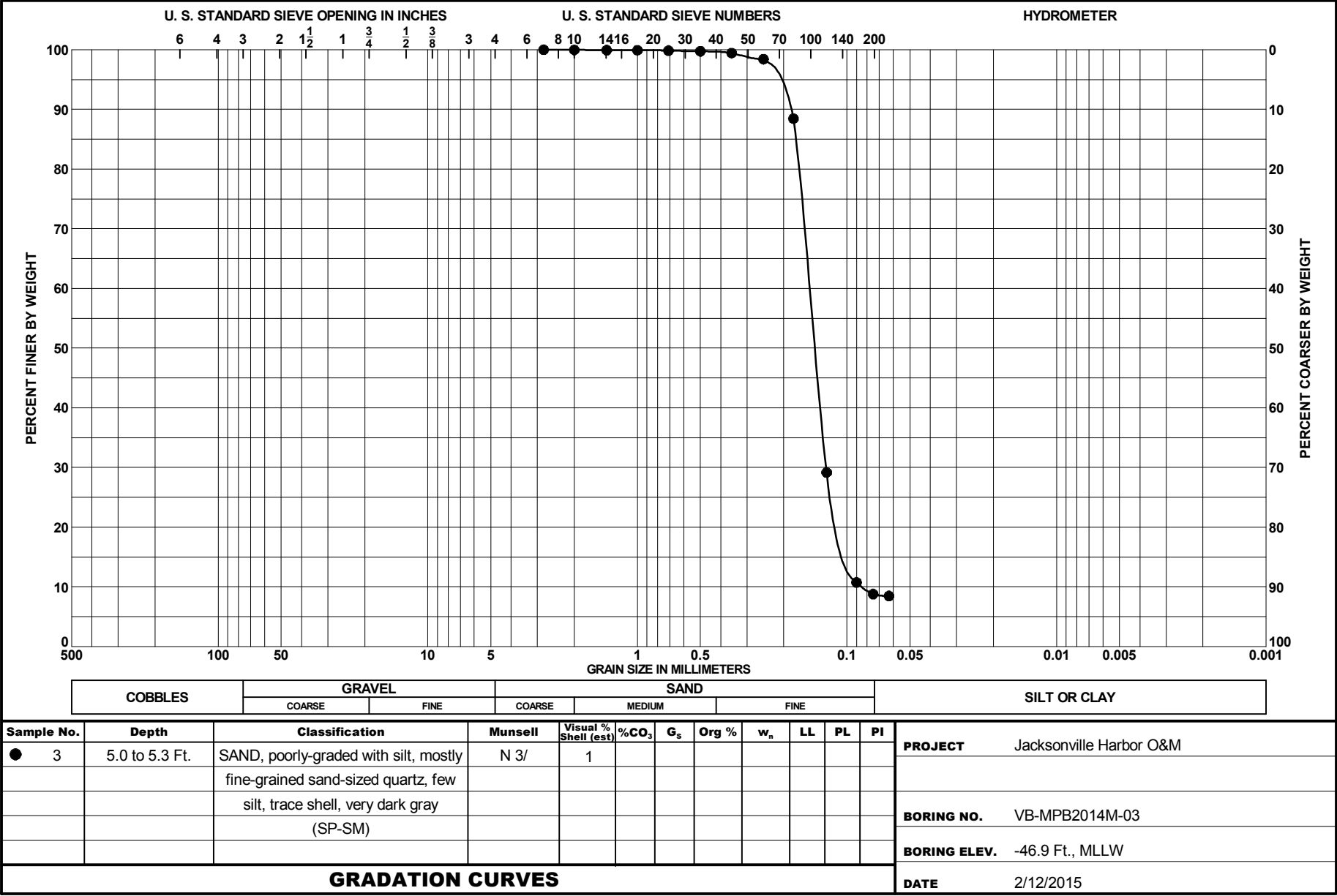
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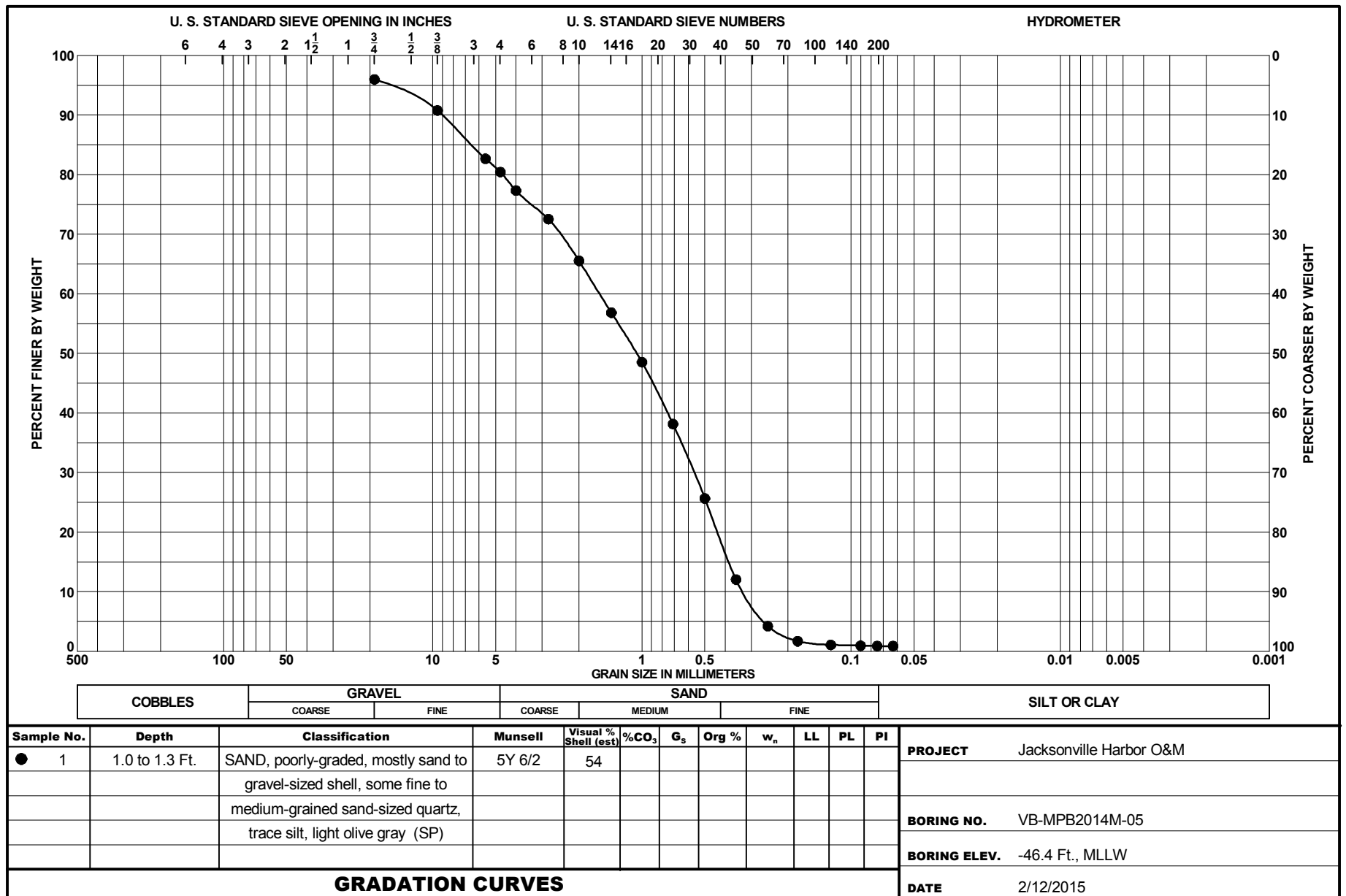


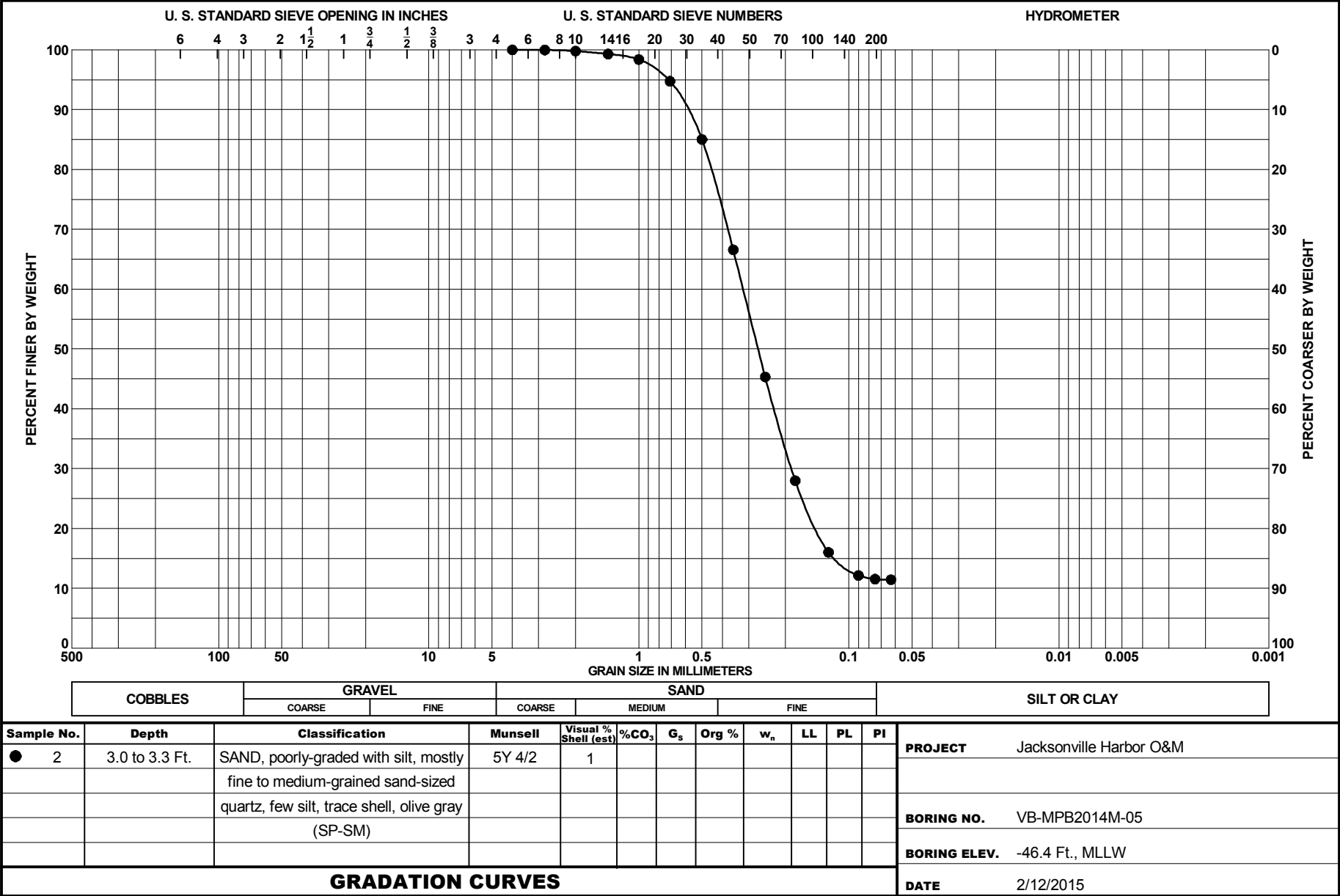


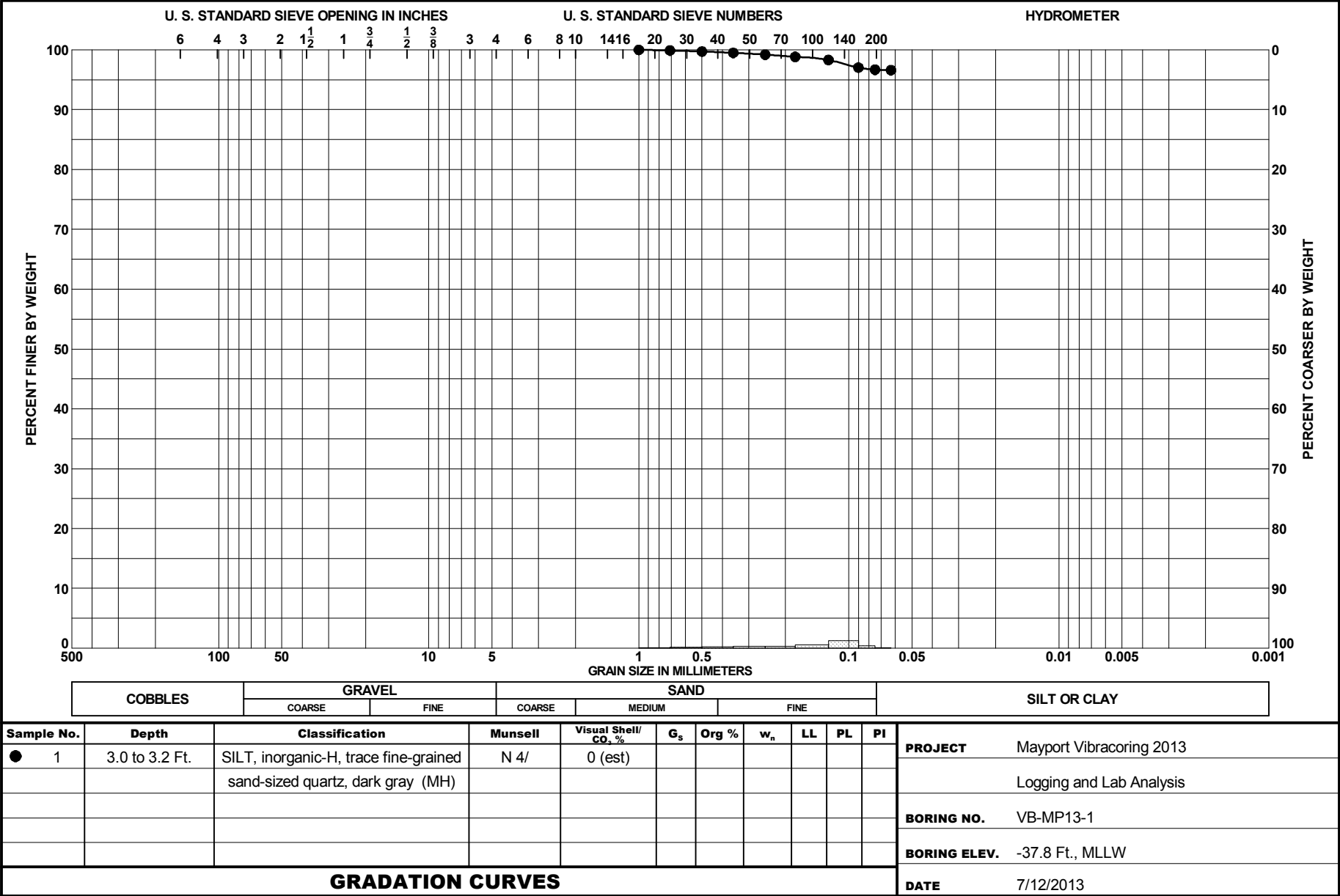


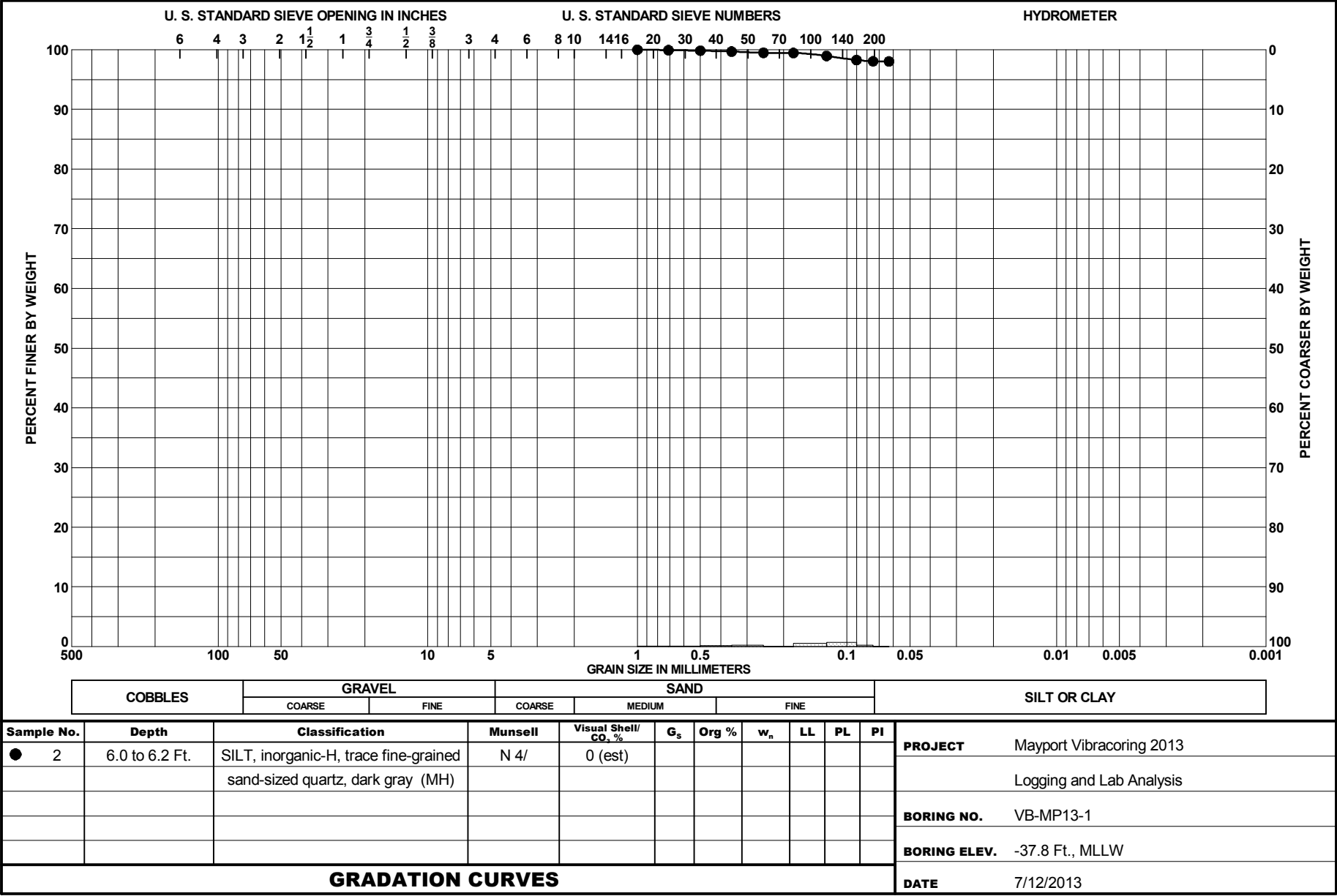


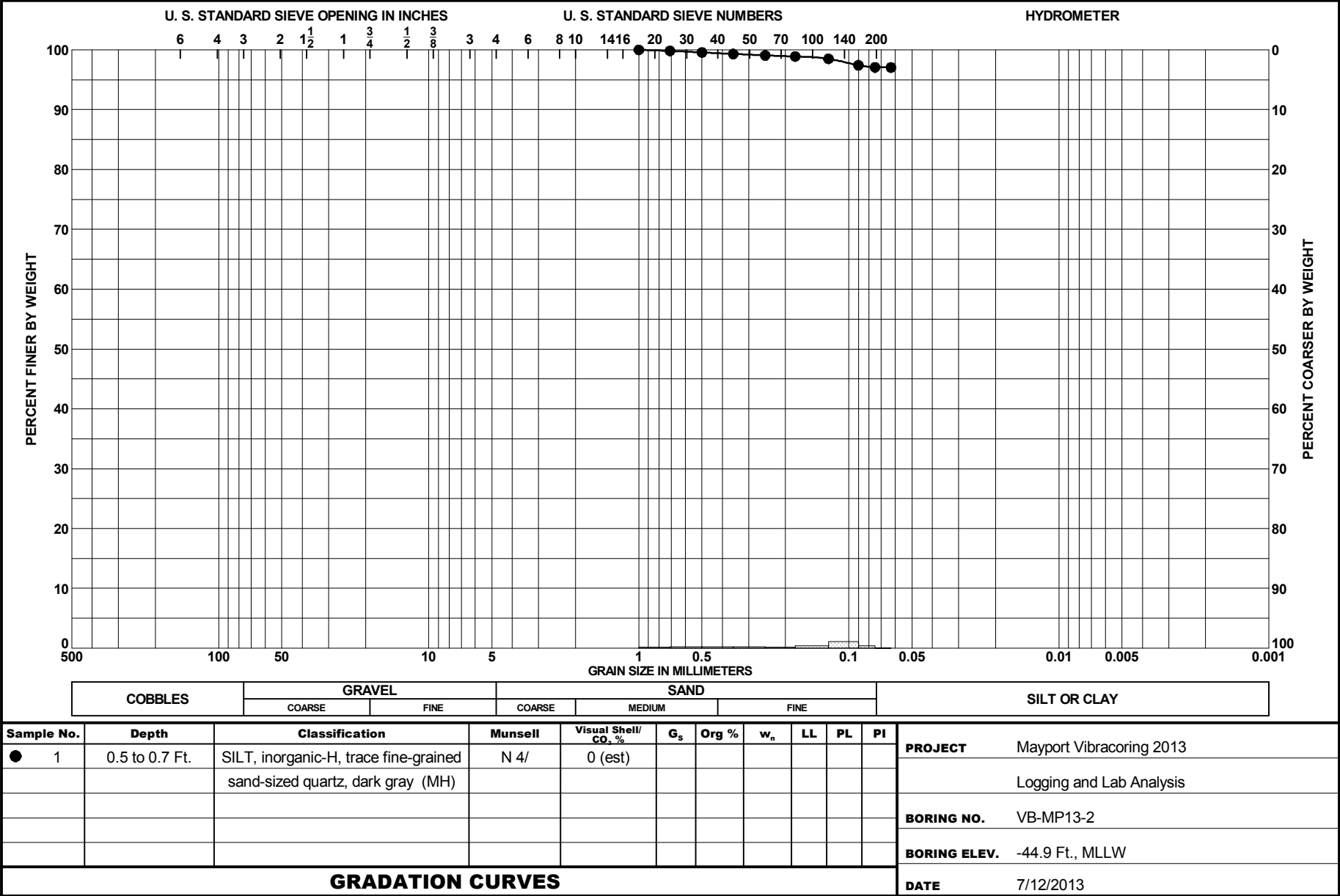


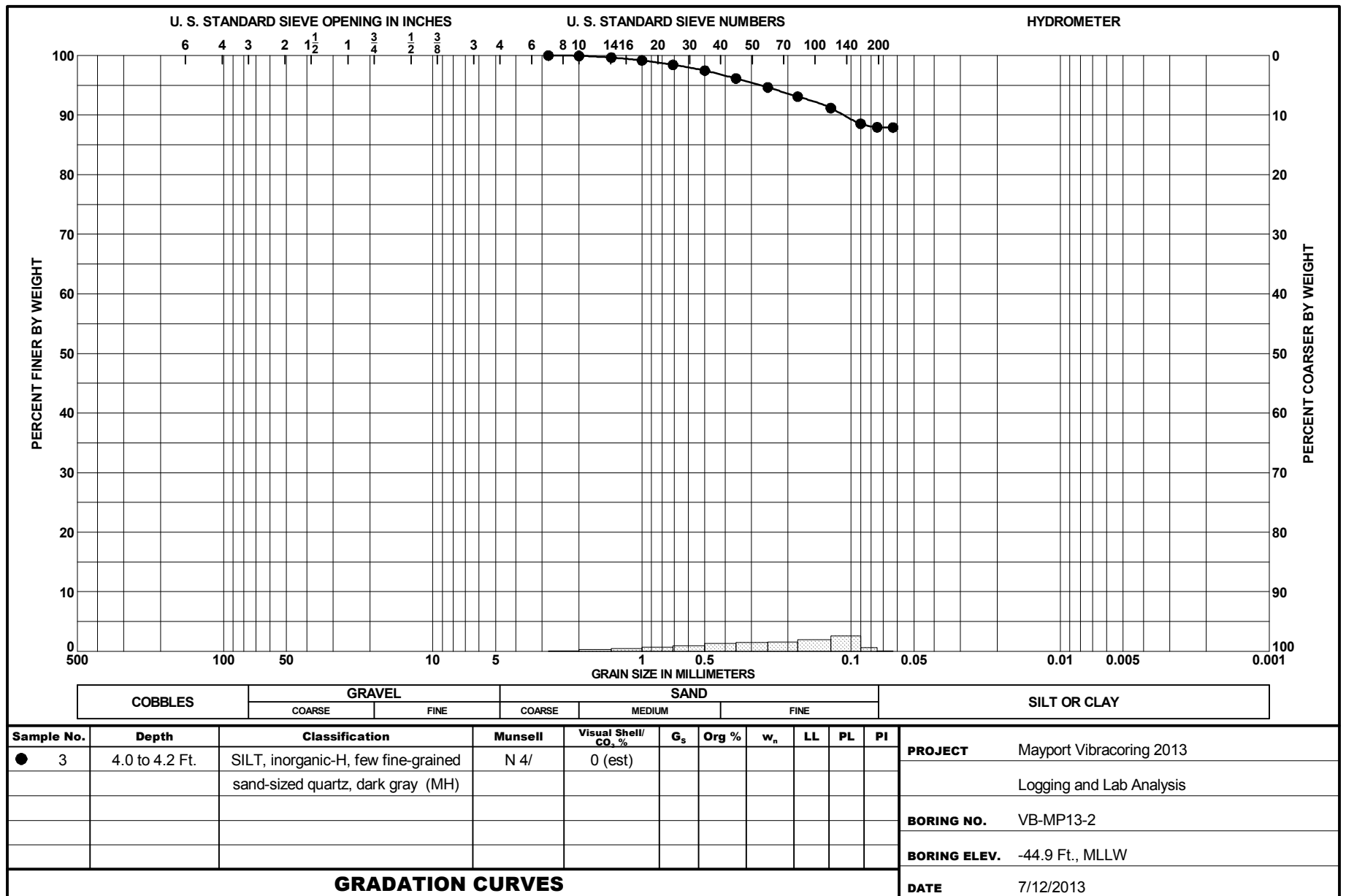


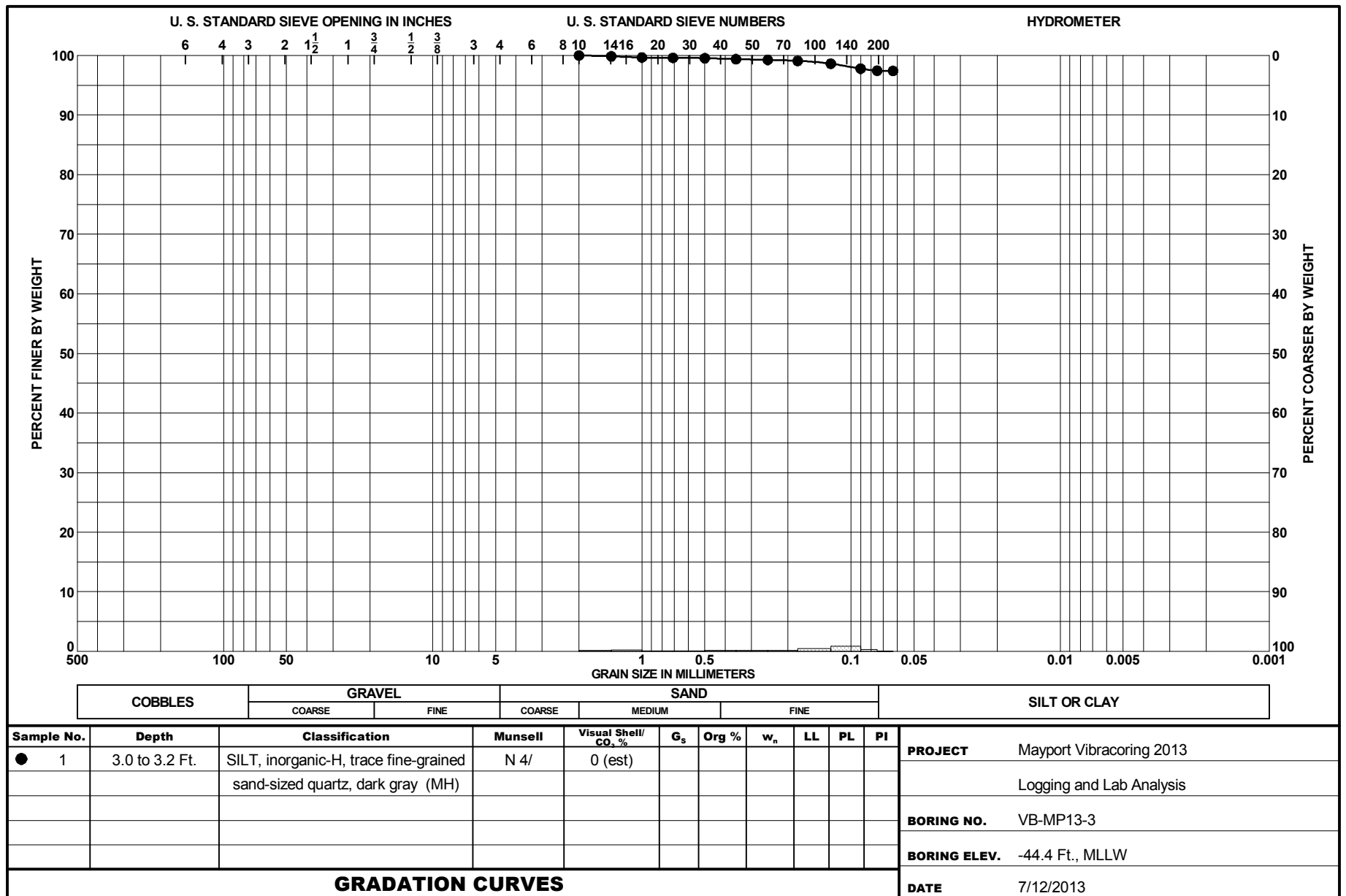


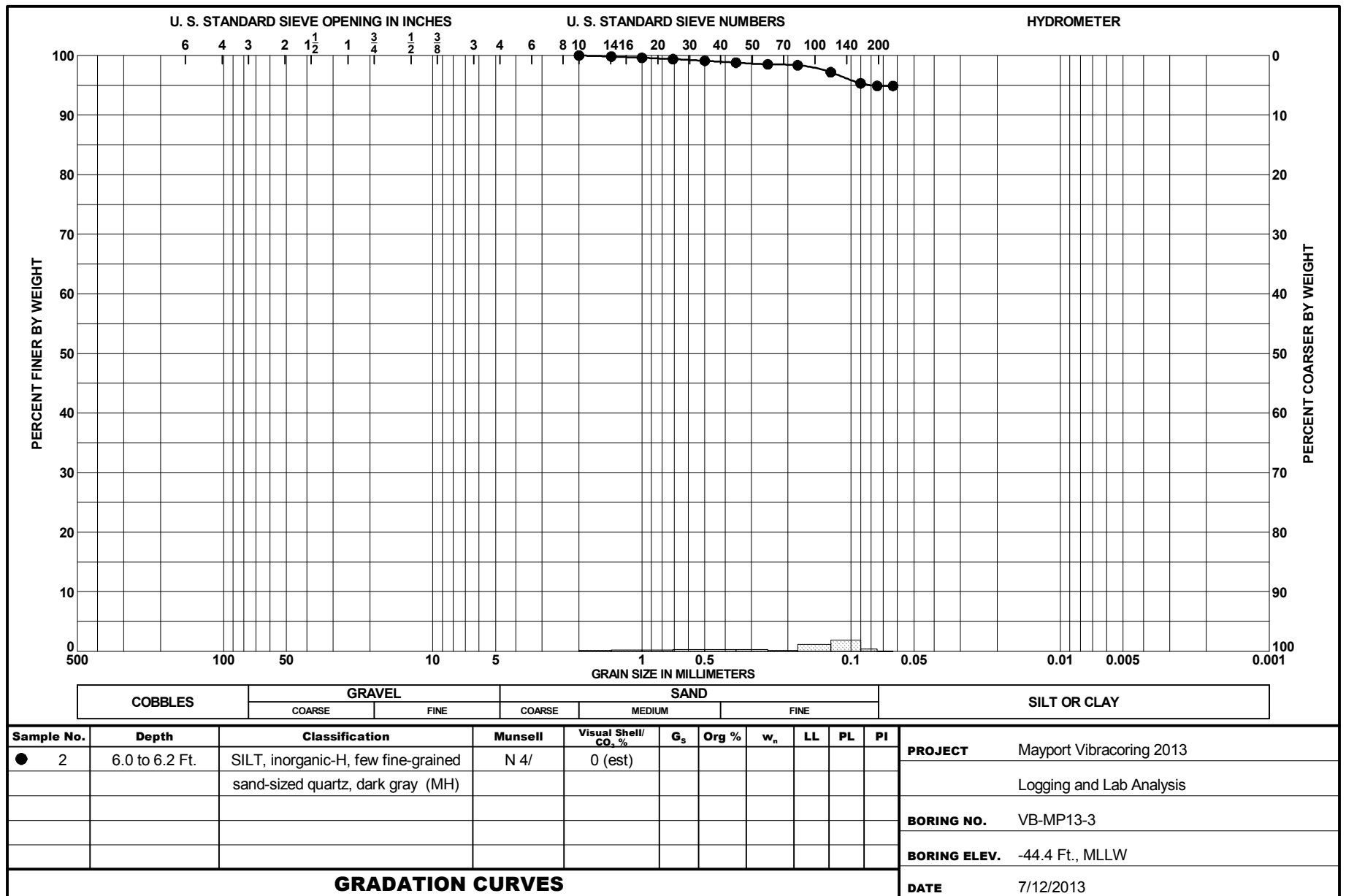


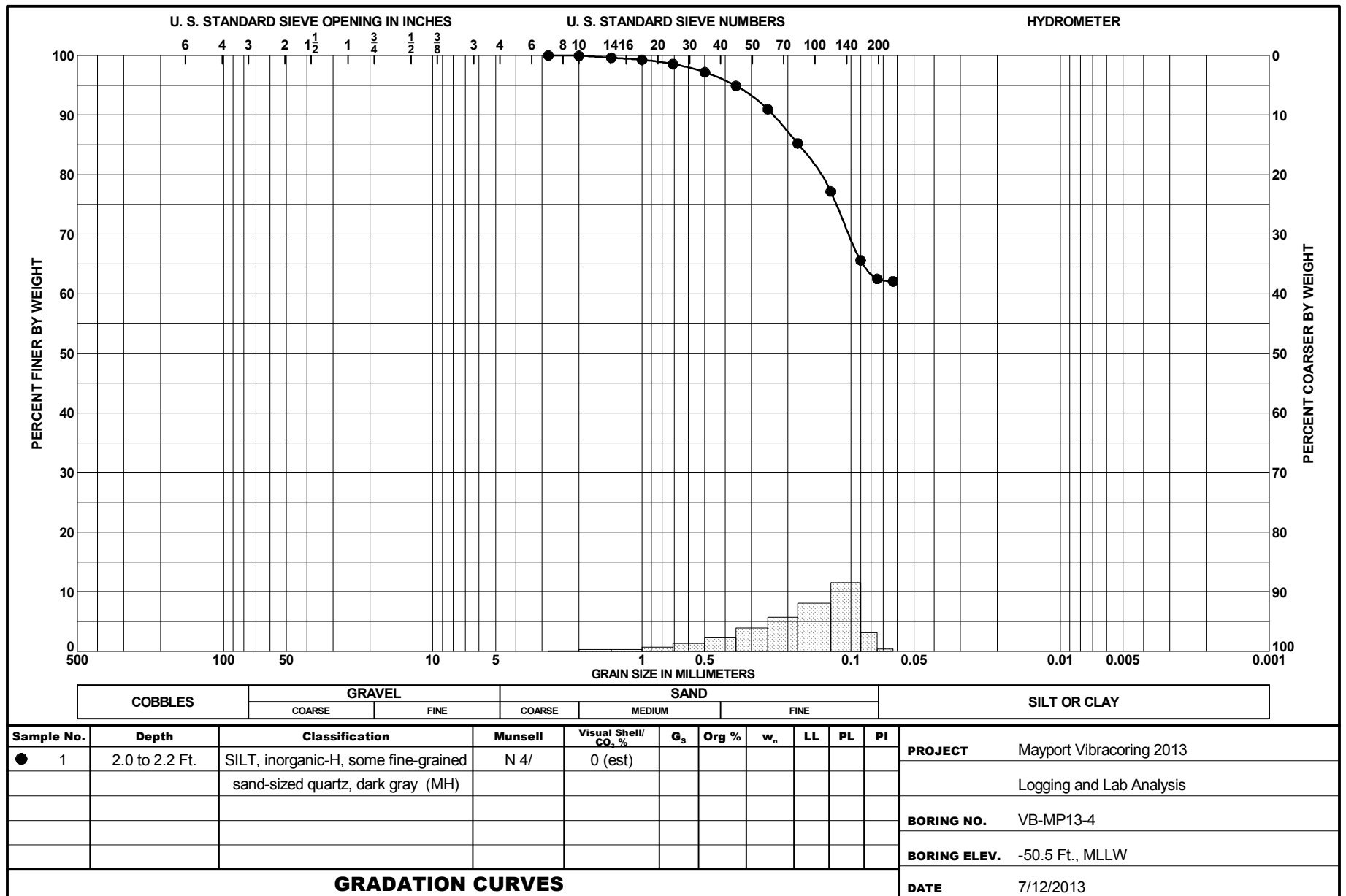


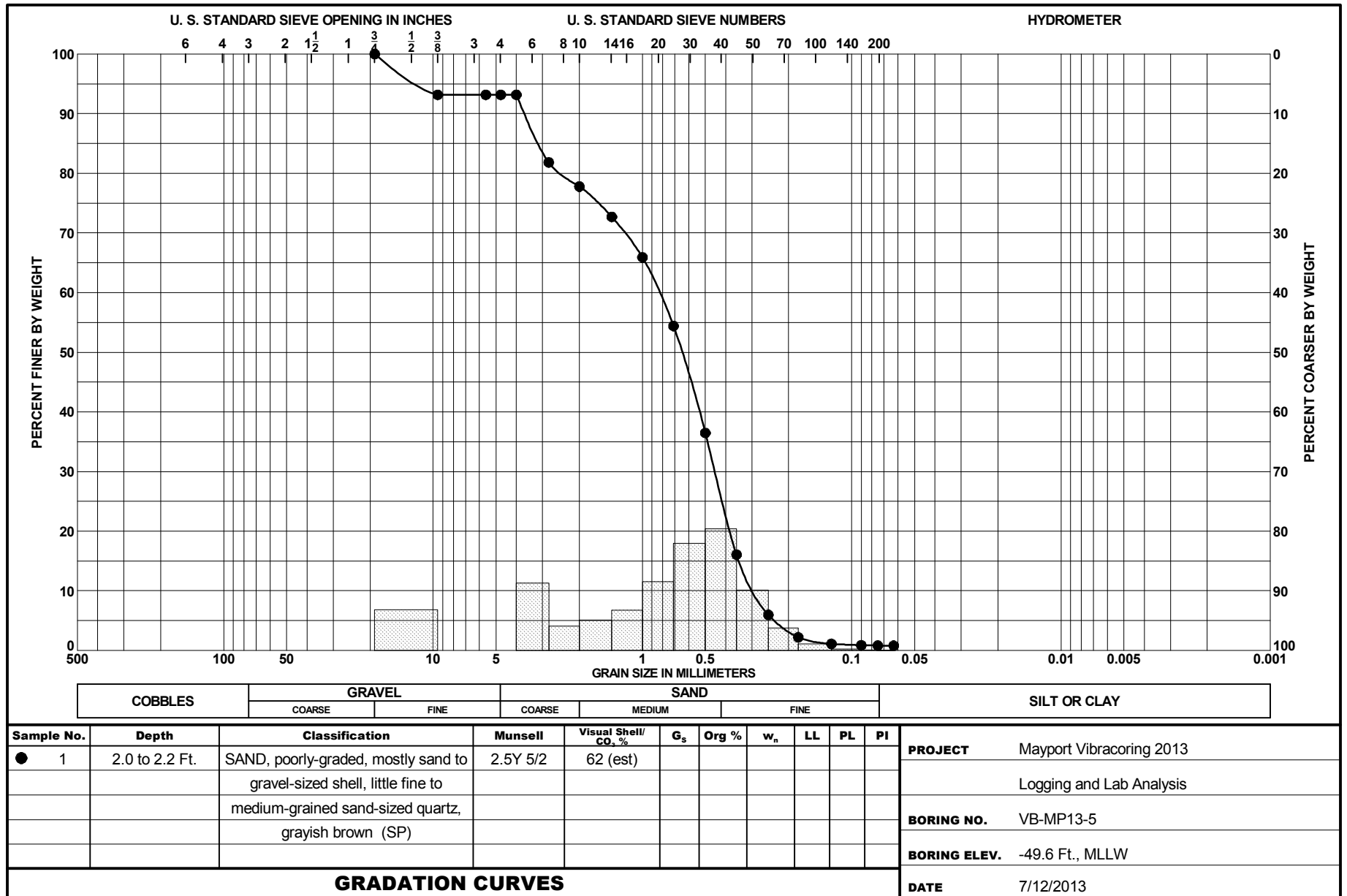


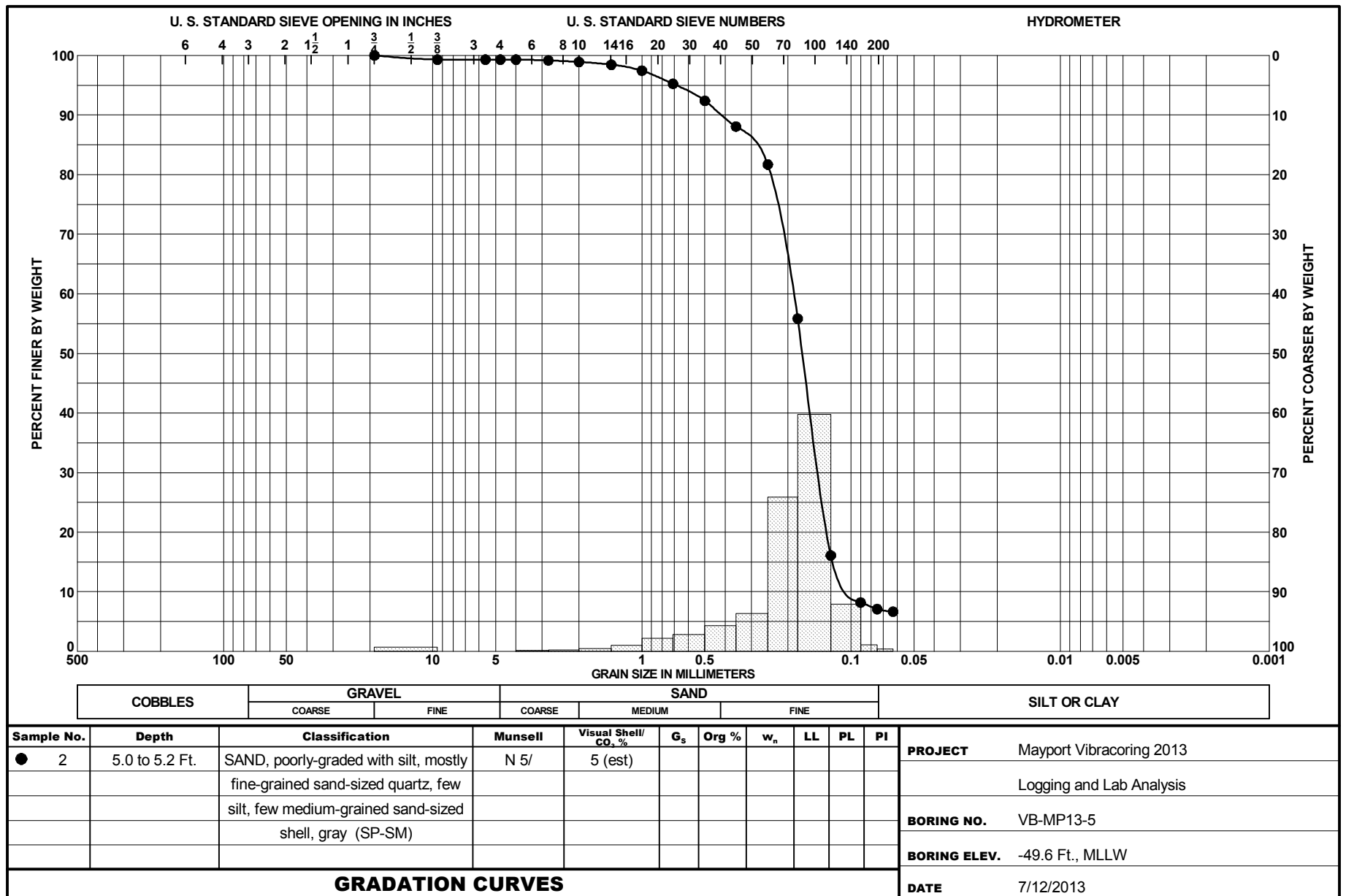


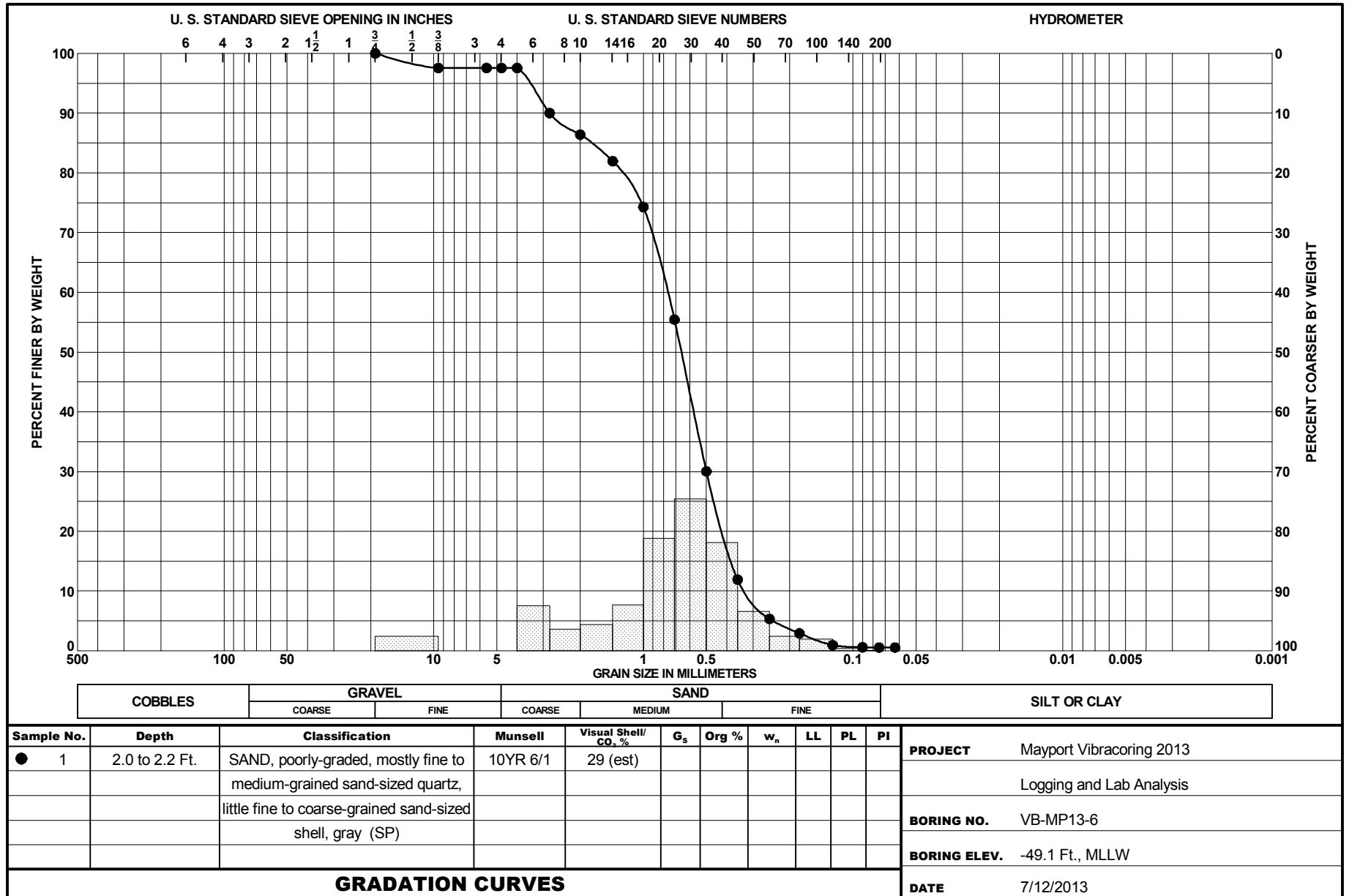


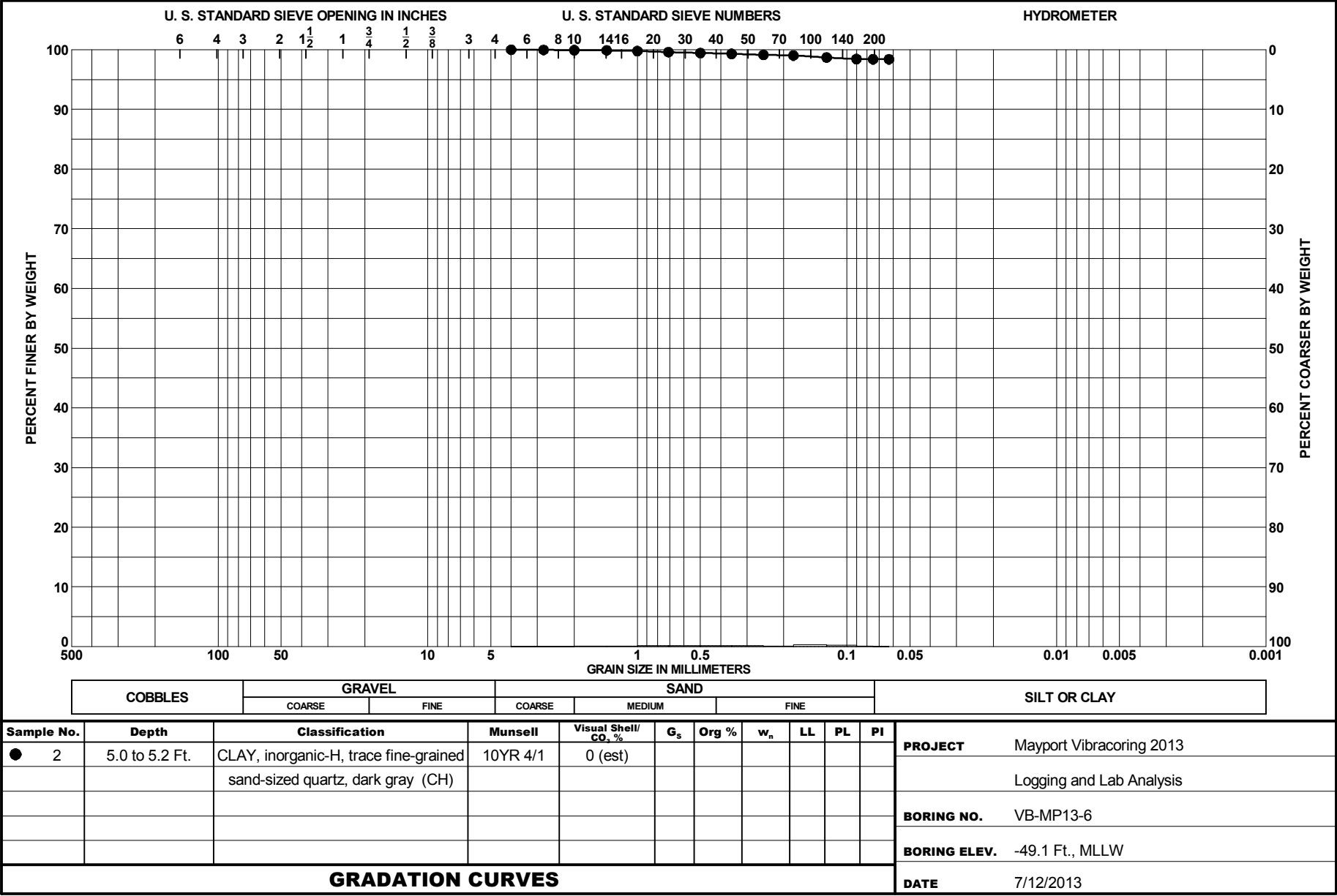


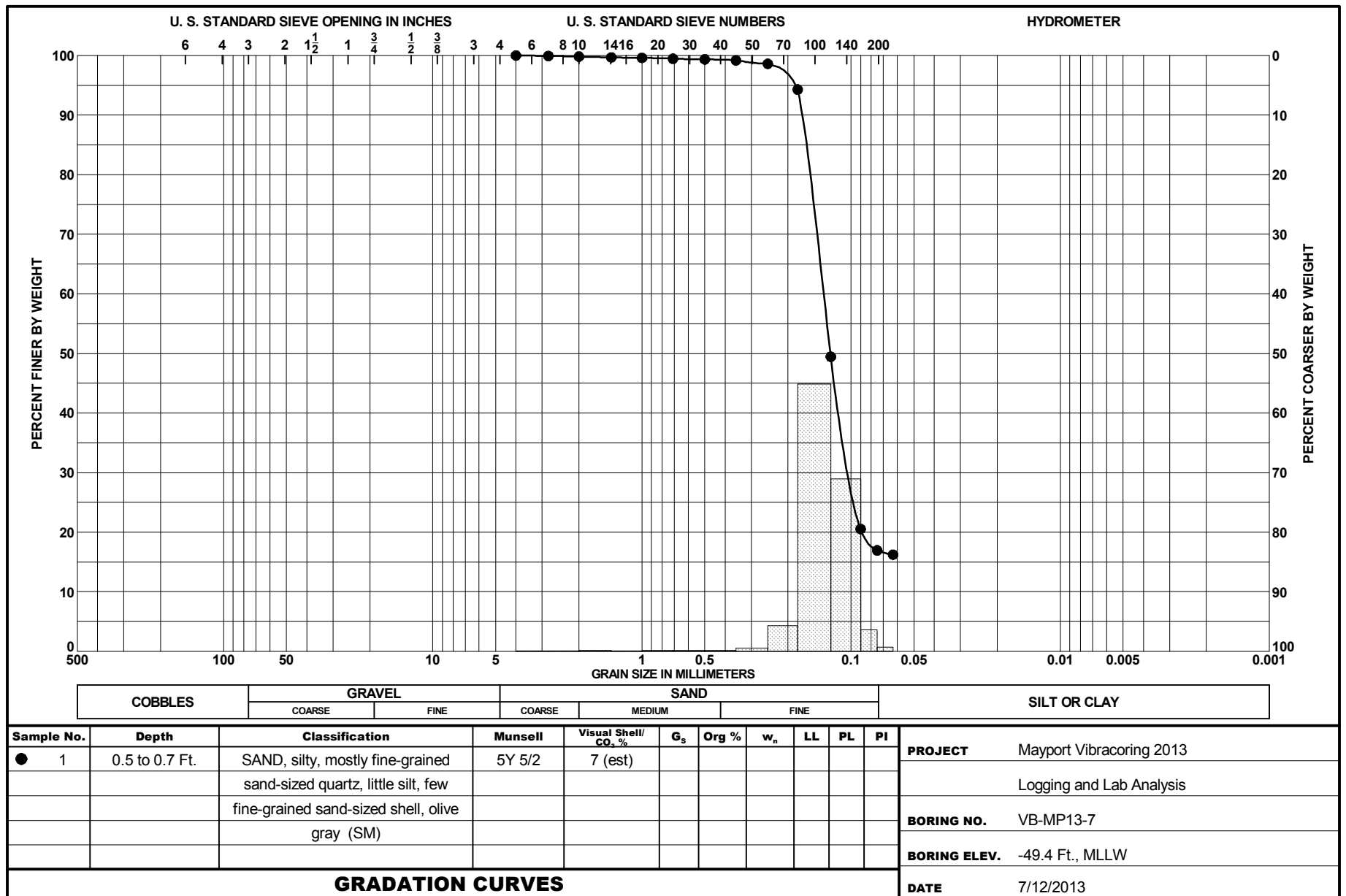


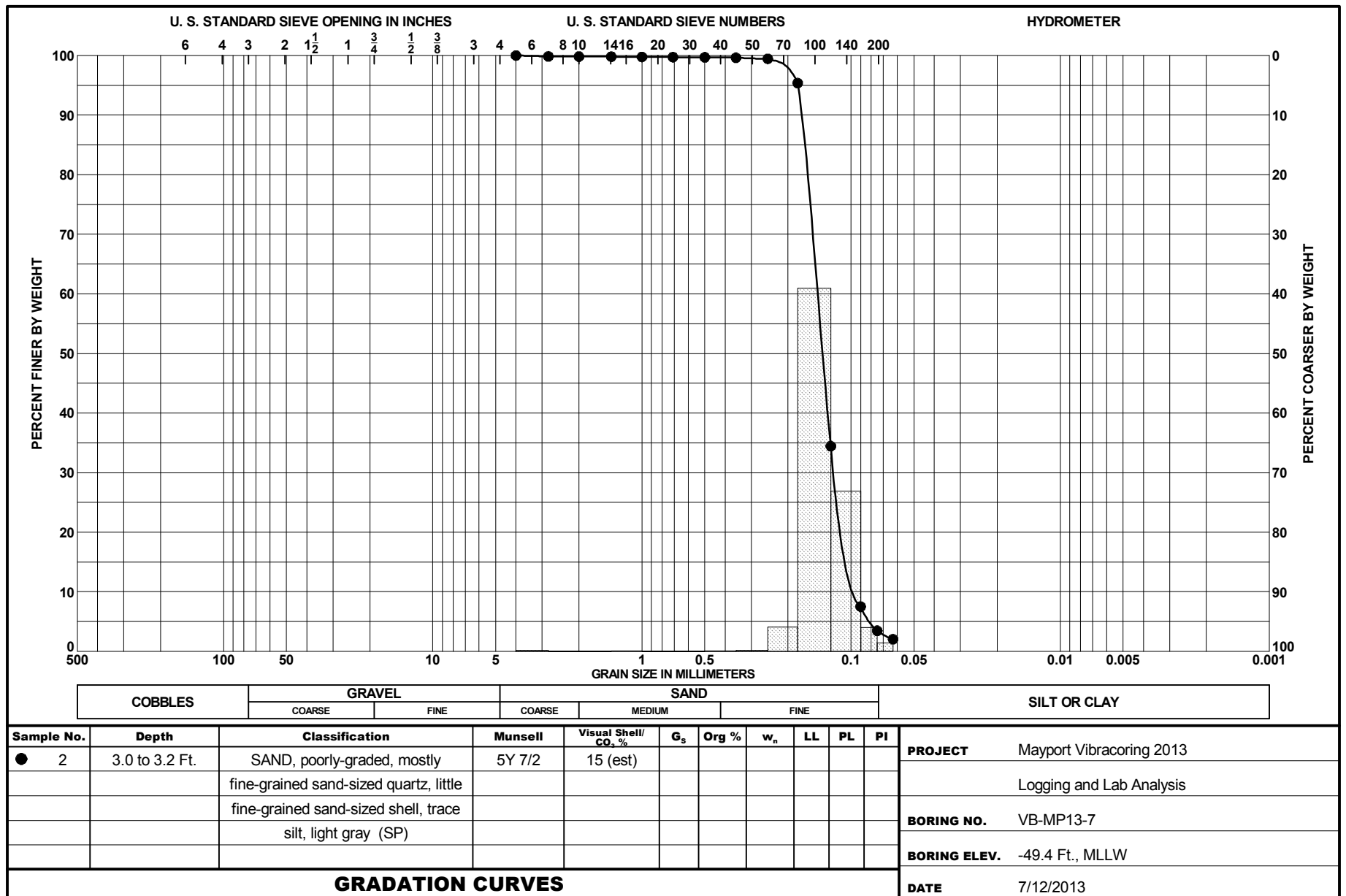




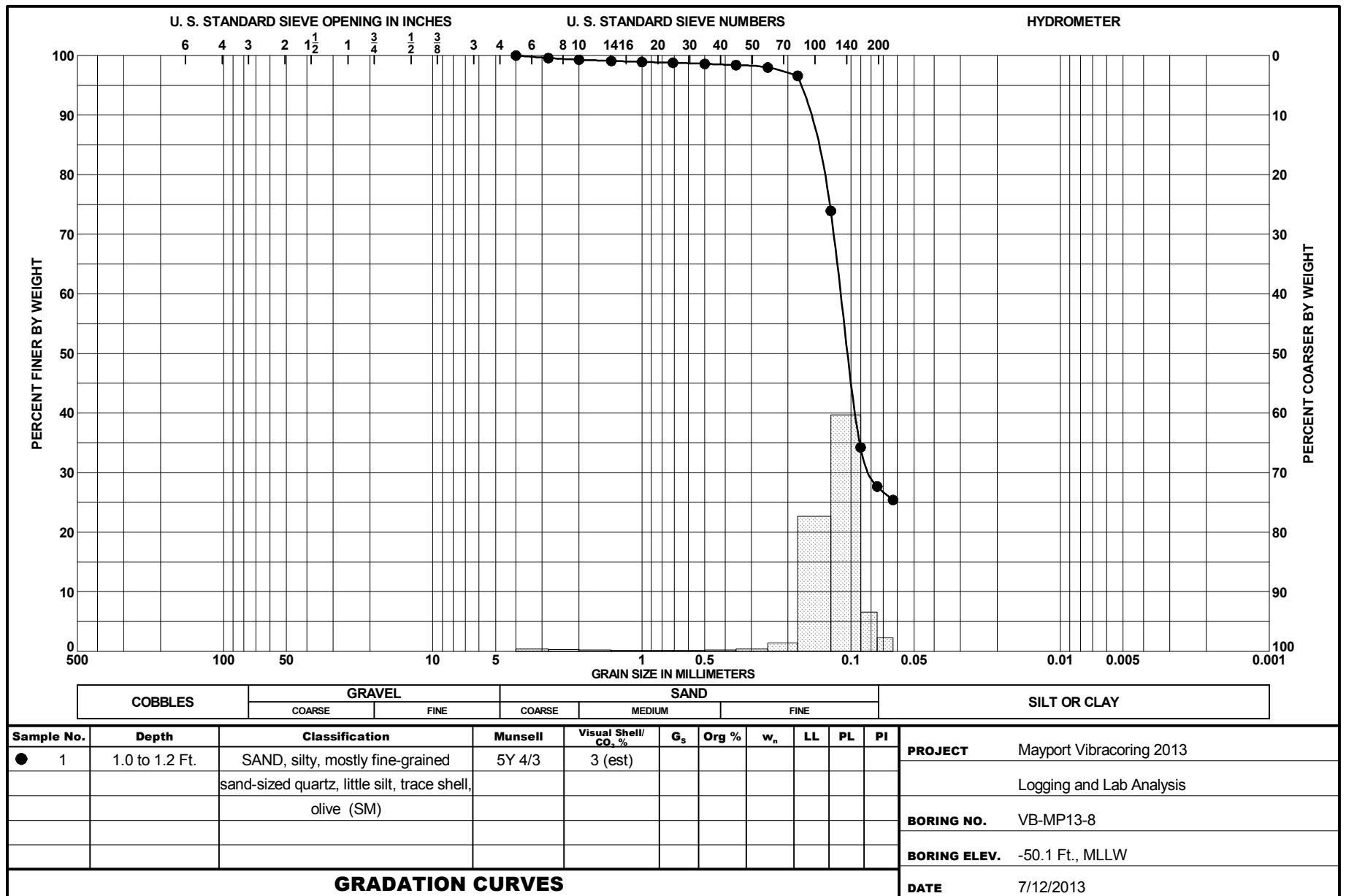


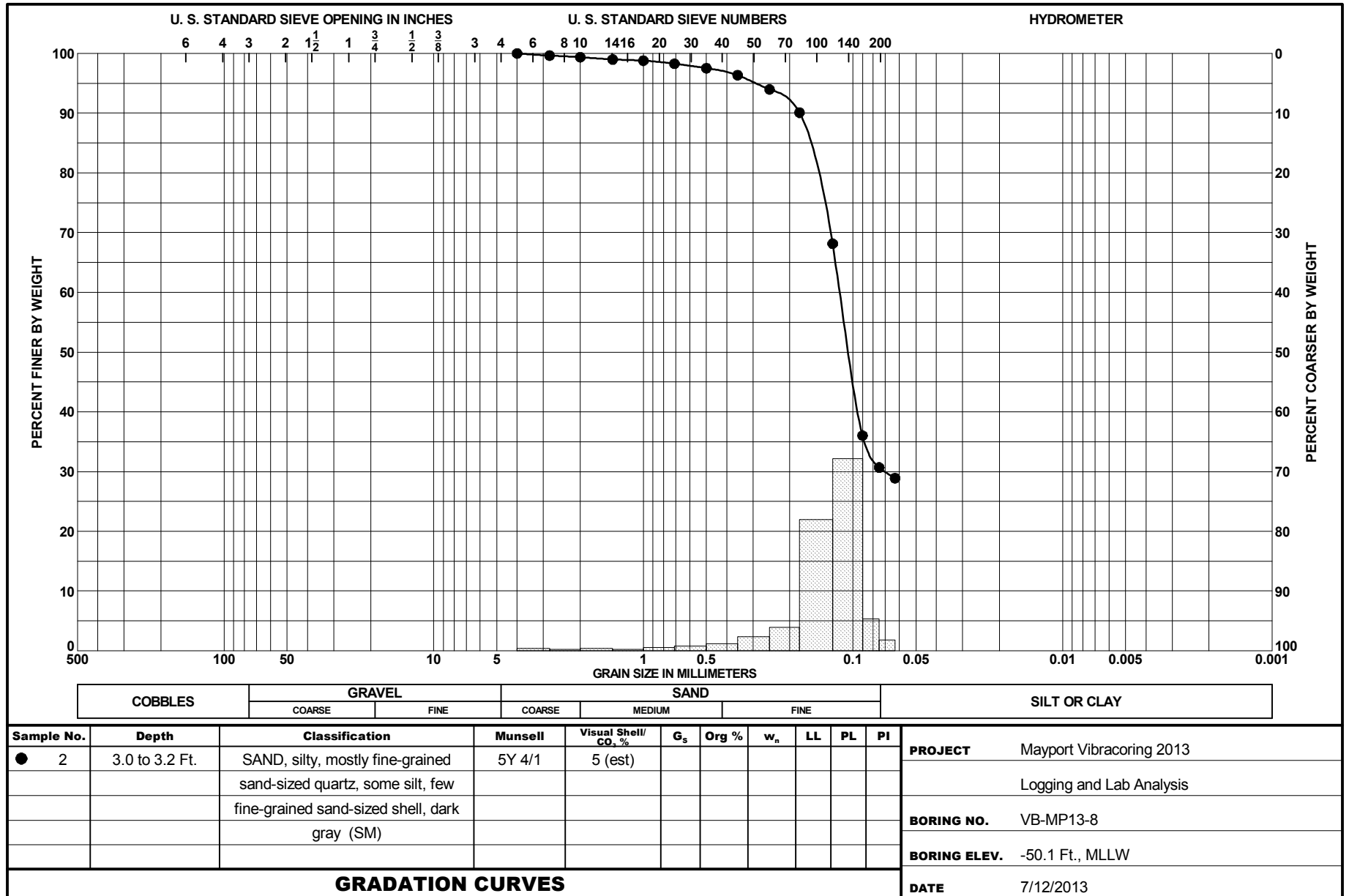






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